

CONVEX COUNTEREXAMPLES TO THE SCHIFFER AND POMPEIU CONJECTURES IN DIMENSIONS THREE, FOUR, SIX, EIGHT, TEN AND FOURTEEN

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ABSTRACT. In dimensions 3, 4, 6, 8, 10, 14 we construct bounded convex non-ball domains with real-analytic spherical boundaries. Each admits a nonconstant solution of $\Delta u + u = 0$ with $u = 1$ and $\nabla u = 0$ on the boundary. Green's identity makes their indicator Fourier transforms vanish on the unit sphere, disproving the Schiffer and Pompeiu conjectures even within the convex class in these dimensions. The three-dimensional domain is axisymmetric, invariant under $O(2)$ on the last two coordinates and reflection of the first, and is obtained by lifting a conformal map of its meridian section. In dimensions $n = 2m + 2$, $m = 1, 2, 3, 4, 6$, the domains lie in the rank-two compact Lie algebras $\mathfrak{u}(2)$, $\mathfrak{so}(4)$, $\mathfrak{su}(3)$, $\mathfrak{so}(5)$ and $\mathfrak{g}_2$. Harish-Chandra's radial part formula reduces these cases to planar Helmholtz problems, and Kostant's theorem reduces convexity to the Cartan section. Exact solutions near dyadic approximations follow from computer-assisted contractions in weighted polynomial coefficient spaces, with interval bounds for finite blocks and analytic bounds for every infinite tail. A second, strictly star-shaped but nonconvex, example in dimension three is retained as an independent result. To our knowledge, apart from the earlier version of this work, these are the first counterexamples in dimensions at least three and the first known convex counterexamples in any dimension.

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1. INTRODUCTION

1.1. The Pompeiu problem and Schiffer's conjecture. Let $n \geq 2$. A bounded domain $\Omega \subset \mathbb{R}^n$ has the *Pompeiu property* if the only continuous function $f: \mathbb{R}^n \rightarrow \mathbb{C}$ such that

$$\int_{\sigma(\Omega)} f(x) dx = 0 \quad \text{for every rigid motion } \sigma \text{ of } \mathbb{R}^n$$

is $f \equiv 0$. The question of which domains have this property goes back to Pompeiu [43–45]. Balls do not have it: if $J_{n/2}(kR) = 0$ for some $k > 0$, then the Fourier transform of $\mathbf{1}_{B_R}$ vanishes on the sphere $\{|\xi| = k\}$, and $f(x) = e^{ikx_1}$ integrates to zero over every ball of radius R . Brown, Schreiber and Taylor [8] characterised the failure of the Pompeiu property through the zero set of the Fourier–Laplace transform of the indicator function; in particular, a bounded domain fails the property whenever $\widehat{\mathbf{1}_\Omega}$ vanishes identically on a sphere $\{|\xi| = k\}$ with $k > 0$ (this direction is elementary, see Section 14). The *Pompeiu conjecture* asserts that a bounded domain with Lipschitz boundary homeomorphic to S^{n-1} fails the Pompeiu property only if it is a ball. Some topological assumption is necessary: for every $k > 0$ the function $R \mapsto R^{n/2} J_{n/2}(kR)$ takes equal values at suitable distinct radii $R_1 < R_2$, so the spherical shell $\{R_1 < |x| < R_2\}$ fails the Pompeiu property, and such shells are simply connected when $n \geq 3$.

Conjecture 1 (Schiffer). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary homeomorphic to S^{n-1} . If, for some $\mu > 0$, there is a nonconstant solution of*

$$(1.1) \quad \Delta u + \mu u = 0 \quad \text{in } \Omega, \quad u = \text{const} \quad \text{and} \quad \partial_\nu u = 0 \quad \text{on } \partial\Omega,$$

then Ω is a ball.

On a ball B_R , the radial Neumann eigenfunctions $u = r^{1-n/2} J_{n/2-1}(\sqrt{\mu} r)$ solve (1.1) precisely when $J_{n/2}(\sqrt{\mu} R) = 0$. Formulations of the conjecture in the literature differ in their regularity and topological hypotheses (smooth boundary, simply connected domain, connected boundary); the domains constructed below satisfy all of them. Williams [55] showed that, for bounded domains with Lipschitz boundary homeomorphic to a sphere, the failure of the Pompeiu property is equivalent to the solvability of (1.1) for some $\mu > 0$, building on [8]; the direction from (1.1) to the failure of the Pompeiu property is a consequence of Green's identity. In the planar case Schiffer's problem appears as Problem 80 in Yau's list [56]. We refer to Zalcman's survey [57] for the classical bibliography.

A large literature proves the conjectured rigidity under additional hypotheses: for domains carrying solutions for infinitely many eigenvalues [4, 5], for low eigenvalues [3, 15, 18], for perturbations of balls [2, 9, 37], for planar convex domains [7, 41], in particular at high frequency [16], and under further conditions [17, 35]. We compare our examples with these results in Section 17.

1.2. Previous constructions. In August 2026 the planar conjectures were disproved independently by two groups. Colbrook and Stepaniants [14] constructed a single explicit ten-fold symmetric noncircular domain $\Omega \subset \mathbb{R}^2$ with real-analytic boundary for which (1.1) holds with $\sqrt{\mu} \in (31.967007261, 31.967007293)$. Their proof pulls the problem back to the unit disc by a conformal map, which turns it into a cubic equation on a space of disk-polynomial coefficients, and establishes the existence of an exact solution near an explicit approximation by a computer-assisted contraction argument. Cao-Labora and de Dios Pont [10] constructed infinitely many N -fold symmetric planar counterexamples, for N sufficiently large, by a bifurcation argument for a relaxed problem in which N is a real parameter, with a branch length that is uniform in N . Colbrook, Sadeghi and Stepaniants [13] adapted the conformal method to disprove the planar Berenstein conjecture. Dolha [21] gave a further computer-assisted planar counterexample, with seven-fold dihedral symmetry, in a public software repository, and Singh [52] used the counterexample of [14] to obtain noncircular homogeneous dielectric inclusions that are exactly non-scattering for an explicit incident wave in the two-dimensional scalar Maxwell reduction. Earlier constructions of Schiffer-type domains in related settings include domains in the flat cylinder $\mathbb{R}^N \times \mathbb{R}/2\pi\mathbb{Z}$ and on S^2 [24], contractible domains on the half-sphere [11], doubly connected planar domains on whose two boundary components the eigenfunction takes different constants [23], and noncircular planar solutions of the related problem with nonzero constant Neumann data [54]. Schiffer-type domains in round spheres bounded by isoparametric hypersurfaces were found earlier [46, 50]; their boundaries are in general not topological spheres. Apart from the earlier version [28] of this paper (see below), to the best of our knowledge no counterexample to either conjecture was previously known for bounded domains in $\mathbb{R}^n$, $n \geq 3$, with boundary homeomorphic to a sphere.

1.3. Main results. Write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^{n-1}$ for $n \in \{3, 4\}$, and $x = (x', x'') \in \mathbb{R}^3 \times \mathbb{R}^3$ for points of $\mathbb{R}^6$.

Theorem 1.1 (Convex example in dimension three). *There exist a bounded convex domain $\Omega \subset \mathbb{R}^3$ and a nonconstant real-valued function u , real analytic on an open neighbourhood of $\overline{\Omega}$, such that*

$$(1.2) \quad \Delta u + u = 0 \quad \text{in } \Omega, \quad u = 1 \quad \text{and} \quad \nabla u = 0 \quad \text{on } \partial\Omega.$$

The domain is not a ball, is strictly star-shaped with respect to the origin, and has compact real-analytic boundary diffeomorphic to S^2 . It is invariant under $O(2)$ acting on x' and the reflection $x_1 \mapsto -x_1$.

For $n \in \{8, 10, 14\}$ let $(\mathfrak{k}, G)$ be $(\mathfrak{su}(3), SU(3))$, $(\mathfrak{so}(5), SO(5))$ and $(\mathfrak{g}_2, G_2)$, respectively. We identify $\mathfrak{k}$ with $\mathbb{R}^n$ by a linear isometry for an $\text{Ad}(G)$ -invariant inner product.

Theorem 1.2 (Convex counterexamples). *Let $n \in \{3, 4, 6, 8, 10, 14\}$. There exist a bounded convex domain $\Omega \subset \mathbb{R}^n$ and a nonconstant real-valued function u , real analytic on an open neighbourhood of $\overline{\Omega}$, such that (1.2) holds. Moreover:*

- (i) Ω is not a ball;
- (ii) Ω is strictly star-shaped with respect to the origin, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} ;
- (iii) for $n = 3$, Ω is invariant under $O(2)$ on x' and $x_1 \mapsto -x_1$; for $n = 4$, Ω is invariant under the action of $O(3)$ on x' and under $x_1 \mapsto -x_1$; for $n = 6$, Ω is invariant under the action of $O(3) \times O(3)$ on (x', x'') and under the exchange $(x', x'') \mapsto (x'', x')$; for $n \in \{8, 10, 14\}$, $\Omega \subset \mathfrak{k}$ is invariant under the adjoint action of G and under $x \mapsto -x$.

In particular, Conjecture 1 fails in each listed dimension, even within the class of convex domains.

The $n = 3$ case is Theorem 1.1; the other cases are proved in Part 1.

Rescaling Ω and u gives, for every $\mu > 0$, a counterexample with eigenvalue μ in (1.1).

Corollary 1.3 (Pompeiu). *Let $n \in \{3, 4, 6, 8, 10, 14\}$ and let $\Omega \subset \mathbb{R}^n$ be the domain of Theorem 1.1 or Theorem 1.2. The Fourier transform of $\mathbf{1}_\Omega$ vanishes on the unit sphere of $\mathbb{R}^n$. Consequently $\int_{\sigma(\Omega)} \cos x_1 dx = 0$ for every rigid motion σ of $\mathbb{R}^n$, the domain Ω fails the Pompeiu property, and the Pompeiu conjecture fails in $\mathbb{R}^n$, even within the class of convex domains.*

In particular, the Schiffer and Pompeiu conjectures fail within the class of bounded convex domains with real-analytic boundary in all six dimensions. To our knowledge, apart from the earlier version of this work, Theorem 1.1 gives the first such counterexample in dimension three, while Theorem 1.2 gives the first known convex counterexamples in Euclidean space. Neither [14] nor [10] asserts convexity; the examples of [14, 21] have certified negative boundary curvature at some points. The convex three-dimensional example is distinct from the nonconvex example of Proposition 13.27, which is retained below. The convex planar question remains open (Section 18). Theorem 1.1 and Corollary 1.3 also contradict several published statements about the three-dimensional problem (Section 17).

Remark 1.4 (The domains). In dimensions 4 to 14 the domains are obtained from a planar domain $D = \psi(\mathbb{D}) \subset \mathbb{R}^2 \cong \mathbb{C}$, the image of the unit disc under a map ψ with real Taylor coefficients; the three-dimensional domain is the image of the unit ball of $\mathbb{R}^3$ under the axisymmetric extension of such a map.

Dimension three. Here $\Omega = \Theta(B^3)$, where B^3 is the unit ball and

$$\Theta(x_1, x') = \left(\text{Re } \psi(x_1 + i|x'|), \frac{\text{Im } \psi(x_1 + i|x'|)}{|x'|} x' \right), \quad \psi(w) = \sum_{j \text{ odd}} c_j w^j.$$

The exact coefficient vector lies in the ball

$$\sum_{j \text{ odd}} 8b j \left(\frac{21}{20} \right)^j |c_j - c_j^\circ| \leq 10^{-7}, \quad b = 7626062690497999 \cdot 2^{-46},$$

about the dyadic centre in `anc/r3_convex/candidate.json`. Its linear coefficient is approximately 108.372869; the nearby radial Schiffer ball has radius $j_{3/2,34} \approx 108.37572$, the thirty-fourth positive root of $\tan x = x$. The non-ball witness is $c_{19} \neq 0$. An independent example near $j_{3/2,15} \approx 48.67414$ is strictly star-shaped but nonconvex (Proposition 13.27).

Dimension four. Here $\Omega = \{(x_1, x') : (x_1, |x'|) \in D\}$ and $\psi(w) = \sum_{j \text{ odd}} c_j w^j$. The coefficients satisfy

$$\sum_{j \text{ odd}} (c_1^\circ)^2 (j+1) \left(\frac{21}{20}\right)^j |c_j - c_j^\circ| \leq \frac{1}{50000},$$

where $c_1^\circ, c_3^\circ, \dots, c_{41}^\circ$ are explicit dyadic rationals listed in the verification script (Section 15) and $c_j^\circ = 0$ for $j > 41$; one has $c_1^\circ \approx 14.840043$ and $c_3^\circ \approx 0.0912$, $c_5^\circ \approx 0.0942$, $c_7^\circ \approx 0.0964$. For orientation we record some numerical observations about the centre ψ° , which are not used in the proofs. The boundary of D is a radial graph whose radius lies between 14.7213 and 15.1343. The ball of $\mathbb{R}^4$ that carries a radial solution of (1.2) with the same eigenvalue and closest radius has radius $j_{2,4} \approx 14.79595$, the fourth positive zero of J_2 ; thus the relative deviation of $\partial\Omega$ from that sphere lies between -0.5% and $+2.3\%$. Expanding the boundary radius of Ω in zonal spherical harmonics on S^3 , the dominant nonconstant term is the zonal harmonic of degree six, with relative amplitude about 2.2×10^{-2} at the poles.

Dimension six. Here $\Omega = \{(x', x'') : (|x'|, |x''|) \in D\}$ and $\psi(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$. The coefficients satisfy

$$\sum_{j \equiv 1 \pmod{4}} \frac{(c_1^\circ)^3}{2} (j+2) \left(\frac{11}{10}\right)^{j+1} |c_j - c_j^\circ| \leq \frac{1}{5000},$$

where $c_1^\circ, c_5^\circ, \dots, c_{57}^\circ$ are explicit dyadic rationals listed in the ancillary data file `center_r6.json` (Section 15.3) and $c_j^\circ = 0$ for $j > 57$; one has $c_1^\circ \approx 28.864751$, $c_5^\circ \approx -0.0852970$ and $c_9^\circ \approx -0.0837734$, while $|c_j^\circ| < 4 \cdot 10^{-4}$ for $j \geq 13$. Numerically, the boundary of D is a radial graph whose radius lies between 28.6958 and 28.9596, and the ball of $\mathbb{R}^6$ with the same volume as Ω has radius 28.908339. The ball of $\mathbb{R}^6$ that carries a radial solution of (1.2) with the same eigenvalue and closest radius has radius $j_{3,8} \approx 28.908351$, the eighth positive zero of J_3 ; the relative deviation of $\partial\Omega$ from that sphere lies between -0.74% and $+0.18\%$, so that the radius of $\partial\Omega$ varies by about 0.92% . Expanding the boundary radius in the $O(3) \times O(3)$ -invariant spherical harmonics on S^5 , which are the functions $U_k(|\omega'|^2 - |\omega''|^2)$ of degree $2k$ (Section 16.2), the dominant nonconstant term has degree eight, with relative amplitude about 7.2×10^{-3} .

Dimensions eight, ten and fourteen. Let $m = 3, 4, 6$, respectively. Here $\Omega = \text{Ad}(G)D$, where $D = \Omega \cap \mathfrak{t}$ is the section of Ω by a Cartan subalgebra $\mathfrak{t} \cong \mathbb{C}$ (Section 1.4), and $\psi(w) = \sum_{j \equiv 1 \pmod{2m}} c_j w^j$ lies in an explicit weighted ℓ^1 ball around explicit dyadic coefficients c_j° (Section 12). One has $c_1^\circ \approx 20.787029, 38.152926, 88.497305$ and $c_{2m+1}^\circ \approx -0.0774840, -0.0138325, 0.0463087$, respectively. Numerically, the boundary of D is a radial graph whose radius lies between 20.7086 and 20.8635, between 38.1390 and 38.1667, and between 88.4477 and 88.5402. The balls that carry a radial solution of (1.2) with the closest radii have radii $j_{4,5} \approx 20.826933$, $j_{5,10} \approx 38.159869$ and $j_{7,25} \approx 88.474363$, and the relative deviations of $\partial\Omega$ from these spheres lie between -0.57% and $+0.18\%$, between -0.055% and $+0.018\%$, and between -0.030% and $+0.074\%$. In each case the dominant nonconstant term of the radius of D is a multiple of $\cos(2m\theta)$, which corresponds to the $\text{Ad}(G)$ -invariant spherical harmonic of degree $2m$; its restriction to the unit circle of $\mathfrak{t}$ is a multiple of $1 + 2\cos(2m\theta)$.

1.4. Compact Lie algebras of rank two. The five convex domains are instances of a single construction. Let $\mathfrak{k}$ be a compact Lie algebra of rank two, with an Ad -invariant inner product, let $\mathfrak{t} \subset \mathfrak{k}$ be a Cartan subalgebra, identified with $\mathbb{C}$ by a linear isometry, and let W be the Weyl group. If $\mathfrak{k}$ is not abelian, then W is the dihedral group of order $2m$ with $m \in \{1, 2, 3, 4, 6\}$, there are m positive roots, $\dim \mathfrak{k} = 2m + 2$, and, up to isomorphism, $\mathfrak{k}$ is $\mathfrak{u}(2)$, $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $\mathfrak{su}(3)$, $\mathfrak{so}(5) \cong \mathfrak{sp}(2)$ or $\mathfrak{g}_2$. The product ϖ of the positive roots is a W -anti-invariant harmonic polynomial of degree m ; in suitable coordinates $\varpi(z)$ is a nonzero multiple of $\text{Im } z^m$. For an

Ad-invariant C^2 function u with restriction $f = u|_{\mathfrak{t}}$, a classical formula of Harish-Chandra [29] (see [30, Ch. II]) states that

$$(1.3) \quad (\Delta_{\mathfrak{t}} u)|_{\mathfrak{t}} = \varpi^{-1} \Delta_{\mathfrak{t}}(\varpi f)$$

at the regular points of $\mathfrak{t}$. Consequently, Ad-invariant solutions of (1.2) on $\Omega \subset \mathfrak{t}$ correspond to W -anti-invariant solutions $F = \varpi f$ of the planar problem

$$(1.4) \quad \Delta F + F = 0 \text{ in } D, \quad F = \varpi \text{ and } \nabla F = \nabla \varpi \text{ on } \partial D,$$

on the Cartan section $D = \Omega \cap \mathfrak{t}$. There is no potential term, so the conformal method of [14] applies to (1.4). Conversely, if D is W -invariant, contains 0 and has a positive real-analytic radial function, then $\Omega = \text{Ad}(G)D$ is a bounded domain whose boundary is a real-analytic hypersurface diffeomorphic to S^{2m+1} , and an analytic solution of (1.4) lifts to a solution of (1.2) on Ω (Section 12); some hypothesis on D is needed here, since for an annulus D the lifted boundary has two components. By Kostant's convexity theorem [39], in the form of [40, Theorems 2.2 and 3.6], Ω is convex if and only if D is. For $\mathfrak{t} = \mathfrak{u}(2) \cong \mathbb{R} \times \mathbb{R}^3$ the adjoint group acts as $SO(3)$ on the second factor and ϖ is a multiple of y ; for $\mathfrak{t} = \mathfrak{so}(4) \cong \mathbb{R}^3 \times \mathbb{R}^3$ it acts as $SO(3) \times SO(3)$ and ϖ is a multiple of $y_1 y_2$. These are the reductions of Sections 2 and 11, which we present there in elementary form. For the abelian algebra $\mathbb{R}^2$ one has $\varpi = 1$, and (1.4) is the planar Schiffer problem itself. Among splittings $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with symmetry group $O(d_1) \times O(d_2)$, the meridian problem has no potential term exactly for $n \in \{2, 4, 6\}$ (Remark 2.5), which are the cases $m \in \{0, 1, 2\}$ above; in dimension three no such reduction is available, and Theorem 1.1 is proved by a different formulation.

The ball $B_\rho \subset \mathfrak{t}$ carries the radial solution $u = r^{-m} J_m(r)$ of (1.2) exactly when $J_{m+1}(\rho) = 0$; in $\mathfrak{t}$ it corresponds to $F = J_m(r) \sin m\theta$. An Ad-invariant spherical harmonic of degree ℓ , which exists for $\ell \in m\mathbb{Z}_{\geq 0}$, corresponds to the planar mode $J_{m+\ell}(r) \sin(m+\ell)\theta$, and the linearisation of the Schiffer problem at B_ρ in this direction is degenerate exactly when $J_{m+\ell}(\rho) = 0$. By Siegel's theorem [51] this never happens for $\ell \geq 2$; the case $\ell = 1$, which occurs only for $m = 1$, corresponds to translations and is excluded by the symmetry $x_1 \mapsto -x_1$. So there is no bifurcation from balls in these symmetry classes; our solutions are found instead near near-resonances, where $J_{m+\ell}(\rho)$ is small (Section 16.2 and Table 1).

1.5. Strategy of the proof. Locating the solutions. All six domains lie close to balls at near-resonant eigenvalues. Let ρ be a zero of $J_{n/2}$, so that the ball of radius ρ in $\mathbb{R}^n$ carries a radial solution of (1.2), and let $\ell \geq 2$ be the degree of a spherical harmonic in the symmetry class. The linearisation at the ball in the direction of this harmonic is degenerate exactly when $J_{\ell+n/2-1}(\rho) = 0$, which never happens, but when $J_{\ell+n/2-1}(\rho)$ is small and $n \geq 3$, a formal Lyapunov–Schmidt expansion has, for the harmonics considered here, a nonzero quadratic term, and it predicts a non-ball solution at a distance from the ball comparable to $J_{\ell+n/2-1}(\rho)$ (Section 16.2). In the plane the corresponding quadratic term vanishes. Table 1 lists the near-resonances at which our domains lie. The numerical centres were obtained by continuation from these predictions; their provenance plays no role in the proofs.

Dimensions $n = 2m+2$: conformal fixed-disc formulation. Following [14] we write $D = \psi(\mathbb{D})$ and $V = F \circ \psi$, so that $\Delta V + |\psi'|^2 V = 0$ in $\mathbb{D}$, and $V - \varpi \circ \psi$ has vanishing value and normal derivative on $\partial\mathbb{D}$. We write $V = Kg + \varpi \circ \psi$, where K is an explicit inverse of the Laplacian with vanishing Cauchy data on its natural range. For $n = 4$, where $\varpi = y$, this gives the cubic equation

$$(1.5) \quad \mathcal{F}(g, \psi) = g + |\psi'|^2 (Kg + \text{Im } \psi) = 0,$$

n	symmetry	ρ	ℓ	$J_{\ell+n/2-1}(\rho)$
3	$O(2) \times \mathbb{Z}_2$	$j_{3/2,34} \approx 108.37572$	18	$J_{37/2}(\rho) \approx 1.25 \cdot 10^{-4}$
4	$\mathfrak{u}(2)$	$j_{2,4} \approx 14.79595$	6	$J_7(\rho) \approx -4.9 \cdot 10^{-3}$
6	$\mathfrak{so}(4)$	$j_{3,8} \approx 28.90835$	8	$J_{10}(\rho) \approx -3.0 \cdot 10^{-3}$
8	$\mathfrak{su}(3)$	$j_{4,5} \approx 20.82693$	6	$J_9(\rho) \approx -3.3 \cdot 10^{-3}$
10	$\mathfrak{so}(5)$	$j_{5,10} \approx 38.15987$	8	$J_{12}(\rho) \approx -4.4 \cdot 10^{-4}$
14	$\mathfrak{g}_2$	$j_{7,25} \approx 88.47436$	12	$J_{18}(\rho) \approx -9.0 \cdot 10^{-4}$

TABLE 1. Near-resonances: the Schiffer ball B_ρ of $\mathbb{R}^n$ at eigenvalue one and the degree ℓ of the dominant spherical harmonic in the symmetry class.

and in general the polynomial equation

$$(1.6) \quad g + |\psi'|^2 (Kg + \varpi \circ \psi) = 0$$

of degree $m + 2$ in the unknowns. Compared with the equation $g + |p|^2(1 + Kg) = 0$ of [14], the constant boundary datum is replaced by $\varpi \circ \psi$, so that the unknown map enters the data as well; the eigenvalue is absorbed into the scale of ψ . The symmetry class consists of maps $\psi(w) = \sum_{j \equiv 1 \pmod{2m}} c_j w^j$ with real coefficients, which is slightly more symmetric than the Weyl group requires, and g lies in the sine sector of angular frequencies $\equiv m \pmod{2m}$. The part of the linearisation acting on high shape modes has a harmonic part which is a difference operator of step $2m$ (upper bidiagonal for $m = 1$), and we invert it exactly. We prove that (1.6) has an exact zero in an explicit ball around a numerically computed centre, using a Newton–Kantorovich (radii polynomial) argument in weighted ℓ^1 spaces of disk-polynomial coefficients. The approximate inverse consists of a finite matrix, of size 525, 465, 200, 259 and 610 for $n = 4, 6, 8, 10, 14$, inverted and checked in ball arithmetic, together with an exact inverse on the tail. All infinitely many columns of the defect operator are covered: finitely many are evaluated without truncation in interval arithmetic, and the remaining ones are bounded by analytic estimates. From the zero we obtain a univalent map ψ , a strictly convex planar domain D (by a lower bound for $\operatorname{Re}(1 + w\psi''/\psi')$ on the closed disc, uniform over the existence ball), a solution of (1.4) that extends analytically across ∂D , and finally the domain Ω and the function u , by the rotations of Sections 2 and 11 for $n = 4, 6$ and by the adjoint lift of Section 12 for $n = 8, 10, 14$.

Dimension three. Here there is no reduction to the planar Laplacian (Remark 2.5), and we pull the three-dimensional problem back to the unit ball B^3 . Let ψ be a map as in Remark 1.4, with odd powers and real coefficients, and Θ its axisymmetric extension. The differential of Θ is a similarity with factor $|\psi'|$ in the meridian directions and multiplication by $q = \operatorname{Im} \psi(w) / \operatorname{Im} w$ in the azimuthal direction, where $w = x_1 + i|x'|$. Hence, for an axisymmetric function $U = u \circ \Theta$ on B^3 , problem (1.2) becomes

$$(1.7) \quad \operatorname{div}(q \nabla U) + q|\psi'|^2 U = 0 \text{ in } B^3, \quad U = 1 \text{ and } \nabla U = 0 \text{ on } \partial B^3.$$

We write $U = 1 + K_3 g$, where K_3 is an explicit inverse of the Laplacian of $\mathbb{R}^3$ with vanishing Cauchy data on ∂B^3 , acting on the axisymmetric ball polynomials $r^\ell P_\ell(x_1/r) P_s^{(0, \ell+1/2)}(2r^2-1)$ with ℓ even and $s \geq 1$. Since q is a real-analytic function of x_1 and $|x'|^2$ that depends linearly on the coefficients of ψ , (1.7) becomes a quartic polynomial equation in the unknown field coefficients and shape coefficients, with no division by the unknown q . The disk-polynomial algebra of [14] is replaced by weighted ℓ^1 norms adapted to the Legendre–Chebyshev connection coefficients; multiplication estimates follow from the positivity of the Gaunt coefficients and of

Jacobi connection coefficients. The approximate inverse consists of a finite block of dimension 7371, with 7280 field and 91 shape unknowns, and an explicit inverse of the principal boundary part on high shape modes, given by a convergent Neumann series. The Newton–Kantorovich argument again covers all columns, finitely many by interval evaluation and all others by analytic estimates. Univalence and positivity of q follow from a lower bound for $\operatorname{Re} \psi'$ on a disc of radius $101/100$, uniform over the existence ball, which also shows that Θ is a real-analytic diffeomorphism near $\overline{B^3}$; strict star-shapedness follows from a lower bound for $\operatorname{Re}(w\psi'/\psi)$ on the unit circle, and strict convexity from a positive lower bound for $\operatorname{Re}(1 + w\psi''/\psi')$ on the whole closed disc and throughout the projected shape ball. The field $u = U \circ \Theta^{-1}$ extends analytically across the axis and across $\partial\Omega$ by elliptic regularity.

Computations. The four-dimensional finite computations are performed by a single self-contained script using ball arithmetic [25, 34]; it runs in about three minutes on one core. The six-dimensional ones are split into five stage scripts, whose interval receipts are combined by a final verification script; the complete reproduction also takes about three minutes on one core. The computations for $n = 8, 10, 14$ are organised in the same way, with 128-bit ball arithmetic and bounded stage scripts, and take about a quarter of an hour of processor time in total. The three-dimensional verification is a single self-contained script that recomputes the approximate inverse and all interval bounds from embedded frozen data; it took about 1.8 hours with five single-threaded processes in the recorded replay. Section 15 describes what the scripts check and how to reproduce the computations.

Relation with an earlier version. Theorem 1.2 for $n \in \{4, 6\}$, numerical solutions in dimensions three and five, among them a numerical approximation of the additional AX3 domain of Proposition 13.27, and the observation that the planar reduction extends to $\mathfrak{su}(3)$, $\mathfrak{so}(5)$ and $\mathfrak{g}_2$ appeared in the preprint [28]. The present paper supersedes it.

Organisation. The paper has three parts. Part 1 treats the planar reductions. Sections 2–10 contain the proof of Theorem 1.2 for $n = 4$: the reduction to the plane (Section 2), the conformal formulation (Section 3), disk polynomials (Section 4), the operator K (Section 5), the cubic equation (Section 6), the approximate inverse (Section 7), the defect bounds with all tail estimates (Section 8), the certified bounds and the exact zero (Section 9), and the reconstruction of the domain (Section 10). Section 11 contains the proof for $n = 6$, which uses most lemmas of the four-dimensional proof verbatim, and Section 12 the Lie-theoretic reduction, the lifting and convexity lemmas, and the proof for $n = 8, 10, 14$. Part 2 (Section 13) proves Theorem 1.1 and records the independent nonconvex example. In Part 3, Section 14 proves Corollary 1.3, Section 15 describes the computer-assisted parts and the ancillary files, Section 16 contains remarks on the bifurcation mechanism, on further numerical solutions and on curved spaces, Section 17 compares our examples with rigidity theorems and with other claims, and Section 18 lists open problems.

Part 1. Planar reductions: convex counterexamples in dimensions four, six, eight, ten and fourteen

2. REDUCTION TO A PLANAR PROBLEM

Sections 2–10 contain the proof for $n = 4$; the proof for $n = 6$ is given in Section 11. Write points of the meridian plane as $(x_1, y) \in \mathbb{R}^2$, identified with $z = x_1 + iy \in \mathbb{C}$, and let $\Pi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$, $\Pi(x_1, x') = (x_1, |x'|)$. The map Π is continuous and its image is the closed half-plane $\{y \geq 0\}$.

Lemma 2.1. *Let f be a C^2 function on an open subset of $\{y > 0\}$ and $F(x_1, y) = yf(x_1, y)$. Then*

$$F_{x_1x_1} + F_{yy} = y \left(f_{x_1x_1} + f_{yy} + \frac{2}{y} f_y \right).$$

The expression in parentheses, evaluated at $(x_1, |x'|)$, is the Laplacian of the function $x \mapsto f(x_1, |x'|)$ at every point with $x' \neq 0$.

Proof. $(yf)_{yy} = yf_{yy} + 2f_y$. The second statement is the formula $\partial_y^2 + \frac{2}{y}\partial_y$ for the Laplacian of $\mathbb{R}^3$ acting on radial functions. $\square$

Lemma 2.2. *Let $N \subset \mathbb{R}^2$ be open and symmetric under $(x_1, y) \mapsto (x_1, -y)$, and let F be real analytic on N and odd in y . Then there is a real-analytic function G , defined on an open set containing $\{(x_1, y^2) : (x_1, y) \in N\}$, such that $F(x_1, y) = yG(x_1, y^2)$ on N . Consequently $x \mapsto G(x_1, |x'|^2)$ is real analytic on the open set $\Pi^{-1}(N)$, and it equals $F(x_1, |x'|)/|x'|$ where $x' \neq 0$.*

Proof. For $t > 0$ with $(x_1, \sqrt{t}) \in N$ put $G(x_1, t) = F(x_1, \sqrt{t})/\sqrt{t}$; this is real analytic. Let $(a, 0) \in N$. For some $\delta > 0$, $F(x_1, y) = \sum_{i,k \geq 0} c_{ik}(x_1 - a)^i y^k$ converges absolutely for $|x_1 - a| < \delta$, $|y| < \delta$, and $c_{ik} = 0$ for even k by oddness. Then $G(x_1, t) = \sum_{i,j} c_{i,2j+1}(x_1 - a)^i t^j$ converges for $|x_1 - a| < \delta$, $|t| < \delta^2$ and agrees with the previous definition where $0 < t < \delta^2$. Two such local definitions agree on their overlap, which is a connected product of intervals containing points with $t > 0$. This defines G . The last assertion follows because $x \mapsto (x_1, |x'|^2)$ is polynomial. $\square$

Proposition 2.3 (From the plane to $\mathbb{R}^4$). *Let $D \subset \mathbb{R}^2$ be a bounded domain, symmetric under $y \mapsto -y$, whose boundary is a Jordan curve meeting the axis $\{y = 0\}$ in exactly two points. Let F be real analytic on an open set $N \supset \bar{D}$ symmetric under $y \mapsto -y$, odd in y , and such that*

$$(2.1) \quad \Delta F + F = 0 \text{ in } D, \quad F = y \text{ and } \nabla F = e_y \text{ on } \partial D.$$

Put $\Omega = \Pi^{-1}(D)$ and let u be the function $x \mapsto G(x_1, |x'|^2)$ of Lemma 2.2. Then Ω is a bounded open set, u is real analytic on the neighbourhood $\Pi^{-1}(N)$ of $\bar{\Omega}$, u is nonconstant, and (1.2) holds.

Proof. Ω is open and bounded because Π is continuous and proper. By continuity $\bar{\Omega} \subset \Pi^{-1}(\bar{D}) \subset \Pi^{-1}(N)$, and $\partial\Omega \subset \Pi^{-1}(\partial D)$, since a point $x \in \bar{\Omega}$ with $\Pi(x) \in D$ lies in the open set Ω .

Off the axis $\{x' = 0\}$, $u(x) = f(x_1, |x'|)$ with $f = F/y$, so Lemma 2.1 gives $|x'|(\Delta u + u)(x) = (\Delta F + F)(\Pi(x)) = 0$ in $\Omega \setminus \{x' = 0\}$. By continuity of $\Delta u + u$ the equation holds in Ω .

Let $x \in \partial\Omega$ with $y = |x'| > 0$. Then $\Pi(x) \in \partial D$, so $f = F/y = 1$ there. Since $\nabla F = e_y$ on ∂D , we have $f_{x_1} = F_{x_1}/y = 0$ and $f_y = (F_y - f)/y = 0$, hence $\nabla u(x) = (f_{x_1}, f_y x'/|x'|) = 0$. The set $\partial\Omega \cap \{x' \neq 0\} = \Pi^{-1}(\partial D \cap \{y > 0\})$ is dense in $\partial\Omega$, since ∂D meets the axis in only two points and each of them is a limit of points of $\partial D \cap \{y > 0\}$. By continuity $u = 1$ and $\nabla u = 0$ on all of $\partial\Omega$. Finally, a constant solution of $\Delta u + u = 0$ vanishes identically, which contradicts $u = 1$ on $\partial\Omega$. $\square$

The reduction is reversible in the analytic class, which explains why nothing is lost by working in the plane.

Proposition 2.4 (From $\mathbb{R}^4$ to the plane). *Let $\Omega \subset \mathbb{R}^4$ be a bounded domain invariant under $O(3)$ acting on x' , and let u be real analytic on a neighbourhood of $\bar{\Omega}$ and solve (1.2). Put $D = \{(x_1, y) : (x_1, |y|, 0, 0) \in \Omega\}$ and $\tilde{u}(x) = \int_{O(3)} u(x_1, Qx') dQ$ (Haar probability measure).*

Then $\tilde{u}$ also solves (1.2), it has the form $\tilde{u}(x) = f(x_1, |x'|)$ with f real analytic and even in y , and $F = yf$ solves (2.1).

Proof. The Laplacian commutes with rotations, and the boundary conditions are rotation invariant; hence $\tilde{u}$ solves (1.2). The function $f(x_1, y) = \tilde{u}(x_1, y, 0, 0)$ is real analytic and even in y (use the rotation $x' \mapsto -x'$), and $\tilde{u}(x) = f(x_1, |x'|)$ by invariance. Lemma 2.1 gives $\Delta F + F = 0$ in $D \cap \{y \neq 0\}$, and by analyticity in D . On $\partial D \cap \{y > 0\}$ we have $f = 1$ and $\nabla f = 0$, hence $F = y$ and $\nabla F = fe_y + y\nabla f = e_y$; continuity extends this to the points of ∂D on the axis. $\square$

Remark 2.5 (Splittings without potential). For a function on $\mathbb{R}^d$ that depends only on $y = |\xi|$, $\xi \in \mathbb{R}^d$, the Laplacian is $\partial_y^2 + (d-1)y^{-1}\partial_y$, and for $F = y^{(d-1)/2}f$ one has

$$\partial_y^2 F = y^{(d-1)/2} \left(f_{yy} + \frac{d-1}{y} f_y \right) + \frac{(d-1)(d-3)}{4} \frac{F}{y^2}.$$

Let $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ and let $u(x) = f(y_1, y_2)$ be invariant under $O(d_1) \times O(d_2)$, where y_i is the modulus of the i -th component of x (for $d_i = 1$ one may use the coordinate itself). Then $F = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2} f$ satisfies

$$F_{y_1 y_1} + F_{y_2 y_2} = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2} \Delta u + \left(\frac{(d_1-1)(d_1-3)}{4y_1^2} + \frac{(d_2-1)(d_2-3)}{4y_2^2} \right) F.$$

The inverse-square potential vanishes exactly when $d_1, d_2 \in \{1, 3\}$, that is, for $n = d_1 + d_2 \in \{2, 4, 6\}$. In these cases the weight $\varphi = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2}$ is one of the harmonic polynomials 1, y , $y_1 y_2$, and the boundary conditions $u = 1$, $\nabla u = 0$ become $F = \varphi$, $\nabla F = \nabla \varphi$. For $n = 4$ ($d_1 = 1, d_2 = 3$) this is Lemma 2.1, and for $n = 6$ ($d_1 = d_2 = 3$) it is Lemma 11.1 below. The conformal method uses in an essential way that the planar operator is the Laplacian, whose conformal pull-back is a multiple of the Laplacian. For $n = 3$ and $n = 5$ every splitting into two blocks contains a block of dimension two or four, and the meridian problem contains a first-order drift term, or equivalently an inverse-square potential; a different fixed-domain formulation is needed, and for $n = 3$ it is given in Section 13. Symmetry classes of adjoint type, which give in addition the dimensions 8, 10 and 14, are treated in Section 12.

Let $x_1 + iy = re^{i\phi}$ be planar polar coordinates, and let U_ℓ denote the Chebyshev polynomial of the second kind, so that $U_\ell(\cos \phi)$ is the zonal spherical harmonic of degree ℓ on S^3 . Under $u = F/y$, the planar Helmholtz solution $F = J_m(r) \sin m\phi$, with m odd, corresponds to the four-dimensional Helmholtz solution

$$u = r^{-1} J_m(r) U_{m-1}(\cos \phi).$$

For instance, $F = J_1(r) \sin \phi$ gives the radial solution $r^{-1} J_1(r)$, whose derivative is $-r^{-1} J_2(r)$; this is why the Schiffer balls of $\mathbb{R}^4$ at eigenvalue one have radii $j_{2,k}$, the positive zeros of J_2 .

3. THE CONFORMAL FIXED-DISC FORMULATION

Let $\mathbb{D} = \{w \in \mathbb{C} : |w| < 1\}$, $w = re^{i\theta}$. The following lemma pulls (2.1) back to the disc; it is the analogue of the reformulation in [14].

Lemma 3.1. *Let ψ be holomorphic and injective on a neighbourhood of $\overline{\mathbb{D}}$, with $\psi' \neq 0$ there, and let $D = \psi(\mathbb{D})$. Let $V \in C^1(\overline{\mathbb{D}})$ be real valued and satisfy*

$$(3.1) \quad \Delta V + |\psi'|^2 V = 0 \quad \text{in } \mathcal{D}'(\mathbb{D}), \quad V - \operatorname{Im} \psi = 0 \quad \text{and} \quad \nabla(V - \operatorname{Im} \psi) = 0 \quad \text{on } \partial \mathbb{D}.$$

Then $F = V \circ \psi^{-1} \in C^1(\overline{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = y$, $\nabla F = e_y$ on ∂D . If moreover ψ is odd with real Taylor coefficients, $V(\bar{w}) = -V(w)$

and $V(-\bar{w}) = V(w)$, then D is symmetric under both coordinate reflections, F is odd in y and even in x_1 .

Proof. For $\varphi \in C_c^\infty(D)$, the function $\varphi \circ \psi$ belongs to $C_c^\infty(\mathbb{D})$. Conformal invariance of the Dirichlet integral and the Jacobian $|\psi'|^2$ give

$$\int_D \nabla F \cdot \nabla \varphi \, dA = \int_{\mathbb{D}} \nabla V \cdot \nabla (\varphi \circ \psi) \, dA = \int_{\mathbb{D}} |\psi'|^2 V(\varphi \circ \psi) \, dA = \int_D F \varphi \, dA,$$

where the middle equality is the distributional equation for $V \in C^1$. The function $V - \text{Im } \psi = (F - y) \circ \psi$ has zero value and gradient on $\partial\mathbb{D}$; since the Jacobian matrix of ψ is invertible on $\partial\mathbb{D}$, the chain rule gives $F - y = 0$ and $\nabla(F - y) = 0$ on ∂D . If ψ is odd with real coefficients, then $\psi(\bar{w}) = \overline{\psi(w)}$ and $\psi(-\bar{w}) = -\overline{\psi(w)}$, which gives the symmetries of D and, with the stated parities of V , those of F . $\square$

We shall look for V of the form $V = Kg + \text{Im } \psi$, where g is an unknown function on $\mathbb{D}$ and K is an inverse of the Laplacian with zero Cauchy data on $\partial\mathbb{D}$ (Section 5). Since $\text{Im } \psi$ is harmonic, (3.1) becomes $g + |\psi'|^2(Kg + \text{Im } \psi) = 0$, which is (1.5). We normalise the eigenvalue to one; the size of D is then an unknown, encoded in the coefficient c_1 of ψ . The symmetry class removes the degeneracies of the problem: the scaling is fixed by the eigenvalue, translations along the axis are excluded by the symmetry $x_1 \mapsto -x_1$, and the Möbius automorphisms of the disc are excluded by requiring $\psi(0) = 0$ with real Taylor coefficients.

4. DISK POLYNOMIALS AND COEFFICIENT SPACES

4.1. Disk polynomials. Write $w = re^{i\theta}$ and $t = r^2$. For $m \in \mathbb{Z}$ and an integer $s \geq 0$ let

$$(4.1) \quad \Phi_{m,s}(w) = r^{|m|} P_s^{(0,|m|)}(2r^2 - 1) e^{im\theta},$$

where $P_s^{(\alpha,\beta)}$ is the Jacobi polynomial in the standard normalisation, so that $P_s^{(0,n)}(1) = 1$ and $\Phi_{m,s} = e^{im\theta}$ on $\partial\mathbb{D}$. Explicitly,

$$(4.2) \quad P_s^{(0,n)}(2t - 1) = \sum_{k=0}^s a_{s,k}^{(n)} t^k, \quad a_{s,k}^{(n)} = (-1)^{s-k} \frac{(n+s+k)!}{k!(s-k)!(n+k)!},$$

and, equivalently, $P_s^{(0,n)}(2t - 1) = \frac{1}{s!} t^{-n} \frac{d^s}{dt^s} \{t^{n+s}(t-1)^s\}$. We set $a_{s,k}^{(n)} = 0$ for $k \notin \{0, \dots, s\}$ and $\Phi_{m,-1} = 0$. Each $\Phi_{m,s}$ is a polynomial in $w, \bar{w}$ of degree $|m| + 2s$; these are the disk (Zernike) polynomials [38]. Complex conjugation gives $\overline{\Phi_{m,s}} = \Phi_{-m,s}$.

Lemma 4.1. *The disk polynomials have the following properties.*

- (a) For $n, s \geq 0$, the polynomial $P_s^{(0,n)}(2t - 1)$ is orthogonal in $L^2([0, 1], t^n dt)$ to all polynomials of degree less than s .
- (b) The functions $\Phi_{m,s}$ are pairwise orthogonal in $L^2(\mathbb{D})$, and $\int_{\mathbb{D}} |\Phi_{m,s}|^2 dA = \pi/(|m| + 2s + 1)$.
- (c) $|\Phi_{m,s}(w)| \leq 1$ for $|w| \leq 1$.

Proof. (a) Integrate the Rodrigues formula by parts s times against $t^n \cdot t^i$, $i < s$; all boundary terms vanish because the derivatives of order less than s of $t^{n+s}(t-1)^s$ vanish at $t = 0$ and $t = 1$.

(b) Angular integration separates different m . For equal m with $n = |m|$, $\int_{\mathbb{D}} \Phi_{m,s} \overline{\Phi_{m,s'}} dA = \pi \int_0^1 t^n P_s^{(0,n)} P_{s'}^{(0,n)} dt$, which vanishes for $s \neq s'$ by (a). For $s = s'$, by (a) only the leading

coefficient $a_{s,s}^{(n)} = \frac{(n+2s)!}{s!(n+s)!}$ of one factor contributes, and s integrations by parts in the Rodrigues formula give $\int_0^1 t^{n+s} P_s^{(0,n)} dt = \int_0^1 t^{n+s} (1-t)^s dt = \frac{(n+s)! s!}{(n+2s+1)!}$. The product is $1/(n+2s+1)$.

(c) Let $|w| \leq 1$, $h = \sqrt{1-|w|^2}$, and consider the unitary matrix $U = \begin{pmatrix} w & -h \\ h & \bar{w} \end{pmatrix}$, acting on the space of homogeneous polynomials of degree $N = n+2s$ in two variables e_1, e_2 , with the $U(2)$ -invariant inner product in which the monomials are orthogonal. The induced operator is unitary, so its diagonal matrix coefficients with respect to the normalised monomials have modulus at most one. The coefficient of $e_1^{n+s} e_2^s$ in $(we_1 + he_2)^{n+s} (-he_1 + \bar{w}e_2)^s$ is

$$\sum_{k=0}^s \binom{n+s}{k} \binom{s}{k} w^{n+s-k} \bar{w}^{s-k} (-h^2)^k = e^{in\theta} r^n \sum_{k=0}^s \binom{n+s}{k} \binom{s}{k} t^{s-k} (t-1)^k.$$

By Leibniz's rule applied to the Rodrigues formula, the last sum equals $P_s^{(0,n)}(2t-1)$. This proves (c) for $m = n \geq 0$; conjugation gives $m < 0$. $\square$

Fix $\rho > 1$ (we shall take $\rho = 21/20$). Let $\mathcal{B}_\rho$ be the space of complex sequences $f = (f_{m,s})_{m \in \mathbb{Z}, s \geq 0}$ with

$$\|f\|_\rho = \sum_{m,s} \rho^{|m|} |f_{m,s}| < \infty.$$

By Lemma 4.1(c), $\sum f_{m,s} \Phi_{m,s}$ converges absolutely and uniformly on $\overline{\mathbb{D}}$, with $\sup_{\overline{\mathbb{D}}} |f| \leq \|f\|_\rho$; by Lemma 4.1(b) the coefficients are determined by the function. We identify f with this continuous function. $\mathcal{B}_\rho$ is a Banach space, and every bounded linear operator on it (or between such weighted ℓ^1 spaces) is determined by its values on the basis vectors.

Lemma 4.2 (Multiplication by w and $\bar{w}$). *Let $n = |m|$. Then*

$$(4.3) \quad w \Phi_{m,s} = \begin{cases} \frac{n+s+1}{n+2s+1} \Phi_{m+1,s} + \frac{s}{n+2s+1} \Phi_{m+1,s-1}, & m \geq 0, \\ \frac{n+s}{n+2s+1} \Phi_{m+1,s} + \frac{s+1}{n+2s+1} \Phi_{m+1,s+1}, & m < 0, \end{cases}$$

and $\bar{w} \Phi_{m,s} = \overline{w \Phi_{-m,s}}$. In both cases the coefficients are nonnegative and sum to one. Consequently $\|wf\|_\rho \leq \rho \|f\|_\rho$ and $\|\bar{w}f\|_\rho \leq \rho \|f\|_\rho$, and for every holomorphic $p(w) = \sum_{a \geq 0} p_a w^a$ with $N_\rho(p) := \sum_a \rho^a |p_a| < \infty$,

$$(4.4) \quad \|pf\|_\rho \leq N_\rho(p) \|f\|_\rho, \quad \|\bar{p}f\|_\rho \leq N_\rho(p) \|f\|_\rho.$$

Moreover, if $m \geq 0$ then $w^N \Phi_{m,s}$ is a convex combination of $\Phi_{m+N,s'}$ with $s' \leq s$.

Proof. For $m = n \geq 0$ both sides of the first formula are $r^{n+1} e^{i(n+1)\theta}$ times a polynomial in t , and the claim is the identity $a_{s,k}^{(n)} = \frac{n+s+1}{n+2s+1} a_{s,k}^{(n+1)} + \frac{s}{n+2s+1} a_{s-1,k}^{(n+1)}$. Dividing by $a_{s,k}^{(n+1)}$ (which is nonzero for $0 \leq k \leq s$) and using (4.2), it reduces to $(n+s+1)(n+s+k+1) - s(s-k) = (n+k+1)(n+2s+1)$. For $m = -n < 0$ the claim is $a_{s,k-1}^{(n)} = \frac{n+s}{n+2s+1} a_{s,k}^{(n-1)} + \frac{s+1}{n+2s+1} a_{s+1,k}^{(n-1)}$; dividing by $a_{s,k}^{(n-1)}$ for $0 \leq k \leq s$ it reduces to $(n+s)(s+1-k) - (s+1)(n+s+k) = -k(n+2s+1)$, and the case $k = s+1$ is checked directly. The formula for $\bar{w}$ follows from $\overline{\Phi_{m,s}} = \Phi_{-m,s}$. Since $|m \pm 1| \leq |m| + 1$, the weight grows by at most a factor ρ , which gives the bounds for w and $\bar{w}$; (4.4) follows by iteration and absolute summation. The last assertion follows from the case $m \geq 0$ of (4.3). $\square$

Lemma 4.3 (Radial index under monomials). *Let $\alpha, \beta \geq 0$, $m \in \mathbb{Z}$, $s \geq 0$. Then $w^\alpha \bar{w}^\beta \Phi_{m,s}$ is a finite linear combination of $\Phi_{m+\alpha-\beta,s'}$ with $s' \geq s - \max(\alpha, \beta)$.*

Proof. Put $m' = m + \alpha - \beta$. The product equals $r^{\alpha+\beta+|m|} P_s^{(0,|m|)}(2t-1) e^{im'\theta}$, so, by Lemma 4.1(b) its coefficient on $\Phi_{m',s'}$ is a multiple of

$$\int_0^1 t^{|m|} P_s^{(0,|m|)}(2t-1) t^e P_{s'}^{(0,|m'|)}(2t-1) dt, \quad e = \frac{\alpha + \beta + |m'| - |m|}{2}.$$

Since $|m'| - |m| \equiv \alpha + \beta \pmod{2}$ and $||m'| - |m|| \leq |\alpha - \beta|$, e is an integer with $0 \leq \min(\alpha, \beta) \leq e \leq \max(\alpha, \beta)$. The second factor is a polynomial of degree $e + s'$, so the integral vanishes if $e + s' < s$ by Lemma 4.1(a). $\square$

Lemma 4.4 (Monomials). *Let $\alpha \geq \beta \geq 0$. Then $w^\alpha \bar{w}^\beta = \sum_{s=0}^\beta \kappa_s \Phi_{\alpha-\beta,s}$ with $\kappa_s \geq 0$, $\sum_s \kappa_s = 1$ and $\kappa_0 = \frac{\alpha - \beta + 1}{\alpha + 1}$.*

Proof. $\bar{w}^\beta = \Phi_{-\beta,0}$. Applying (4.3) α times preserves nonnegativity and total mass one, raises the radial index by at most one in each of the first β steps and never afterwards. The harmonic coefficient is the L^2 projection onto $r^{\alpha-\beta} e^{i(\alpha-\beta)\theta}$, namely $\int_0^1 r^{2\alpha+1} dr / \int_0^1 r^{2(\alpha-\beta)+1} dr$. $\square$

4.2. The sine sector. For odd $n \geq 1$ and $s \geq 0$ let

$$S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta = \frac{\Phi_{n,s} - \Phi_{-n,s}}{2i}.$$

Let $\mathcal{S}$ be the real Banach space of $f = \sum_{n \text{ odd}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$; this is also its norm as an element of $\mathcal{B}_\rho$. The elements of $\mathcal{S}$ are real-valued, odd under $w \mapsto \bar{w}$ and even under $w \mapsto -\bar{w}$. Conversely, since $\Phi_{m,s}(\bar{w}) = \Phi_{-m,s}(w)$ and $\Phi_{m,s}(-\bar{w}) = (-1)^m \Phi_{-m,s}(w)$, every real-valued $f \in \mathcal{B}_\rho$ with these two symmetries belongs to $\mathcal{S}$. The functions $S_{n,0} = r^n \sin n\theta$ are harmonic; we call $f_{n,0}$ the harmonic coefficients and put

$$\mathcal{G} = \{g \in \mathcal{S} : g_{n,0} = 0 \text{ for all } n\}.$$

5. THE COMPATIBLE INVERSE OF THE LAPLACIAN

For odd $n \geq 1$ and $s \geq 1$ put $q = n + 2s$ and define

$$(5.1) \quad KS_{n,s} = \frac{S_{n,s-1}}{4q(q+1)} - \frac{S_{n,s}}{2q(q+2)} + \frac{S_{n,s+1}}{4(q+1)(q+2)}.$$

Lemma 5.1. *For odd $n \geq 1$ and $s \geq 1$:*

- (a) $\Delta KS_{n,s} = S_{n,s}$;
- (b) $KS_{n,s} = 0$ and $\nabla KS_{n,s} = 0$ on $\partial\mathbb{D}$;
- (c) $KS_{n,s} = \frac{(1-r^2)^2}{4s(s+1)} r^n P_{s-1}^{(2,n)}(2r^2 - 1) \sin n\theta$;
- (d) *the absolute sum of the coefficients in (5.1) is $1/(q(q+2))$. Hence K extends to a bounded operator $K: \mathcal{G} \rightarrow \mathcal{S}$ with $\|K\| = 1/15$.*

Proof. Write A, B, C for the three coefficients in (5.1) and $a_{s,k} = a_{s,k}^{(n)}$.

(a) Since $\Delta(r^{n+2k} \sin n\theta) = 4k(n+k)r^{n+2k-2} \sin n\theta$, the claim is equivalent to $Aa_{s-1,k+1} + Ba_{s,k+1} + Ca_{s+1,k+1} = a_{s,k}/(4(k+1)(n+k+1))$ for $0 \leq k \leq s$. By (4.2) the three ratios $a_{s-1,k+1}/a_{s,k}$, $a_{s,k+1}/a_{s,k}$, $a_{s+1,k+1}/a_{s,k}$ equal $(s-k)(s-k-1)$, $-(n+s+k+1)(s-k)$ and $(n+s+k+2)(n+s+k+1)$, each divided by $(k+1)(n+k+1)$, and a direct computation gives $A(s-k)(s-k-1) - B(n+s+k+1)(s-k) + C(n+s+k+2)(n+s+k+1) = \frac{1}{4}$.

(b) At $r = 1$ each $r^n P_t^{(0,n)}(2r^2 - 1)$ equals one, and $A + B + C = 0$. Its radial derivative at $r = 1$ is $\beta_t = n + 2t(t+n+1)$, because $\frac{d}{dx} P_t^{(0,n)}(1) = t(t+n+1)/2$. Since $\beta_s - \beta_{s-1} = 2q$ and

$\beta_{s+1} - \beta_s = 2(q+2)$, we get $A\beta_{s-1} + B\beta_s + C\beta_{s+1} = -2qA + 2(q+2)C = 0$. The tangential derivative vanishes because the boundary value vanishes identically.

(c) By (b), $KS_{n,s} = r^n R(t) \sin n\theta$ where R is a polynomial of degree $s+1$ with $R(1) = 0$ and $nR(1) + 2R'(1) = 0$, so $R(t) = (1-t)^2 Q(t)$ with $\deg Q = s-1$. As a combination of $P_{s-1}^{(0,n)}, P_s^{(0,n)}, P_{s+1}^{(0,n)}$, the polynomial R is orthogonal in $L^2(t^n dt)$ to polynomials of degree less than $s-1$ (Lemma 4.1(a)); hence Q is orthogonal in $L^2(t^n(1-t)^2 dt)$ to them and is a multiple of $P_{s-1}^{(2,n)}(2t-1)$. Comparing leading coefficients, $C a_{s+1,s+1}$ for R and $\frac{(n+2s)!}{(s-1)!(n+s+1)!}$ for $P_{s-1}^{(2,n)}(2t-1)$, gives the factor $1/(4s(s+1))$.

(d) $A - B + C = \frac{(q+2)+2(q+1)+q}{4q(q+1)(q+2)} = \frac{1}{q(q+2)}$. The operator K does not change the angular index, hence not the weight. Since $q \geq 3$, the supremum of $1/(q(q+2))$ is $1/15$, attained at $(n, s) = (1, 1)$. $\square$

Lemma 5.2. *For $g \in \mathcal{G}$, the function $W = Kg$ belongs to $C^1(\overline{\mathbb{D}})$, satisfies $\Delta W = g$ in $\mathcal{D}'(\mathbb{D})$, and $W = 0, \nabla W = 0$ on $\partial\mathbb{D}$.*

Proof. Let g_N be the truncations of g to finitely many coefficients and $W_N = Kg_N$, a polynomial with $\Delta W_N = g_N$ and zero Cauchy data by Lemma 5.1. The Dirichlet Green function of the disc is $G(w, \zeta) = \frac{1}{2\pi} \log \left| \frac{w-\zeta}{1-w\bar{\zeta}} \right|$, and $W_N(w) = \int_{\mathbb{D}} G(w, \zeta) g_N(\zeta) dA(\zeta)$. We have $|\nabla_w G(w, \zeta)| \leq \frac{1}{2\pi} (|w-\zeta|^{-1} + |\zeta| |1-w\bar{\zeta}|^{-1})$. For $|w| \leq 1$, $\int_{\mathbb{D}} |w-\zeta|^{-1} dA(\zeta) \leq 4\pi$, since $\mathbb{D}$ lies in the disc of radius 2 about w . If $|w| < 1/2$ then $|\zeta|/|1-w\bar{\zeta}| \leq 2$; if $|w| \geq 1/2$ then $|\zeta|/|1-w\bar{\zeta}| \leq 2/|\zeta - 1/\bar{w}|$ and $\mathbb{D}$ lies in the disc of radius 3 about $1/\bar{w}$; in both cases the integral of this term is at most 12π . Hence

$$\sup_{\mathbb{D}} |\nabla(W_N - W_L)| \leq 8 \sup_{\mathbb{D}} |g_N - g_L| \leq 8 \|g_N - g_L\|_{\rho}, \quad \sup_{\mathbb{D}} |W_N - W_L| \leq \frac{1}{15} \|g_N - g_L\|_{\rho}.$$

Thus (W_N) is Cauchy in $C^1(\overline{\mathbb{D}})$; its limit is Kg (the coefficients converge in $\mathcal{S}$), the zero Cauchy data pass to the limit, and so does the distributional equation. $\square$

Remark 5.3. Conversely, if $W \in C^1(\overline{\mathbb{D}})$ has $\Delta W = g \in \mathcal{S}$ in $\mathcal{D}'(\mathbb{D})$ and zero Cauchy data, then $g \in \mathcal{G}$ and $W = Kg$: the extension of W by zero is C^1 and satisfies $\Delta W = g \mathbf{1}_{\mathbb{D}}$ in $\mathcal{D}'(\mathbb{R}^2)$, and pairing with the harmonic polynomials $r^n \sin n\theta$ gives $g_{n,0} = 0$; then $W - Kg$ is harmonic with zero boundary values. This is why the equation is posed on $\mathcal{G}$.

6. THE CUBIC EQUATION

From now on $\rho = 21/20$, $M = 41$ and $S = 24$. The centre $x^\circ = (g^\circ, c^\circ)$ consists of 504 coefficients $g_{n,s}^\circ$, $n \in \{1, 3, \dots, 41\}$, $1 \leq s \leq 24$, and 21 coefficients c_j° , $j \in \{1, 3, \dots, 41\}$; all other coefficients of x° vanish. Each of these 525 numbers is an explicitly given dyadic rational (a binary64 number interpreted exactly). We write $b = c_1^\circ \approx 14.840043$ and

$$\omega_j = b^2(j+1)\rho^j \quad (j \text{ odd}).$$

Let X be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}$ and $c = (c_j)_{j \text{ odd} \geq 1}$ real, normed by

$$\|(g, c)\|_X = \|g\|_{\rho} + \sum_{j \text{ odd}} \omega_j |c_j|.$$

To c we associate $\psi_c(w) = \sum_j c_j w^j$, which is holomorphic in $|w| < \rho$ and continuous up to $|w| = \rho$ because $|c_j| \rho^j \leq \omega_j |c_j| / (2b^2)$. Note that $\text{Im } \psi_c = \sum_j c_j S_{j,0} \in \mathcal{S}$. Define

$$(6.1) \quad \mathcal{F}(g, c) = g + |\psi'_c|^2 (Kg + \text{Im } \psi_c).$$

Lemma 6.1. $\mathcal{F}$ is a continuous cubic polynomial map $X \rightarrow \mathcal{S}$. If $\mathcal{F}(g, c) = 0$, then $V = Kg + \text{Im } \psi_c$ belongs to $C^1(\mathbb{D})$, is odd under $w \mapsto \bar{w}$ and even under $w \mapsto -\bar{w}$, and satisfies (3.1) with $\psi = \psi_c$.

Proof. $\psi'_c = \sum_j j c_j w^{j-1}$ has real coefficients and even powers, and $N_\rho(\psi'_c) = \sum_j j \rho^{j-1} |c_j| \leq \|c\|/(b^2 \rho)$, where $\|c\| = \sum \omega_j |c_j|$. Thus $|\psi'_c|^2$ is real and invariant under $w \mapsto \bar{w}$ and $w \mapsto -\bar{w}$, and multiplication by it maps $\mathcal{S}$ into $\mathcal{S}$ with norm at most $N_\rho(\psi'_c)^2$ by (4.4). Together with $\|Kg\|_\rho \leq \|g\|_\rho/15$ and $\|\text{Im } \psi_c\|_\rho = \sum \rho^j |c_j|$, this shows that (6.1) is a sum of bounded multilinear maps of degree at most three. If $\mathcal{F}(g, c) = 0$, then Lemma 5.2 gives $V \in C^1(\mathbb{D})$, $V - \text{Im } \psi_c = Kg$ with zero Cauchy data, and $\Delta V = g = -|\psi'_c|^2 V$ in $\mathcal{D}'(\mathbb{D})$; the last identity holds pointwise because both sides are uniformly convergent disk-polynomial series with the same coefficients. The parities hold because $V \in \mathcal{S}$. $\square$

6.1. Expansion at the centre. Write $p = \psi'_{c^\circ} = \sum_{a=0,2,\dots,40} p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, so $p_0 = b$, and put

$$P = N_\rho(p), \quad V^\circ = Kg^\circ + \text{Im } \psi_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X$ let $\delta V = K\delta g + \text{Im } \psi_\eta$ and, for shape perturbations η, η_1, η_2 ,

$$L(\eta) = \bar{p} \psi'_\eta + p \overline{\psi'_\eta} = 2 \text{Re}(\bar{p} \psi'_\eta), \quad Q(\eta_1, \eta_2) = \text{Re}(\psi'_{\eta_1} \overline{\psi'_{\eta_2}}).$$

Since $|p + \psi'_\eta|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$, expanding (6.1) gives

$$(6.2) \quad \mathcal{F}(x^\circ + h) = \mathcal{F}(x^\circ) + D\mathcal{F}(x^\circ)h + B_2(h, h) + B_3(h, h, h),$$

where

$$(6.3) \quad \begin{aligned} D\mathcal{F}(x^\circ)h &= \delta g + |p|^2 K\delta g + L(\eta)V^\circ + |p|^2 \text{Im } \psi_\eta, \\ B_2(h_1, h_2) &= \frac{1}{2} \{L(\eta_1) \delta V_2 + L(\eta_2) \delta V_1\} + Q(\eta_1, \eta_2)V^\circ, \\ B_3(h_1, h_2, h_3) &= \frac{1}{3} \{Q(\eta_1, \eta_2) \delta V_3 + Q(\eta_1, \eta_3) \delta V_2 + Q(\eta_2, \eta_3) \delta V_1\}. \end{aligned}$$

There are no terms of higher order.

Lemma 6.2. Let $\kappa = 1/15$, $\vartheta_1 = 1/(b^2 \rho)$ and $\vartheta_0 = 1/(2b^2)$. Then for all $h_i \in X$,

$$\|B_2(h_1, h_2)\|_\rho \leq c_2 \|h_1\|_X \|h_2\|_X, \quad \|B_3(h_1, h_2, h_3)\|_\rho \leq c_3 \|h_1\|_X \|h_2\|_X \|h_3\|_X,$$

with $c_2 = \max\{P\vartheta_1\kappa, 2P\vartheta_1\vartheta_0 + \vartheta_1^2 V_0\}$ and $c_3 = \vartheta_1^2 \max\{\kappa, \vartheta_0\}$.

Proof. Write $a_i = \|\delta g_i\|_\rho$ and $e_i = \sum_j \omega_j |\eta_{i,j}|$. Termwise, $N_\rho(\psi'_{\eta_i}) \leq \vartheta_1 e_i$ (as $j/(j+1) \leq 1$), $\|\text{Im } \psi_{\eta_i}\|_\rho \leq \vartheta_0 e_i$ (as $j+1 \geq 2$), and $\|K\delta g_i\|_\rho \leq \kappa a_i$, so $\|\delta V_i\|_\rho \leq \kappa a_i + \vartheta_0 e_i$. By (4.4), $\|L(\eta_i)f\|_\rho \leq 2P\vartheta_1 e_i \|f\|_\rho$ and $\|Q(\eta_i, \eta_k)f\|_\rho \leq \vartheta_1^2 e_i e_k \|f\|_\rho$. Hence $\|B_2(h_1, h_2)\|_\rho \leq P\vartheta_1\kappa(e_1 a_2 + e_2 a_1) + (2P\vartheta_1\vartheta_0 + \vartheta_1^2 V_0)e_1 e_2 \leq c_2(a_1 + e_1)(a_2 + e_2)$. Similarly every monomial $e_i e_k a_l$ in the bound for B_3 has coefficient $\vartheta_1^2 \kappa/3$ and $e_1 e_2 e_3$ has coefficient $\vartheta_1^2 \vartheta_0$; each is at most its coefficient in $c_3 \prod_i (a_i + e_i)$. $\square$

7. THE CONTRACTION ARGUMENT AND THE APPROXIMATE INVERSE

7.1. A Newton–Kantorovich theorem.

Proposition 7.1. Let $\mathcal{F}: X \rightarrow \mathcal{S}$ have the form (6.2) with symmetric bounded B_2 and B_3 , and let $A: \mathcal{S} \rightarrow X$ be bounded, linear and injective. Suppose that

$$\|A\mathcal{F}(x^\circ)\|_X \leq Y, \quad \|I - A D\mathcal{F}(x^\circ)\|_{X \rightarrow X} \leq Z, \quad \|A\|_{c_2} \leq C_2, \quad \|A\|_{c_3} \leq C_3,$$

and that $r > 0$ satisfies

$$(7.1) \quad Y + (Z - 1)r + C_2 r^2 + C_3 r^3 < 0, \quad Z - 1 + 2C_2 r + 3C_3 r^2 < 0.$$

Then $\mathcal{F}$ has exactly one zero in the closed ball $\overline{B}_X(x^\circ, r)$.

Proof. Let $\mathcal{N}(x) = x - A\mathcal{F}(x)$. For $\|h\|_X \leq r$, (6.2) gives $\mathcal{N}(x^\circ + h) - x^\circ = -A\mathcal{F}(x^\circ) + (I - AD\mathcal{F}(x^\circ))h - AB_2(h, h) - AB_3(h, h, h)$, whose norm is at most $Y + Zr + C_2r^2 + C_3r^3 < r$. For h_1, h_2 in the ball, $B_2(h_1, h_1) - B_2(h_2, h_2) = B_2(h_1 - h_2, h_1 + h_2)$ and $B_3(h_1, h_1, h_1) - B_3(h_2, h_2, h_2) = B_3(h_1 - h_2, h_1, h_1) + B_3(h_2, h_1 - h_2, h_1) + B_3(h_2, h_2, h_1 - h_2)$, so $\mathcal{N}$ is Lipschitz on the ball with constant $Z + 2C_2r + 3C_3r^2 < 1$. By the contraction mapping principle $\mathcal{N}$ has a unique fixed point in the ball, and since A is injective the fixed points of $\mathcal{N}$ are exactly the zeros of $\mathcal{F}$. $\square$

In the weighted ℓ^1 spaces X and $\mathcal{S}$, operator norms are computed column by column.

Lemma 7.2. *Let E and E' be weighted ℓ^1 spaces with bases (e_i) and (e'_κ) and weights (λ_i) and (λ'_κ) , and let $T: E \rightarrow E'$ be bounded. Then $\|T\| = \sup_i \|Te_i\|_{E'}/\lambda_i$.*

Proof. The inequality $\geq$ follows by testing on e_i/λ_i , and $\leq$ by the triangle inequality and absolute summation, using continuity of T and density of finitely supported vectors. $\square$

We shall apply Lemma 7.2 with the bases $\{e_{(n,s)} : s \geq 1\} \cup \{e_j\}$ of X , of weights ρ^n and ω_j , and $\{S_{n,s} : s \geq 0\}$ of $\mathcal{S}$, of weights ρ^n . We call the basis vectors of X *inputs* and those of $\mathcal{S}$ *rows*.

7.2. Finite block and tail. The *finite inputs* are the 504 coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and the 21 shape coordinates c_j with $j \leq M$. The *finite rows* are the 504 rows (n, s) with $n \leq M$, $1 \leq s \leq S$, and the 21 harmonic rows $(n, 0)$ with $n \leq M$. All other inputs and rows are *tail* inputs and rows. We write $X = X_f \oplus X_t$ and $\mathcal{S} = \mathcal{S}_f \oplus \mathcal{S}_t$ accordingly; $\dim X_f = \dim \mathcal{S}_f = 525$. Let M_f be the 525×525 matrix of the restriction of $D\mathcal{F}(x^\circ)$ to finite inputs and finite rows.

On tail inputs we use the following explicit model of $D\mathcal{F}(x^\circ)$, obtained by replacing ψ° by bw and V° by $b \operatorname{Im} w$:

$$(7.2) \quad T(\delta g, \eta) = \delta g + b^2 \sum_{j > M} \eta_j \{ (j+1)S_{j,0} - (j-1)S_{j-2,0} - S_{j-2,1} \}.$$

Lemma 7.3. *For odd $j \geq 3$, $L_b(w^j) b \operatorname{Im} w + b^2 \operatorname{Im} w^j = b^2 \{ (j+1)S_{j,0} - (j-1)S_{j-2,0} - S_{j-2,1} \}$, where $L_b(\eta) = 2 \operatorname{Re}(b\psi'_\eta)$.*

Proof. $L_b(w^j) b \operatorname{Im} w = 2b^2 j r^j \cos((j-1)\theta) \sin \theta = b^2 j r^j \{ \sin j\theta - \sin(j-2)\theta \}$. Moreover $r^2 \Phi_{j-2,0} = \frac{1}{j} \Phi_{j-2,1} + \frac{j-1}{j} \Phi_{j-2,0}$, because $P_1^{(0,n)}(2t-1) = (n+2)t - (n+1)$ by (4.2); hence $j r^j \sin(j-2)\theta = S_{j-2,1} + (j-1)S_{j-2,0}$. $\square$

Write $Te = (Be, T_{tt}e)$ for tail inputs e , with B the part in finite rows and T_{tt} the part in tail rows. Only the shape input $j = M+2$ reaches finite rows:

$$(7.3) \quad B(\delta g, \eta) = -b^2 \eta_{M+2} ((M+1)e_{(M,0)} + e_{(M,1)}).$$

Put $\tau = (1 - \rho^{-2})^{-1}$.

Lemma 7.4. *$T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is*

$$(7.4) \quad \eta_n = \sum_{\substack{m \geq n \\ m \text{ odd}}} \frac{y_{m,0}}{b^2(m+1)} \quad (n > M), \quad \delta g_{n,s} = y_{n,s} + \mathbf{1}_{s=1} b^2 \eta_{n+2} \quad ((n, s) \text{ a tail } g\text{-input}).$$

Moreover $\|T_{tt}\| \leq 1 + \rho^{-2}$ and $\|T_{tt}^{-1}\| \leq \max\{1, \tau + \tau/((M+3)\rho^2)\}$.

Proof. By (7.2), the tail harmonic rows read $b^2(n+1)(\eta_n - \eta_{n+2}) = y_{n,0}$ for $n \geq M+2$, the tail rows $(n, 1)$ read $\delta g_{n,1} - b^2\eta_{n+2} = y_{n,1}$, and all other tail rows are $\delta g_{n,s} = y_{n,s}$. Since $\sum \omega_n |\eta_n| < \infty$ forces $\eta_n \rightarrow 0$, summation of the harmonic equations gives (7.4), and a solution with $y = 0$ vanishes; conversely (7.4) solves the equations whenever the series converge. For the bounds we use Lemma 7.2. The shape input e_j has three nonzero entries, $b^2(j+1)$ in row $(j, 0)$, $-b^2(j-1)$ in row $(j-2, 0)$ and $-b^2$ in row $(j-2, 1)$; its weighted norm divided by ω_j is $1 + j\rho^{-2}/(j+1) \leq 1 + \rho^{-2}$. Tail g -inputs are mapped to the identical rows. For T_{tt}^{-1} , a nonharmonic tail row is mapped to the identical input. The normalised harmonic row $\rho^{-m}S_{m,0}$ is mapped to $\eta_m = \rho^{-m}/(b^2(m+1))$ for $M < n \leq m$ and $\delta g_{n-2,1} = b^2\eta_n$ for $M+4 \leq n \leq m$, with norm

$$\sum_{M < n \leq m} \frac{(n+1)\rho^n}{(m+1)\rho^m} + \sum_{M+4 \leq n \leq m} \frac{\rho^{n-2}}{(m+1)\rho^m} \leq \tau + \frac{\tau}{(M+3)\rho^2}. \quad \square$$

7.3. The approximate inverse. Let $\hat{A}$ be a 525×525 matrix with dyadic rational entries (in the computation, the midpoints of an interval enclosure of M_f^{-1}), regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, and suppose that

$$(7.5) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1.$$

Define $A: \mathcal{S} = \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X = X_f \oplus X_t$ by

$$(7.6) \quad A = \begin{pmatrix} \hat{A} & -\hat{A}BT_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix}.$$

Lemma 7.5. *Under (7.5), A is bounded and injective, and $ATe = e$ for every tail input e .*

Proof. By (7.5) and the Neumann series, $\hat{A}M_f$ is invertible, so $\hat{A}$ is invertible. The operator $\begin{pmatrix} \hat{A}^{-1} & B \\ 0 & T_{tt} \end{pmatrix}$ is a left inverse of A . For a tail input e , $ATe = (\hat{A}Be - \hat{A}BT_{tt}^{-1}T_{tt}e, T_{tt}^{-1}T_{tt}e) = (0, e)$. $\square$

Let $e_{(n,s)}$ denote the finite rows as elements of $\mathcal{S}_f$, and define

$$\|R\| = \max_{(n,s) \text{ finite row}} \frac{\|\hat{A}e_{(n,s)}\|_{X_f}}{\rho^n}, \quad H = \|\hat{A}(e_{(M,1)} + (M+1)e_{(M,0)})\|_{X_f},$$

and, for odd $n \geq M+2$,

$$(7.7) \quad h_n = \frac{H + \sum_{j=M+2, M+4, \dots, n} (j+1)\rho^j + \sum_{j=M+4, M+6, \dots, n} \rho^{j-2}}{(n+1)\rho^n}.$$

Lemma 7.6. *Let $N_H \geq M+2$ be odd. Then A maps every residual supported in tail rows with norm at most*

$$A_t := \max \left\{ 1, \max_{M+2 \leq n \leq N_H} h_n, \tau + \frac{\tau}{(N_H+3)\rho^2} + \frac{H}{(N_H+3)\rho^{N_H+2}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. We use Lemma 7.2. A finite row $e_{(n,s)}$ is mapped to $\hat{A}e_{(n,s)}$. A nonharmonic tail row is mapped to the identical tail input by (7.4), and B vanishes on it. The normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to the shape coordinates $\eta_j = \rho^{-n}/(b^2(n+1))$, $M+2 \leq j \leq n$, to $\delta g_{j-2,1} = b^2\eta_j$, $M+4 \leq j \leq n$, and, by (7.3), to the finite part $b^2\eta_{M+2}\hat{A}((M+1)e_{(M,0)} + e_{(M,1)})$; the sum of the three norms is exactly h_n . For $n \geq N_H+2$ we bound $(j+1)/(n+1) \leq 1$ in the first sum, sum the geometric series $\sum_{i \geq 0} \rho^{-2i} = \tau$, and use $n+1 \geq N_H+3$ in the other two terms. $\square$

8. THE DEFECT BOUNDS

In this section we explain how the bounds Y and Z of Proposition 7.1 are obtained for the operator A of (7.6). By Lemma 7.2, $\|I - AD\mathcal{F}(x^\circ)\|$ is the supremum over all inputs e of $\|(I - AD\mathcal{F}(x^\circ))e\|_X$ divided by the weight of e . We partition the inputs into five classes and bound each class separately; every input belongs to exactly one class, and no column is estimated by sampling. Throughout, $d = \deg p = 40$, and $\Lambda = M + d = 81$.

8.1. The residual and the finite columns. The residual $\mathcal{F}(x^\circ)$ is a polynomial in $w, \bar{w}$. Using (4.3) and (5.1), its expansion in the basis $S_{n,s}$ is computed without truncation (it has 2286 coefficients) in interval arithmetic. Only finitely many tail rows occur, so $A\mathcal{F}(x^\circ)$ is computed without truncation from (7.6), (7.3) and (7.4), and Y is an upper bound of its norm.

For each of the 525 finite inputs e , $D\mathcal{F}(x^\circ)e$ is again a polynomial by (6.3), and $(I - AD\mathcal{F}(x^\circ))e$ is computed in the same way. We denote by Z_f an upper bound for the maximum of the normalised norms of these columns.

8.2. Tail g -columns. Let $n_{\max} = M + 2d + 2 = 123$ and $s_{\max} = S + d + 3 = 67$.

Lemma 8.1. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{F}(x^\circ))e = -A(|p|^2 K S_{n,s})$. Moreover, writing $p = p_{\leq 16} + p_{> 16}$ for the splitting of p into terms of degree ≤ 16 and > 16 , and $P_s = N_\rho(p_{\leq 16})$, $P_t = N_\rho(p_{> 16})$,*

$$(8.1) \quad \frac{\|(I - AD\mathcal{F}(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq 16}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| (2P_s P_t + P_t^2)}{q(q+2)}.$$

Proof. By (6.3), $D\mathcal{F}(x^\circ)e = S_{n,s} + |p|^2 K S_{n,s}$, and $Te = S_{n,s}$, so $AS_{n,s} = ATe = e$ by Lemma 7.5. Since $|p|^2 - |p_{\leq 16}|^2 = \overline{p_{\leq 16}} p_{> 16} + \overline{p_{> 16}} p_{\leq 16} + |p_{> 16}|^2$, (4.4) and Lemma 5.1(d) give $\|(|p|^2 - |p_{\leq 16}|^2) K S_{n,s}\|_\rho \leq (2P_s P_t + P_t^2) \rho^n / (q(q+2))$. $\square$

For the $62 \cdot 67 - 21 \cdot 24 = 3650$ tail g -inputs with $n \leq n_{\max}$ and $s \leq s_{\max}$, the first term on the right of (8.1) is computed without truncation; let $Z_{g,\partial}$ bound the maximum of the right-hand side over these inputs.

Lemma 8.2. *For every tail g -input $e_{(n,s)}$ with $n > n_{\max}$ or $s > s_{\max}$,*

$$\frac{\|(I - AD\mathcal{F}(x^\circ))e_{(n,s)}\|_X}{\rho^n} \leq Z_{g,\infty} := \max \left\{ \frac{P^2}{137 \cdot 139}, \frac{A_t P^2}{127 \cdot 129} \right\}.$$

Proof. We have $|p|^2 = \sum_{a,a'} p_a p_{a'} w^a \bar{w}^{a'}$ with $a, a' \leq d$. By (5.1) and Lemma 4.3, every component of $|p|^2 K S_{n,s}$ has radial index at least $s - 1 - d$ and angular frequency of modulus at least $n - d$. Also $\||p|^2 K S_{n,s}\|_\rho \leq P^2 \rho^n / (q(q+2))$ with $q = n + 2s$.

If $s > s_{\max}$, all radial indices are at least $s - 1 - d \geq S + 3$, so all components lie in nonharmonic tail rows, on which A is the identity (Lemma 7.6); and $q \geq 1 + 2(s_{\max} + 1) = 137$.

If $s \leq s_{\max}$ and $n > n_{\max}$, all frequencies are at least $n - d \geq n_{\max} + 2 - d > M$, so all components lie in tail rows, on which A has norm at most A_t ; and $q \geq n_{\max} + 2 + 2 = 127$. $\square$

8.3. Shape columns. For the 480 shape inputs e_j with $M < j \leq j_* = 1001$, $(I - AD\mathcal{F}(x^\circ))e_j$ is computed without truncation as in Section 8.1; let $Z_{\text{sh},\partial}$ bound their normalised norms. For $j > j_*$ we prove a uniform bound.

Let $W^\circ = Kg^\circ$, so that $V^\circ = W^\circ + \text{Im } \psi_{c^\circ}$. By (6.3) and Lemma 7.3, for odd $j > M$,

$$D\mathcal{F}(x^\circ)e_j = 2 \text{Re} \left[w^{j-1} \left(j \bar{p} V^\circ + \frac{|p|^2 w}{2i} \right) \right], \quad T e_j = 2 \text{Re} \left[w^{j-1} \left(j b^2 \text{Im } w + \frac{b^2 w}{2i} \right) \right].$$

Since $ATe_j = e_j$, we get $(I - AD\mathcal{F}(x^\circ))e_j = -A\Delta_j$ with

$$(8.2) \quad \Delta_j = 2 \operatorname{Re} [w^{j-1}(j\Gamma_1 + \Gamma_0)] + 2 \operatorname{Re} [jw^{j-1}\bar{p}W^\circ],$$

where $\Gamma_1 = \bar{p} \operatorname{Im} \psi_{c^\circ} - b^2 \operatorname{Im} w$ and $\Gamma_0 = (|p|^2 - b^2)w/(2i)$. Writing $\operatorname{Im} \psi_{c^\circ} = (\psi_{c^\circ} - \overline{\psi_{c^\circ}})/(2i)$ and $|p|^2 = \sum p_a p_{a'} w^a \bar{w}^{a'}$, and using $p_0 = c_1^\circ = b$,

$$2i\Gamma_1 = \sum_{(a,l) \neq (0,1)} p_a c_l^\circ (w^l \bar{w}^a - \bar{w}^{a+l}), \quad 2i\Gamma_0 = \sum_{(a,a') \neq (0,0)} p_a p_{a'} w^{a+1} \bar{w}^{a'},$$

with $a, a' \in \{0, 2, \dots, 40\}$ and $l \in \{1, 3, \dots, 41\}$. After combining equal monomials, $2i\Gamma_1 = \sum_{\alpha, \beta} q_{\alpha\beta}^1 w^\alpha \bar{w}^\beta$ and $2i\Gamma_0 = \sum_{\alpha, \beta} q_{\alpha\beta}^0 w^\alpha \bar{w}^\beta$ with finitely many real coefficients. The differences $k = \alpha - \beta$ that occur are odd and lie in $[k_-, k_+] = [-81, 41]$. For odd $k \in [k_-, k_+]$ define

$$\begin{aligned} \bar{q}_k &= \sum_{\alpha-\beta=k} q_{\alpha\beta}^1, & \varepsilon_k &= \frac{1}{j_*+1} \sum_{\alpha-\beta=k} ((1+\beta)|q_{\alpha\beta}^1| + |q_{\alpha\beta}^0|), & \bar{\eta}_k &= \frac{1}{b^2} \sum_{l \geq k} \bar{q}_l, \\ \delta_k &= \frac{1}{b^2} \sum_{l \geq k} \left[\varepsilon_l \left(1 + \frac{|l-1|}{j_*+k_-} \right) + \frac{|\bar{q}_l| |l-1|}{j_*+k_-} \right], & E_k &= |\bar{\eta}_k| + \delta_k, \end{aligned}$$

where the sums over l run over odd $l \leq k_+$. By Lemma 5.1(c), $W^\circ = (1 - |w|^2)^2 Q^\circ$ with

$$Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)} (2r^2 - 1) \sin n\theta,$$

and Q° has a finite disk-polynomial expansion, which we compute from the connection formulas (they follow from [20, Eq. 18.9.5] and the symmetry $P_k^{(\alpha,\beta)}(-x) = (-1)^k P_k^{(\beta,\alpha)}(x)$, [20, Table 18.6.1])

$$P_k^{(a+1,n)} = \frac{2k+a+n+1}{k+a+n+1} P_k^{(a,n)} + \frac{k+n}{k+a+n+1} P_{k-1}^{(a+1,n)} \quad (a = 0, 1).$$

The function $w^\Lambda \bar{p}Q^\circ$ has only nonnegative frequencies, since the frequencies of $\bar{p}Q^\circ$ are at least $-M - d$; write $w^\Lambda \bar{p}Q^\circ = \sum_{m \geq 0, t \geq 0} \gamma_{m,t}^W \Phi_{m,t}$ (a finite sum, computed in interval arithmetic). Finally let

$$\begin{aligned} U_1 &= \sum_k \left(1 + \frac{\max(0, k-1)}{j_*+1} \right) \rho^{k-1} E_k, & U_2 &= E_{k_-} \rho^{k_- - 3} \tau, \\ U_3 &= \frac{1}{j_*+1} \left(\sum_k \rho^{k-3} E_k + E_{k_-} \rho^{k_- - 5} \tau \right), & U_4 &= \frac{H E_{k_-}}{(j_*+1) \rho^{j_*}}, \\ U_5 &= \sum_{\alpha, \beta} \frac{(|q_{\alpha\beta}^1| + |q_{\alpha\beta}^0|/j_*) \beta \rho^{\alpha-\beta-1}}{b^2(j_* + \alpha)}, \\ U_W &= \frac{8}{b^2(j_* - \Lambda)^2} \sum_{m,t} |\gamma_{m,t}^W| (2t+1)(2t+3) \rho^{m-1-\Lambda}, \end{aligned}$$

where k runs over odd integers in $[k_-, k_+]$.

Lemma 8.3. *For integers $m, t, N \geq 0$,*

$$\|(1 - |w|^2)^2 w^N \Phi_{m,t}\|_\rho \leq \frac{4(2t+1)(2t+3)}{(m+N+1)^2} \rho^{m+N}.$$

Proof. Applying (4.3) for w and then for $\bar{w}$ gives $|w|^2\Phi_{m,t} = \lambda_t^+\Phi_{m,t+1} + (1 - \lambda_t^+ - \lambda_t^-)\Phi_{m,t} + \lambda_t^-\Phi_{m,t-1}$ with

$$\lambda_t^+ = \frac{(t+1)(m+t+1)}{(m+2t+1)(m+2t+2)} \leq \frac{t+1}{m+1}, \quad \lambda_t^- = \frac{t(m+t)}{(m+2t)(m+2t+1)} \leq \frac{t}{m+1}.$$

Hence $(1 - |w|^2)\Phi_{m,t}$ has the same frequency and absolute coefficient sum $2(\lambda_t^+ + \lambda_t^-) \leq 2(2t+1)/(m+1)$, and its radial indices are at most $t+1$. Applying this twice, $\|(1 - |w|^2)^2\Phi_{m,t}\|_\rho \leq 4(2t+1)(2t+3)\rho^m/(m+1)^2$. By Lemma 4.2, $w^N\Phi_{m,t}$ is a convex combination of $\Phi_{m+N,t'}$ with $t' \leq t$; apply the previous bound to each term. $\square$

Proposition 8.4. *Assume $j_* + k_- - 1 > M + 2$ and $j_* > 2M + d + 2$. Then for every odd $j > j_*$,*

$$\frac{\|(I - AD\mathcal{F}(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := \mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3 + \mathbf{U}_4 + \mathbf{U}_5 + A_t \mathbf{U}_W.$$

Proof. Fix an odd $j > j_*$. For real γ , $2\operatorname{Re}[w^{j-1}\gamma w^\alpha \bar{w}^\beta/(2i)] = \gamma \operatorname{Im}(w^{j-1+\alpha}\bar{w}^\beta)$. For every monomial occurring, $k = \alpha - \beta \geq k_-$ and $j - 1 + k > M + 2 \geq 1$, so by Lemma 4.4 $\operatorname{Im}(w^{j-1+\alpha}\bar{w}^\beta) = \sum_s \kappa_s S_{j-1+k,s}$ with nonnegative coefficients summing to one and harmonic coefficient $\kappa_0 = 1 - \beta/(j + \alpha)$. Accordingly we split $\Delta_j = \Delta^H + \Delta^N + \Delta^W$ into the harmonic part and the nonharmonic part of the first term of (8.2), and the second term Δ^W . All rows occurring in Δ^H and Δ^N have frequency at least $j - 1 + k_- > M + 2$, so they are tail rows.

The harmonic part. The coefficient of Δ^H in row $(j - 1 + k, 0)$ is $y_k = \sum_{\alpha-\beta=k} (jq_{\alpha\beta}^1 + q_{\alpha\beta}^0)(1 - \frac{\beta}{j+\alpha})$. Since $\frac{j}{j+1}(1 - \frac{\beta}{j+\alpha}) - 1 = -\frac{1}{j+1} - \frac{j\beta}{(j+1)(j+\alpha)}$ and $0 \leq 1 - \frac{\beta}{j+\alpha} \leq 1$, we have $|y_k/(j+1) - \bar{q}_k| \leq \varepsilon_k$. By (7.4), the shape part of $A\Delta^H$ is $\eta_{j-1+k} = b^{-2} \sum_{l \geq k} y_l/(j+l)$ in the window $k_- \leq k \leq k_+$, and $\eta_n = \eta_{j-1+k_-}$ for $M < n < j - 1 + k_-$. Writing $\frac{y_l}{j+l} = \frac{y_l}{j+1}(1 + \frac{1-l}{j+1})$ and using $|\frac{1-l}{j+1}| \leq \frac{|l-1|}{j_*+k_-}$, we get $|\eta_{j-1+k} - \bar{\eta}_k| \leq \delta_k$, hence $|\eta_{j-1+k}| \leq E_k$, and $|\eta_n| \leq E_{k_-}$ below the window. We now bound the parts of $A\Delta^H$, divided by $\omega_j = b^2(j+1)\rho^j$.

- Shape coordinates in the window: $\omega_{j-1+k}/\omega_j = (1 + \frac{k-1}{j+1})\rho^{k-1}$, which gives at most $\mathbf{U}_1$.
- Shape coordinates below the window: $\sum_{n \leq j-3+k_-} (n+1)\rho^n/((j+1)\rho^j) \leq \rho^{k_- - 3}\tau$, which gives at most $\mathbf{U}_2$.
- The coupled coordinates $\delta g_{n-2,1} = b^2\eta_n$, of weight ρ^{n-2} : at most $\mathbf{U}_3$, by the same two computations and $j+1 > j_* + 1$.
- The finite part $b^2\eta_{M+2}\hat{A}((M+1)e_{(M,0)} + e_{(M,1)})$: its norm is at most $b^2HE_{k_-}$, which gives at most $\mathbf{U}_4$ since $j > j_*$.

The nonharmonic part. Δ^N lies in nonharmonic tail rows, on which A is the identity, and its normalised norm is at most

$$\sum_{\alpha,\beta} \frac{|jq_{\alpha\beta}^1 + q_{\alpha\beta}^0|}{j+1} \cdot \frac{\beta}{j+\alpha} \cdot \frac{\rho^{j-1+\alpha-\beta}}{b^2\rho^j} \leq \mathbf{U}_5.$$

The part Δ^W . We have $jw^{j-1}\bar{p}W^\circ = j(1 - |w|^2)^2w^{j-1-\Lambda}(w^\Lambda\bar{p}Q^\circ)$ with $j - 1 - \Lambda \geq 0$. Since conjugation preserves $\|\cdot\|_\rho$, $\|2\operatorname{Re} f\|_\rho \leq 2\|f\|_\rho$, and Lemma 8.3 with $N = j - 1 - \Lambda$ gives

$$\frac{\|\Delta^W\|_\rho}{\omega_j} \leq \frac{2j}{b^2(j+1)\rho^j} \sum_{m,t} |\gamma_{m,t}^W| \frac{4(2t+1)(2t+3)\rho^{m+j-1-\Lambda}}{(m+j-\Lambda)^2} \leq \mathbf{U}_W.$$

All frequencies in Δ^W are at least $j - 1 - \Lambda > M$ because $j_* > 2M + d + 2$, so $\|A\Delta^W\|_X \leq A_t\|\Delta^W\|_\rho$ by Lemma 7.6.

Adding the three contributions proves the claim. $\square$

Remark 8.5. Each of ε_k , δ_k , U_3 , U_5 is $O(1/j_*)$, $U_4 = O(\rho^{-j_*}/j_*)$ and $U_W = O(j_*^{-2})$. Hence, as $j_* \rightarrow \infty$, the majorant $Z_{\text{sh},\infty}$ tends to $\sum_k \rho^{k-1} |\bar{\eta}_k| + |\bar{\eta}_{k-}| \rho^{k-3} \tau \approx 0.19723$; for comparison, the directly computed normalised column at $j = 1003$ is about 0.19752.

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - \text{ADF}(x^\circ)\|_{X \rightarrow X}$.

9. CERTIFIED BOUNDS AND THE EXACT ZERO

The quantities defined in Sections 6–8 are finite expressions in the 525 dyadic centre coefficients and in $\rho = 21/20$, apart from the entries of $\hat{A}$, which are dyadic numbers produced by the computation and are then treated as exact data. They were evaluated in ball arithmetic as described in Section 15. Table 2 lists the results, rounded outward.

Quantity	Where defined	Certified bound
$b = c_1^\circ$	Section 6	14.8400430096918... (exact dyadic)
$P = N_\rho(p)$	Section 6	< 16.833325
$V_0 = \ V^\circ\ _\rho$	Section 6	< 134.261058
$\ I - \hat{A}M_f\ $	(7.5)	$< 1.343 \cdot 10^{-34}$
$\ R\ $	Lemma 7.6	< 675.571877
H	Lemma 7.6	< 1966.719238
A_t ($N_H = 443$)	Lemma 7.6	< 10.777973
$\ A\ $	Lemma 7.6	< 675.571877
Y	Section 8.1	$< 6.297631 \cdot 10^{-6}$
Z_f (525 columns)	Section 8.1	< 0.180814
$2P_s P_t + P_t^2$	Lemma 8.1	< 0.056966
$Z_{g,\partial}$ (3650 columns, maximum at (43, 1))	Lemma 8.1	< 0.313479
$Z_{g,\infty}$	Lemma 8.2	< 0.186417
$Z_{\text{sh},\partial}$ (480 columns, maximum at $j = 43$)	Section 8	< 0.360262
U_1	Proposition 8.4	< 0.217542
U_2	Proposition 8.4	$< 4.45704 \cdot 10^{-4}$
U_3	Proposition 8.4	$< 1.97149 \cdot 10^{-4}$
U_4	Proposition 8.4	$< 3.018 \cdot 10^{-24}$
U_5	Proposition 8.4	$< 9.23594 \cdot 10^{-4}$
$\sum_{m,t} \gamma_{m,t}^W (2t+1)(2t+3) \rho^{m-1-\Lambda}$	Proposition 8.4	< 19930.219512
$U_W, A_t U_W$	Proposition 8.4	$< 8.55375 \cdot 10^{-4}, < 9.219199 \cdot 10^{-3}$
$Z_{\text{sh},\infty}$ (all $j > 1001$)	Proposition 8.4	< 0.228328
Z	Section 8	< 0.360262
$C_2 = \ A\ _{c_2}$	Lemma 6.2	< 3.278617
$C_3 = \ A\ _{c_3}$	Lemma 6.2	$< 8.42289 \cdot 10^{-4}$

TABLE 2. Certified bounds, rounded outward, for $M = 41$, $S = 24$, $\rho = 21/20$, $j_* = 1001$. The complete interval enclosures are recorded in the certificate file (Section 15).

Theorem 9.1. *Let $r = 1/50000$. There is exactly one $x^* = (g^*, c^*) \in X$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{F}(x^*) = 0$.*

Proof. The conditions $j_* + k_- - 1 = 919 > M + 2$ and $j_* = 1001 > 2M + d + 2 = 124$ of Proposition 8.4 hold, and (7.5) holds by Table 2; hence A is bounded and injective (Lemma 7.5). By Table 2, Proposition 7.1 applies with the rational majorants $Y = 7 \cdot 10^{-6}$, $Z = 37/100$, $C_2 = 4$ and $C_3 = 10^{-3}$. For these values and $r = 1/50000$, exact rational arithmetic gives

$$Y + (Z - 1)r + C_2r^2 + C_3r^3 = -5.5983999999920 \cdot 10^{-6} < 0,$$

$$Z - 1 + 2C_2r + 3C_3r^2 = -0.6298399999988 < 0. \quad \square$$

Remark 9.2 (The centre). The centre x° was obtained numerically, and its provenance plays no role in the proof. A spectral collocation computation in spherical coordinates, started from the ball of radius $j_{2,4}$ perturbed by the zonal harmonic of degree six (see Section 16), produced an approximate solution of (1.2); its meridian domain was mapped conformally to the disc, and Newton's method was applied to the Galerkin truncation of (6.1) to the 21 shape coefficients $c_1, c_3, \dots, c_{41}$ and the 504 coefficients of g described above. The resulting binary64 numbers were then frozen and are interpreted as exact dyadic rationals.

10. RECONSTRUCTION OF THE DOMAIN FOR $n = 4$

Let $x^* = (g^*, c^*)$ be the zero of Theorem 9.1, $\psi^* = \psi_{c^*}$, and $\delta c = c^* - c^\circ$. Then $\sum_j \omega_j |\delta c_j| \leq r$, and consequently

$$(10.1) \quad \begin{aligned} |c_1^* - b| &\leq \frac{r}{2b^2\rho}, & \sum_j j\rho^{j-1}|\delta c_j| &\leq \frac{r}{b^2\rho}, \\ \sum_{j \geq 3} |\delta c_j| &\leq \frac{r}{4b^2\rho^3}, & \sum_j j(j-1)|\delta c_j| &\leq \frac{r\rho}{b^2(\rho-1)^2}, \end{aligned}$$

because $j\rho^{j-1}/\omega_j \leq 1/(b^2\rho)$, $1/\omega_j \leq 1/(4b^2\rho^3)$ for $j \geq 3$, and $j(j-1)/\omega_j \leq j\rho^{-j}/b^2 \leq \sum_{i \geq 1} i\rho^{-i}/b^2 = \rho/(b^2(\rho-1)^2)$. In particular ψ^* is holomorphic in $|w| < \rho$.

10.1. Univalence, star-shapedness and the non-disc property.

Proposition 10.1. *The map ψ^* is injective on the disc $\{|w| < 41/40\}$, and $\psi^{*'} has no zeros there. The domain $D = \psi^*(\mathbb{D})$ is bounded by the real-analytic Jordan curve $\psi^*(\partial\mathbb{D})$, it is strictly star-shaped with respect to 0, symmetric under $x_1 \mapsto -x_1$ and $y \mapsto -y$, and it is not a disc.$*

Proof. Let $R = 41/40 < \rho$, $b_- = b - r/(2b^2\rho) \leq c_1^*$, and

$$s_1 = \sum_{j \geq 3} j|c_j^\circ| + \frac{r}{b^2\rho}, \quad s_0 = \sum_{j \geq 3} |c_j^\circ| + \frac{r}{4b^2\rho^3}.$$

By (10.1), $\sum_{j \geq 3} j|c_j^*| \leq s_1$ and $\sum_{j \geq 3} |c_j^*| \leq s_0$. The computation (Section 15) certifies

$$(10.2) \quad \operatorname{Re} \psi^{*'}(w) \geq b_- - \sum_{j \geq 3} j|c_j^\circ|R^{j-1} - \frac{r}{b^2\rho} > 13.081358 \quad (|w| \leq R),$$

$$(10.3) \quad \frac{s_1}{b_-} < 0.104547, \quad \frac{s_1 + s_0}{b_- - s_0} < 0.126890, \quad |c_3^*| \geq |c_3^\circ| - \frac{r}{4b^2\rho^3} > 0.0911958,$$

where in (10.2) we used $j|\delta c_j|R^{j-1} \leq j|\delta c_j|\rho^{j-1}$ and (10.1).

Univalence. For $w_1 \neq w_2$ in the convex disc $|w| < R$, $\frac{\psi^*(w_2) - \psi^*(w_1)}{w_2 - w_1} = \int_0^1 \psi^{*'}(w_1 + \sigma(w_2 - w_1)) d\sigma$ has positive real part by (10.2).

Star-shapedness. For $|w| = 1$ we have $|\psi^*(w) - c_1^*w| \leq s_0$, $|w\psi^{*'}(w) - \psi^*(w)| \leq \sum_{j \geq 3} (j-1)|c_j^*| \leq s_1$ and $|\psi^*(w)| \geq b_- - s_0 > 0$, hence $|w\psi^{*'}/\psi^* - 1| < 0.127$ by (10.3). Therefore

$\frac{d}{d\theta} \arg \psi^*(e^{i\theta}) = \operatorname{Re} (w\psi^{*'}(w)/\psi^*(w)) > 0$. The winding number of $\psi^*(e^{i\theta})$ about 0 equals that of $c_1^*e^{i\theta}$, namely one, since $|\psi^*(w) - c_1^*w| \leq s_0 < c_1^*$ on $\partial\mathbb{D}$. Hence the argument increases by exactly 2π , and ∂D is a radial graph $\{R_D(\phi)e^{i\phi}\}$ with R_D positive and real analytic; D is strictly star-shaped.

Symmetry. ψ^* is odd with real coefficients, so $\psi^*(\bar{w}) = \overline{\psi^*(w)}$ and $\psi^*(-w) = -\psi^*(w)$.

Not a disc. If D were a disc, the two reflection symmetries would force its centre to be 0, and $w \mapsto \psi^*(w)/\psi^{*'}(0)$ would be a conformal map of $\mathbb{D}$ onto a disc centred at 0 fixing 0 with derivative one there; by the Schwarz lemma applied to it and to its inverse, it is the identity, so ψ^* is linear. This contradicts $c_3^* \neq 0$. $\square$

10.2. Convexity.

Proposition 10.2. *For $|w| \leq 1$,*

$$\operatorname{Re} \left(1 + \frac{w\psi^{*''}(w)}{\psi^{*'}(w)} \right) > 0.414074.$$

Consequently D is strictly convex, and $\Omega = \Pi^{-1}(D)$ is convex.

Proof. Let $A_1 = \sum_{j \geq 3} j|c_j^o|$, $A_2 = \sum_{j \geq 3} j(j-1)|c_j^o|$, $E_1 = r/(b^2\rho)$ and $E_2 = r\rho/(b^2(\rho-1)^2)$. By (10.1), for $|w| \leq 1$,

$$|\psi^{*'}(w)| \geq c_1^* - \sum_{j \geq 3} j|c_j^*| \geq b - A_1 - E_1 =: L, \quad |\psi^{*''}(w)| \leq \sum_{j \geq 3} j(j-1)|c_j^*| \leq A_2 + E_2 =: U.$$

The computation certifies $A_1 < 1.5514770$, $A_2 < 7.7860764$, $E_1 < 8.6491 \cdot 10^{-8}$, $E_2 < 3.81425 \cdot 10^{-5}$, hence $L > 13.288565$, $U < 7.786115$ and $\operatorname{Re}(1 + w\psi^{*''}/\psi^{*'}) \geq 1 - U/L > 0.414074$.

Let $z(\theta) = \psi^*(e^{i\theta})$. Then $z'(\theta) = ie^{i\theta}\psi^{*'}(e^{i\theta}) \neq 0$ and

$$\frac{d}{d\theta} \arg z'(\theta) = \operatorname{Re} \left(1 + \frac{w\psi^{*''}(w)}{\psi^{*'}(w)} \right) \Big|_{w=e^{i\theta}} > 0.$$

Thus the tangent direction of the simple closed analytic curve ∂D turns strictly monotonically, through a total angle 2π (the winding number of $\psi^{*'}(e^{i\theta})$ about 0 is zero, as $\psi^{*'}$ has no zeros in $\overline{\mathbb{D}}$), and ∂D bounds a strictly convex domain; this is the classical analytic convexity criterion for conformal maps of the disc, cf. [22, Chapter 2].

Now let $(x_1, x'), (z_1, z') \in \Omega$, $\sigma \in [0, 1]$, and put $y_\sigma = \sigma|x'| + (1-\sigma)|z'|$. By convexity of D , $(\sigma x_1 + (1-\sigma)z_1, \pm y_\sigma) \in D$, and by convexity again the vertical segment joining these two points lies in D . Since $|\sigma x' + (1-\sigma)z'| \leq y_\sigma$, the point $\sigma(x_1, x') + (1-\sigma)(z_1, z')$ lies in Ω . $\square$

10.3. Analytic regularity and continuation.

Lemma 10.3. *Let $U \subset \mathbb{R}^2$ be open and $F \in C(U)$ with $\int_U F(\Delta\varphi + \varphi) = 0$ for all $\varphi \in C_c^\infty(U)$. Then F is real analytic in U .*

Proof. Let $\mathcal{H}(x_1, y, \zeta) = e^\zeta F(x_1, y)$ on $U \times \mathbb{R}$. For $\Phi \in C_c^\infty(U \times \mathbb{R})$, integration by parts in ζ gives $\int \mathcal{H} \Delta_3 \Phi = \int_U F(\Delta\varphi + \varphi) = 0$ with $\varphi = \int e^\zeta \Phi(\cdot, \zeta) d\zeta \in C_c^\infty(U)$. By Weyl's lemma the continuous function $\mathcal{H}$ is harmonic, hence real analytic, and so is $F = \mathcal{H}(\cdot, \cdot, 0)$. $\square$

Lemma 10.4. *Let $D \subset \mathbb{R}^2$ be a bounded domain whose boundary is a real-analytic regular Jordan curve, and let $F \in C^1(\overline{D})$ satisfy $\int_D (\nabla F \cdot \nabla \varphi - F\varphi) = 0$ for all $\varphi \in C_c^\infty(D)$, together with $F = y$ and $\nabla F = e_y$ on ∂D . Then F extends to a real-analytic solution of $\Delta F + F = 0$ on an open neighbourhood of $\overline{D}$.*

Proof. Local Cauchy problem. Fix $z_0 \in \partial D$ and a real-analytic regular parametrisation γ of ∂D near z_0 with $\gamma(0) = z_0$; extend γ holomorphically to a disc $\{|a| < a_0\}$ on which $\gamma' \neq 0$ and γ is injective, and put $\gamma^\dagger(b) = \overline{\gamma(\bar{b})}$. In the complex coordinates $\zeta = x_1 + iy$, $\chi = x_1 - iy$ the Laplacian is $4\partial_\zeta\partial_\chi$, and the substitution $\zeta = \gamma(a)$, $\chi = \gamma^\dagger(b)$ turns $4\partial_\zeta\partial_\chi F + F = 0$ into $\partial_a\partial_b F = -kF$ with $k(a, b) = \gamma'(a)\gamma'^\dagger(b)/4$. The real plane corresponds to $b = \bar{a}$, and ∂D to the real diagonal $a = b \in \mathbb{R}$. The function $F_0(a, b) = (\gamma(a) - \gamma^\dagger(b))/(2i)$ is the holomorphic extension of y and satisfies $\partial_a\partial_b F_0 = 0$. On the closed bidisc $\{|a|, |b| \leq \varepsilon\}$ consider

$$(\mathcal{T}W)(a, b) = - \int_{[b, a]} \int_{[t, b]} k(t, v) (W(t, v) + F_0(t, v)) dv dt$$

along straight segments, which stay in the bidisc by convexity. $\mathcal{T}$ maps the Banach space of functions continuous on the closed bidisc and holomorphic in its interior into itself, and $\|\mathcal{T}W - \mathcal{T}\tilde{W}\|_\infty \leq 2\varepsilon^2 K_0 \|W - \tilde{W}\|_\infty$ with $K_0 = \sup |k|$, since $\int_{[b, a]} |t - b| |dt| = |a - b|^2/2 \leq 2\varepsilon^2$. For $2\varepsilon^2 K_0 < 1$ the contraction mapping principle gives a fixed point W , and differentiation of the iterated integral gives $W_{ab} = -k(W + F_0)$ and $W = W_a = W_b = 0$ on the diagonal $a = b$. Hence $\tilde{F}(x_1, y) = (F_0 + W)(\gamma^{-1}(x_1 + iy), (\gamma^\dagger)^{-1}(x_1 - iy))$ is a real-analytic (possibly complex-valued) solution of $\Delta\tilde{F} + \tilde{F} = 0$ on a disc U about z_0 , with $\tilde{F} = y$ and $\nabla\tilde{F} = e_y$ on $\partial D \cap U$.

Matching. Shrink U so that ∂D divides it into two connected pieces. The function $Z = F - \tilde{F}$ on $D \cap U$, extended by zero to $U \setminus D$, is C^1 on U , because its value and gradient vanish on $\partial D \cap U$. It is a weak solution of $\Delta Z + Z = 0$ on U . Indeed, let $\varphi \in C_c^\infty(U)$, and let $\chi_\delta \in C_c^\infty(D)$ be equal to one at distance at least 2δ from ∂D , with $0 \leq \chi_\delta \leq 1$ and $|\nabla\chi_\delta| \leq C/\delta$. Since F and $\tilde{F}$ are weak solutions in $D \cap U$ and $\chi_\delta\varphi \in C_c^\infty(D \cap U)$,

$$\int_U (\nabla Z \cdot \nabla \varphi - Z\varphi) = \int_{D \cap U} (1 - \chi_\delta)(\nabla Z \cdot \nabla \varphi - Z\varphi) - \int_{D \cap U} \varphi \nabla Z \cdot \nabla \chi_\delta.$$

The integrands are supported in the layer $\Sigma_\delta = \{\text{dist}(\cdot, \partial D) < 2\delta\} \cap D \cap \text{supp } \varphi$, whose area is $O(\delta)$. The first integral is $O(\delta)$, and the second is bounded by $C' \sup_{\Sigma_\delta} |\nabla Z|$, which tends to zero because ∇Z is continuous and vanishes on $\partial D \cap U$. By Lemma 10.3 (applied to the real and imaginary parts) Z is real analytic on U ; it vanishes on the open set $U \setminus \bar{D}$, hence on U . Thus $\tilde{F} = F$ on $D \cap U$.

Gluing. Cover ∂D by finitely many discs $B(z_i, r_i/2)$ such that on $B(z_i, r_i)$ a local extension $\tilde{F}_i$ as above exists, and let $0 < \varepsilon \leq \min_i r_i/6$. For z with $\text{dist}(z, \partial D) < \varepsilon$ choose $q \in \partial D$ with $|z - q| < \varepsilon$ and i with $q \in B(z_i, r_i/2)$, and put $\tilde{F}(z) = \tilde{F}_i(z)$. If another choice q', i' is made, then $|q - q'| < 2\varepsilon$, so $B(q', \varepsilon) \subset B(z_i, r_i) \cap B(z_{i'}, r_{i'})$; both $\tilde{F}_i$ and $\tilde{F}_{i'}$ coincide with F on the nonempty open set $B(q', \varepsilon) \cap D$, hence on the disc $B(q', \varepsilon) \ni z$. Thus $\tilde{F}$ is well defined and real analytic on $\{\text{dist}(\cdot, \partial D) < \varepsilon\}$, and together with F it gives the required extension. It is real valued because it is real on D . $\square$

10.4. Proof of Theorem 1.2 for $n = 4$. Let $V^* = Kg^* + \text{Im } \psi^*$. By Lemma 6.1, $V^* \in C^1(\bar{\mathbb{D}})$ satisfies (3.1) with $\psi = \psi^*$ and has the parities required in Lemma 3.1. By Proposition 10.1, ψ^* is injective with nonvanishing derivative on a neighbourhood of $\bar{\mathbb{D}}$. Lemma 3.1 shows that $F = V^* \circ (\psi^*)^{-1} \in C^1(\bar{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = y$ and $\nabla F = e_y$ on ∂D , odd in y and even in x_1 . By Lemmas 10.3 and 10.4, F extends to a real-analytic solution on an open neighbourhood of $\bar{D}$; replacing this neighbourhood by the connected component containing $\bar{D}$ of its intersection with its images under the two reflections, we obtain a symmetric connected neighbourhood N on which, by the identity theorem, F is still odd in y and even in x_1 .

Since D is strictly star-shaped and symmetric, ∂D meets the axis in exactly two points. Proposition 2.3 now shows that $\Omega = \Pi^{-1}(D)$ and $u = F(x_1, |x'|)/|x'|$ (extended analytically to the axis) satisfy (1.2), that u is real analytic on the neighbourhood $\Pi^{-1}(N)$ of $\bar{\Omega}$, and that u is nonconstant.

Geometry of Ω . Let $R_D(\phi)$ be the radial function of D from Proposition 10.1. It is positive, real analytic, 2π -periodic and even. Its Fourier series $R_D(\phi) = \sum_{k \geq 0} \alpha_k \cos k\phi$ has exponentially decaying coefficients, so $\tilde{R}(\tau) = \sum_k \alpha_k T_k(\tau)$, with T_k the Chebyshev polynomials, converges on a complex neighbourhood of $[-1, 1]$ (since $|T_k| \leq \varrho^k$ on the Bernstein ellipse of parameter ϱ) and defines a real-analytic function with $\tilde{R}(\cos \phi) = R_D(\phi)$, positive near $[-1, 1]$. For $x \neq 0$, $\Pi(x)$ has polar angle $\phi = \arccos(x_1/|x|) \in [0, \pi]$ and modulus $|x|$. Hence

$$\Omega = \{0\} \cup \{x \neq 0 : |x| < \tilde{R}(x_1/|x|)\}, \quad \partial\Omega = \{\tilde{R}(\omega_1)\omega : \omega \in S^3\}.$$

Thus Ω is strictly star-shaped with respect to 0, in particular connected. The map $\omega \mapsto \tilde{R}(\omega_1)\omega$ is real analytic and injective on S^3 (its inverse is $x \mapsto x/|x|$), and its differential maps a tangent vector $v \perp \omega$ to $\tilde{R}'(\omega_1)v_1\omega + \tilde{R}(\omega_1)v$, whose component tangent to S^3 is $\tilde{R}(\omega_1)v \neq 0$. Hence it is a real-analytic embedding, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^3 . The symmetries of Ω follow from $\Omega = \Pi^{-1}(D)$ and the symmetry of D under $x_1 \mapsto -x_1$.

Ω is not a ball. If it were, invariance under $x' \mapsto -x'$ and $x_1 \mapsto -x_1$ would put its centre at 0; then $\Pi(\Omega) = D \cap \{y \geq 0\}$ would be the upper half of a disc centred at 0, and by the symmetry $y \mapsto -y$, D would be that disc, contradicting Proposition 10.1. Finally, Ω is convex by Proposition 10.2. This completes the proof of Theorem 1.2 for $n = 4$. $\square$

11. DIMENSION SIX

In this section we prove Theorem 1.2 for $n = 6$. The argument follows Sections 2–10 step by step. We state explicitly which results of those sections are used verbatim, and why, and we give complete proofs of everything that changes. Points of $\mathbb{R}^6$ are written $x = (x', x'') \in \mathbb{R}^3 \times \mathbb{R}^3$, points of the meridian plane $(y_1, y_2) \in \mathbb{R}^2$, identified with $z = y_1 + iy_2 \in \mathbb{C}$, and

$$\Pi_6 : \mathbb{R}^6 \rightarrow \mathbb{R}^2, \quad \Pi_6(x', x'') = (|x'|, |x''|).$$

The map Π_6 is continuous, its image is the closed quadrant $[0, \infty)^2$, and $|\Pi_6(x)| = |x|$.

11.1. Reduction to a planar problem.

Lemma 11.1. *Let f be a C^2 function on an open subset of $\{y_1 > 0, y_2 > 0\}$ and $F = y_1 y_2 f$. Then*

$$F_{y_1 y_1} + F_{y_2 y_2} = y_1 y_2 \left(f_{y_1 y_1} + f_{y_2 y_2} + \frac{2}{y_1} f_{y_1} + \frac{2}{y_2} f_{y_2} \right).$$

The expression in parentheses, evaluated at $\Pi_6(x)$, is the Laplacian of the function $x \mapsto f(|x'|, |x''|)$ at every point with $x' \neq 0$ and $x'' \neq 0$.

Proof. $(y_1 y_2 f)_{y_1 y_1} = y_2 (y_1 f_{y_1 y_1} + 2f_{y_1})$, and similarly for y_2 . The Laplacian of $\mathbb{R}^3 \times \mathbb{R}^3$ is the sum of the Laplacians of the two factors, each of which acts on functions of the modulus y of its variable as $\partial_y^2 + \frac{2}{y} \partial_y$. $\square$

Analyticity across the two axes is obtained by applying the division argument of Lemma 2.2 in each variable.

Lemma 11.2. *Let $N \subset \mathbb{R}^2$ be open and invariant under $(y_1, y_2) \mapsto (-y_1, y_2)$ and $(y_1, y_2) \mapsto (y_1, -y_2)$, and let F be real analytic on N , odd in y_1 and odd in y_2 . Then there is a real-analytic function G , defined on an open set containing $\{(y_1^2, y_2^2) : (y_1, y_2) \in N\}$, such that*

$F(y_1, y_2) = y_1 y_2 G(y_1^2, y_2^2)$ on N . Consequently $x \mapsto G(|x'|^2, |x''|^2)$ is real analytic on the open set $\Pi_6^{-1}(N)$, and it equals $F(\Pi_6(x))/(|x'| |x''|)$ where $x' \neq 0$ and $x'' \neq 0$.

Proof. Let $a = (a_1, a_2) \in N$. Choose $\delta > 0$, with $\delta < |a_i|$ whenever $a_i \neq 0$, such that $F(y) = \sum_{i,k \geq 0} c_{ik} (y_1 - a_1)^i (y_2 - a_2)^k$ converges absolutely for $|y_1 - a_1|, |y_2 - a_2| < \delta$. For $i = 1, 2$ let $I_i = (-\delta^2, \delta^2)$ if $a_i = 0$, and $I_i = (|a_i| - \delta)^2, (|a_i| + \delta)^2) \subset (0, \infty)$ if $a_i \neq 0$, and let σ_i be the sign of a_i . We define G on the box $I_1 \times I_2$.

- If $a_1 = a_2 = 0$, oddness in both variables gives $c_{ik} = 0$ unless i and k are odd, and we put $G(t_1, t_2) = \sum_{i,k} c_{2i+1, 2k+1} t_1^i t_2^k$.
- If $a_1 = 0 \neq a_2$, oddness in y_1 gives $c_{ik} = 0$ for even i , so $F = y_1 H(y_1^2, y_2)$ with $H(t_1, y_2) = \sum_{i,k} c_{2i+1, k} t_1^i (y_2 - a_2)^k$, and we put $G(t_1, t_2) = H(t_1, \sigma_2 \sqrt{t_2}) / (\sigma_2 \sqrt{t_2})$. The case $a_2 = 0 \neq a_1$ is symmetric.
- If $a_1 a_2 \neq 0$, we put $G(t_1, t_2) = F(\sigma_1 \sqrt{t_1}, \sigma_2 \sqrt{t_2}) / (\sigma_1 \sigma_2 \sqrt{t_1 t_2})$.

In each case G is real analytic on $I_1 \times I_2$, because $\sqrt{t_i}$ is real analytic and positive on I_i when $a_i \neq 0$. By the oddness of F in each variable and the invariance of N , $G(t_1, t_2) = F(\sqrt{t_1}, \sqrt{t_2}) / \sqrt{t_1 t_2}$ at the points of the box with $t_1, t_2 > 0$. The intersection of two such boxes is a product of two intervals, each of which is contained in $(0, \infty)$ or symmetric about 0; if it is nonempty, it is connected and contains points with $t_1, t_2 > 0$, at which the two definitions agree. By the identity theorem they agree on the whole intersection, and G is well defined. It satisfies $F = y_1 y_2 G(y_1^2, y_2^2)$ near every point of N . The last assertion follows because $x \mapsto (|x'|^2, |x''|^2)$ is polynomial. $\square$

Proposition 11.3 (From the plane to $\mathbb{R}^6$). *Let $D \subset \mathbb{R}^2$ be a bounded domain, invariant under $(y_1, y_2) \mapsto (-y_1, y_2)$ and $(y_1, y_2) \mapsto (y_1, -y_2)$, whose boundary is a Jordan curve meeting each coordinate axis in exactly two points. Let F be real analytic on an open set $N \supset \overline{D}$ invariant under the same two reflections, odd in y_1 and in y_2 , and such that*

$$(11.1) \quad \Delta F + F = 0 \text{ in } D, \quad F = y_1 y_2 \text{ and } \nabla F = (y_2, y_1) \text{ on } \partial D.$$

Put $\Omega = \Pi_6^{-1}(D)$ and let u be the function $x \mapsto G(|x'|^2, |x''|^2)$ of Lemma 11.2. Then Ω is a bounded open set, u is real analytic on the neighbourhood $\Pi_6^{-1}(N)$ of $\overline{\Omega}$, u is nonconstant, and (1.2) holds.

Proof. We follow the proof of Proposition 2.3. The set Ω is open and bounded because Π_6 is continuous and proper; $\overline{\Omega} \subset \Pi_6^{-1}(\overline{D}) \subset \Pi_6^{-1}(N)$ and $\partial\Omega \subset \Pi_6^{-1}(\partial D)$. Let $E = \{x : x' \neq 0, x'' \neq 0\}$; its complement is the union of two three-dimensional subspaces. On $\Omega \cap E$ we have $u = f \circ \Pi_6$ with $f = F/(y_1 y_2)$, and Lemma 11.1 gives $|x'| |x''| (\Delta u + u)(x) = (\Delta F + F)(\Pi_6(x)) = 0$. Since $\Omega \cap E$ is dense in Ω and $\Delta u + u$ is continuous, $\Delta u + u = 0$ in Ω .

Let $x \in \partial\Omega \cap E$. Then $\Pi_6(x) \in \partial D$ lies in the open quadrant, and $f = F/(y_1 y_2) = 1$ there. From $F_{y_1} = y_2 f + y_1 y_2 f_{y_1} = y_2$ and the analogous identity for y_2 we get $f_{y_1} = f_{y_2} = 0$, hence $\nabla u(x) = (f_{y_1} x' / |x'|, f_{y_2} x'' / |x''|) = 0$.

It remains to show that $\partial\Omega \cap E$ is dense in $\partial\Omega$. By the reflection symmetries, the two points of ∂D on the axis $\{y_1 = 0\}$ are $(0, \pm a)$ with $a > 0$. The reflection $y_1 \mapsto -y_1$ fixes them and maps $\{y_1 > 0\}$ to $\{y_1 < 0\}$, so it exchanges the two arcs of ∂D between them, and one of these arcs lies in $\{y_1 > 0\}$. Hence $(0, a)$ is a limit of points of ∂D in the open quadrant; the same holds for the points of ∂D on the other axis. Every point of $\Pi_6^{-1}(\partial D)$ lies outside Ω and is a limit of points of Ω , so $\Pi_6^{-1}(\partial D) = \partial\Omega$, and lifting the above sequences by Π_6 shows that $\partial\Omega \cap E$ is dense in $\partial\Omega$. By continuity $u = 1$ and $\nabla u = 0$ on $\partial\Omega$. Finally, a constant solution of $\Delta u + u = 0$ vanishes identically. $\square$

As in Proposition 2.4, averaging over $O(3) \times O(3)$ shows that the reduction is reversible in the analytic class. Let $y_1 + iy_2 = re^{i\phi}$. Since $y_1 y_2 = \frac{1}{2} r^2 \sin 2\phi$ and $\sin m\phi = \sin 2\phi U_{m/2-1}(\cos 2\phi)$ for even m , the planar Helmholtz solution $F = J_m(r) \sin m\phi$, with $m \equiv 2 \pmod{4}$, corresponds to the six-dimensional Helmholtz solution

$$u = 2r^{-2} J_m(r) U_{m/2-1}(\cos 2\phi), \quad \cos 2\phi = \frac{|x'|^2 - |x''|^2}{|x|^2}.$$

For $m = 2$ this is the radial solution $2r^{-2} J_2(r)$, whose derivative is $-2r^{-2} J_3(r)$; the Schiffer balls of $\mathbb{R}^6$ at eigenvalue one have radii $j_{3,k}$, the positive zeros of J_3 .

11.2. Symmetry sector and conformal formulation. For a real sequence $c = (c_j)_{j \equiv 1 \pmod{4}}$ put $\psi_c(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$. Since $i^j = i$ for $j \equiv 1 \pmod{4}$,

$$\psi_c(\bar{w}) = \overline{\psi_c(w)}, \quad \psi_c(-\bar{w}) = -\overline{\psi_c(w)}, \quad \psi_c(i\bar{w}) = i \overline{\psi_c(w)}.$$

Hence, if ψ_c is injective on $\mathbb{D}$, then $D = \psi_c(\mathbb{D})$ is invariant under $z \mapsto \bar{z}$, $z \mapsto -\bar{z}$ and $z \mapsto i\bar{z}$, that is, under $y_2 \mapsto -y_2$, $y_1 \mapsto -y_1$ and the exchange $(y_1, y_2) \mapsto (y_2, y_1)$.

Lemma 11.4. *Let ψ be holomorphic and injective on a neighbourhood of $\bar{\mathbb{D}}$, with $\psi' \neq 0$ there, let $D = \psi(\mathbb{D})$ and $h_\psi = \text{Im}(\psi^2)/2$. Let $V \in C^1(\bar{\mathbb{D}})$ be real valued and satisfy*

$$(11.2) \quad \Delta V + |\psi'|^2 V = 0 \text{ in } \mathcal{D}'(\mathbb{D}), \quad V - h_\psi = 0 \text{ and } \nabla(V - h_\psi) = 0 \text{ on } \partial\mathbb{D}.$$

Then $F = V \circ \psi^{-1} \in C^1(\bar{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = y_1 y_2$, $\nabla F = (y_2, y_1)$ on ∂D . If moreover $\psi = \psi_c$ as above and $V(\bar{w}) = -V(w)$, $V(-\bar{w}) = -V(w)$, $V(i\bar{w}) = V(w)$, then F is odd in y_1 and in y_2 and invariant under $(y_1, y_2) \mapsto (y_2, y_1)$.

Proof. Since $y_1 y_2 = \text{Im}(z^2)/2$, we have $h_\psi = (y_1 y_2) \circ \psi$ and $V - h_\psi = (F - y_1 y_2) \circ \psi$. With this change, the proof of Lemma 3.1 applies verbatim, and the parities of F follow from those of V and the displayed symmetries of ψ_c . $\square$

For $n \equiv 2 \pmod{4}$, $n \geq 2$, and $s \geq 0$ let $S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta$. Fix $\rho = 11/10$ in this section, and let $\mathcal{S}_6$ be the real Banach space of $f = \sum_{n \equiv 2 \pmod{4}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$, which is also its norm in $\mathcal{B}_\rho$. Since $\Phi_{m,s}(\bar{w}) = \Phi_{-m,s}(w)$, $\Phi_{m,s}(-\bar{w}) = (-1)^m \Phi_{-m,s}(w)$ and $\Phi_{m,s}(i\bar{w}) = i^m \Phi_{-m,s}(w)$, the real-valued elements of $\mathcal{B}_\rho$ with the three parities of Lemma 11.4 are exactly the elements of $\mathcal{S}_6$: the first two parities force m to be even, and the third one forces $i^m = -1$. Let $\mathcal{G}_6 = \{g \in \mathcal{S}_6 : g_{n,0} = 0 \text{ for all } n\}$, and define K on $\mathcal{G}_6$ by (5.1), now for $n \equiv 2 \pmod{4}$.

Lemma 11.5 (Results used verbatim). *The following statements hold, with the proofs given in the previous sections.*

- (a) *Lemmas 4.1, 4.2, 4.3 and 4.4, and the bounds (4.4). They are stated for all $m \in \mathbb{Z}$ and $s \geq 0$.*
- (b) *Lemma 5.1(a)–(c), for every $n \geq 1$ and $s \geq 1$. Its proof uses only the explicit coefficients $a_{s,k}^{(n)}$ of (4.2) and Lemma 4.1(a), and never the parity of n . The absolute column sum of (5.1) is $1/(q(q+2))$ with $q = n + 2s$, as in Lemma 5.1(d); in the present sector $q \geq 4$, so $K: \mathcal{G}_6 \rightarrow \mathcal{S}_6$ is bounded with $\|K\| = 1/24$, attained at $(n, s) = (2, 1)$.*
- (c) *Lemma 5.2 and Remark 5.3, with $\mathcal{G}_6$, $\mathcal{S}_6$ and the harmonic polynomials $r^n \sin n\theta$, $n \equiv 2 \pmod{4}$ (and with $1/24$ in place of $1/15$ in the uniform bound).*
- (d) *Lemma 7.2, Proposition 7.1 in the quartic form of Proposition 11.8 below, and Lemma 7.5 for the operator A defined in (11.13) below; the proof of Lemma 7.5 uses only the block structure of A .*

- (e) Lemma 8.3, which is stated for all $m, t, N \geq 0$, and the connection formulas of Section 8 for the Jacobi polynomials $P_k^{(1,n)}$ and $P_k^{(2,n)}$, which hold for every $n \geq 0$.
- (f) Lemma 10.3, and Lemma 10.4 with the Cauchy data (y, e_y) replaced by $(y_1 y_2, (y_2, y_1))$. The only property of the datum used in the proof of Lemma 10.4 is that it is a harmonic polynomial: in the coordinates $\zeta = y_1 + iy_2$, $\chi = y_1 - iy_2$ one has $y_1 y_2 = (\zeta^2 - \chi^2)/(4i)$, whose holomorphic extension $F_0(a, b) = (\gamma(a)^2 - \gamma^\dagger(b)^2)/(4i)$ again satisfies $\partial_a \partial_b F_0 = 0$, and the rest of the proof is unchanged.

11.3. The quartic equation. In the rest of this section $\rho = 11/10$, $M = 58$ and $S = 30$. The centre $x^\circ = (g^\circ, c^\circ)$ consists of 450 coefficients $g_{n,s}^\circ$, $n \in \{2, 6, \dots, 58\}$, $1 \leq s \leq 30$, and 15 coefficients c_j° , $j \in \{1, 5, \dots, 57\}$; all other coefficients of x° vanish. Each of these 465 numbers is an explicitly given dyadic rational (a binary64 number interpreted exactly), listed in the ancillary file `center_r6.json`. We write $b = c_1^\circ \approx 28.864751$, $\alpha = b^3/2$ and

$$\omega_j = \alpha(j+2)\rho^{j+1} \quad (j \equiv 1 \pmod{4}).$$

Let X_6 be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}_6$ and $c = (c_j)_{j \equiv 1 \pmod{4}}$ real, normed by $\|(g, c)\|_X = \|g\|_\rho + \|c\|$, $\|c\| = \sum_j \omega_j |c_j|$. With

$$\theta_0 = \frac{1}{3\alpha\rho}, \quad \theta_1 = \frac{1}{\alpha\rho^2},$$

we have $N_\rho(\psi_c) \leq \theta_0 \|c\|$ and $N_\rho(\psi'_c) \leq \theta_1 \|c\|$, because $\rho^j/\omega_j = 1/(\alpha(j+2)\rho)$ and $j\rho^{j-1}/\omega_j = j/(\alpha(j+2)\rho^2)$. In particular ψ_c is holomorphic in $|w| < \rho$ and continuous up to $|w| = \rho$. Put

$$h_c = \frac{1}{2} \operatorname{Im}(\psi_c^2) = \frac{1}{2} \sum_{j,k} c_j c_k S_{j+k,0} \in \mathcal{S}_6$$

(note that $j+k \equiv 2 \pmod{4}$), and define

$$(11.3) \quad \mathcal{F}_6(g, c) = g + |\psi'_c|^2 (Kg + h_c).$$

Lemma 11.6. $\mathcal{F}_6$ is a continuous polynomial map of degree four from X_6 to $\mathcal{S}_6$. If $\mathcal{F}_6(g, c) = 0$, then $V = Kg + h_c$ belongs to $C^1(\overline{\mathbb{D}})$, has the three parities of Lemma 11.4, and satisfies (11.2) with $\psi = \psi_c$.

Proof. $\psi'_c = \sum_j j c_j w^{j-1}$ has real coefficients and only powers divisible by four. Hence $|\psi'_c|^2$ is real and invariant under $w \mapsto \bar{w}$ and $w \mapsto iw$, multiplication by it preserves the three parities, and by (4.4) it maps $\mathcal{S}_6$ into $\mathcal{S}_6$ with norm at most $N_\rho(\psi'_c)^2$. Since $w^k = \Phi_{k,0}$ has norm ρ^k and $\|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$, we have $\|h_c\|_\rho \leq N_\rho(\psi_c)^2/2$. Together with $\|K\| = 1/24$, this shows that (11.3) is a sum of bounded multilinear maps of degree at most four. The rest of the proof is that of Lemma 6.1, using Lemma 11.5(c) and the fact that h_c is harmonic. $\square$

11.4. Expansion at the centre. Write $p = \psi'_{c^\circ} = \sum_{a \in \{0,4,\dots,56\}} p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, so $p_0 = b$ and $d = \deg p = 56$, and put

$$P = N_\rho(p), \quad C = N_\rho(\psi_{c^\circ}), \quad V^\circ = Kg^\circ + h_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X_6$ let

$$L(\eta) = 2 \operatorname{Re}(\bar{p} \psi'_\eta), \quad Q(\eta_1, \eta_2) = \operatorname{Re}(\psi'_{\eta_1} \overline{\psi'_{\eta_2}}), \quad \ell(\eta) = \operatorname{Im}(\psi_{c^\circ} \psi_\eta), \quad I(\eta_1, \eta_2) = \operatorname{Im}(\psi_{\eta_1} \psi_{\eta_2}),$$

and $\delta V = K\delta g + \ell(\eta)$. Since $|p + \psi'_\eta|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$ and $h_{c^\circ + \eta} = h_{c^\circ} + \ell(\eta) + \frac{1}{2}I(\eta, \eta)$, expanding (11.3) gives

$$(11.4) \quad \mathcal{F}_6(x^\circ + h) = \mathcal{F}_6(x^\circ) + D\mathcal{F}_6(x^\circ)h + B_2(h, h) + B_3(h, h, h) + B_4(h, h, h, h),$$

where

$$(11.5) \quad \begin{aligned} D\mathcal{F}_6(x^\circ)h &= \delta g + |p|^2(K\delta g + \ell(\eta)) + L(\eta)V^\circ, \\ B_2(h, h) &= \frac{1}{2}|p|^2 I(\eta, \eta) + L(\eta)\delta V + Q(\eta, \eta)V^\circ, \\ B_3(h, h, h) &= \frac{1}{2}L(\eta)I(\eta, \eta) + Q(\eta, \eta)\delta V, \quad B_4(h, h, h, h) = \frac{1}{2}Q(\eta, \eta)I(\eta, \eta), \end{aligned}$$

and B_2, B_3, B_4 denote the symmetric multilinear maps with these diagonals.

Lemma 11.7. *Let $\kappa = 1/24$. For all $h_i \in X_6$ and $k = 2, 3, 4$, $\|B_k(h_1, \dots, h_k)\|_\rho \leq c_k \|h_1\|_X \cdots \|h_k\|_X$, where*

$$c_2 = \max\{2P\theta_1\kappa, 2P\theta_1 C\theta_0 + \frac{1}{2}P^2\theta_0^2 + \theta_1^2 V_0\}, \quad c_3 = \max\{\theta_1^2\kappa, P\theta_1\theta_0^2 + \theta_1^2 C\theta_0\}, \quad c_4 = \frac{1}{2}\theta_1^2\theta_0^2.$$

Proof. Write $a_i = \|\delta g_i\|_\rho$ and $e_i = \|\eta_i\|$. As in the proof of Lemma 6.2, $N_\rho(\psi'_{\eta_i}) \leq \theta_1 e_i$, $N_\rho(\psi_{\eta_i}) \leq \theta_0 e_i$, $\|\ell(\eta_i)\|_\rho \leq C\theta_0 e_i$, $\|I(\eta_i, \eta_k)\|_\rho \leq \theta_0^2 e_i e_k$, $\|\delta V_i\|_\rho \leq \kappa a_i + C\theta_0 e_i$, $\|L(\eta_i)f\|_\rho \leq 2P\theta_1 e_i \|f\|_\rho$ and $\|Q(\eta_i, \eta_k)f\|_\rho \leq \theta_1^2 e_i e_k \|f\|_\rho$. Hence

$$\|B_2(h_1, h_2)\|_\rho \leq P\theta_1\kappa(e_1 a_2 + e_2 a_1) + (\frac{1}{2}P^2\theta_0^2 + 2P\theta_1 C\theta_0 + \theta_1^2 V_0)e_1 e_2 \leq c_2(a_1 + e_1)(a_2 + e_2).$$

In the bound for B_3 , every monomial $e_i e_k a_l$ has coefficient $\theta_1^2\kappa/3$ and $e_1 e_2 e_3$ has coefficient $P\theta_1\theta_0^2 + \theta_1^2 C\theta_0$; in the bound for B_4 , $e_1 e_2 e_3 e_4$ has coefficient $\frac{1}{2}\theta_1^2\theta_0^2$. Each is at most its coefficient in $c_k \prod_i (a_i + e_i)$. $\square$

Proposition 11.8. *Let $\mathcal{F}_6: X_6 \rightarrow \mathcal{S}_6$ have the form (11.4) with symmetric bounded B_2, B_3, B_4 , and let $A: \mathcal{S}_6 \rightarrow X_6$ be bounded, linear and injective. Suppose that $\|A\mathcal{F}_6(x^\circ)\|_X \leq Y$, $\|I - A D\mathcal{F}_6(x^\circ)\|_{X \rightarrow X} \leq Z$ and $\|A\| c_k \leq C_k$ for $k = 2, 3, 4$, and that $r > 0$ satisfies*

$$(11.6) \quad Y + (Z - 1)r + C_2 r^2 + C_3 r^3 + C_4 r^4 < 0, \quad Z - 1 + 2C_2 r + 3C_3 r^2 + 4C_4 r^3 < 0.$$

Then $\mathcal{F}_6$ has exactly one zero in the closed ball $\overline{B}_{X_6}(x^\circ, r)$.

Proof. This is the proof of Proposition 7.1, with one more term: the difference $B_4(h_1, \dots, h_1) - B_4(h_2, \dots, h_2)$ is a sum of four terms, each of which contains $h_1 - h_2$ once and three factors h_1 or h_2 , so that the norm of its image under A is at most $4C_4 r^3 \|h_1 - h_2\|_X$. $\square$

11.5. The principal part of the linearisation. In the model on high shape modes the map is replaced by the disc $\psi = bw$ and g by 0. We need the following radial expansion; here $S_{m,s} = r^m P_s^{(0,m)}(2r^2 - 1) \sin m\theta$ for any integer $m \geq 0$.

Lemma 11.9. *For every integer $m \geq 0$,*

$$r^{m+4} \sin m\theta = \frac{m+1}{m+3} S_{m,0} + \frac{2}{m+4} S_{m,1} + \frac{2}{(m+3)(m+4)} S_{m,2}.$$

Proof. Dividing by $r^m \sin m\theta$ and writing $t = r^2 = 1 + u$, the claim is $t^2 = \frac{m+1}{m+3} P_0 + \frac{2}{m+4} P_1 + \frac{2}{(m+3)(m+4)} P_2$ with $P_s = P_s^{(0,m)}(2t - 1)$. By (4.2), $P_0 = 1$, $P_1 = 1 + (m+2)u$ and $P_2 = 1 + 2(m+3)u + \frac{1}{2}(m+3)(m+4)u^2$. The right-hand side is therefore $c_0 + c_1 u + c_2 u^2$ with $c_2 = 1$, $c_1 = \frac{2(m+2)}{m+4} + \frac{4}{m+4} = 2$ and $c_0 = \frac{(m+1)(m+4)+2(m+3)+2}{(m+3)(m+4)} = 1$, which is $(1+u)^2 = t^2$. $\square$

Proposition 11.10. *For every $x = (g, c) \in X_6$ and every $j \equiv 1 \pmod{4}$,*

$$(11.7) \quad D\mathcal{F}_6(x)e_j = 2 \operatorname{Re}(jw^{j-1}\overline{\psi'_c})(Kg + h_c) + |\psi'_c|^2 \operatorname{Im}(\psi_c w^j).$$

Let $x_b = (0, b e_1)$, so that $\psi_{x_b}(w) = bw$. For every $j \equiv 1 \pmod{4}$ with $j \geq 5$,

$$(11.8) \quad T_j := D\mathcal{F}_6(x_b)e_j = b^3 \left\{ \frac{j+2}{2} S_{j+1,0} - \frac{j-2}{2} S_{j-3,0} - \frac{j}{j+1} S_{j-3,1} - \frac{1}{j+1} S_{j-3,2} \right\}.$$

Proof. Replacing c by $c + \varepsilon e_j$ changes ψ_c by εw^j ; since $\partial_\varepsilon |\psi'_c + \varepsilon j w^{j-1}|^2 = 2 \operatorname{Re}(j w^{j-1} \overline{\psi'_c})$ and $\partial_\varepsilon \frac{1}{2} \operatorname{Im}((\psi_c + \varepsilon w^j)^2) = \operatorname{Im}(\psi_c w^j)$ at $\varepsilon = 0$, we obtain (11.7). At x_b we have $g = 0$, $\psi'_c = b$ and $h_c = \frac{1}{2} b^2 \operatorname{Im}(w^2)$, so

$$T_j = b^3 [j \operatorname{Re}(w^{j-1}) \operatorname{Im}(w^2) + \operatorname{Im}(w^{j+1})] = b^3 [j r^{j+1} \cos((j-1)\theta) \sin 2\theta + r^{j+1} \sin((j+1)\theta)].$$

Since $2 \cos((j-1)\theta) \sin 2\theta = \sin((j+1)\theta) - \sin((j-3)\theta)$,

$$T_j = b^3 \left[\frac{j+2}{2} r^{j+1} \sin((j+1)\theta) - \frac{j}{2} r^{j+1} \sin((j-3)\theta) \right].$$

The first term is $\frac{j+2}{2} S_{j+1,0}$. Lemma 11.9 with $m = j-3 \geq 2$ gives

$$-\frac{j}{2} r^{j+1} \sin((j-3)\theta) = -\frac{j}{2} \left[\frac{j-2}{j} S_{j-3,0} + \frac{2}{j+1} S_{j-3,1} + \frac{2}{j(j+1)} S_{j-3,2} \right],$$

which is the rest of (11.8). $\square$

Both frequencies $j+1$ and $j-3$ in (11.8) are $\equiv 2 \pmod{4}$, so $T_j \in \mathcal{S}_6$. The harmonic part of T_j is $\alpha(j+2)$ in row $(j+1, 0)$ and $-\alpha(j-2)$ in row $(j-3, 0)$; this replaces the upper bidiagonal structure of Lemma 7.3 by a difference operator of step four.

11.6. Tail model and approximate inverse. The *finite inputs* are the 450 coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and the 15 shape coordinates c_j with $j \leq M-1$. The *finite rows* are the 450 rows (n, s) with $n \leq M$, $1 \leq s \leq S$, and the 15 harmonic rows $(n, 0)$ with $n \leq M$. All other inputs and rows are *tail* inputs and rows, and we write $X_6 = X_f \oplus X_t$, $\mathcal{S}_6 = \mathcal{S}_f \oplus \mathcal{S}_t$; $\dim X_f = \dim \mathcal{S}_f = 465$. Let M_f be the 465×465 matrix of the restriction of $D\mathcal{F}_6(x^\circ)$ to finite inputs and finite rows. On tail inputs we use the model

$$(11.9) \quad T(\delta g, \eta) = \delta g + \sum_{j \geq M+3} \eta_j T_j.$$

Write $Te = (Be, T_{tt}e)$ for tail inputs e , with B the part in finite rows. By (11.8) with $j = M+3 = 61$, only the shape input $j = M+3$ reaches finite rows:

$$(11.10) \quad B(\delta g, \eta) = -\eta_{M+3} \left(\alpha(M+1) e_{(M,0)} + \frac{b^3(M+3)}{M+4} e_{(M,1)} + \frac{b^3}{M+4} e_{(M,2)} \right).$$

Put $\tau_4 = (1 - \rho^{-4})^{-1}$.

Lemma 11.11. $T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection with $\|T_{tt}\| \leq 1 + \rho^{-4}$. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is given by

$$(11.11) \quad \eta_{n-1} = \sum_{k \geq 0} \frac{y_{n+4k,0}}{\alpha(n+4k+1)} \quad (n \geq M+4, n \equiv 2 \pmod{4}),$$

$$\delta g_{n,1} = y_{n,1} + \frac{b^3(n+3)}{n+4} \eta_{n+3}, \quad \delta g_{n,2} = y_{n,2} + \frac{b^3}{n+4} \eta_{n+3} \quad (n \geq M+4),$$

and $\delta g_{n,s} = y_{n,s}$ for all other tail g -inputs.

Proof. By (11.9) and (11.8), the tail harmonic rows read $\alpha(n+1)(\eta_{n-1} - \eta_{n+3}) = y_{n,0}$ for $n \geq M+4$, the tail rows $(n, 1)$ and $(n, 2)$ with $n \geq M+4$ read $\delta g_{n,1} - \frac{b^3(n+3)}{n+4} \eta_{n+3} = y_{n,1}$ and $\delta g_{n,2} - \frac{b^3}{n+4} \eta_{n+3} = y_{n,2}$, and all other tail rows read $\delta g_{n,s} = y_{n,s}$. Since $\sum \omega_j |\eta_j| < \infty$ forces $\eta_j \rightarrow 0$, summation of the harmonic equations gives (11.11), and a solution with $y = 0$ vanishes; conversely (11.11) solves the equations whenever the series converge, which is the case for every $y \in \mathcal{S}_t$ by the proof of Lemma 11.12 below (with $H = 0$). For $j \geq M+7$ the column $T_{tt}e_j$ has the four entries $\alpha(j+2)$, $-\alpha(j-2)$, $-b^3 j/(j+1)$ and $-b^3/(j+1)$ in

the rows $(j+1, 0)$, $(j-3, 0)$, $(j-3, 1)$ and $(j-3, 2)$; its weighted norm divided by ω_j is $1 + j\rho^{-4}/(j+2) \leq 1 + \rho^{-4}$. For $j = M+3$ only the first entry lies in a tail row, and tail g -inputs are mapped to the identical rows. Lemma 7.2 gives the bound for $\|T_{tt}\|$. $\square$

Let $\hat{A}$ be a 465×465 matrix with dyadic rational entries, regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, such that

$$(11.12) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1,$$

and define

$$(11.13) \quad A = \begin{pmatrix} \hat{A} & -\hat{A}BT_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix} : \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X_f \oplus X_t.$$

By Lemma 7.5 (Lemma 11.5(d)), A is bounded and injective, and $ATe = e$ for every tail input e . In the computation, $\hat{A}$ is a fixed binary64 approximation of M_f^{-1} , stored in the ancillary file `inverse_r6_M58_S30.npy`, whose entries are interpreted as exact dyadic rationals; only (11.12) is used.

Lemma 11.12. *Let*

$$\|R\| = \max_{(n,s) \text{ finite row}} \frac{\|\hat{A}e_{(n,s)}\|_X}{\rho^n},$$

$$H = \left\| \hat{A} \left(\alpha(M+1)e_{(M,0)} + \frac{b^3(M+3)}{M+4}e_{(M,1)} + \frac{b^3}{M+4}e_{(M,2)} \right) \right\|_X,$$

and, for $n \geq M+4$ with $n \equiv 2 \pmod{4}$,

$$(11.14) \quad h_n = \frac{1}{(n+1)\rho^n} \left(\sum_{\substack{M+4 \leq k \leq n \\ k \equiv 2(4)}} (k+1)\rho^k + 2 \sum_{\substack{M+4 \leq k \leq n-4 \\ k \equiv 2(4)}} \rho^k + \frac{H}{\alpha} \right).$$

Let $N_H \geq M+4$ with $N_H \equiv 2 \pmod{4}$. Then A maps every residual supported in tail rows with norm at most

$$A_t := \max \left\{ 1, \max_{M+4 \leq n \leq N_H} h_n, \tau_4 + \frac{2\tau_4\rho^{-4}}{N_H+5} + \frac{H}{\alpha(N_H+5)\rho^{N_H+4}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. We use Lemma 7.2. A finite row $e_{(n,s)}$ is mapped to $\hat{A}e_{(n,s)}$. A nonharmonic tail row is mapped to the identical tail input by Lemma 11.11, and B vanishes on it. The normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to the shape coordinates $\eta_{k-1} = \rho^{-n}/(\alpha(n+1))$, $M+4 \leq k \leq n$, of weights $\omega_{k-1} = \alpha(k+1)\rho^k$; to the coordinates $\delta g_{k,1}$ and $\delta g_{k,2}$, $M+4 \leq k \leq n-4$, of weight ρ^k and total absolute value $b^3\rho^{-n}/(\alpha(n+1)) = 2\rho^{-n}/(n+1)$; and, by (11.10) and (11.13), to the finite part $\eta_{M+3}\hat{A}(\alpha(M+1)e_{(M,0)} + \dots)$, of norm $H\rho^{-n}/(\alpha(n+1))$. The sum of the three norms is exactly h_n . For $n \geq N_H+4$ we bound $(k+1)/(n+1) \leq 1$ in the first sum and sum the geometric series $\sum_{i \geq 0} \rho^{-4i} = \tau_4$; the second term is at most $2\tau_4\rho^{-4}/(n+1)$, and the third one decreases in n . $\square$

11.7. The defect bounds. We bound $\|I - A\mathcal{F}_6(x^\circ)\|$ as in Section 8, by partitioning the inputs into five classes; every input belongs to exactly one class, and no column is estimated by sampling.

The residual and the finite columns. $\mathcal{F}_6(x^\circ)$ is a polynomial in $w, \bar{w}$; its expansion in the basis $S_{n,s}$ (it has 2601 coefficients) is computed without truncation in interval arithmetic, using (4.3) and (5.1). Only finitely many tail rows occur, so $A\mathcal{F}_6(x^\circ)$ is computed without truncation

from (11.13), (11.10) and Lemma 11.11, and Y is an upper bound of its norm. For each of the 465 finite inputs e , $(I - AD\mathcal{F}_6(x^\circ))e$ is computed in the same way, and Z_f bounds the maximum of the normalised norms of these columns.

Tail g -columns.

Lemma 11.13. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{F}_6(x^\circ))e = -A(|p|^2 K S_{n,s})$. Let $p = p_{\leq 24} + p_{> 24}$ be the splitting of p into terms of degree ≤ 24 and > 24 , and $P_s = N_\rho(p_{\leq 24})$, $P_t = N_\rho(p_{> 24})$, $\delta_a = 2P_s P_t + P_t^2$. Then*

$$(11.15) \quad \frac{\|(I - AD\mathcal{F}_6(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq 24}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| \delta_a}{q(q+2)}.$$

Moreover, the right-hand side is at most $(A_t P_s^2 + \|A\| \delta_a)/(88 \cdot 90)$ if $n \geq M + 28 = 86$, and at most $(P_s^2 + \|A\| \delta_a)/(114 \cdot 116)$ if $s \geq S + 26 = 56$.

Proof. By (11.5), $D\mathcal{F}_6(x^\circ)e = S_{n,s} + |p|^2 K S_{n,s}$, and $Te = S_{n,s}$, so $AS_{n,s} = e$. The splitting is estimated as in Lemma 8.1. The function $|p_{\leq 24}|^2 = \sum_{a,a' \leq 24} p_a p_{a'} w^a \bar{w}^{a'}$ changes angular frequencies by at most 24 in modulus, and $\||p_{\leq 24}|^2 K S_{n,s}\|_\rho \leq P_s^2 \rho^n / (q(q+2))$. If $n \geq 86$, all components of $|p_{\leq 24}|^2 K S_{n,s}$ have frequencies of modulus at least $n - 24 \geq 62 > M$, so they lie in tail rows, on which A has norm at most A_t (Lemma 11.12); and $q \geq 88$. If $s \geq 56$, then by (5.1) and Lemma 4.3 all components have radial index at least $s - 1 - 24 \geq 31 > S$, so they lie in nonharmonic tail rows, on which A is the identity; and $q \geq 2 + 2 \cdot 56 = 114$. $\square$

The tail g -inputs with $n \leq 82$ and $s \leq 55$ are the $21 \cdot 55 - 450 = 705$ *near* ones; for them the first term of (11.15) is computed without truncation, and $Z_{g,\partial}$ bounds the maximum of the right-hand side. The remaining tail g -inputs are covered by the last statement of Lemma 11.13; let $Z_{g,\infty}$ be the larger of the two bounds.

Near shape columns. For the 85 shape inputs e_j with $M + 3 = 61 \leq j \leq 397$, we have $(I - AD\mathcal{F}_6(x^\circ))e_j = -A\Delta_j$ with $\Delta_j = D\mathcal{F}_6(x^\circ)e_j - T_j$, because $ATe_j = e_j$; Δ_j is a polynomial, and these columns are computed without truncation. Let $Z_{\text{sh},\partial}$ bound their normalised norms.

Far shape columns. Let $W^\circ = Kg^\circ$ and $h^\circ = h_{c^\circ}$. By (11.7) at x° and at x_b , for every $j \equiv 1 \pmod{4}$, $j \geq 5$,

$$(11.16) \quad \Delta_j = 2 \operatorname{Re}(jw^{j-1}\Gamma_1) + \operatorname{Im}(w^j\Gamma_0) + 2 \operatorname{Re}(jw^{j-1}\bar{p}W^\circ),$$

where

$$\Gamma_1 = \bar{p}h^\circ - \frac{1}{2}b^3 \operatorname{Im}(w^2), \quad \Gamma_0 = |p|^2 \psi_{c^\circ} - b^3 w.$$

By Lemma 5.1(c) (Lemma 11.5(b)), $W^\circ = (1 - |w|^2)^2 Q^\circ$ with

$$Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)} (2r^2 - 1) \sin n\theta,$$

and the connection formulas of Section 8 give the finite expansion of Q° in the basis $S_{n,t}$; its connection coefficients are positive rational numbers. Let $\Lambda = M + d = 114$. The function $w^\Lambda \bar{p}Q^\circ$ has only nonnegative frequencies; write $w^\Lambda \bar{p}Q^\circ = \sum_{m \geq 0, t \geq 0} \gamma_{m,t}^W \Phi_{m,t}$ (a finite sum, computed in interval arithmetic). Let $j_* = 401$ and

$$U_\Gamma = \frac{1}{\alpha} \left(\frac{2\|\Gamma_1\|_\rho}{\rho^2} + \frac{\|\Gamma_0\|_\rho}{\rho(j_* + 2)} \right), \quad U_W = \frac{8}{\alpha(j_* - \Lambda)^2} \sum_{m,t} |\gamma_{m,t}^W| (2t+1)(2t+3) \rho^{m-\Lambda-2},$$

where $\|\cdot\|_\rho$ is the norm of $\mathcal{B}_\rho$.

Proposition 11.14. *For every $j \equiv 1 \pmod{4}$ with $j \geq j_*$,*

$$\frac{\|(I - AD\mathcal{F}_6(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := A_t(\mathbf{U}_\Gamma + \mathbf{U}_W).$$

Proof. The frequencies of $\bar{p}$ lie in $[-d, 0] = [-56, 0]$, those of h° in $[-114, 114]$, since h° is a combination of $S_{j+k,0}$ with $j, k \leq 57$, and those of W° and Q° in $[-M, M]$. Hence all frequencies of $w^{j-1}\Gamma_1$, $w^j\Gamma_0$ and $w^{j-1}\bar{p}W^\circ$ are at least $j-1-170 \geq 230 > M$. Taking real and imaginary parts only changes the signs of frequencies, so all components of $\Delta_j \in \mathcal{S}_6$ lie in tail rows, and $\|A\Delta_j\|_X \leq A_t\|\Delta_j\|_\rho$ by Lemma 11.12. By Lemma 4.2 and $\|\text{Re } f\|_\rho, \|\text{Im } f\|_\rho \leq \|f\|_\rho$,

$$\frac{\|2\text{Re}(jw^{j-1}\Gamma_1)\|_\rho}{\omega_j} \leq \frac{2j\rho^{j-1}\|\Gamma_1\|_\rho}{\alpha(j+2)\rho^{j+1}} \leq \frac{2\|\Gamma_1\|_\rho}{\alpha\rho^2}, \quad \frac{\|\text{Im}(w^j\Gamma_0)\|_\rho}{\omega_j} \leq \frac{\|\Gamma_0\|_\rho}{\alpha\rho(j_*+2)}.$$

Finally $jw^{j-1}\bar{p}W^\circ = j(1 - |w|^2)^2w^{j-1-\Lambda}(w^\Lambda\bar{p}Q^\circ)$ with $j-1-\Lambda \geq 0$, and Lemma 8.3 with $N = j-1-\Lambda$ gives

$$\frac{\|2\text{Re}(jw^{j-1}\bar{p}W^\circ)\|_\rho}{\omega_j} \leq \frac{2j}{\alpha(j+2)\rho^{j+1}} \sum_{m,t} |\gamma_{m,t}^W| \frac{4(2t+1)(2t+3)\rho^{m+j-1-\Lambda}}{(m+j-\Lambda)^2} \leq \mathbf{U}_W.$$

Adding the three contributions proves the claim. $\square$

Remark 11.15. This far-shape bound is simpler and cruder than Proposition 8.4: it does not exploit the cancellation between $D\mathcal{F}_6(x^\circ)e_j$ and T_j in the harmonic rows, and $\mathbf{U}_\Gamma$ does not tend to zero as $j_* \rightarrow \infty$. It suffices here because Γ_1 and Γ_0 measure the deviation of the centre from the disc, which is small compared with $\alpha = b^3/2 \approx 1.2 \cdot 10^4$; the directly computed normalised column at $j = 397$ is about 0.082, while $Z_{\text{sh},\infty} < 0.465$ (Table 3).

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - AD\mathcal{F}_6(x^\circ)\|_{X \rightarrow X}$.

11.8. Certified bounds and the exact zero. The quantities defined above are finite expressions in the 465 dyadic centre coefficients, the entries of $\hat{A}$ and $\rho = 11/10$. They were evaluated in ball arithmetic at 128 bits as described in Section 15.3. Table 3 lists the results, rounded outward.

Theorem 11.16. *Let $r = 1/5000$. There is exactly one $x^* = (g^*, c^*) \in X_6$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{F}_6(x^*) = 0$.*

Proof. The support condition of Proposition 11.14, $j_* - 1 - 170 = 230 > M$, holds, and (11.12) holds by Table 3; hence A is bounded and injective. By Table 3, Proposition 11.8 applies with the rational majorants $Y = 53 \cdot 10^{-6}$, $Z = 47/100$, $C_2 = 251/1000$, $C_3 = 3 \cdot 10^{-7}$ and $C_4 = 3 \cdot 10^{-15}$. For these values and $r = 1/5000$, exact rational arithmetic gives

$$Y + (Z - 1)r + C_2r^2 + C_3r^3 + C_4r^4 = -5.29899599999975999999999952 \cdot 10^{-5} < 0,$$

$$Z - 1 + 2C_2r + 3C_3r^2 + 4C_4r^3 = -0.5298995999999639999999999904 < 0. \quad \square$$

Remark 11.17 (The centre for $n = 6$). The centre x° was obtained numerically, and its provenance plays no role in the proof. A Fourier–Bessel collocation in $\mathbb{R}^6$, with u expanded in the regular Helmholtz solutions $r^{-2}J_{\ell+2}(\rho_*r)U_{\ell/2}(\cos 2\phi)$, $\ell \equiv 0 \pmod{4}$, at the ball wavenumber $\rho_* = j_{3,8}$ and seeded with the second-order amplitude predicted by the mechanism of Section 16.2, produced a numerical solution of (1.2) with Cauchy defects of order 10^{-13} . Its

Quantity	Where defined	Certified bound
$b = c_1^\circ$	Section 11	28.864750643300045... (exact dyadic)
P, C	Lemma 11.7	$< 31.143514, < 32.088747$
V_0	Lemma 11.7	< 7841.683303
$\ I - \hat{A}M_f\ $	(11.12)	$< 6.714 \cdot 10^{-11}$
$\ R\ $	Lemma 11.12	< 1402.313800
H	Lemma 11.12	$< 5.848182 \cdot 10^8$
A_t ($N_H = 458$)	Lemma 11.12	< 3.164016
$\ A\ $	Lemma 11.12	< 1402.314
Y	Section 11	$< 5.243310 \cdot 10^{-5}$
Z_f (465 columns)	Section 11	< 0.073227
P_s, P_t	Lemma 11.13	$< 31.143246, < 2.675771 \cdot 10^{-4}$
$\delta_a = 2P_s P_t + P_t^2$	Lemma 11.13	< 0.016667
$Z_{g,\partial}$ (705 columns, max. at (62, 1))	Lemma 11.13	< 0.313291
$Z_{g,\infty}$	Lemma 11.13	< 0.390424
$Z_{\text{sh},\partial}$ (85 columns, max. at $j = 61$)	Section 11	< 0.387180
$\ \Gamma_1\ _\rho, \ \Gamma_0\ _\rho$	Proposition 11.14	$< 844.817923, < 4110.324378$
$\sum_{m,t} \gamma_{m,t}^W (2t+1)(2t+3)\rho^{m-\Lambda-2}$	Proposition 11.14	< 3721574.315769
$\mathbf{U}_\Gamma, \mathbf{U}_W$	Proposition 11.14	$< 0.11689839, < 0.03005935$
$Z_{\text{sh},\infty}$ (all $j \geq 401$)	Proposition 11.14	< 0.464977
Z	Section 11	< 0.464977
$C_2 = \ A\ c_2$	Lemma 11.7	< 0.250135
$C_3 = \ A\ c_3$	Lemma 11.7	$< 2.7601 \cdot 10^{-7}$
$C_4 = \ A\ c_4$	Lemma 11.7	$< 2.1035 \cdot 10^{-15}$

TABLE 3. Certified bounds for $n = 6$, rounded outward, for $M = 58$, $S = 30$, $\rho = 11/10$, $j_* = 401$. The complete interval enclosures are recorded in the certificate `certificate_r6.json` and in the seven interval receipts (Section 15.3); P, C and V_0 are intermediate values of the final verification script, which uses $\|A\| \leq 1402.314$ in C_2, C_3, C_4 .

meridian domain was mapped conformally to the disc by Theodorsen's method, and Newton's method was applied to the Galerkin truncation of (11.3) to the 15 shape coefficients $c_1, c_5, \dots, c_{57}$ and the 450 coefficients of g described above, with residual about 10^{-10} in double precision. The resulting binary64 numbers were then frozen and are interpreted as exact dyadic rationals. (The wavenumber ρ_* used to locate the centre is unrelated to the weight $\rho = 11/10$ of the proof.)

11.9. Reconstruction, geometry and convexity. Let $x^* = (g^*, c^*)$ be the zero of Theorem 11.16, $\psi^* = \psi_{c^*}$ and $\delta c = c^* - c^\circ$. Then $\sum_j \omega_j |\delta c_j| \leq r$, and consequently

$$(11.17) \quad |c_1^* - b| \leq \frac{r}{\omega_1}, \quad \sum_j j \rho^{j-1} |\delta c_j| \leq \theta_1 r, \quad \sum_j \rho^j |\delta c_j| \leq \theta_0 r.$$

Proposition 11.18. *The map ψ^* is injective on the disc $\{|w| < 21/20\}$, and $\psi^{*'} has no zeros there. The domain $D = \psi^*(\mathbb{D})$ is bounded by the real-analytic Jordan curve $\psi^*(\partial\mathbb{D})$, it is strictly star-shaped with respect to 0, invariant under $z \mapsto \bar{z}$, $z \mapsto -\bar{z}$ and $z \mapsto i\bar{z}$, and it is not a disc.$*

Proof. Let $R = 21/20 < \rho$, $b_- = b - r/\omega_1 \leq c_1^*$, and

$$s_1 = \sum_{j \geq 5} j |c_j^\circ| R^{j-1} + \theta_1 r, \quad s_0 = \sum_{j \geq 5} |c_j^\circ| + \theta_0 r.$$

By (11.17) and $R < \rho$, $\sum_{j \geq 5} j |c_j^*| R^{j-1} \leq s_1$ and $\sum_{j \geq 5} |c_j^*| \leq s_0$. The computation (Section 15.3) certifies

$$(11.18) \quad b_- - s_1 > 27.2138856, \quad \frac{s_1 + s_0}{b_- - s_0} < 0.0634442, \quad |c_5^*| \geq |c_5^\circ| - \frac{r}{\omega_5} > 0.08529699.$$

Thus $\operatorname{Re} \psi'^*(w) \geq c_1^* - \sum_{j \geq 5} j |c_j^*| R^{j-1} > 27.21$ for $|w| \leq R$, which gives univalence on the convex disc $|w| < R$ exactly as in Proposition 10.1. For $|w| = 1$ we have $|\psi^*(w) - c_1^* w| \leq s_0$, $|w \psi'^*(w) - \psi^*(w)| \leq s_1$ and $|\psi^*(w)| \geq b_- - s_0 > 0$, hence $|w \psi'^*/\psi^* - 1| < 0.0635$; the argument of Proposition 10.1 shows that D is strictly star-shaped and that ∂D is a radial graph with positive real-analytic radius. The symmetries were shown at the beginning of this section. If D were a disc, its symmetries would force its centre to be 0, and the Schwarz lemma argument of Proposition 10.1 would show that ψ^* is linear, contradicting $c_5^* \neq 0$. $\square$

Proposition 11.19. *Let c be any real sequence indexed by $j \equiv 1 \pmod{4}$ with $\sum_j \omega_j |c_j - c_j^\circ| \leq r$, for instance $c = c^*$. Then for $|w| \leq 1$,*

$$|\psi'_c(w)| > 27.67533225, \quad |\psi''_c(w)| < 7.86831783, \quad \operatorname{Re} \left(1 + \frac{w \psi''_c(w)}{\psi'_c(w)} \right) > 0.71569200.$$

Consequently $D = \psi^*(\mathbb{D})$ is strictly convex, and $\Omega = \Pi_6^{-1}(D)$ is convex.

Proof. Let $A_1 = \sum_{j \geq 5} j |c_j^\circ|$, $A_2 = \sum_{j \geq 5} j(j-1) |c_j^\circ|$, $\delta = c - c^\circ$, and

$$E_1 = r \max \left\{ \frac{1}{\omega_1}, \frac{1}{\alpha \rho^6} \right\}, \quad E_2 = \frac{r}{\alpha \rho e \log \rho}.$$

We have $|\delta_1| + \sum_{j \geq 5} j |\delta_j| = \sum_j \kappa_j \omega_j |\delta_j| \leq r \sup_j \kappa_j$ with $\kappa_1 = 1/\omega_1$ and $\kappa_j = j/\omega_j = j/((j+2)\alpha\rho^{j+1}) < 1/(\alpha\rho^6)$ for $j \geq 5$; hence this sum is at most E_1 . For $j \geq 5$, $j(j-1)/\omega_j < j\rho^{-j}/(\alpha\rho) \leq 1/(\alpha\rho e \log \rho)$, since $x \mapsto x\rho^{-x}$ attains its maximum $1/(e \log \rho)$ on $(0, \infty)$ at $x = 1/\log \rho$; hence $\sum_{j \geq 5} j(j-1) |\delta_j| \leq E_2$. Both suprema are over all j , so the infinite tail of c is covered. For $|w| \leq 1$ it follows that

$$|\psi'_c(w)| \geq c_1 - \sum_{j \geq 5} j |c_j| \geq b - A_1 - E_1 =: L, \quad |\psi''_c(w)| \leq \sum_{j \geq 5} j(j-1) |c_j| \leq A_2 + E_2 =: U.$$

The computation certifies $A_1 < 1.18941839$, $A_2 < 7.86831777$, $E_1 < 9.39 \cdot 10^{-9}$ and $E_2 < 5.84 \cdot 10^{-8}$, hence $L > 27.67533225$, $U < 7.86831783$ and $\operatorname{Re}(1 + w \psi''_c/\psi'_c) \geq 1 - U/L > 0.71569200$.

For $c = c^*$, the argument of Proposition 10.2 (the analytic convexity criterion) shows that D is strictly convex. Now let $x = (x', x'')$, $z = (z', z'') \in \Omega$ and $\sigma \in [0, 1]$, and put $m = (m_1, m_2) = \sigma \Pi_6(x) + (1 - \sigma) \Pi_6(z)$. By convexity $m \in D$, and $m_1, m_2 \geq 0$. Since D is invariant under both reflections, it contains the four points $(\pm m_1, \pm m_2)$, and by convexity their convex hull, the rectangle $[-m_1, m_1] \times [-m_2, m_2]$. By the triangle inequality in each factor $\mathbb{R}^3$, $|\sigma x' + (1 - \sigma) z'| \leq m_1$ and $|\sigma x'' + (1 - \sigma) z''| \leq m_2$, so $\Pi_6(\sigma x + (1 - \sigma) z) \in D$, that is, $\sigma x + (1 - \sigma) z \in \Omega$. $\square$

11.10. Proof of Theorem 1.2 for $n = 6$. Let $V^* = Kg^* + h_{c^*}$. By Lemma 11.6, $V^* \in C^1(\overline{\mathbb{D}})$ satisfies (11.2) with $\psi = \psi^*$ and has the parities required in Lemma 11.4. By Proposition 11.18, ψ^* is injective with nonvanishing derivative on a neighbourhood of $\overline{\mathbb{D}}$. Lemma 11.4 shows that $F = V^* \circ (\psi^*)^{-1} \in C^1(\overline{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = y_1 y_2$ and $\nabla F = (y_2, y_1)$ on ∂D , odd in y_1 and in y_2 , and invariant under the exchange of y_1 and y_2 . By Lemma 10.3 and Lemma 10.4 in the form of Lemma 11.5(f), F extends to a real-analytic solution on an open neighbourhood of $\overline{D}$; replacing this neighbourhood by the connected component containing $\overline{D}$ of its intersection with its images under the two reflections and the exchange, we obtain a neighbourhood N with these symmetries on which, by the identity theorem, F keeps its parities.

Since D is strictly star-shaped and symmetric, ∂D meets each axis in exactly two points. Proposition 11.3 shows that $\Omega = \Pi_6^{-1}(D)$ and $u = F(|x'|, |x''|)/(|x'| |x''|)$, extended analytically to $\{x' = 0\} \cup \{x'' = 0\}$, satisfy (1.2), that u is real analytic on the neighbourhood $\Pi_6^{-1}(N)$ of $\overline{\Omega}$, and that u is nonconstant.

Geometry of Ω . Let $R_D(\phi)$ be the radial function of D . It is positive and real analytic, even (by $z \mapsto \bar{z}$) and invariant under $\phi \mapsto \pi - \phi$ (by $z \mapsto -\bar{z}$), hence π -periodic, and its Fourier series has the form $R_D(\phi) = \sum_{k \geq 0} \alpha_k \cos 2k\phi$, with exponentially decaying coefficients (by the exchange symmetry $\alpha_k = 0$ for odd k , which we do not need). As in the case $n = 4$, $\tilde{R}(\tau) = \sum_k \alpha_k T_k(\tau)$ converges on a complex neighbourhood of $[-1, 1]$ and defines a real-analytic function with $\tilde{R}(\cos 2\phi) = R_D(\phi)$, positive near $[-1, 1]$. For $x \neq 0$ we have $\Pi_6(x) = |x|(\cos \phi, \sin \phi)$ with $\phi \in [0, \pi/2]$ and $\cos 2\phi = (|x'|^2 - |x''|^2)/|x|^2$. Hence

$$\Omega = \{0\} \cup \left\{ x \neq 0 : |x| < \tilde{R}\left(\frac{|x'|^2 - |x''|^2}{|x|^2}\right) \right\},$$

$$\partial\Omega = \left\{ \tilde{R}(|\omega'|^2 - |\omega''|^2) \omega : \omega = (\omega', \omega'') \in S^5 \right\}.$$

Thus Ω is strictly star-shaped with respect to 0, in particular connected. The map $\omega \mapsto \tilde{R}(|\omega'|^2 - |\omega''|^2) \omega$ is real analytic and injective on S^5 , and, exactly as for $n = 4$, its differential maps a tangent vector $v \perp \omega$ to a vector whose component tangent to S^5 is $\tilde{R}(|\omega'|^2 - |\omega''|^2) v \neq 0$. Hence $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^5 . The symmetries of Ω follow from $\Omega = \Pi_6^{-1}(D)$ and the invariance of D under the exchange of coordinates.

Ω is not a ball. If it were, its centre would be fixed by every isometry of $\mathbb{R}^6$ preserving Ω , in particular by $O(3) \times O(3)$, whose only fixed point is 0. Then $\Pi_6(\Omega) = D \cap [0, \infty)^2$ would be a quarter of a disc centred at 0, and by the reflection symmetries D would be that disc, contradicting Proposition 11.18. Finally, Ω is convex by Proposition 11.19. This completes the proof of Theorem 1.2 for $n = 6$. $\square$

12. RANK-TWO LIE ALGEBRAS

The symmetry classes used for $n = 4$ and $n = 6$ are, up to finite index and the additional reflections, the adjoint representations of the compact Lie algebras $\mathfrak{u}(2)$ and $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and the weights y and $y_1 y_2$ are the products of their positive roots. In this section we carry out the reduction for an arbitrary compact Lie algebra of rank two, and we use it to prove the following theorem for the three simple ones.

Theorem 12.1. *Let $(\mathfrak{k}, G)$ be one of $(\mathfrak{su}(3), SU(3))$, $(\mathfrak{so}(5), SO(5))$ and $(\mathfrak{g}_2, G_2)$, so that $n = \dim \mathfrak{k}$ equals 8, 10 and 14, respectively, and equip $\mathfrak{k}$ with an $\text{Ad}(G)$ -invariant inner product, so that $\mathfrak{k} \cong \mathbb{R}^n$ as a Euclidean space. There exist a bounded convex domain $\Omega \subset \mathfrak{k}$ and a nonconstant real-valued function u , real analytic on an open neighbourhood of $\overline{\Omega}$, such that (1.2) holds. Moreover, Ω is not a ball, it is strictly star-shaped with respect to 0, its boundary*

is a compact real-analytic hypersurface diffeomorphic to S^{n-1} , and Ω is invariant under $\text{Ad}(G)$ and under $X \mapsto -X$. Consequently the Fourier transform of $\mathbf{1}_\Omega$ vanishes on the unit sphere, Ω fails the Pompeiu property, and the Schiffer and Pompeiu conjectures fail in $\mathbb{R}^8$, $\mathbb{R}^{10}$ and $\mathbb{R}^{14}$ within the class of bounded convex domains with real-analytic boundary.

Theorem 12.1 contains the cases $n = 8, 10, 14$ of Theorem 1.2 and of Corollary 1.3. Only the adjoint action of G enters, so G may be replaced by any compact connected group with Lie algebra $\mathfrak{k}$, for instance $\text{Spin}(5)$ in place of $SO(5)$. For a simple $\mathfrak{k}$ the invariant inner product is unique up to a positive factor; changing it amounts to a dilation, which is absorbed by rescaling Ω and u . For $\mathfrak{k} = \mathfrak{su}(3)$, identified with the traceless Hermitian 3×3 matrices via $X \mapsto iX$, the domain is a spectral set: Ω consists of the matrices whose eigenvalue vector, a point of the plane $\{\lambda_1 + \lambda_2 + \lambda_3 = 0\} \cong \mathbb{R}^2$, lies in a fixed planar domain.

The proof follows the pattern of the cases $n = 4, 6$. Sections 12.1–12.4 reduce the problem, for every compact Lie algebra of rank two, to a planar Helmholtz problem on a Cartan section D with the Cauchy data of $\text{Im } \tau^m$, and show that the lifted domain is analytic, spherical and convex whenever D is. Sections 12.5–12.8 solve the planar problems for $m = 3, 4, 6$ by the conformal fixed-disc method, with a computer-assisted Newton–Kantorovich argument in a symmetry class that contains the cases $m = 1, 2$ of Sections 3–11 as special cases. Section 12.9 proves Theorem 12.1, Section 15.5 describes the computations, and Section 12.10 explains why the dimensions 4, 6, 8, 10, 14 are the only ones that this reduction can reach.

12.1. The radial part of the Laplacian. Let G be a compact connected Lie group with Lie algebra $\mathfrak{k}$ and an $\text{Ad}(G)$ -invariant inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{k}$. We identify $\mathfrak{k}$ with $\mathbb{R}^n$, $n = \dim \mathfrak{k}$, by an orthonormal basis, so that $\Delta_{\mathfrak{k}}$ is the Euclidean Laplacian. Let $\mathbb{T} \subset G$ be a maximal torus with Lie algebra $\mathfrak{t}$, and let $W = N_G(\mathbb{T})/\mathbb{T}$ be the Weyl group, which acts on $\mathfrak{t}$ by orthogonal maps. There is an orthogonal decomposition

$$\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{k}_\alpha, \quad \dim \mathfrak{k}_\alpha = 2,$$

indexed by a set Φ^+ of positive real roots, which are linear forms on $\mathfrak{t}$, such that $\text{ad}(\tau)$ acts on $\mathfrak{k}_\alpha$ as $\alpha(\tau)J_\alpha$ for $\tau \in \mathfrak{t}$, where J_α is a rotation through a right angle. We use the following standard facts [6, Ch. IV–V], [32, Ch. 1]: every $X \in \mathfrak{k}$ is $\text{Ad}(G)$ -conjugate to an element of $\mathfrak{t}$; two elements of $\mathfrak{t}$ are $\text{Ad}(G)$ -conjugate if and only if they are W -conjugate; the roots in Φ^+ are pairwise non-proportional; W is generated by the orthogonal reflections s_α in the walls $\ker \alpha$, $\alpha \in \Phi^+$; and the function

$$\varpi(\tau) = \prod_{\alpha \in \Phi^+} \alpha(\tau)$$

satisfies $\varpi \circ w = \det(w) \varpi$ for $w \in W$. Let $\mathfrak{t}_{\text{reg}} = \{\tau \in \mathfrak{t} : \varpi(\tau) \neq 0\}$. For $\tau \in \mathfrak{t}_{\text{reg}}$ the kernel of the skew-symmetric map $\text{ad } \tau$ is $\mathfrak{t}$, hence $[\mathfrak{k}, \tau] = \mathfrak{t}^\perp$.

Proposition 12.2 (Harish-Chandra). *The polynomial ϖ is harmonic on $\mathfrak{t}$. Let $U \subset \mathfrak{k}$ be open and $\text{Ad}(G)$ -invariant, let $u \in C^2(U)$ be $\text{Ad}(G)$ -invariant, and let $f = u|_{U \cap \mathfrak{t}}$. Then for every $\tau \in U \cap \mathfrak{t}_{\text{reg}}$,*

$$(12.1) \quad \nabla u(\tau) = \nabla_{\mathfrak{t}} f(\tau) \in \mathfrak{t}, \quad (\Delta_{\mathfrak{k}} u)(\tau) = \frac{\text{div}_{\mathfrak{t}}(\varpi^2 \nabla_{\mathfrak{t}} f)}{\varpi^2}(\tau) = \frac{\Delta_{\mathfrak{t}}(\varpi f)}{\varpi}(\tau).$$

The identity (12.1) is the flat case of Harish-Chandra’s radial-part formula [29]; see also [30, Ch. II]. We include the short proof.

Proof. The Laplacian commutes with orthogonal maps, so $\Delta\varpi$ is a W -anti-invariant polynomial of degree $|\Phi^+| - 2$. An anti-invariant polynomial vanishes on each wall $\ker \alpha$, since s_α fixes it pointwise, and is therefore divisible by α ; the roots being pairwise non-proportional, it is divisible by ϖ . Hence $\Delta\varpi = 0$.

For $Y \in \mathfrak{k}$ the function $s \mapsto u(\text{Ad}(\exp sY)\tau)$ is constant, so $\nabla u(\tau) \perp [Y, \tau]$; since $[\mathfrak{k}, \tau] = \mathfrak{t}^\perp$, $\nabla u(\tau) \in \mathfrak{t}$, and it equals $\nabla_{\mathfrak{t}} f(\tau)$. Let $\mathfrak{m} = \mathfrak{t}^\perp$ and $\Psi: G/\mathbb{T} \times \mathfrak{t}_{\text{reg}} \rightarrow \mathfrak{k}$, $\Psi(g\mathbb{T}, \tau) = \text{Ad}(g)\tau$. Its differential maps the tangent vector $(\frac{d}{ds}g \exp(sY)\mathbb{T}|_{s=0}, \tau')$, $Y \in \mathfrak{m}$, $\tau' \in \mathfrak{t}$, to $\text{Ad}(g)([Y, \tau] + \tau')$. As $[Y, \tau] \in \mathfrak{m} \perp \mathfrak{t}$ and $|[Y, \tau]|^2 = \sum_\alpha \alpha(\tau)^2 |Y_\alpha|^2$, where Y_α is the component of Y in $\mathfrak{k}_\alpha$, the map Ψ is a local diffeomorphism, and the pull-back of the Euclidean metric is the orthogonal sum of $|d\tau|^2$ and a metric on $G/\mathbb{T}$ whose volume density is $\prod_\alpha \alpha(\tau)^2 = \varpi(\tau)^2$ times a fixed smooth density. Since $u \circ \Psi(g\mathbb{T}, \tau) = f(\tau)$, the coordinate formula $\Delta = |\gamma|^{-1/2} \partial_i (|\gamma|^{1/2} \gamma^{ij} \partial_j)$ for this block-diagonal metric γ gives $(\Delta_{\mathfrak{k}} u)(\text{Ad}(g)\tau) = \varpi^{-2} \text{div}_{\mathfrak{t}}(\varpi^2 \nabla_{\mathfrak{t}} f)(\tau)$. Finally $\varpi^{-2} \text{div}(\varpi^2 \nabla f) = \Delta f + 2\varpi^{-1} \nabla \varpi \cdot \nabla f = \varpi^{-1}(\Delta(\varpi f) - f \Delta \varpi) = \varpi^{-1} \Delta(\varpi f)$. $\square$

Because ϖ is harmonic, (12.1) conjugates the Laplacian of $\mathfrak{k}$, acting on invariant functions, to the flat Laplacian of $\mathfrak{t}$ without any potential term; the normalisation of ϖ cancels.

Rank two. From now on $\mathfrak{k}$ has rank two, $\dim \mathfrak{t} = 2$. Let $m = |\Phi^+| \geq 1$. Then W is a finite group generated by the reflections in the m distinct lines $\ker \alpha$, that is, a dihedral group of order $2m$, which we denote by $I_2(m)$ (so $I_2(1)$ is generated by a single reflection). Since W preserves the lattice $\ker(\exp|_{\mathfrak{t}})$, its rotations have order 1, 2, 3, 4 or 6, so $m \in \{1, 2, 3, 4, 6\}$ [32], and $n = \dim \mathfrak{k} = 2 + 2m$. We identify $\mathfrak{t}$ with $\mathbb{C}$, $\tau = x + iy = |\tau|e^{i\theta}$, by an orthonormal basis such that the real axis is a wall. The walls are then the lines $e^{ik\pi/m}\mathbb{R}$, and

$$W = \{\tau \mapsto e^{2\pi ik/m} \tau, \tau \mapsto e^{2\pi ik/m} \bar{\tau} : 0 \leq k < m\}.$$

Since $\prod_{k=0}^{m-1} \sin(\theta - k\pi/m) = 2^{1-m} \sin m\theta$, the product of the positive roots is a nonzero multiple of $\text{Im } \tau^m$; as the normalisation of ϖ plays no role, we put

$$\varpi(\tau) = \text{Im } \tau^m = |\tau|^m \sin m\theta.$$

The W -invariant polynomials $p_1(\tau) = |\tau|^2$ and $p_2(\tau) = \text{Re } \tau^m$ separate W -orbits: $p(\tau) = p(\tau')$ for $p = (p_1, p_2)$ if and only if $\tau' \in W\tau$. By Chevalley's restriction theorem [30, Ch. II] there are $\text{Ad}(G)$ -invariant polynomials $\mathbf{P}_1(X) = |X|^2$ and $\mathbf{P}_2$ on $\mathfrak{k}$ with $\mathbf{P}_j|_{\mathfrak{t}} = p_j$; we write $\mathbf{P} = (\mathbf{P}_1, \mathbf{P}_2)$. A direct computation, using $\partial_x p_2 = m \text{Re } \tau^{m-1}$ and $\partial_y p_2 = -m \text{Im } \tau^{m-1}$, gives

$$(12.2) \quad \det Dp(\tau) = -2m \varpi(\tau).$$

The compact Lie algebras of rank two with $m \geq 1$ are listed in Table 4.

$\mathfrak{k}$	n	roots	W	m	treated in
$\mathfrak{u}(2) \cong \mathbb{R} \oplus \mathfrak{su}(2)$	4	A_1	$I_2(1) \cong \mathbb{Z}_2$	1	Sections 2–10
$\mathfrak{su}(2) \oplus \mathfrak{su}(2) \cong \mathfrak{so}(4)$	6	$A_1 \times A_1$	$I_2(2) \cong \mathbb{Z}_2^2$	2	Section 11
$\mathfrak{su}(3)$	8	A_2	$I_2(3) \cong S_3$	3	this section
$\mathfrak{so}(5) \cong \mathfrak{sp}(2)$	10	$B_2 = C_2$	$I_2(4)$	4	this section
$\mathfrak{g}_2$	14	G_2	$I_2(6)$	6	this section

TABLE 4. The compact Lie algebras of rank two with at least one root. In each case $n = 2 + 2m$, and in suitable orthonormal coordinates $\tau \in \mathfrak{t} \cong \mathbb{C}$ the product of the positive roots is a multiple of $\text{Im } \tau^m$.

Remark 12.3 (Dimensions four and six). For $\mathfrak{k} = \mathfrak{u}(2)$, the group $\text{Ad}(U(2))$ acts on $\mathfrak{su}(2) \cong \mathbb{R}^3$ as $SO(3)$ and trivially on the centre; $\mathfrak{t}$ is spanned by the centre and a line in $\mathfrak{su}(2)$, and in the coordinates $\tau = x_1 + iy$ of Section 2 one has $W = \{1, y \mapsto -y\}$, $\varpi = y$ and $p = (x_1^2 + y^2, x_1)$. For $\mathfrak{k} = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $\text{Ad}(G) = SO(3) \times SO(3)$ acts on $\mathbb{R}^3 \times \mathbb{R}^3$, W is generated by the reflections in the two axes of $\tau = y_1 + iy_2$, $\varpi = y_1 y_2 = \frac{1}{2} \text{Im } \tau^2$ and $p_2 = y_1^2 - y_2^2$. Accordingly, the maps Π and Π_6 of Sections 2 and 11 send X to its conjugate in the closed Weyl chamber; Lemmas 2.1 and 11.1 are the cases $m = 1, 2$ of Proposition 12.2; Lemmas 2.2 and 11.2 combine the cases $m = 1, 2$ of Lemmas 12.4 and 12.5 below; the groups $O(3) \times \mathbb{Z}_2$ and $(O(3) \times O(3)) \rtimes \mathbb{Z}_2$ used there contain $\text{Ad}(G)$ and realise, in addition, the enlarged symmetry $I_2(2m)$ of Section 12.4; and (6.1) and (11.3) are the cases $m = 1, B = 1$ and $m = 2, B = 2$ of the equation (12.7) below, with the same weights, the same principal parts (Lemma 7.3, Proposition 11.10) and the same tail inverses (Lemmas 7.4 and 11.11). For $m \geq 3$ the direct arguments of Sections 10 and 11 for the geometry and the convexity of the lifted domain are replaced by Lemmas 12.6 and 12.8, and the non-ball argument uses that $\mathfrak{k}$ is simple.

12.2. Lifting from the Cartan plane. In this subsection $\mathfrak{k}$ has rank two, with the notation introduced above. A function F on a W -invariant subset of $\mathfrak{t}$ is called *anti-invariant* if $F \circ w = \det(w)F$ for all $w \in W$.

Lemma 12.4 (Anti-invariant division). *Let $N \subset \mathfrak{t}$ be open and W -invariant, and let F be real analytic and anti-invariant on N . Then $F = \varpi q$ with q real analytic and W -invariant on N .*

Proof. Let $\ell_1, \dots, \ell_m$ be the walls and λ_k a unit linear form with $\ell_k = \ker \lambda_k$, so that $\varpi = c\lambda_1 \cdots \lambda_m$ with $c \neq 0$. Since the reflection in ℓ_k lies in W and has determinant -1 , F vanishes on $N \cap \ell_k$. Let $U \subset N$ be an open disc centred at a point τ_0 . If τ_0 lies on ℓ_k , write points of U in orthonormal coordinates (λ, μ) with $\lambda = \lambda_k$; since U contains the point $(0, \mu)$ together with (λ, μ) , Hadamard's formula

$$f(\lambda, \mu) = \lambda \int_0^1 \partial_\lambda f(\sigma \lambda, \mu) d\sigma$$

shows that every real-analytic f on U vanishing on $\ell_k \cap U$ is λ_k times a real-analytic function on U . If $\tau_0 \neq 0$, then τ_0 lies on at most one wall, and dividing F by that λ_k (if any) and by the other factors of ϖ , which do not vanish near τ_0 , shows that F/ϖ extends analytically to a neighbourhood of τ_0 . If $\tau_0 = 0$, then every wall passes through the centre of U . We divide F successively by $\lambda_1, \dots, \lambda_m$: after dividing by $\lambda_1, \dots, \lambda_{k-1}$, the quotient vanishes on $(\ell_k \cap U) \setminus \{0\}$, where the divisors are nonzero, hence on $\ell_k \cap U$ by continuity, and the next division is possible. Hence $q = F/\varpi$, defined on $N \cap \mathfrak{t}_{\text{reg}}$, extends to a real-analytic function on N . It is W -invariant, because F and ϖ are both anti-invariant. $\square$

For a W -invariant open set $N \subset \mathfrak{t}$ let $\text{Ad}(G)N = \{\text{Ad}(g)\tau : g \in G, \tau \in N\}$. For a W -invariant function q on N we define $q^\sharp$ on $\text{Ad}(G)N$ by $q^\sharp(X) = q(\tau)$ for any $\tau \in N$ conjugate to X ; this is well defined, since the elements of $\mathfrak{t}$ conjugate to X form one W -orbit.

Lemma 12.5 (Invariant lift). *Let $N \subset \mathfrak{t}$ be open and W -invariant, and let q be a W -invariant function on N .*

- (a) $\text{Ad}(G)N$ is open in $\mathfrak{k}$, $\text{Ad}(G)N \cap \mathfrak{t} = N$, and if q is continuous, so is $q^\sharp$.
- (b) If q is real analytic, then $q^\sharp$ is real analytic on $\text{Ad}(G)N \setminus \{0\}$. In particular, for $\tau \in N \cap \mathfrak{t}_{\text{reg}}$, (12.1) holds for $u = q^\sharp$ and $f = q$.

Proof. (a) If $\tau' \in \mathfrak{t}$ is conjugate to some $\tau \in N$, then $\tau' \in W\tau \subset N$; this gives $\text{Ad}(G)N \cap \mathfrak{t} = N$. Let $X_j \rightarrow X$ with $X = \text{Ad}(g)\tau$, $\tau \in N$, and write $X_j = \text{Ad}(g_j)\tau_j$ with $\tau_j \in \mathfrak{t}$. Given any subsequence, a further subsequence has $g_j \rightarrow g'$ and $\tau_j \rightarrow \tau'$ (as $|\tau_j| = |X_j|$ is bounded), and

then τ' is conjugate to X , so $\tau' \in N$. Hence $\tau_j \in N$ for large j , and $q^\sharp(X_j) = q(\tau_j) \rightarrow q(\tau') = q^\sharp(X)$ if q is continuous. Both assertions follow because the subsequence was arbitrary.

(b) Let $X_0 \in \text{Ad}(G)N \setminus \{0\}$, and let $\tau_0 \in N$ be conjugate to X_0 , so $\tau_0 \neq 0$. We first find an open disc $U \ni \tau_0$ in N and a real-analytic function Q on an open set containing $p(U)$ with $q = Q \circ p$ on U . If $\varpi(\tau_0) \neq 0$, then p is a local analytic diffeomorphism at τ_0 by (12.2), and $Q = q \circ (p|_U)^{-1}$ for a small U . Otherwise τ_0 lies on exactly one wall ℓ . In orthonormal coordinates (λ, μ) with $\ell = \{\lambda = 0\}$, the functions p and q are even in λ near τ_0 , because the reflection in ℓ belongs to W . As in the proof of Lemma 2.2, $p = \tilde{P}(\lambda^2, \mu)$ and $q = \tilde{q}(\lambda^2, \mu)$ near τ_0 with $\tilde{P}$ and $\tilde{q}$ real analytic near $(0, \mu_0)$. Then $\det Dp = 2\lambda \det D\tilde{P}(\lambda^2, \mu)$ in these coordinates, while $\det Dp = \pm 2m\varpi$ by (12.2), and near τ_0 one has $\varpi = \lambda\varpi_1$ with $\varpi_1(\tau_0) \neq 0$, since only one factor of ϖ vanishes at τ_0 . Hence $\det D\tilde{P}(0, \mu_0) \neq 0$, $\tilde{P}$ is a local analytic diffeomorphism, and $Q = \tilde{q} \circ \tilde{P}^{-1}$ has the required property.

By the argument in (a), every X close to X_0 is conjugate to some $\tau \in U$ (apply an element of W to a limit point conjugate to X_0). For such X , $q^\sharp(X) = q(\tau) = Q(p(\tau)) = Q(\mathbf{P}(X))$. Since $\mathbf{P}$ is polynomial, $q^\sharp$ is real analytic near X_0 . The last assertion follows from Proposition 12.2. $\square$

Lemma 12.6 (Lifting). *Let $\mathfrak{k}$ have rank two. Let $R: \mathbb{R} \rightarrow (0, \infty)$ be real analytic with $R(\theta) = R(-\theta) = R(\theta + 2\pi/m)$, and let*

$$D = \{re^{i\theta} : 0 \leq r < R(\theta)\} \subset \mathfrak{t}.$$

Let $N \supset \overline{D}$ be a W -invariant open subset of $\mathfrak{t}$, and let F be real analytic and anti-invariant on N and satisfy

$$(12.3) \quad \Delta F + F = 0 \text{ in } D, \quad F = \varpi \text{ and } \nabla F = \nabla \varpi \text{ on } \partial D.$$

Let q be the function of Lemma 12.4, so that $F = \varpi q$, let $\Omega = \text{Ad}(G)D$ and $u = q^\sharp$. Then Ω is a bounded domain, strictly star-shaped with respect to 0, with $\Omega \cap \mathfrak{t} = D$ and $\partial\Omega = \text{Ad}(G)\partial D$, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} . The function u is real analytic on the open neighbourhood $\text{Ad}(G)N$ of $\overline{\Omega}$, it is nonconstant, and (1.2) holds.

Proof. The domain. The function R is even and $2\pi/m$ -periodic, so its Fourier series has the form $R(\theta) = \sum_{k \geq 0} a_k \cos(km\theta)$, with $|a_k| \leq C\varrho^{-k}$ for some $\varrho > 1$ because R is real analytic. As $|T_k| \leq \varrho_1^k$ on the Bernstein ellipse of parameter ϱ_1 , the series $R_T(s) = \sum_k a_k T_k(s)$, with T_k the Chebyshev polynomials, converges on a complex neighbourhood of $[-1, 1]$, and $R_T(\cos m\theta) = R(\theta)$. If $X \neq 0$ is conjugate to $\tau = |\tau|e^{i\theta}$, then $\mathbf{P}_2(X)/|X|^m = \text{Re } \tau^m/|\tau|^m = \cos m\theta \in [-1, 1]$. Hence

$$\mathbf{R}(X) = R_T(\mathbf{P}_2(X)/|X|^m), \quad X \in \mathfrak{k} \setminus \{0\},$$

is real analytic, positive, $\text{Ad}(G)$ -invariant and homogeneous of degree zero, and $\mathbf{R}(\tau) = R(\theta)$ for $\tau = |\tau|e^{i\theta}$. Consequently

$$\Omega = \{0\} \cup \{X \neq 0 : |X| < \mathbf{R}(X)\}, \quad \partial\Omega = \{\mathbf{R}(\omega)\omega : \omega \in S^{n-1}\} = \text{Ad}(G)\partial D,$$

and $\Omega \cap \mathfrak{t} = D$. Thus Ω is open, bounded and strictly star-shaped with respect to 0, hence connected. Exactly as for $n = 4$ (Section 10), the map $\omega \mapsto \mathbf{R}(\omega)\omega$ is an injective real-analytic immersion of S^{n-1} with inverse $X \mapsto X/|X|$, so $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} . Moreover $\overline{\Omega} = \text{Ad}(G)\overline{D} \subset \text{Ad}(G)N$, and $\text{Ad}(G)N$ is open by Lemma 12.5(a).

The equation. By Lemma 12.5, u is continuous on $\text{Ad}(G)N$ and real analytic on $\text{Ad}(G)N \setminus \{0\}$. For $\tau \in D \cap \mathfrak{t}_{\text{reg}}$, (12.1) gives $\Delta u(\tau) = \varpi^{-1} \Delta(\varpi q)(\tau) = \varpi^{-1} \Delta F(\tau) = -u(\tau)$; since Δ commutes with the orthogonal maps $\text{Ad}(g)$, $\Delta u + u = 0$ on $\Omega \cap \mathfrak{t}_{\text{reg}}$, where $\mathfrak{t}_{\text{reg}} = \text{Ad}(G)\mathfrak{t}_{\text{reg}}$. The polynomial ϖ^2 is W -invariant, so by Chevalley's theorem $\mathfrak{t}_{\text{reg}}$ is the complement of the

zero set of an $\text{Ad}(G)$ -invariant polynomial, which is not identically zero; hence $\mathfrak{k}_{\text{reg}}$ is dense, and by continuity $\Delta u + u = 0$ on $\Omega \setminus \{0\}$. The point 0 lies in Ω , because $R > 0$. Let $U_0 \subset \Omega$ be a ball about 0, $\varphi \in C_c^\infty(U_0)$, and $\chi_\varepsilon(X) = \chi(|X|/\varepsilon)$ with $\chi \in C^\infty(\mathbb{R})$, $\chi = 0$ on $(-\infty, 1]$ and $\chi = 1$ on $[2, \infty)$. Then $\int u(\Delta + 1)(\chi_\varepsilon \varphi) = 0$, and $(\Delta + 1)(\chi_\varepsilon \varphi) - \chi_\varepsilon(\Delta + 1)\varphi = 2\nabla \chi_\varepsilon \cdot \nabla \varphi + \varphi \Delta \chi_\varepsilon$ is supported in $\{|X| \leq 2\varepsilon\}$ and bounded by $C\varepsilon^{-2}$. Since u is bounded near 0, letting $\varepsilon \rightarrow 0$ gives $\int_{U_0} u(\Delta + 1)\varphi = 0$ ($n \geq 3$). The proof of Lemma 10.3 applies verbatim in $\mathbb{R}^n$ and shows that the continuous weak solution u is real analytic on U_0 . Hence u is real analytic on $\text{Ad}(G)N$ and $\Delta u + u = 0$ in Ω .

The boundary conditions. Let $\tau \in \partial D$ with $\varpi(\tau) \neq 0$. Then $u(\tau) = q(\tau) = F(\tau)/\varpi(\tau) = 1$, and by (12.1) $\nabla u(\tau) = \nabla q(\tau) = \varpi^{-1}(\nabla F - q\nabla \varpi)(\tau) = 0$. By invariance, $u = 1$ and $\nabla u = 0$ on $\text{Ad}(G)(\partial D \cap \mathfrak{k}_{\text{reg}})$. Since ∂D meets the m walls in $2m$ points, $\partial D \cap \mathfrak{k}_{\text{reg}}$ is dense in ∂D , and since $(g, \tau) \mapsto \text{Ad}(g)\tau$ maps $G \times \partial D$ continuously onto $\partial \Omega$, the set $\text{Ad}(G)(\partial D \cap \mathfrak{k}_{\text{reg}})$ is dense in $\partial \Omega$. By continuity $u = 1$ and $\nabla u = 0$ on $\partial \Omega$. A constant solution of $\Delta u + u = 0$ vanishes, which contradicts $u = 1$ on $\partial \Omega$. $\square$

Remark 12.7 (The radial-graph hypothesis). Some hypothesis on D beyond W -invariance and analyticity of ∂D is necessary. For the annulus $D = \{a < |\tau| < b\}$, which is W -invariant with real-analytic boundary, $\text{Ad}(G)D = \{a < |X| < b\}$ is a spherical shell, whose boundary consists of two spheres. Spherical shells do fail the Pompeiu property (Section 1), so a topological hypothesis of this kind cannot be dispensed with in any statement of this type. The positive analytic radial function in Lemma 12.6 is what makes $\partial \Omega$ a sphere, including at the points where ∂D meets the walls. In our applications it is supplied by strict convexity together with $0 \in D$ (Proposition 12.21).

12.3. Convexity. Kostant's convexity theorem [39] is usually stated for a real reductive Lie algebra $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{p}$ with Cartan decomposition, the group H acting on $\mathfrak{p}$, and a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$; in this form it reads: for $a \in \mathfrak{a}$, the orthogonal projection of $H \cdot a$ onto $\mathfrak{a}$ is the convex hull of the orbit of a under the Weyl group of $(\mathfrak{g}, \mathfrak{a})$ [40, Theorem 2.2]. The adjoint representation of a compact $\mathfrak{k}$ is of this type. Let $\mathfrak{g} = \mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \oplus i\mathfrak{k}$, regarded as a real Lie algebra, with the Cartan involution $X + iY \mapsto X - iY$, so that $\mathfrak{h} = \mathfrak{k}$ and $\mathfrak{p} = i\mathfrak{k}$. The map $X \mapsto iX$ is a G -equivariant linear isometry of $\mathfrak{k}$ onto $\mathfrak{p}$ (with $\langle iX, iY \rangle := \langle X, Y \rangle$), because $\text{Ad}(g)$ is complex linear on $\mathfrak{k}_{\mathbb{C}}$. It maps $\mathfrak{t}$ onto the maximal abelian subspace $\mathfrak{a} = i\mathfrak{t}$, and the Weyl group $N_G(\mathfrak{a})/Z_G(\mathfrak{a}) = N_G(\mathbb{T})/\mathbb{T}$ of $(\mathfrak{g}, \mathfrak{a})$ acts on $\mathfrak{a} \cong \mathfrak{t}$ as W [36, Ch. VI]. Hence, for a compact semisimple $\mathfrak{k}$ and $\tau \in \mathfrak{t}$, the orthogonal projection $\text{pr}: \mathfrak{k} \rightarrow \mathfrak{t}$ satisfies

$$(12.4) \quad \text{pr}(\text{Ad}(G)\tau) = \text{conv}(W\tau).$$

Lemma 12.8 (Convexity transfer). *Let $\mathfrak{k}$ be compact and semisimple, of any rank, let $D \subset \mathfrak{t}$ be W -invariant, and $\Omega = \text{Ad}(G)D$. Then $\Omega \cap \mathfrak{t} = D$, and Ω is convex if and only if D is convex.*

Proof. If $\tau' \in \mathfrak{t}$ is conjugate to $\tau \in D$, then $\tau' \in W\tau \subset D$; hence $\Omega \cap \mathfrak{t} = D$, and if Ω is convex, so is D . Conversely, let D be convex. By (12.4), $\text{pr}(\Omega) = \bigcup_{\tau \in D} \text{conv}(W\tau) = D$, since $W\tau \subset D$ and $\tau \in \text{conv}(W\tau)$. The convex hull $\text{conv} \Omega$ is $\text{Ad}(G)$ -invariant, as each $\text{Ad}(g)$ is linear, and $\text{pr}(\text{conv} \Omega) = \text{conv}(\text{pr} \Omega) = D$. As pr is the identity on $\mathfrak{t}$, $\text{conv} \Omega \cap \mathfrak{t} \subset D$. Every element of $\text{conv} \Omega$ is conjugate to an element of $\text{conv} \Omega \cap \mathfrak{t}$, hence $\text{conv} \Omega \subset \text{Ad}(G)D = \Omega$, and Ω is convex. $\square$

This is the case of adjoint representations of [40, Theorem 3.6], which holds for arbitrary invariant sets. No closedness or openness of D is needed.

12.4. The reduced problem and the balls. By Lemmas 12.6 and 12.8, Theorem 12.1 follows once we find a W -invariant domain D with a positive real-analytic radial function, which is convex and not a disc, and an anti-invariant real-analytic F on a W -invariant neighbourhood of $\bar{D}$ that satisfies (12.3). The reduction is reversible in the analytic class: if $\Omega = \text{Ad}(G)D$ and an $\text{Ad}(G)$ -invariant u , real analytic on a neighbourhood of $\bar{\Omega}$, satisfy (1.2), then $F = \varpi \cdot u|_{\mathfrak{t}}$ is anti-invariant and satisfies (12.3), by (12.1); any solution can be made invariant by averaging over G , as in Proposition 2.4.

Balls. Let $D = \{|\tau| < R\}$, so that Ω is the ball of radius R . The function $F = J_m(|\tau|) \sin m\theta$ solves $\Delta F + F = 0$ and is anti-invariant, and $F/\varpi = |\tau|^{-m} J_m(|\tau|)$ is the regular radial Helmholtz solution in $\mathbb{R}^{2m+2}$. Since $J'_m(r) = \frac{m}{r} J_m(r) - J_{m+1}(r)$, a multiple of F satisfies (12.3) if and only if $J_{m+1}(R) = 0$ (then $J_m(R) \neq 0$, and the multiple is $R^m F/J_m(R)$), in agreement with the Schiffer balls of $\mathbb{R}^n$ at eigenvalue one, whose radii are the zeros of $J_{n/2} = J_{m+1}$.

The enlarged symmetry. We shall look for D invariant under the dihedral group

$$I_2(2m) = \langle \tau \mapsto \bar{\tau}, \tau \mapsto e^{i\pi/m} \tau \rangle \supset W,$$

and for F with $F(\bar{\tau}) = -F(\tau)$ and $F(e^{i\pi/m} \tau) = -F(\tau)$; then $F \circ w = \det(w)F$ for $w \in W$, and ϖ transforms in the same way. The regular Helmholtz solutions in this class are the series $\sum_{k \geq 0} a_k J_{(2k+1)m}(r) \sin((2k+1)m\theta)$. Since $-1 \in I_2(2m)$, the lifted domain is centrally symmetric; for $m = 3$ the rotation through $\pi/3$ is the composition of -1 with an element of W , while for $m = 4, 6$ one has $-1 \in W$ already. For $m = 1, 2$ the enlarged symmetry is that of Sections 2 and 11.

Near-resonances. By Proposition 12.2, an $\text{Ad}(G)$ -invariant harmonic polynomial h of degree ℓ on $\mathfrak{k}$ corresponds to the anti-invariant harmonic polynomial $\varpi \cdot h|_{\mathfrak{t}}$ of degree $\ell + m$. Anti-invariance under $\tau \mapsto \bar{\tau}$ and invariance under the rotation through $2\pi/m$ force this polynomial to be a multiple of $\text{Im } \tau^{\ell+m}$ with $m \mid \ell$. Thus the invariant spherical harmonics restrict on $\mathfrak{t}$ to multiples of $U_{\ell/m}(\cos m\theta)$, where U_k is the Chebyshev polynomial of the second kind, and the planar mode $J_{m+\ell}(r) \sin((m+\ell)\theta)$ corresponds to the Helmholtz solution $r^{-m} J_{m+\ell}(r) U_{\ell/m}(\cos m\theta)$ in $\mathbb{R}^{2m+2}$; the enlarged symmetry allows $\ell \in 2m\mathbb{Z}$. The linearisation of the Schiffer problem at the ball of radius R , $J_{m+1}(R) = 0$, in the direction of such a harmonic is degenerate if and only if $J_{m+\ell}(R) = 0$, which is impossible for $\ell \geq 2$ by Siegel's theorem [51], as in Section 16.2. (The case $\ell = 1$ occurs only for $m = 1$, where it corresponds to translations along the centre of $\mathfrak{u}(2)$.) The centres used below were located at the near-resonances

$$(m, R, \ell) = (3, j_{4,5}, 6), \quad (4, j_{5,10}, 8), \quad (6, j_{7,25}, 12),$$

with $j_{4,5} \approx 20.826933$, $j_{5,10} \approx 38.159869$, $j_{7,25} \approx 88.474363$ and small divisors $J_9(j_{4,5}) \approx -3.3 \cdot 10^{-3}$, $J_{12}(j_{5,10}) \approx -4.4 \cdot 10^{-4}$, $J_{18}(j_{7,25}) \approx -9.0 \cdot 10^{-4}$ (Remark 12.20).

12.5. The conformal formulation for general m . Fix $m \geq 1$ and $\rho > 1$. For a real sequence $c = (c_j)_{j \equiv 1 \pmod{2m}}$, $j \geq 1$, put $\psi_c(w) = \sum_j c_j w^j$. Since $e^{i\pi j/m} = e^{i\pi/m}$ for $j \equiv 1 \pmod{2m}$,

$$(12.5) \quad \psi_c(\bar{w}) = \overline{\psi_c(w)}, \quad \psi_c(e^{i\pi/m} w) = e^{i\pi/m} \psi_c(w);$$

hence, if ψ_c is injective on $\bar{\mathbb{D}}$, then $D = \psi_c(\mathbb{D})$ is invariant under $I_2(2m)$.

For $n \equiv m \pmod{2m}$, $n \geq m$, and $s \geq 0$ let $S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta$, and let $\mathcal{S}_m$ be the real Banach space of $f = \sum_{n \equiv m \pmod{2m}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$, which is also its norm in $\mathcal{B}_\rho$. Since $\Phi_{k,s}(\bar{w}) = \Phi_{-k,s}(w)$ and $\Phi_{k,s}(e^{i\pi/m} w) = e^{ik\pi/m} \Phi_{k,s}(w)$, and $e^{ik\pi/m} = -1$ exactly when $k \equiv m \pmod{2m}$, the real-valued elements $f \in \mathcal{B}_\rho$ with $f(\bar{w}) = -f(w)$ and $f(e^{i\pi/m} w) = -f(w)$ are exactly the elements of $\mathcal{S}_m$. Let $\mathcal{G}_m = \{g \in \mathcal{S}_m :$

$g_{n,0} = 0$ for all n }, and define K on $\mathcal{G}_m$ by (5.1). For $m = 1$ and $m = 2$ these are the spaces $\mathcal{S}, \mathcal{G}$ of Section 4 and $\mathcal{S}_6, \mathcal{G}_6$ of Section 11. Put

$$\kappa_m = \frac{1}{(m+2)(m+4)}.$$

Lemma 12.9 (Results used verbatim). *The statements of Lemma 11.5(a)–(f) hold with $\mathcal{S}_6, \mathcal{G}_6$ and the condition $n \equiv 2 \pmod{4}$ replaced by $\mathcal{S}_m, \mathcal{G}_m$ and $n \equiv m \pmod{2m}$, with $1/24$ replaced by κ_m , and with the Cauchy datum $(y_1 y_2, (y_2, y_1))$ in (f) replaced by $(\varpi, \nabla \varpi)$. In particular $K: \mathcal{G}_m \rightarrow \mathcal{S}_m$ is bounded with $\|K\| = \kappa_m$, the supremum of $1/(q(q+2))$ over $q = n+2s \geq m+2$ being attained at $(n, s) = (m, 1)$.*

Proof. Only (f) needs a comment. In the coordinates $\zeta = x + iy, \chi = x - iy$ one has $\varpi = (\zeta^m - \chi^m)/(2i)$, whose holomorphic extension $F_0(a, b) = (\gamma(a)^m - \gamma^\dagger(b)^m)/(2i)$ in the proof of Lemma 10.4 satisfies $\partial_a \partial_b F_0 = 0$; the rest of that proof is unchanged. $\square$

Let $B > 0$ (a fixed *field scale*) and $b > 0$ (in the application, $b = c_1^\circ$), and put

$$\alpha = \frac{b^{m+1}}{B}, \quad \omega_j = \alpha(j+m)\rho^{j+m-1}, \quad \theta_0 = \frac{1}{\alpha(m+1)\rho^{m-1}}, \quad \theta_1 = \frac{1}{\alpha\rho^m}.$$

Let X_m be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}_m$ and $c = (c_j)_{j \equiv 1 \pmod{2m}}$ real, normed by $\|(g, c)\|_X = \|g\|_\rho + \|c\|$, $\|c\| = \sum_j \omega_j |c_j|$. Since $\rho^j/\omega_j = 1/(\alpha(j+m)\rho^{m-1})$ and $j\rho^{j-1}/\omega_j = j/(\alpha(j+m)\rho^m)$,

$$(12.6) \quad N_\rho(\psi_c) \leq \theta_0 \|c\|, \quad N_\rho(\psi'_c) \leq \theta_1 \|c\|.$$

The powers of w in ψ_c^m are $\equiv m \pmod{2m}$, so $h_c = \text{Im}(\psi_c^m)/B$ is a harmonic element of $\mathcal{S}_m$. Define

$$(12.7) \quad \mathcal{H}_m(g, c) = g + |\psi'_c|^2(Kg + h_c), \quad h_c = \frac{\text{Im}(\psi_c^m)}{B}.$$

For $m = 1, B = 1$ this is (6.1), with $\omega_j = b^2(j+1)\rho^j$, $\theta_0 = \vartheta_0$ and $\theta_1 = \vartheta_1$; for $m = 2, B = 2$ it is (11.3), with the weights of Section 11.

Lemma 12.10. *$\mathcal{H}_m$ is a continuous polynomial map of degree $m+2$ from X_m to $\mathcal{S}_m$. If $\mathcal{H}_m(g, c) = 0$, then $V = Kg + h_c$ belongs to $C^1(\mathbb{D}) \cap \mathcal{S}_m$, and it satisfies $\Delta V + |\psi'_c|^2 V = 0$ in $\mathcal{D}'(\mathbb{D})$ and $V - h_c = 0, \nabla(V - h_c) = 0$ on $\partial\mathbb{D}$.*

Proof. ψ'_c has real coefficients and only powers divisible by $2m$, so $|\psi'_c|^2$ is real, invariant under $w \mapsto \bar{w}$ and $w \mapsto e^{i\pi/m}w$, and multiplication by it maps $\mathcal{S}_m$ into itself with norm at most $N_\rho(\psi'_c)^2$ by (4.4). Moreover $\|h_c\|_\rho \leq N_\rho(\psi_c)^m/B$, since $w^k = \Phi_{k,0}$ has norm ρ^k and N_ρ is submultiplicative. With (12.6) and $\|K\| = \kappa_m$, (12.7) is a sum of bounded multilinear maps of degree at most $m+2$. The rest is the proof of Lemma 6.1, using Lemma 12.9. $\square$

Lemma 12.11. *Let ψ be holomorphic and injective on a neighbourhood of $\mathbb{D}$ with $\psi' \neq 0$ there, $D = \psi(\mathbb{D})$, and let $V \in C^1(\mathbb{D})$ be real valued with $\Delta V + |\psi'|^2 V = 0$ in $\mathcal{D}'(\mathbb{D})$ and $V - h = 0, \nabla(V - h) = 0$ on $\partial\mathbb{D}$, where $h = \text{Im}(\psi^m)/B$. Then $F = BV \circ \psi^{-1} \in C^1(\bar{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F\varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = \varpi, \nabla F = \nabla \varpi$ on ∂D . If moreover $\psi = \psi_c$ and $V \in \mathcal{S}_m$, then $F(\bar{\tau}) = -F(\tau)$ and $F(e^{i\pi/m}\tau) = -F(\tau)$.*

Proof. Since $\varpi \circ \psi = \text{Im} \psi^m = Bh$, we have $B(V - h) = (F - \varpi) \circ \psi$, and the proof of Lemma 3.1 applies. The parities follow from (12.5) and the characterisation of $\mathcal{S}_m$. $\square$

Expansion at a centre. Let $x^\circ = (g^\circ, c^\circ) \in X_m$ have finitely many nonzero coefficients, $b = c_1^\circ$, and put

$$p = \psi'_{c^\circ}, \quad P = N_\rho(p), \quad C = N_\rho(\psi_{c^\circ}), \quad V^\circ = Kg^\circ + h_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X_m$, $|\psi'_{c^\circ + \eta}|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$ with $L(\eta) = 2 \operatorname{Re}(\bar{p} \psi'_\eta)$ and $Q(\eta, \eta') = \operatorname{Re}(\psi'_\eta \overline{\psi'_{\eta'}})$, and $h_{c^\circ + \eta} = h_{c^\circ} + \sum_{k=1}^m H_k(\eta)$ with $H_k(\eta) = \binom{m}{k} \operatorname{Im}(\psi_{c^\circ}^{m-k} \psi_\eta^k) / B$. Hence

$$(12.8) \quad D\mathcal{H}_m(x^\circ)h = \delta g + |p|^2(K\delta g + H_1(\eta)) + L(\eta)V^\circ,$$

and $\mathcal{H}_m(x^\circ + h) = \mathcal{H}_m(x^\circ) + D\mathcal{H}_m(x^\circ)h + \sum_{q=2}^{m+2} B_q(h, \dots, h)$, where the degree- q part consists of the terms $|p|^2 H_q(\eta)$ ($q \leq m$), $L(\eta)H_{q-1}(\eta)$ ($2 \leq q \leq m+1$), $Q(\eta, \eta)H_{q-2}(\eta)$ ($3 \leq q \leq m+2$), $L(\eta)K\delta g + Q(\eta, \eta)V^\circ$ ($q = 2$) and $Q(\eta, \eta)K\delta g$ ($q = 3$), and B_q is the symmetric q -linear map with this diagonal.

Lemma 12.12. *Let $h_k = \binom{m}{k} C^{m-k} \theta_0^k / B$ for $1 \leq k \leq m$, and $h_k = 0$ otherwise. For $2 \leq q \leq m+2$ and all $h_i \in X_m$, $\|B_q(h_1, \dots, h_q)\|_\rho \leq c_q \|h_1\|_X \cdots \|h_q\|_X$, where*

$$(12.9) \quad c_q = P^2 h_q + 2P\theta_1 h_{q-1} + \theta_1^2 h_{q-2} + \mathbf{1}_{q=2}(2P\theta_1 \kappa_m + \theta_1^2 V_0) + \mathbf{1}_{q=3} \theta_1^2 \kappa_m.$$

Proof. By (12.6), (4.4), $\|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$ and submultiplicativity of N_ρ ,

$$\|L(\eta)f\|_\rho \leq 2P\theta_1 \|\eta\| \|f\|_\rho, \quad \|Q(\eta, \eta')f\|_\rho \leq \theta_1^2 \|\eta\| \|\eta'\| \|f\|_\rho, \quad \|K\delta g\|_\rho \leq \kappa_m \|\delta g\|_\rho,$$

and the symmetric k -linear map with diagonal H_k is bounded by h_k , because it is a symmetrised product of k multiplications by ψ_η , each bounded by θ_0 , and of the fixed factor $\binom{m}{k} \psi_{c^\circ}^{m-k} / B$. Each degree- q term listed above is a product of q such linear factors, and the symmetrisation of a product of linear maps is bounded by the product of their norms. Adding the bounds of the terms of degree q gives c_q . $\square$

Proposition 12.13. *Let $A: \mathcal{S}_m \rightarrow X_m$ be bounded, linear and injective. Suppose that $\|A\mathcal{H}_m(x^\circ)\|_X \leq Y$, $\|I - AD\mathcal{H}_m(x^\circ)\|_{X \rightarrow X} \leq Z$ and $\|A\| c_q \leq C_q$ for $2 \leq q \leq m+2$, and that $r > 0$ satisfies*

$$(12.10) \quad Y + (Z-1)r + \sum_{q=2}^{m+2} C_q r^q < 0, \quad Z-1 + \sum_{q=2}^{m+2} q C_q r^{q-1} < 0.$$

Then $\mathcal{H}_m$ has exactly one zero in the closed ball $\overline{B}_{X_m}(x^\circ, r)$.

Proof. This is the proof of Proposition 11.8: $B_q(h_1, \dots, h_1) - B_q(h_2, \dots, h_2)$ is a sum of q terms, each containing $h_1 - h_2$ once, so its image under A has norm at most $q C_q r^{q-1} \|h_1 - h_2\|_X$ on the ball. $\square$

12.6. The principal part and the approximate inverse.

Proposition 12.14. *For every $x = (g, c) \in X_m$ and $j \equiv 1 \pmod{2m}$,*

$$D\mathcal{H}_m(x)e_j = 2 \operatorname{Re}(jw^{j-1} \overline{\psi'_c})(Kg + h_c) + \frac{m}{B} |\psi'_c|^2 \operatorname{Im}(\psi_c^{m-1} w^j).$$

Let $x_b = (0, b e_1)$, so that $\psi_{x_b}(w) = bw$. For $n \geq 0$ write

$$(12.11) \quad r^{n+2m} \sin n\theta = \sum_{s=0}^m q_s(n) S_{n,s};$$

then $q_s(n) \geq 0$, $\sum_s q_s(n) = 1$ and $q_0(n) = (n+1)/(n+m+1)$. For every $j \equiv 1 \pmod{2m}$ with $j \geq 2m+1$,

$$(12.12) \quad T_j := D\mathcal{H}_m(x_b)e_j = \alpha \left\{ (j+m)S_{j+m-1,0} - j \sum_{s=0}^m q_s(j-m-1)S_{j-m-1,s} \right\}.$$

Proof. The first formula follows by differentiating $|\psi'_c + \varepsilon j w^{j-1}|^2$ and $\text{Im}((\psi_c + \varepsilon w^j)^m)/B$ at $\varepsilon = 0$. The statements on q_s are Lemma 4.4 for $w^{n+m}\bar{w}^m = r^{n+2m}e^{in\theta}$, after taking imaginary parts. At x_b we have $Kg + h_c = b^m \text{Im}(w^m)/B$ and $\psi'_c = b$, so

$$\begin{aligned} T_j &= \alpha [2j \text{Re}(w^{j-1}) \text{Im}(w^m) + m \text{Im}(w^{j+m-1})] \\ &= \alpha r^{j+m-1} [(j+m) \sin((j+m-1)\theta) - j \sin((j-m-1)\theta)], \end{aligned}$$

using $2 \cos((j-1)\theta) \sin(m\theta) = \sin((j+m-1)\theta) - \sin((j-m-1)\theta)$. With $n = j-m-1 \geq m$, $n \equiv m \pmod{2m}$, (12.11) gives (12.12). $\square$

The harmonic part of T_j is $\alpha(j+m)$ in row $(j+m-1, 0)$ and $-\alpha(j-m)$ in row $(j-m-1, 0)$, a difference operator of step $2m$; the nonharmonic part lies in the rows $(j-m-1, s)$, $1 \leq s \leq m$, with total mass $\alpha j \sum_{s \geq 1} q_s(j-m-1) = \alpha m$. For $m = 1, 2$, (12.12) reduces to Lemma 7.3 and to (11.8) (Lemma 11.9 gives q_s for $m = 2$).

Finite block and tail. Let $M \equiv m \pmod{2m}$ and $S \geq m$. The *finite inputs* are the coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and c_j with $j \leq M-m+1$; the *finite rows* are (n, s) with $n \leq M$, $1 \leq s \leq S$, and $(n, 0)$ with $n \leq M$. All others are *tail inputs* and rows, $X_m = X_f \oplus X_t$, $\mathcal{S}_m = \mathcal{S}_f \oplus \mathcal{S}_t$, and $\dim X_f = \dim \mathcal{S}_f = (S+1)(M+m)/(2m)$. Let M_f be the matrix of $D\mathcal{H}_m(x^\circ)$ on finite inputs and finite rows. On tail inputs we use the model $T(\delta g, \eta) = \delta g + \sum_{j \geq M+m+1} \eta_j T_j$. By (12.12), only η_{M+m+1} reaches finite rows, since $S \geq m$:

$$(12.13) \quad B_f(\delta g, \eta) = -\eta_{M+m+1} \alpha \left\{ (M+1)e_{(M,0)} + (M+m+1) \sum_{s=1}^m q_s(M)e_{(M,s)} \right\}.$$

Write $Te = (B_f e, T_{tt} e)$ for tail inputs e , and put $\tau_m = (1 - \rho^{-2m})^{-1}$.

Lemma 12.15. $T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection with $\|T_{tt}\| \leq 1 + \rho^{-2m}$. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is

$$(12.14) \quad \eta_{n-m+1} = \sum_{k \geq 0} \frac{y_{n+2mk,0}}{\alpha(n+2mk+1)}, \quad \delta g_{n,s} = y_{n,s} + \alpha(n+m+1)q_s(n)\eta_{n+m+1} \quad (1 \leq s \leq m),$$

for $n \geq M+2m$, $n \equiv m \pmod{2m}$, and $\delta g_{n,s} = y_{n,s}$ for all other tail g -inputs.

Proof. By (12.12), the tail harmonic rows read $\alpha(n+1)(\eta_{n-m+1} - \eta_{n+m+1}) = y_{n,0}$ for $n \geq M+2m$, the rows (n, s) , $1 \leq s \leq m$, $n \geq M+2m$, read $\delta g_{n,s} - \alpha(n+m+1)q_s(n)\eta_{n+m+1} = y_{n,s}$, and all other tail rows read $\delta g_{n,s} = y_{n,s}$. The rest is the proof of Lemma 11.11: $\eta_j \rightarrow 0$ gives uniqueness and (12.14), and the column $T_{tt}e_j$, $j \geq M+3m+1$, has weighted norm $\omega_j(1 + j\rho^{-2m}/(j+m)) \leq (1 + \rho^{-2m})\omega_j$. $\square$

Let $\hat{A}$ be a matrix with dyadic entries, regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, with

$$(12.15) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1,$$

and define $A = \begin{pmatrix} \hat{A} - \hat{A}B_f T_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix}: \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X_f \oplus X_t$. By Lemma 7.5 (Lemma 12.9), A is bounded and injective, and $ATe = e$ for every tail input e .

Lemma 12.16. *Let $\|R\| = \max_{(n,s) \text{ finite row}} \|\hat{A}e_{(n,s)}\|_X / \rho^n$,*

$$H = \left\| \hat{A} \alpha \left((M+1)e_{(M,0)} + (M+m+1) \sum_{s=1}^m q_s(M) e_{(M,s)} \right) \right\|_X,$$

and, for $n \geq M+2m$, $n \equiv m \pmod{2m}$,

$$(12.16) \quad h_n = \frac{1}{(n+1)\rho^n} \left(\sum_{\substack{M+2m \leq k \leq n \\ k \equiv m \pmod{2m}}} (k+1)\rho^k + m \sum_{\substack{M+2m \leq k \leq n-2m \\ k \equiv m \pmod{2m}}} \rho^k + \frac{H}{\alpha} \right).$$

Let $N_H \geq M+2m$, $N_H \equiv m \pmod{2m}$. Then A maps every residual supported in tail rows with norm at most

$$A_t := \max \left\{ 1, \max_{M+2m \leq n \leq N_H} h_n, \tau_m + \frac{m\tau_m\rho^{-2m}}{N_H+2m+1} + \frac{H}{\alpha(N_H+2m+1)\rho^{N_H+2m}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. As in Lemma 11.12: the normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to $\eta_{k-m+1} = \rho^{-n}/(\alpha(n+1))$ for $M+2m \leq k \leq n$, of weights $\omega_{k-m+1} = \alpha(k+1)\rho^k$; to the coordinates $\delta g_{k,s}$, $1 \leq s \leq m$, $M+2m \leq k \leq n-2m$, of weight ρ^k and total absolute value $\alpha m \rho^{-n}/(\alpha(n+1))$, because $\sum_{s \geq 1} (k+m+1)q_s(k) = m$; and to the finite part $\eta_{M+m+1}\hat{A}\alpha(\dots)$, of norm $H\rho^{-n}/(\alpha(n+1))$. The sum is h_n . For $n \geq N_H+2m$ we use $(k+1)/(n+1) \leq 1$ and sum the geometric series. $\square$

12.7. The defect bounds. We bound $\|I - AD\mathcal{H}_m(x^\circ)\|$ by Lemma 7.2, splitting the inputs into five classes; every input belongs to exactly one class, and no column is estimated by sampling. Let $d = \deg p = M - m$ and $L = M + d$.

Residual and finite columns. $\mathcal{H}_m(x^\circ)$ and the columns $D\mathcal{H}_m(x^\circ)e$, e a finite input, are polynomials in $w, \bar{w}$; they are expanded in the basis $S_{n,s}$ without truncation, A is applied exactly using (12.13) and (12.14), and Y and Z_f bound $\|A\mathcal{H}_m(x^\circ)\|_X$ and the normalised norms of the finite columns of $I - AD\mathcal{H}_m(x^\circ)$.

Lemma 12.17. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{H}_m(x^\circ))e = -A(|p|^2 K S_{n,s})$. Let $p = p_{\leq d_s} + p_{> d_s}$ be the splitting of p into the terms of degree $\leq d_s$ and $> d_s$, where $d_s \equiv 0 \pmod{2m}$ is fixed, $P_s = N_\rho(p_{\leq d_s})$, $P_t = N_\rho(p_{> d_s})$ and $\delta_a = 2P_s P_t + P_t^2$. Then*

$$(12.17) \quad \frac{\|(I - AD\mathcal{H}_m(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq d_s}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| \delta_a}{q(q+2)}.$$

Moreover, the right-hand side is at most $(A_t P_s^2 + \|A\| \delta_a)/((n_f + 2)(n_f + 4))$ if $n \geq n_f := M + d_s + 2m$, and at most $(P_s^2 + \|A\| \delta_a)/((m + 2s_f)(m + 2s_f + 2))$ if $s \geq s_f := S + d_s + 2$.

Proof. This is the proof of Lemma 11.13. Multiplication by $|p_{\leq d_s}|^2$ changes frequencies by at most d_s , so for $n \geq n_f$ all components lie in rows of frequency at least $M + 2m$, which are tail rows; and by (5.1) and Lemma 4.3, for $s \geq s_f$ all radial indices are at least $s - 1 - d_s > S$, so all components lie in nonharmonic tail rows, on which A is the identity. Finally $q \geq n_f + 2$, respectively $q \geq m + 2s_f$. $\square$

The tail g -inputs with $n \leq M + d_s$ and $s \leq S + d_s + 1$ are the *near* ones; for them the first term of (12.17) is computed without truncation, and $Z_{g,\partial}$ bounds the maximum of the right-hand side. The remaining tail g -inputs are covered by the last statement of Lemma 12.17, and $Z_{g,\infty}$ is the larger of the two bounds.

Shape columns. For a tail shape input e_j , $j \geq M+m+1$, we have $(I-AD\mathcal{H}_m(x^\circ))e_j = -A\Delta_j$ with $\Delta_j = D\mathcal{H}_m(x^\circ)e_j - T_j$. For the *near* shape inputs $M+m+1 \leq j \leq j_* - 2m$, Δ_j is a polynomial, the columns are computed without truncation, and $Z_{\text{sh},\partial}$ bounds their normalised norms. Let $W^\circ = Kg^\circ$, $h^\circ = h_{c^\circ}$ and $h_{be_1} = b^m \text{Im}(w^m)/B$. By Proposition 12.14,

$$(12.18) \quad \Delta_j = 2 \text{Re}(jw^{j-1}\Gamma_1) + m \text{Im}(w^j\Gamma_0) + 2 \text{Re}(jw^{j-1}\bar{p}W^\circ),$$

where

$$\Gamma_1 = \bar{p}h^\circ - bh_{be_1}, \quad \Gamma_0 = \frac{|p|^2\psi_{c^\circ}^{m-1}}{B} - \alpha w^{m-1}.$$

By Lemma 5.1(c), $W^\circ = (1-|w|^2)^2Q^\circ$ with $Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)}(2r^2-1) \sin n\theta$, whose finite expansion in the basis $S_{n,t}$ is given by the connection formulas of Section 8, with positive rational coefficients. The function $w^L\bar{p}Q^\circ$ has only nonnegative frequencies; write $w^L\bar{p}Q^\circ = \sum_{k,t \geq 0} \gamma_{k,t}^W \Phi_{k,t}$, $\Gamma_1 = \sum \gamma_{k,t}^1 \Phi_{k,t}$ and $\Gamma_0 = \sum \gamma_{k,t}^0 \Phi_{k,t}$ (finite sums, computed in interval arithmetic). For $j_* \equiv 1 \pmod{2m}$ with $j_* > L$ let

$$\begin{aligned} N_1^+ &= \sum_{k+j_*-1 > M} \rho^{|k|} |\gamma_{k,t}^1|, & N_0^+ &= \sum_{k+j_* > M} \rho^{|k|} |\gamma_{k,t}^0|, \\ \Sigma_W^+ &= \sum_{k+j_*-1-L > M} \frac{|\gamma_{k,t}^W| (2t+1)(2t+3) \rho^{k-L-m}}{(k+j_*-L)^2}, \end{aligned}$$

and let N_1^-, N_0^-, Σ_W^- be the corresponding sums over the remaining indices.

Proposition 12.18. *For every $j \equiv 1 \pmod{2m}$ with $j \geq j_* > L$,*

$$\frac{\|(I-AD\mathcal{H}_m(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := U'_\Gamma + U'_W,$$

where

$$U'_\Gamma = \frac{1}{\alpha} \left(\frac{2(A_t N_1^+ + \|A\| N_1^-)}{\rho^m} + \frac{m(A_t N_0^+ + \|A\| N_0^-)}{\rho^{m-1}(j_*+m)} \right), \quad U'_W = \frac{8}{\alpha} (A_t \Sigma_W^+ + \|A\| \Sigma_W^-).$$

If moreover $j_*-1-((m+1)M-m^2) > M$, then $N_1^- = N_0^- = \Sigma_W^- = 0$ and $Z_{\text{sh},\infty} \leq A_t(U_\Gamma + U_W)$ with

$$U_\Gamma = \frac{1}{\alpha} \left(\frac{2\|\Gamma_1\|_\rho}{\rho^m} + \frac{m\|\Gamma_0\|_\rho}{\rho^{m-1}(j_*+m)} \right), \quad U_W = \frac{8}{\alpha(j_*-L)^2} \sum_{k,t} |\gamma_{k,t}^W| (2t+1)(2t+3) \rho^{k-L-m}.$$

Proof. Fix $j \geq j_*$. By Lemma 4.2, $w^{j-1}\Phi_{k,t}$ is a combination of $\Phi_{k+j-1,t'}$ of norm at most $\rho^{j-1+|k|}$, and taking real or imaginary parts only changes the signs of frequencies. Hence the contribution of $\gamma_{k,t}^1$ to $2 \text{Re}(jw^{j-1}\Gamma_1)$ has norm at most $2j\rho^{j-1}\rho^{|k|}|\gamma_{k,t}^1|$ and lies in rows of frequency $|k+j-1|$; if $k+j-1 > M$ these are tail rows, on which A has norm at most A_t (Lemma 12.16), and otherwise we use $\|A\| \geq A_t$. Dividing by $\omega_j = \alpha(j+m)\rho^{j+m-1}$ and using $j/(j+m) \leq 1$ gives the first term of U'_Γ . The term $m \text{Im}(w^j\Gamma_0)$ is treated in the same way, with $m\rho^j/\omega_j = m/(\alpha(j+m)\rho^{m-1}) \leq m/(\alpha(j_*+m)\rho^{m-1})$. Finally $jw^{j-1}\bar{p}W^\circ = j(1-|w|^2)^2w^{j-1-L}(w^L\bar{p}Q^\circ)$ with $j-1-L \geq 0$, and by Lemma 8.3 with $N = j-1-L$ the contribution of $\gamma_{k,t}^W$ has frequency $k+j-1-L$ and normalised norm at most

$$\frac{2j}{\alpha(j+m)\rho^{j+m-1}} |\gamma_{k,t}^W| \frac{4(2t+1)(2t+3)\rho^{k+j-1-L}}{(k+j-L)^2} \leq \frac{8|\gamma_{k,t}^W|(2t+1)(2t+3)\rho^{k-L-m}}{\alpha(k+j-L)^2}.$$

As j increases, every output frequency increases, so the sets of indices counted with A_t can only grow, and every factor above decreases; hence the bound at j_* covers all $j \geq j_*$. For the

last statement: the frequencies of $\bar{p}$, h° , ψ_{c° and Q° lie in $[-d, 0]$, $[-m(M - m + 1), m(M - m + 1)]$, $[0, M - m + 1]$ and $[-M, M]$, so the frequencies of Γ_1 , Γ_0 and $\bar{p}Q^\circ$ are all at least $-(m(M - m + 1) + d) = -((m + 1)M - m^2)$, and $L = 2M - m \leq (m + 1)M - m^2$. Under the stated condition all output frequencies therefore exceed M , so that $N_1^- = N_0^- = \Sigma_W^- = 0$, and $(k + j_* - L)^2 \geq (j_* - L)^2$ since $k \geq 0$. $\square$

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - AD\mathcal{H}_m(x^\circ)\|_{X \rightarrow X}$.

12.8. Certified bounds and the exact zeros. For $m = 3, 4, 6$ we take $\rho = 11/10$ and the data of Table 5. Each centre x° consists of the finite inputs listed there, given as binary64 numbers interpreted as exact dyadic rationals. The field scale B is such a number as well: $B = 1$ for $m = 3$, and $B \approx j_{m+1,k}^m$ is the binary64 number with hexadecimal representation `0x1.02d7fa4b541bbp+21` for $m = 4$ and `0x1.beb0441320216p+38` for $m = 6$. The finite inverse $\hat{A}$ is a fixed binary64 approximation of M_f^{-1} whose entries are interpreted as exact dyadic rationals; only (12.15) is used. All quantities of Sections 12.5–12.7 were evaluated in ball arithmetic at 128 bits (Section 15.5). Table 6 lists the results, rounded outward.

	$n = 8$	$n = 10$	$n = 14$
$(\mathfrak{k}, m)$	$(\mathfrak{su}(3), 3)$	$(\mathfrak{so}(5), 4)$	$(\mathfrak{g}_2, 6)$
(M, S)	$(45, 24)$	$(52, 36)$	$(114, 60)$
finite $g_{n,s}$	8×24	7×36	10×60
finite c_j	$c_1, c_7, \dots, c_{43}$	$c_1, c_9, \dots, c_{49}$	$c_1, c_{13}, \dots, c_{109}$
$\dim X_f$	200	259	610
B	1	≈ 2120447.2868	$\approx 4.7962817 \cdot 10^{11}$
$b = c_1^\circ$	20.78702921089377...	38.15292567121707...	88.49730472771703...
$d_s; n_f, s_f$	24; 75, 50	24; 84, 62	60; 186, 122
$j_*; L$	223; 87	305; 100	241; 222
N_H	645	852	1314
r	1/30000	10^{-7}	1/6000

TABLE 5. Parameters of the three computer-assisted proofs ($\rho = 11/10$). The finite shape inputs are c_j , $j \equiv 1 \pmod{2m}$, $j \leq M - m + 1$; the finite g -inputs are $g_{n,s}$, $n \equiv m \pmod{2m}$, $n \leq M$, $1 \leq s \leq S$. B and b are exact dyadic rationals.

Theorem 12.19. *Let $m \in \{3, 4, 6\}$, and let M , S , B , x° and r be as in Table 5. There is exactly one $x^* = (g^*, c^*) \in X_m$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{H}_m(x^*) = 0$.*

Proof. The condition (12.15) holds by Table 6, so A is bounded and injective. The hypotheses of Proposition 12.18 hold: for $m = 3$ and $m = 4$, $j_* - 1 - ((m + 1)M - m^2)$ equals $222 - 171 = 51 > 45$ and $304 - 244 = 60 > 52$, and for $m = 6$, $j_* = 241 > L = 222$. By Table 6, Proposition 12.13 applies with the following rational majorants:

- $m = 3$: $Y = 153 \cdot 10^{-7}$, $Z = 43/100$, $C_2 = 1/125$, $C_3 = 5 \cdot 10^{-10}$, $C_4 = 2 \cdot 10^{-18}$, $C_5 = 3 \cdot 10^{-26}$;
- $m = 4$: $Y = 10^{-8}$, $Z = 56/100$, $C_2 = 396$, $C_3 = 72/1000$, $C_4 = 4 \cdot 10^{-7}$, $C_5 = 2 \cdot 10^{-11}$, $C_6 = 4 \cdot 10^{-16}$;

Quantity	$n = 8$	$n = 10$	$n = 14$
P	< 21.788460	< 38.428146	< 91.291342
C	< 23.020116	< 42.001369	< 97.545293
V_0	< 131547.591	< 28.746248	< 79.018365
$\ I - \hat{A}M_f\ $	$< 2.2858 \cdot 10^{-14}$	$< 5.7259 \cdot 10^{-12}$	$< 5.0709 \cdot 10^{-9}$
$\ R\ $	< 963.7779	< 10194.7512	< 11766.8786
H	$< 1.4540263 \cdot 10^9$	< 718348.086	$< 1.2223739 \cdot 10^9$
A_t	< 2.302038	< 2.014454	< 1.661175
$\ A\ $ (ceiling)	964	10195	11767
Y	$< 1.519444 \cdot 10^{-5}$	$< 9.591861 \cdot 10^{-9}$	$< 4.592200 \cdot 10^{-5}$
Z_f	< 0.052262	< 0.060605	< 0.120542
P_s	< 21.788460	< 38.428143	< 91.291236
P_t	$< 1.4978 \cdot 10^{-7}$	$< 2.6609 \cdot 10^{-6}$	$< 1.05815 \cdot 10^{-4}$
δ_a	$< 6.5267 \cdot 10^{-6}$	$< 2.04502 \cdot 10^{-4}$	< 0.0193200
near g -columns; max. at	396; (51, 1)	358; (60, 1)	1215; (126, 1)
$Z_{g,\partial}$	< 0.194306	< 0.462762	< 0.576826
far bound, $n \geq n_f$	< 0.179735	< 0.393351	< 0.393946
far bound, $s \geq s_f$	< 0.043897	< 0.088871	< 0.135896
near shape columns; max. at	29; $j = 49$	31; $j = 57$	10; $j = 121$
$Z_{sh,\partial}$	< 0.284628	< 0.552897	< 0.600896
$\ \Gamma_1\ _\rho$ resp. N_1^+	< 14093.696	< 0.466786	< 5.147419
N_1^-	—	—	$< 3.3948 \cdot 10^{-8}$
$\ \Gamma_0\ _\rho$ resp. N_0^+	< 21422.774	< 0.692754	< 7.696736
N_0^-	—	—	0
Σ_W^+	—	—	< 3.293822
Σ_W^-	—	—	$< 1.0394 \cdot 10^{-17}$
U_Γ resp. U'_Γ	< 0.114684	< 0.016902	< 0.111093
U_W resp. U'_W	< 0.067068	< 0.088733	< 0.493856
$Z_{sh,\infty}$ (all $j \geq j_*$)	< 0.418572	< 0.212795	< 0.604948
Z	< 0.418572	< 0.552897	< 0.604948
C_2	< 0.0072192	< 395.40418	< 210.53260
C_3	$< 4.8836 \cdot 10^{-10}$	< 0.071133	< 0.0060610
C_4	$< 1.5491 \cdot 10^{-18}$	$< 3.2353 \cdot 10^{-7}$	$< 1.9135 \cdot 10^{-9}$
C_5	$< 2.1152 \cdot 10^{-26}$	$< 1.7470 \cdot 10^{-11}$	$< 2.2718 \cdot 10^{-14}$
C_6	—	$< 3.7236 \cdot 10^{-16}$	$< 1.5942 \cdot 10^{-19}$
C_7	—	—	$< 6.1326 \cdot 10^{-25}$
C_8	—	—	$< 9.9963 \cdot 10^{-31}$

TABLE 6. Certified bounds, rounded outward. The row $\|A\|$ gives the rational ceiling for $\max\{\|R\|, A_t\}$ used in $C_q = \|A\| c_q$; the tail estimates use this ceiling ($n = 8$) or the certified enclosure of $\|R\|$ ($n = 10, 14$). For $n = 8, 10$ the far shape bound is the second one of Proposition 12.18, $Z_{sh,\infty} = A_t(U_\Gamma + U_W)$ (for $n = 8$ with the ceiling $A_t \leq 2.303$); for $n = 14$ it is the first one, $Z_{sh,\infty} = U'_\Gamma + U'_W$. The maximum defining A_t is attained by the remainder term for $n = 8$, and at $n = M + 2m$ for $n = 10, 14$. P , C and V_0 are intermediate values of the final verification scripts; the complete enclosures of the other quantities are recorded in the certificates and receipts (Section 15.5).

- $m = 6$: $Y = 46 \cdot 10^{-6}$, $Z = 61/100$, $C_2 = 211$, $C_3 = 61 \cdot 10^{-4}$, $C_4 = 2 \cdot 10^{-9}$, $C_5 = 3 \cdot 10^{-14}$, $C_6 = 2 \cdot 10^{-19}$, $C_7 = 7 \cdot 10^{-25}$, $C_8 = 2 \cdot 10^{-30}$.

For these values and $r = 1/30000$, 10^{-7} , $1/6000$, respectively, exact rational arithmetic gives for the two expressions in (12.10)

$$\begin{aligned} & -3.699991111111111 \dots \cdot 10^{-6} \quad \text{and} \quad -0.5699994666666666 \dots \quad (m = 3), \\ & -3.399603999999992 \dots \cdot 10^{-8} \quad \text{and} \quad -0.4399207999999978 \dots \quad (m = 4), \\ & -1.31388886064814 \dots \cdot 10^{-5} \quad \text{and} \quad -0.319666661583333 \dots \quad (m = 6), \end{aligned}$$

all negative. $\square$

Remark 12.20 (The centres). The centres were obtained numerically, and their provenance plays no role in the proof. A Fourier–Bessel collocation of (12.3) in the enlarged symmetry class, at the ball wavenumber $j_{m+1,k}$ of Section 12.4 and seeded with the near-resonant mode $\ell = 2m$, produced numerical solutions with Cauchy defects of order 10^{-12} or smaller. Their domains were mapped conformally to the disc by Theodorsen’s method, and Newton’s method was applied to the Galerkin truncation of (12.7). For $m = 4, 6$ the unscaled field has boundary data of size about $j_{m+1,k}^m$, and double-precision Newton iterations stagnated at residuals of about 10^{-6} and 2; dividing the field by the frozen constant B , which does not change the Cauchy data of $F = BV \circ \psi^{-1}$, removed this obstruction. For $m = 6$ a first centre with $M = 54$ gave $Y \approx 1.7$, the defect being concentrated in the omitted angular frequency 66; this led to $M = 114$. For orientation we record some numerical observations about the centres, not used in the proofs. The boundary radius of D lies between 20.7086 and 20.8635 ($m = 3$), 38.1390 and 38.1667 ($m = 4$), 88.4477 and 88.5402 ($m = 6$), so that $\partial\Omega$ deviates from the sphere of radius $j_{m+1,k}$ by between -0.57% and $+0.18\%$, -0.055% and $+0.018\%$, -0.030% and $+0.074\%$, respectively; the minimum of $\operatorname{Re}(1 + w\psi''/\psi')$ on $\partial\mathbb{D}$ is about 0.832, 0.973 and 0.896, close to the certified lower bounds of Table 7.

12.9. Geometry and proof of Theorem 12.1.

Proposition 12.21. *Let $m \in \{3, 4, 6\}$, let c be a real sequence indexed by $j \equiv 1 \pmod{2m}$ with $\sum_j \omega_j |c_j - c_j^\circ| \leq r$ (for instance $c = c^*$), and put $R_e = 21/20$, $E_1 = r\theta_1$, $E_2 = r/(\alpha\rho^{m-1}e \log \rho)$,*

$$\Lambda_e = b - \sum_{j>1} j |c_j^\circ| R_e^{j-1} - E_1, \quad L_1 = b - \sum_{j>1} j |c_j^\circ| - E_1, \quad U_2 = \sum_{j>1} j(j-1) |c_j^\circ| + E_2.$$

Then $\operatorname{Re} \psi'_c \geq \Lambda_e$ on $\{|w| \leq R_e\}$, $|\psi'_c| \geq L_1$ and $|\psi''_c| \leq U_2$ on $\overline{\mathbb{D}}$, and $|c_{2m+1}| \geq |c_{2m+1}^\circ| - r/\omega_{2m+1}$. The computation certifies the bounds of Table 7; in particular, for $|w| \leq 1$,

$$\operatorname{Re} \left(1 + \frac{w\psi''_c(w)}{\psi'_c(w)} \right) \geq 1 - \frac{U_2}{L_1} > 0.$$

Consequently $\psi^ = \psi_{c^*}$ is injective with nonvanishing derivative on $\{|w| < 21/20\}$, and $D = \psi^*(\mathbb{D})$ is a strictly convex domain bounded by a real-analytic Jordan curve, invariant under $I_2(2m)$, not a disc, with $0 \in D$ and $\partial D = \{R(\theta)e^{i\theta}\}$ for a positive, real-analytic, W -invariant function R .*

Proof. Let $\delta = c - c^\circ$. Since $j\rho^{j-1}/\omega_j \leq \theta_1$ for all j , we have $\sum_j j\rho^{j-1}|\delta_j| \leq E_1$; since, for $j \geq 2$, $j(j-1)/\omega_j \leq j\rho^{-j}/(\alpha\rho^{m-1}) \leq 1/(\alpha\rho^{m-1}e \log \rho)$, as $x \mapsto x\rho^{-x}$ has maximum $1/(e \log \rho)$ on $(0, \infty)$, we have $\sum_{j>1} j(j-1)|\delta_j| \leq E_2$; and $|\delta_{2m+1}| \leq r/\omega_{2m+1}$. Both suprema are over all j , so the infinite tail of c is covered. For $|w| \leq R_e < \rho$ this gives $\operatorname{Re} \psi'_c(w) \geq c_1 - \sum_{j>1} j |c_j| R_e^{j-1} \geq \Lambda_e$, and similarly $|\psi'_c| \geq L_1$ and $|\psi''_c| \leq U_2$ on $\overline{\mathbb{D}}$, so $\operatorname{Re}(1 + w\psi''_c/\psi'_c) \geq 1 - |\psi''_c|/|\psi'_c| \geq 1 - U_2/L_1$.

For $c = c^*$, univalence on the convex disc $\{|w| < R_e\}$ follows from $\operatorname{Re} \psi^{*'} > 0$ as in Proposition 10.1, and strict convexity of D from the analytic convexity criterion, as in Proposition 10.2.

Quantity	$n = 8$	$n = 10$	$n = 14$
Λ_e	> 20.037195	> 37.965074	> 87.134069
L_1	> 20.231942	> 38.026654	> 87.809995
U_2	< 3.408445	< 1.024680	< 9.301964
$1 - U_2/L_1$	> 0.83153153	> 0.97305365	> 0.89406714
$ c_{2m+1} $	$ c_7 > 0.07748402$	$ c_9 > 0.01383247$	$ c_{13} > 0.04630866$

TABLE 7. Certified geometric bounds of Proposition 12.21, valid for every c in the ball of Theorem 12.19.

The symmetries follow from (12.5), and $0 = \psi^*(0) \in D$. A line tangent to ∂D at some point is a supporting line of the convex set D , so it cannot pass through the interior point 0; hence $\frac{d}{d\phi} \arg \psi^*(e^{i\phi}) \neq 0$, every ray from 0 meets ∂D exactly once and transversally, and ∂D is the graph of a positive real-analytic radial function R , which is W -invariant because D is. If D were a disc, its centre would be fixed by the rotation through π/m , hence equal to 0, and the Schwarz lemma argument of Proposition 10.1 would show that ψ^* is linear, contradicting $c_{2m+1}^* \neq 0$. $\square$

Proof of Theorem 12.1. Let $m = 3, 4, 6$ for $\mathfrak{k} = \mathfrak{su}(3), \mathfrak{so}(5), \mathfrak{g}_2$, respectively, let $x^* = (g^*, c^*)$ be the zero of Theorem 12.19, $\psi^* = \psi_{c^*}$ and $V^* = Kg^* + h_{c^*}$. By Lemma 12.10, V^* satisfies the hypotheses of Lemma 12.11 with $\psi = \psi^*$, and by Proposition 12.21 ψ^* is injective with nonvanishing derivative on a neighbourhood of $\overline{\mathbb{D}}$. Hence $F = BV^* \circ (\psi^*)^{-1} \in C^1(\overline{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = \varpi$ and $\nabla F = \nabla \varpi$ on ∂D , and $F(\bar{\tau}) = -F(\tau)$, $F(e^{i\pi/m}\tau) = -F(\tau)$. By Lemma 10.3 and Lemma 10.4 in the form of Lemma 12.9, F extends to a real-analytic solution on an open neighbourhood of $\overline{D}$. Replacing this neighbourhood by the connected component containing $\overline{D}$ of its intersection with its images under $I_2(2m)$, we obtain an $I_2(2m)$ -invariant neighbourhood N on which, by the identity theorem, F keeps its parities; in particular F is anti-invariant under W .

By Proposition 12.21, $D = \{re^{i\theta} : r < R(\theta)\}$ with R positive, real analytic and W -invariant. Lemma 12.6 shows that $\Omega = \text{Ad}(G)D$ is a bounded domain, strictly star-shaped with respect to 0, whose boundary is a compact real-analytic hypersurface diffeomorphic to S^{n-1} , and that $u = (F/\varpi)^\sharp$ is a nonconstant real-analytic function on the neighbourhood $\text{Ad}(G)N$ of $\overline{\Omega}$ satisfying (1.2). Since D is convex, Ω is convex by Lemma 12.8. It is $\text{Ad}(G)$ -invariant by construction, and $-\Omega = \text{Ad}(G)(-D) = \Omega$ because $-1 \in I_2(2m)$.

If Ω were a ball, its centre would be fixed by $\text{Ad}(G)$. As $\mathfrak{k}$ is simple, the only $\text{Ad}(G)$ -fixed vector is 0, so Ω would be a ball centred at 0 and $D = \Omega \cap \mathfrak{t}$ a disc, contradicting Proposition 12.21. Finally, the proof of Corollary 1.3 in Section 14 uses only (1.2) and the regularity of u and $\partial\Omega$, and applies verbatim: $\widehat{\mathbf{1}}_\Omega$ vanishes on the unit sphere, and $\int_{\sigma(\Omega)} \cos x_1 dx = 0$ for every rigid motion σ of $\mathfrak{k} \cong \mathbb{R}^n$. $\square$

12.10. Which dimensions arise.

Remark 12.22. The reduction of Proposition 12.2 lowers the dimension to the rank of $\mathfrak{k}$ and never produces a potential term, because ϖ is harmonic; it is planar exactly when $\mathfrak{k}$ has rank two. Since $\mathfrak{k} = \mathfrak{z}(\mathfrak{k}) \oplus [\mathfrak{k}, \mathfrak{k}]$ with $[\mathfrak{k}, \mathfrak{k}]$ semisimple, a compact Lie algebra of rank two is isomorphic to $\mathbb{R}^2$, to $\mathbb{R} \oplus \mathfrak{su}(2) \cong \mathfrak{u}(2)$, to $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, or to one of the simple algebras $\mathfrak{su}(3)$, $\mathfrak{so}(5)$, $\mathfrak{g}_2$ of types A_2 , $B_2 = C_2$, G_2 [6, 36]. Leaving aside the abelian case, in which there is nothing to reduce, the dimensions $n = 2 + 2m$ are 4, 6, 8, 10, 14, corresponding to $m \in \{1, 2, 3, 4, 6\}$, the orders of rotations allowed by the crystallographic restriction. Thus $\{4, 6, 8, 10, 14\}$ is the

complete list of dimensions obtainable by exact planar, potential-free reductions of adjoint type, and Theorems 1.2 and 12.1 cover all of them. The classes $O(d_1) \times O(d_2)$ of Remark 2.5 give no further dimensions: the admissible ones, $(d_1, d_2) = (1, 3)$ and $(3, 3)$, are the adjoint classes of $\mathfrak{u}(2)$ and $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. For $m = 5$ or $m \geq 7$, the planar problem (12.3) with $\varpi = \operatorname{Im} \tau^m$ still makes sense and could be treated by the method of this section, but the dihedral group $I_2(m)$ is not crystallographic, and the problem is not the reduction of a problem in a Euclidean space of this type. More generally, for the isotropy representation of a Riemannian symmetric space of rank two whose restricted roots all have the same multiplicity k , the radial part of the Laplacian is $f \mapsto J^{-1} \operatorname{div}(J \nabla f)$ with $J = |\varpi|^k$ [30, Ch. II], and conjugation by $J^{1/2}$ leaves the potential $\frac{k}{2}(\frac{k}{2} - 1)|\nabla \varpi|^2/\varpi^2$, which vanishes only for $k = 2$, the adjoint case.

Part 2. Convex and additional counterexamples in dimension three

13. DIMENSION THREE

In this section we prove Theorem 1.1. Points of the physical space $\mathbb{R}^3$ are written $X = (X_1, X') \in \mathbb{R} \times \mathbb{R}^2$, so that X_1 and X' are the coordinates x_1 and x' of Theorem 1.1; the letter x is used below for points of a reference ball. The convex domain and its dyadic centre are described in Remark 1.4. The independent earlier example is stated at the end of this section.

Strategy. For $n = 3$ there is no reduction to the planar Laplacian of the kind used for $n = 4$ and $n = 6$ (Remark 2.5): for $O(2)$ -invariant functions the meridian problem contains the drift term $y^{-1} \partial_y$. We therefore keep the three-dimensional Laplacian and pull back only the meridian plane by a conformal map ψ of the unit disc, which we lift to a diffeomorphism Θ of the unit ball $B \subset \mathbb{R}^3$ (Section 13.1). With $p = \psi'$ and $q = \operatorname{Im} \psi/y$, the pulled-back problem is

$$\operatorname{div}(q \nabla U) + q|p|^2 U = 0 \text{ in } B, \quad U = 1 \text{ and } \nabla U = 0 \text{ on } \partial B.$$

We write $U = 1 + K_3 g$, where K_3 is an explicit inverse of the Laplacian of $\mathbb{R}^3$ on axisymmetric functions with vanishing Cauchy data on ∂B (Section 13.4), expand g in zonal solid harmonics times Jacobi polynomials (Section 13.2), and obtain a quartic equation $\mathcal{F}(g, c) = 0$ for g and the Taylor coefficients c of ψ (Section 13.5). The unknown coefficient q is not divided out, so that the equation is polynomial. The approximate inverse of the linearisation consists of a 7371×7371 matrix, the identity on high field modes, and the exact inverse of a Toeplitz operator that describes the linearisation on high shape modes (Section 13.6). The defect bounds (Section 13.7) cover all columns of the linearisation: the finite parts of 7280 columns and the complete columns of $457 + 310$ further inputs are evaluated in interval arithmetic, while the output tails of the former and all remaining columns are covered by explicit envelopes (91 radial classes, fourteen near angular classes, one uniform angular class and one far shape class), whose validity is proved there. Compared with $n = 4$ and $n = 6$, the equation has the variable coefficient q in its principal part, so that the model of the linearisation on high shape modes is a Toeplitz operator whose symbol is built from q and ψ' rather than a bidiagonal operator; the output space measures harmonic components through their boundary traces; and the computation is much larger, taking about 1.8 hours with five processes (Section 15.6).

13.1. Axisymmetric functions and the pull-back. Throughout this section $R = 21/20$; the notation is independent of that of Sections 2–12. Let $B \subset \mathbb{R}^3$ be the open unit ball and $B_R = \{|x| < R\}$. We write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^2$ for points of the reference space, X for points of the physical space, and

$$y = |x'|, \quad w = x_1 + iy, \quad r = |x| = |w|, \quad t = r^2, \quad \xi = x_1/r.$$

A function on a rotation-invariant subset of $\mathbb{R}^3$ is *axisymmetric* if it is invariant under the action of $O(2)$ on x' ; it is then a function of (x_1, y) , and we use both descriptions. Polynomials in x_1 and y^2 are polynomials on $\mathbb{R}^3$. We write T_n , U_n , P_ℓ and $C_n^{(2)}$ for the Chebyshev polynomials of the first and second kind, the Legendre polynomials and the Gegenbauer polynomials of index two, with $U_{-1} = 0$.

Let $\mathcal{E}$ be the Banach space of real sequences $c = (c_j)_{j \text{ odd} \geq 1}$ with $\|c\|_{\mathcal{E}} = \sum_j j R^j |c_j| < \infty$. For $c \in \mathcal{E}$ let $\psi_c(w) = \sum_j c_j w^j$; the series for ψ_c and ψ'_c converge absolutely and uniformly on $|w| \leq R$. Put

$$p_c = \psi'_c(x_1 + iy), \quad q_c = \frac{\operatorname{Im} \psi_c(x_1 + iy)}{y} \quad (y \neq 0), \quad q_c(x_1, 0) = \operatorname{Re} \psi'_c(x_1).$$

Since $\operatorname{Im}(w^j) = r^j \sin j\theta$ for $w = re^{i\theta}$, $y = r \sin \theta$ and $U_{j-1}(\cos \theta) = \sin j\theta / \sin \theta$,
(13.1)

$$q_c = \sum_j c_j r^{j-1} U_{j-1}(\xi), \quad \partial_{x_1} q_c = \sum_{j \geq 3} j c_j r^{j-2} U_{j-2}(\xi), \quad \frac{\partial_y q_c}{y} = -2 \sum_{j \geq 3} c_j r^{j-3} C_{j-3}^{(2)}(\xi).$$

The second identity follows from $\partial_{x_1} \operatorname{Im}(w^j) = \operatorname{Im}(jw^{j-1})$. For the third, a computation in polar coordinates gives

$$\frac{1}{y} \partial_y \frac{\operatorname{Im} w^j}{y} = r^{j-3} \frac{(j-1) \sin \theta \cos(j-1)\theta - \sin(j-1)\theta \cos \theta}{\sin^3 \theta} = -r^{j-3} U'_{j-2}(\cos \theta),$$

and $U'_{j-2} = 2C_{j-3}^{(2)}$. Moreover, with $p_a = (a+1)c_{a+1}$,

$$(13.2) \quad |p_c|^2 = \sum_{\alpha, \beta \text{ even}} p_\alpha p_\beta \operatorname{Re}(w^\alpha \bar{w}^\beta) = \sum_{\alpha, \beta \text{ even}} p_\alpha p_\beta r^{\alpha+\beta} T_{|\alpha-\beta|}(\xi).$$

Since ψ_c is odd with real coefficients, $\psi_c(\bar{w}) = \overline{\psi_c(w)}$ and $\psi_c(-\bar{w}) = -\overline{\psi_c(w)}$; hence q_c , $y^{-1} \partial_y q_c$ and $|p_c|^2$ are even functions of x_1 , and $\partial_{x_1} q_c$ is odd.

For $c \in \mathcal{E}$ define $\Theta_c: B_R \rightarrow \mathbb{R}^3$ by

$$(13.3) \quad \Theta_c(x) = (\operatorname{Re} \psi_c(x_1 + i|x'|), q_c(x) x').$$

Lemma 13.1. *Let $c \in \mathcal{E}$ and suppose that $\operatorname{Re} \psi'_c > 0$ on a disc $\{|w| < R_0\}$ with $1 < R_0 < R$. Then:*

- (a) q_c , $\partial_{x_1} q_c$, $y^{-1} \partial_y q_c$, $|p_c|^2$ and $\operatorname{Re} \psi_c(x_1 + iy)$, regarded as functions of x , are real analytic on B_{R_0} , and $q_c > 0$ there;
- (b) Θ_c is a real-analytic diffeomorphism of B_{R_0} onto an open subset of $\mathbb{R}^3$, with $\det D\Theta_c = q_c |p_c|^2$; it commutes with the action of $O(2)$ on x' , and $\Theta_c(-x_1, x') = (-X_1, X')$ if $\Theta_c(x_1, x') = (X_1, X')$;
- (c) for every axisymmetric C^1 function U on B_{R_0} , $(\det D\Theta_c)(D\Theta_c^T D\Theta_c)^{-1} \nabla U = q_c \nabla U$.

Proof. (a) The function $F(x_1, y) = \operatorname{Im} \psi_c(x_1 + iy)$ is real analytic on the disc $\{|w| < R_0\}$ and odd in y . By Lemma 2.2 there is a real-analytic G with $F = yG(x_1, y^2)$, and $x \mapsto G(x_1, |x'|^2)$ is real analytic on B_{R_0} ; it equals q_c off the axis, and on the axis $G(x_1, 0) = \partial_y F(x_1, 0) = \operatorname{Re} \psi'_c(x_1)$ by the Cauchy–Riemann equations. Hence q_c is real analytic, and so are $\partial_{x_1} q_c = (\partial_1 G)(x_1, y^2)$ and $y^{-1} \partial_y q_c = 2(\partial_2 G)(x_1, y^2)$. Functions of (x_1, y) that are real analytic and even in y , such as $\operatorname{Re} \psi_c$ and $|\psi'_c|^2$, are treated by applying Lemma 2.2 to their product with y . For $y \neq 0$, the Cauchy–Riemann equations give $q_c = y^{-1} \int_0^y \operatorname{Re} \psi'_c(x_1 + i\varsigma) d\varsigma > 0$, and $q_c = \operatorname{Re} \psi'_c > 0$ on the axis.

(b) Both components of Θ_c are real analytic by (a). For $w_1 \neq w_2$ in the convex disc $\{|w| < R_0\}$, $(\psi_c(w_2) - \psi_c(w_1))/(w_2 - w_1) = \int_0^1 \psi'_c(w_1 + \varsigma(w_2 - w_1)) d\varsigma$ has positive real part, so ψ_c is injective there. Since $q_c > 0$, the second component of $\Theta_c(x)$ has modulus $\text{Im } \psi_c(x_1 + i|x'|)$. Hence $\Theta_c(x) = \Theta_c(z)$ implies $\psi_c(x_1 + i|x'|) = \psi_c(z_1 + i|z'|)$, so $x_1 = z_1$ and $|x'| = |z'|$, and then $q_c(x)x' = q_c(z)z'$ gives $x' = z'$. At a point with $x' \neq 0$ let $e_\rho = x'/|x'|$ and let e_ϕ be the unit vector in the x' -plane orthogonal to e_ρ . On the half-plane spanned by e_1 and e_ρ , Θ_c acts as ψ_c in the coordinates (x_1, y) , and $D\Theta_c$ maps this plane to itself as $|p_c|$ times a rotation; it maps e_ϕ to $q_c e_\phi$, since Θ_c maps the circle of radius y about the axis onto the circle of radius $q_c y$. Hence $\det D\Theta_c = q_c |p_c|^2 > 0$ off the axis, and by continuity on B_{R_0} . By the inverse function theorem Θ_c is a real-analytic diffeomorphism onto its open image. The symmetries follow from those of ψ_c .

(c) In the frame (e_1, e_ρ, e_ϕ) of (b), $D\Theta_c^\top D\Theta_c = \text{diag}(|p_c|^2, |p_c|^2, q_c^2)$, and ∇U lies in the span of e_1 and e_ρ . Off the axis the identity follows, and on the axis by continuity. $\square$

Lemma 13.2. *Let c be as in Lemma 13.1, $\Omega = \Theta_c(B)$, and let $U \in C^1(\overline{B})$ be axisymmetric and satisfy*

$$(13.4) \quad \int_B (q_c \nabla U \cdot \nabla \varphi - q_c |p_c|^2 U \varphi) dx = 0 \quad (\varphi \in C_c^\infty(B)).$$

Then $u = U \circ \Theta_c^{-1} \in C^1(\overline{\Omega})$ satisfies $\int_\Omega (\nabla u \cdot \nabla \varphi - u \varphi) dX = 0$ for all $\varphi \in C_c^\infty(\Omega)$. If $U = 1$ and $\nabla U = 0$ on ∂B , then $u = 1$ and $\nabla u = 0$ on $\partial \Omega$.

Proof. Let $\varphi \in C_c^\infty(\Omega)$ and $\tilde{\varphi} = \varphi \circ \Theta_c \in C_c^\infty(B)$. Since $(\nabla u) \circ \Theta_c = D\Theta_c^{-\top} \nabla U$, the change of variables $X = \Theta_c(x)$ and Lemma 13.1(b),(c) give

$$\int_\Omega (\nabla u \cdot \nabla \varphi - u \varphi) dX = \int_B ((\det D\Theta_c)(D\Theta_c^\top D\Theta_c)^{-1} \nabla U \cdot \nabla \tilde{\varphi} - q_c |p_c|^2 U \tilde{\varphi}) dx,$$

which vanishes by Lemma 13.1(c) and (13.4). The boundary values follow from $(\nabla u) \circ \Theta_c = D\Theta_c^{-\top} \nabla U$. $\square$

13.2. Zonal harmonics, Jacobi polynomials and coefficient spaces. For $\ell \geq 0$ let $H_\ell(x) = r^\ell P_\ell(\xi)$, the axisymmetric harmonic polynomial on $\mathbb{R}^3$, homogeneous of degree ℓ , with $H_\ell(1, 0, 0) = 1$; axisymmetric harmonic homogeneous polynomials of degree ℓ are multiples of H_ℓ [53, Ch. IV]. We have $|H_\ell| \leq 1$ on $\overline{B}$, $H_\ell = P_\ell(\xi)$ on ∂B , and, with $H_{-1} = 0$,

$$(13.5) \quad x_1 H_\ell = \frac{(\ell+1)H_{\ell+1} + \ell t H_{\ell-1}}{2\ell+1}, \quad \partial_{x_1} H_\ell = \ell H_{\ell-1}, \quad (x \cdot \nabla) H_\ell = \ell H_\ell.$$

The first identity is Bonnet's recurrence for P_ℓ multiplied by $r^{\ell+1}$; for the second, $\partial_{x_1} H_\ell$ is an axisymmetric harmonic homogeneous polynomial of degree $\ell-1$, hence a multiple of $H_{\ell-1}$, and the multiple is ℓ because $H_\ell(x_1, 0, 0) = x_1^\ell$.

For real $\beta > -1$ and integers $s \geq 0$ let $J_s^\beta(t) = P_s^{(0,\beta)}(2t-1)$, so that

$$(13.6) \quad J_s^\beta(t) = \sum_{k=0}^s (-1)^{s-k} \frac{\Gamma(\beta+s+k+1)}{k!(s-k)!\Gamma(\beta+k+1)} t^k = \frac{t^{-\beta}}{s!} \frac{d^s}{dt^s} \{t^{\beta+s}(t-1)^s\}, \quad J_s^\beta(1) = 1,$$

and $J_{-1}^\beta = 0$. We write $[t^k]f$ for the coefficient of t^k in a polynomial f , and $\|f\|_{J^\beta,1} = \sum_s |f_s|$ if $f = \sum_s f_s J_s^\beta$. Define the axisymmetric polynomials

$$\Psi_{\ell,s} = H_\ell J_s^{\ell+1/2}(t) \quad (\ell, s \geq 0),$$

of degree $\ell+2s$; thus $\Psi_{\ell,0} = H_\ell$.

Lemma 13.3. *Let $\beta > 0$ and $s, s' \geq 0$.*

(a) *For every integer $m \geq 0$,*

$$\int_0^1 t^{\beta+m} J_s^\beta(t) dt = \frac{m(m-1) \cdots (m-s+1)}{(\beta+m+1)(\beta+m+2) \cdots (\beta+m+s+1)};$$

in particular J_s^β is orthogonal in $L^2([0, 1], t^\beta dt)$ to all polynomials of degree less than s , and $\int_0^1 t^\beta J_s^\beta J_{s'}^\beta dt = \delta_{ss'}/(\beta+2s+1)$.

(b) $J_s^\beta = \frac{\beta+s+1}{\beta+2s+1} J_s^{\beta+1} + \frac{s}{\beta+2s+1} J_{s-1}^{\beta+1}$, *which holds also for $-1 < \beta \leq 0$, and*
 $tJ_s^\beta = \frac{\beta+s}{\beta+2s+1} J_s^{\beta-1} + \frac{s+1}{\beta+2s+1} J_{s+1}^{\beta-1}$. *The coefficients are nonnegative and sum to one, and in the first identity the coefficient of the term with the lower index is at most $1/2$.*

(c) *The coefficient of J_s^β in the expansion of $(1-t)^m$ is $(-1)^s$ times a positive number for $0 \leq s \leq m$, and vanishes for $s > m$. Moreover $\|(1-t)^m\|_{J^\beta, 1} = m!/(\beta/2+1)_m$, which is decreasing in β .*

(d) *For $M \geq 0$, the coefficients of t^M in the basis $(J_s^\beta)_s$ are nonnegative and sum to one, the coefficient of J_0^β is $(\beta+1)/(\beta+M+1)$, and $\|t^M - 1\|_{J^\beta, 1} = 2M/(\beta+M+1)$.*

Proof. (a) Insert the Rodrigues formula (13.6) and integrate by parts s times. The boundary terms vanish: at $t = 1$ the derivatives of order less than s of $t^{\beta+s}(t-1)^s$ vanish, and at $t = 0$ they are $O(t^{\beta+1})$. This gives $\frac{m(m-1) \cdots (m-s+1)}{s!} \int_0^1 t^{\beta+m} (1-t)^s dt$, which is the stated value. For $m < s$ the product vanishes. The leading coefficient of J_s^β is $\Gamma(\beta+2s+1)/(s! \Gamma(\beta+s+1))$, and multiplying it by the value for $m = s$ gives $1/(\beta+2s+1)$.

(b) Divide the coefficient of t^k of each side by $[t^k]J_s^{\beta+1}$, respectively by $[t^k]J_s^{\beta-1}$, using (13.6). For $0 \leq k \leq s$ the first identity reduces to $(\beta+k+1)(\beta+2s+1) = (\beta+s+1)(\beta+s+k+1) - s(s-k)$, and for $1 \leq k \leq s$ the second to $-k(\beta+2s+1) = (\beta+s)(s-k+1) - (s+1)(\beta+s+k)$; both are verified by expansion. In the second identity the cases $k = 0$ and $k = s+1$ are checked directly. The computation for the first identity uses only $\beta > -1$. The remaining assertions are clear, and $s \leq (\beta+2s+1)/2$ for $\beta \geq -1$.

(c) By the orthogonality in (a), the coefficient is $(\beta+2s+1) \int_0^1 t^\beta (1-t)^m J_s^\beta dt$. Integrating by parts s times in the Rodrigues formula gives $\frac{(-1)^s}{s!} m(m-1) \cdots (m-s+1) \int_0^1 t^{\beta+s} (1-t)^m dt$ times $\beta+2s+1$, which proves the sign pattern. Let $\Lambda(f) = \frac{\beta}{2} \int_0^1 t^{\beta/2-1} f(t) dt$. The same integration by parts, now against $t^{-\beta/2-1}$, gives $\Lambda(J_s^\beta) = (-1)^s \frac{\beta}{2} (\frac{\beta}{2}+1)_s B(\frac{\beta}{2}, s+1)/s! = (-1)^s$. Hence $\|(1-t)^m\|_{J^\beta, 1} = \Lambda((1-t)^m) = \frac{\beta}{2} B(\frac{\beta}{2}, m+1) = m!/(\beta/2+1)_m$.

(d) By (b), multiplication by t is the lowering step followed by the raising step (with $\beta-1$ in place of β); both are given by nonnegative matrices with column sums one. Hence $t^M = t^M J_0^\beta$ has nonnegative coefficients summing to $t^M|_{t=1} = 1$. The coefficient of $J_0^\beta = 1$ is $(\beta+1) \int_0^1 t^{\beta+M} dt$. Therefore $\|t^M - 1\|_{J^\beta, 1}$ equals $(1 - \frac{\beta+1}{\beta+M+1})$ plus the sum of the other coefficients, which is the same number. $\square$

Chebyshev and Legendre coordinates. Let $a_j = \binom{2j}{j} 4^{-j}$. Then $a_0 = 1$, $a_{j+1}/a_j = (2j+1)/(2j+2)$, so (a_j) is decreasing, and $a_k \sqrt{k+1/4}$ is nondecreasing in k , because $(2k+1)^2(4k+5) - (2k+2)^2(4k+1) = 1$. Consequently

$$(13.7) \quad \frac{a_\ell}{a_m} \leq \left(\frac{m+1/4}{\ell+1/4} \right)^{1/2} \quad (\ell \leq m), \quad a_{\ell-j} \leq \sqrt{2} a_\ell \quad (0 \leq j \leq \ell/2).$$

Factoring the generating function, $(1 - 2\xi z + z^2)^{-1/2} = (1 - e^{i\theta} z)^{-1/2} (1 - e^{-i\theta} z)^{-1/2}$ for $\xi = \cos \theta$, gives $P_\ell(\cos \theta) = \sum_{j=0}^\ell a_j a_{\ell-j} e^{i(\ell-2j)\theta}$, that is,

$$P_\ell = \sum_n D_{n\ell} T_n, \quad D_{\ell-2j,\ell} = (2 - \delta_{\ell,2j}) a_j a_{\ell-j} \geq 0.$$

Conversely write $T_n = \sum_\ell C_{\ell n} P_\ell$ and $U_n = \sum_\ell v_{\ell n} P_\ell$. Let $\langle \cdot \rangle_R$ be the linear functional on polynomials in ξ with $\langle T_n \rangle_R = R^n$, and put

$$h_\ell = \langle P_\ell \rangle_R = \sum_n D_{n\ell} R^n, \quad \kappa_R = \frac{4}{\sqrt{1 - R^{-2}}} - 1, \quad B_0 = \frac{2\sqrt{2}}{\sqrt{1 - R^{-2}}}.$$

Numerically $\kappa_R \approx 12.118596$ and $B_0 \approx 9.276248$.

Lemma 13.4. (a) $C_{00} = 1$, $C_{nn} = 1/(2a_n)$ for $n \geq 1$, and for $n = \ell + 2j$ with $j \geq 1$,

$$C_{\ell n} = -\frac{n(2\ell + 1) a_j}{2(\ell + j)(2\ell + 2j + 1) a_{\ell+j} (2j - 1)} < 0.$$

(b) For $n = \ell + 2j$, $v_{\ell n} = (2\ell + 1) a_j / ((2\ell + 2j + 1) a_{\ell+j}) > 0$. The Legendre coefficients of $C_n^{(2)}$ are positive.

Proof. We use the connection formula for ultraspherical polynomials [20, §18.18(iii)],

$$C_n^{(\lambda)} = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(\lambda - \mu)_j (\lambda)_{n-j}}{(\mu + 1)_{n-j} j!} \frac{\mu + n - 2j}{\mu} C_{n-2j}^{(\mu)},$$

with $\mu = \frac{1}{2}$, so that $C_\ell^{(\mu)} = P_\ell$ and $(\mu + n - 2j)/\mu = 2\ell + 1$ for $\ell = n - 2j$. We use $(\frac{1}{2})_j = j! a_j$, $(-\frac{1}{2})_j = -j! a_j / (2j - 1)$ and $(\frac{3}{2})_m = (2m + 1) m! a_m$. For $\lambda = 1$, $C_n^{(1)} = U_n$, and the coefficient is $(\frac{1}{2})_j (n - j)! (2\ell + 1) / ((\frac{3}{2})_{n-j} j!)$, which is the stated $v_{\ell n}$. For $\lambda \rightarrow 0$, $C_n^{(\lambda)} / \lambda \rightarrow (2/n) T_n$ for $n \geq 1$ [20, §18.7(iii)] and $(\lambda)_{n-j} / \lambda \rightarrow (n - j - 1)!$, which gives (a). For $\lambda = 2$ all coefficients are positive, since $(\frac{3}{2})_j > 0$. $\square$

Lemma 13.5. For all $\ell, n \geq 0$:

- (a) $1 \leq h_\ell \leq B_0 a_\ell R^\ell$;
- (b) $h_{\ell+1} \leq R h_\ell$, and $h_{\ell-1} \leq R^{-1} \frac{\ell}{\ell-1} h_\ell$ for $\ell \geq 2$;
- (c) $\sum_\ell |C_{\ell n}| h_\ell \leq \kappa_R R^n$;
- (d) $C_{nn} h_n \leq \frac{B_0}{2} R^n$; for $j \geq 1$ and $\ell = n - 2j \geq 0$, $|C_{\ell n}| h_\ell \leq B_0 \frac{a_j R^{-2j}}{2j-1} R^n$ and $v_{\ell n} h_\ell \leq B_0 a_j R^{-2j} R^n$;
- (e) $D_{n\ell} R^n \leq \sqrt{2} R^{n-\ell} h_\ell$.

Proof. (a) $h_\ell \geq \sum_n D_{n\ell} = P_\ell(1) = 1$. By (13.7),

$$h_\ell \leq 2 \sum_{j \leq \ell/2} a_j a_{\ell-j} R^{\ell-2j} \leq 2\sqrt{2} a_\ell R^\ell \sum_j a_j R^{-2j},$$

and $\sum_j a_j z^j = (1 - z)^{-1/2}$.

(b) Write $R h_\ell - h_{\ell+1} = \sum_e \delta_e R^e$, where $e \equiv \ell + 1 \pmod{2}$ runs over $0 \leq e \leq \ell + 1$. Comparing coefficients, $\delta_e = 2a_i(a_{\ell-i} - a_{\ell+1-i}) \geq 0$ for $e \geq 2$, where $i = (\ell + 1 - e)/2$. The lowest exponent $e_0 \in \{0, 1\}$ has a negative coefficient: $\delta_1 = a_{\ell/2}(a_{\ell/2} - 2a_{\ell/2+1}) < 0$ if ℓ is even, and $\delta_0 = -a_{(\ell+1)/2}^2$ if ℓ is odd. For $R = 1$ all h_k equal one, so $\sum_e \delta_e = 0$. Hence $R h_\ell - h_{\ell+1} = \sum_{e > e_0} \delta_e (R^e - R^{e_0}) \geq 0$ for $R \geq 1$. For the second inequality we compare the coefficients of R^e in $R h_{\ell-1}$ and h_ℓ . For $e = \ell - 2i \geq 2$ they are $2a_i a_{\ell-1-i}$ and $2a_i a_{\ell-i}$, whose

ratio is $2m/(2m-1)$ with $m = \ell - i \geq (\ell + 2)/2$, hence at most $(\ell + 2)/(\ell + 1) \leq \ell/(\ell - 1)$. For $e = 1$ (ℓ odd, $i = (\ell - 1)/2$) they are a_i^2 and $2a_i a_{i+1}$, with ratio $(\ell + 1)/(2\ell) \leq \ell/(\ell - 1)$. For ℓ even, h_ℓ has an additional nonnegative coefficient at $e = 0$.

(c) For $n = 0$ the sum is $1 \leq \kappa_R$. For $n \geq 1$, since $h_\ell = \langle P_\ell \rangle_R$, we have $\sum_\ell C_{\ell n} h_\ell = \langle T_n \rangle_R = R^n$. By Lemma 13.4(a) all $C_{\ell n}$ with $\ell < n$ are negative, so $\sum_\ell |C_{\ell n}| h_\ell = 2C_{nn} h_n - R^n \leq (B_0 - 1)R^n \leq \kappa_R R^n$ by (a).

(d) The first bound follows from (a). For $j \geq 1$ put $m = \ell + j$. By (a) and Lemma 13.4, the second bound reduces to $n(2\ell + 1)a_\ell \leq 2m(2m + 1)a_m$. By (13.7) it suffices that $4m^2(2m + 1)^2(\ell + \frac{1}{4}) - n^2(2\ell + 1)^2(m + \frac{1}{4}) \geq 0$; with $m = \ell + j$, $n = \ell + 2j$ this is a polynomial in (ℓ, j) whose coefficients are all nonnegative. The third bound reduces to $(2\ell + 1)a_\ell \leq (2m + 1)a_m$, which follows from (13.7) and the monotonicity of $(2x + 1)^2/(x + \frac{1}{4})$ on $[0, \infty)$.

(e) For $\ell \geq 1$, $h_\ell \geq D_{\ell\ell} R^\ell = 2a_\ell R^\ell$, and $D_{n\ell} \leq 2a_{\ell-j} \leq 2\sqrt{2}a_\ell$ by (13.7). For $\ell = 0$ the claim is trivial. $\square$

Coefficient spaces. Let $\mathcal{L}$ be the Banach space of real families $f = (f_{\ell,s})_{\ell,s \geq 0}$ with $\|f\|_{\mathcal{L}} = \sum_{\ell,s} h_\ell |f_{\ell,s}| < \infty$, identified with the formal series $\sum f_{\ell,s} \Psi_{\ell,s}$, and let $\mathcal{L}^{\text{ev}}$ be its subspace with $f_{\ell,s} = 0$ for odd ℓ . The field space is

$$\mathcal{G} = \{g \in \mathcal{L}^{\text{ev}} : g_{\ell,0} = 0 \text{ for all } \ell\}.$$

For even $n \geq 0$ let $\mathcal{T}_n = \sum_\ell C_{\ell n} H_\ell$; this is the harmonic polynomial with boundary values $T_n(\xi)$ on ∂B , and $|\mathcal{T}_n| \leq 1$ on $\bar{B}$ by the maximum principle. The output space $\mathcal{Y}$ is the Banach space of families $f = ((f_n)_n \text{ even}, (f_{\ell,s})_{\ell \text{ even}, s \geq 1})$, identified with $\sum_n f_n \mathcal{T}_n + \sum_{s \geq 1} f_{\ell,s} \Psi_{\ell,s}$, with

$$\|f\|_{\mathcal{Y}} = \sum_n R^n |f_n| + \sum_{\ell,s \geq 1} h_\ell |f_{\ell,s}|.$$

Thus the harmonic layer of $\mathcal{Y}$ is measured through the Chebyshev coefficients of its boundary trace. Since $H_\ell = \sum_n D_{n\ell} \mathcal{T}_n$ with $D_{n\ell} \geq 0$, the inclusion $\mathcal{L}^{\text{ev}} \rightarrow \mathcal{Y}$ satisfies $\|f\|_{\mathcal{Y}} \leq \|f\|_{\mathcal{L}}$. The unknowns lie in $\mathcal{X} = \mathcal{G} \times \mathcal{E}$, with norm $\|(g, c)\|_{\mathcal{X}} = \|g\|_{\mathcal{L}} + \sigma \|c\|_{\mathcal{E}}$ for a constant $\sigma > 0$ fixed in Section 13.5. All three spaces are weighted ℓ^1 spaces, and Lemma 7.2 applies to operators between them.

Lemma 13.6. (a) *The functions $\Psi_{\ell,s}$ are pairwise orthogonal in $L^2(B)$, $\|\Psi_{\ell,s}\|_{L^2(B)}^2 = 4\pi/((2\ell + 1)(2\ell + 4s + 3))$, and $\Psi_{\ell,s}$ is orthogonal to all polynomials on $\mathbb{R}^3$ of degree less than $\ell + 2s$.*

(b) *The series defining the elements of $\mathcal{L}$ and $\mathcal{Y}$ converge in $L^2(B)$, and the resulting maps $\iota: \mathcal{L} \rightarrow L^2(B)$ and $\iota: \mathcal{Y} \rightarrow L^2(B)$ are continuous, injective and compatible with the inclusion $\mathcal{L}^{\text{ev}} \subset \mathcal{Y}$. The harmonic part $\sum_n f_n \mathcal{T}_n$ of $f \in \mathcal{Y}$ converges uniformly on $\bar{B}$.*

Proof. (a) In spherical coordinates,

$$\int_B \Psi_{\ell,s} \Psi_{\ell',s'} dx = 2\pi \int_{-1}^1 P_\ell P_{\ell'} d\xi \cdot \frac{1}{2} \int_0^1 t^{(\ell+\ell'+1)/2} J_s^{\ell+1/2} J_{s'}^{\ell'+1/2} dt,$$

and Lemma 13.3(a) gives orthogonality and the norm. A polynomial of degree less than $\ell + 2s$ is a sum of terms $|x|^{2m} Y_k$ with Y_k harmonic and homogeneous of degree k , $k + 2m < \ell + 2s$ [53, Ch. IV]. The angular integral of $\Psi_{\ell,s} |x|^{2m} Y_k$ vanishes unless $k = \ell$, and then $m < s$ and the radial integral $\frac{1}{2} \int_0^1 t^{\ell+1/2+m} J_s^{\ell+1/2} dt$ vanishes by Lemma 13.3(a).

(b) Continuity follows from $\|\Psi_{\ell,s}\|_{L^2} \leq (4\pi/3)^{1/2} \leq (4\pi/3)^{1/2} h_\ell$ and $|\mathcal{T}_n| \leq 1 \leq R^n$. On $\mathcal{L}$, injectivity follows from (a). Let $f \in \mathcal{Y}$ with $\iota f = 0$. The $\mathcal{T}_n$ are harmonic polynomials, hence orthogonal to all $\Psi_{\ell,s}$ with $s \geq 1$ by (a); pairing with $\Psi_{\ell,s}$ gives $f_{\ell,s} = 0$ for $s \geq 1$. The

harmonic part $h = \sum_n f_n \mathcal{T}_n$ converges uniformly on $\overline{B}$, so it is harmonic in B , continuous on $\overline{B}$, and vanishes in B ; hence its boundary values $\sum_n f_n \mathcal{T}_n(\cos \theta) = \sum_n f_n \cos n\theta$ vanish, and all $f_n = 0$. $\square$

13.3. Multiplication.

Lemma 13.7 (Adams). *For $a, \ell \geq 0$, $P_a P_\ell = \sum_k G_{a\ell k} P_k$, where $G_{a\ell k} \geq 0$ vanishes unless $|a - \ell| \leq k \leq a + \ell$ and $k \equiv a + \ell \pmod{2}$, and $\sum_k G_{a\ell k} = 1$. Consequently $H_a H_\ell = \sum_k G_{a\ell k} H_k t^{(a+\ell-k)/2}$ and*

$$\sum_k G_{a\ell k} h_k \leq h_a h_\ell.$$

Proof. Adams' formula [1] gives, with $v = (a + \ell + k)/2$ in the support,

$$G_{a\ell k} = (2k + 1) \left[\frac{v!}{(v-a)!(v-\ell)!(v-k)!} \right]^2 \frac{(2v-2a)!(2v-2\ell)!(2v-2k)!}{(2v+1)!} \geq 0;$$

the sum is one by evaluation at $\xi = 1$, and the identity for $H_a H_\ell$ follows after multiplication by $r^{a+\ell}$. Finally, $\sum_k G_{a\ell k} h_k = \langle P_a P_\ell \rangle_R$, and $P_a P_\ell = \sum_{m,n} D_{ma} D_{n\ell} \frac{1}{2} (T_{m+n} + T_{|m-n|})$ with $D \geq 0$, so $\langle P_a P_\ell \rangle_R \leq \sum_{m,n} D_{ma} D_{n\ell} R^{m+n} = h_a h_\ell$. $\square$

Every axisymmetric polynomial m has unique expansions

$$m = \sum_{a,k \geq 0} m_{a,k} H_a t^k = \sum_{n,k \geq 0} \tilde{m}_{n,k} r^{n+2k} T_n(\xi),$$

the first by the Fischer decomposition, the second because $r^n T_n(\xi) = \operatorname{Re}(x_1 + iy)^n$. The homogeneous part of degree d of m consists of the terms with $a+2k = d$, respectively $n+2k = d$; we call $H_a t^k$ a *solid monomial* of degree $a + 2k$. We put

$$\tau(m) = \sum_{a,k} h_a |m_{a,k}|, \quad \|m\|_{\text{ch}} = \sum_{n,k} R^n |\tilde{m}_{n,k}|,$$

and extend both to series with finite norm; such series converge uniformly on $\overline{B}$, since $|H_a t^k| \leq 1$ and $|r^{n+2k} T_n(\xi)| \leq 1$ there.

Lemma 13.8. (a) *Let $H_a t^k$ be a solid monomial and $\ell, s \geq 0$. Then*

$$(13.8) \quad H_a t^k \Psi_{\ell,s} = \sum_v G_{a\ell v} \sum_{\varsigma \geq 0} \pi_\varsigma^{(v)} \Psi_{v,\varsigma},$$

where, for each v , $(\pi_\varsigma^{(v)})_\varsigma$ is a probability vector. Consequently $\|mf\|_{\mathcal{L}} \leq \tau(m) \|f\|_{\mathcal{L}}$ for every $f \in \mathcal{L}$ and every m with $\tau(m) < \infty$.

(b) $\tau(m) \leq \kappa_R \|m\|_{\text{ch}}$. If $m = r^{n+2k} U_n(\xi)$ or $m = r^{n+2k} C_n^{(2)}(\xi)$, then $\tau(m) = \|m\|_{\text{ch}}$.

(c) $\|m_1 m_2\|_{\text{ch}} \leq \|m_1\|_{\text{ch}} \|m_2\|_{\text{ch}}$ and $\|\operatorname{Re}(w^\alpha \overline{w}^\beta)\|_{\text{ch}} \leq R^{\alpha+\beta}$.

(d) For m with $\tau(m) < \infty$ and $f \in \mathcal{L}$, $\iota(mf) = m \iota(f)$.

Proof. (a) By Lemma 13.7, $H_a t^k \Psi_{\ell,s} = \sum_v G_{a\ell v} H_v t^e J_s^{\ell+1/2}$ with $e = (a + \ell - v)/2 + k$. If $v \geq \ell$, we apply the first identity of Lemma 13.3(b) $v - \ell$ times, raising the parameter from $\ell + \frac{1}{2}$ to $v + \frac{1}{2}$, and then write the factor t^e as e multiplications by t , each a lowering step followed by a raising step. If $v < \ell$, we apply the second identity $\ell - v$ times, which uses $\ell - v$ factors t ; this is possible because $e \geq \ell - v$, as $\ell - v \leq a$. The remaining $e - (\ell - v)$ factors t are treated as before. Every step is given by a nonnegative matrix with column sums one, which proves (13.8). Since $\|\Psi_{v,\varsigma}\|_{\mathcal{L}} = h_v$, Lemma 13.7 gives $\|H_a t^k \Psi_{\ell,s}\|_{\mathcal{L}} \leq h_a h_\ell$, and the bound for mf follows by absolute summation.

(b) Since $r^{n+2k}T_n(\xi) = \sum_{\ell} C_{\ell n} H_{\ell} t^{(n-\ell)/2+k}$, Lemma 13.5(c) gives $\tau(r^{n+2k}T_n) \leq \kappa_R R^n$. For U_n the Legendre coefficients are nonnegative (Lemma 13.4(b)), so $\tau(r^{n+2k}U_n) = \sum_{\ell} v_{\ell n} h_{\ell} = \langle U_n \rangle_R$, which is the Chebyshev norm, because $U_n(\cos \theta) = \sum_{|\nu| \leq n, \nu \equiv n \pmod{2}} e^{i\nu\theta}$ has non-negative Chebyshev coefficients. For $C_n^{(2)}$ the same argument applies, since $C_n^{(2)}(\cos \theta) = \sum_{p+p'=n} (p+1)(p'+1)e^{i(p-p')\theta}$.

(c) $T_m T_n = \frac{1}{2}(T_{m+n} + T_{|m-n|})$ and $R^{|m-n|} \leq R^{m+n}$; and $\operatorname{Re}(w^{\alpha} \bar{w}^{\beta}) = r^{\alpha+\beta} T_{|\alpha-\beta|}(\xi)$.

(d) Both sides are continuous in $f \in \mathcal{L}$ with values in $L^2(B)$, by (a) and Lemma 13.6, and m is bounded; they agree when f is a finite combination of the $\Psi_{\ell,s}$ and m is a polynomial, and the general case follows by uniform convergence of m and density. $\square$

Lemma 13.9. *Let $H_a t^k$ be a solid monomial of degree $d = a + 2k$, let $\ell, s \geq 0$, and let $\pi^{(v)}$ be as in (13.8). Then:*

- (a) $\pi_{\zeta}^{(v)} = 0$ unless $s - d \leq \zeta \leq s + d$, and $G_{a\ell v} = 0$ unless $|\ell - a| \leq v \leq \ell + a$;
- (b) $\pi_0^{(v)} \leq \vartheta_d(\ell, s)$, where $\vartheta_d(\ell, s) = (d/(\ell + d + \frac{5}{2}))^s$ if $d \geq s$ (with $0^0 = 1$) and $\vartheta_d(\ell, s) = 0$ if $d < s$;
- (c) for every integer $j \geq 1$, $\sum_{\zeta \leq s-j} \pi_{\zeta}^{(v)} \leq 2^{-d} \sum_{i \geq j} \binom{d}{i}$.

Proof. In the proof of Lemma 13.8(a), only the raising steps can decrease the radial index, each by at most one, and only the lowering steps can increase it, each by at most one. The number of raising steps is $e + v - \ell = (a - \ell + v)/2 + k$ and the number of lowering steps is $e = (a + \ell - v)/2 + k$; since $|\ell - v| \leq a$, both are at most $a + k \leq d$. This proves (a), the statement about v being part of Lemma 13.7.

(b) Since $J_0^{v+1/2} = 1$, Lemma 13.3(a) gives $\pi_0^{(v)} = (v + \frac{3}{2}) \int_0^1 t^{v+1/2+e} J_s^{\ell+1/2} dt$, which is the value in Lemma 13.3(a) with $\beta = \ell + \frac{1}{2}$ and $m = e + v - \ell = (a - \ell + v)/2 + k$, multiplied by $v + \frac{3}{2}$. Here $0 \leq m \leq a + k \leq d$. If $m < s$, in particular if $d < s$, it vanishes. Otherwise we pair the factor $v + \frac{3}{2}$ with $\ell + m + \frac{3}{2}$, which is not smaller because $\ell + m = (a + \ell + v)/2 + k \geq v$, and each factor $m - i$ ($0 \leq i < s$) with $\ell + m + \frac{5}{2} + i$; since $x \mapsto x/(\ell + x + \frac{5}{2})$ is increasing, each quotient $(m - i)/(\ell + m + \frac{5}{2} + i)$ is at most $d/(\ell + d + \frac{5}{2})$.

(c) The expansion coefficients define a Markov chain on the radial index: a raising step at parameter β moves ζ to $\zeta - 1$ with probability $\zeta/(\beta + 2\zeta + 1) \leq \frac{1}{2}$ and otherwise leaves it unchanged, and lowering steps never decrease ζ . Let N_i be the number of decreases among the first i raising steps. Then $\Pr(N_{i+1} \geq j) \leq \Pr(N_i \geq j) + \frac{1}{2} \Pr(N_i = j - 1) = \frac{1}{2} (\Pr(N_i \geq j) + \Pr(N_i \geq j - 1))$, and by induction $\Pr(N_i \geq j) \leq \Pr(\operatorname{Bin}(i, \frac{1}{2}) \geq j)$. Since there are at most d raising steps and the final index is at least $s - N$, the claim follows. (The probabilistic language only organises sums of products of the nonnegative coefficients.) $\square$

13.4. The operator K_3 . For $\ell \geq 0$ and $s \geq 1$ put $\beta = \ell + \frac{1}{2}$, $d = \beta + 2s = \ell + 2s + \frac{1}{2}$, and define

$$(13.9) \quad K_3 \Psi_{\ell,s} = \frac{\Psi_{\ell,s-1}}{4d(d+1)} - \frac{\Psi_{\ell,s}}{2d(d+2)} + \frac{\Psi_{\ell,s+1}}{4(d+1)(d+2)} = H_{\ell} k_{\ell,s}(t),$$

and let $E = x \cdot \nabla = r \partial_r$ be the Euler operator.

Lemma 13.10. *For $\ell \geq 0$ and $s \geq 1$, with β, d as above:*

- (a) $k'_{\ell,s} = (J_s^{\beta+1} - J_{s-1}^{\beta+1})/(4(d+1))$;
- (b) $\Delta K_3 \Psi_{\ell,s} = \Psi_{\ell,s}$;
- (c) $K_3 \Psi_{\ell,s} = 0$ and $\nabla K_3 \Psi_{\ell,s} = 0$ on ∂B ;

(d) with $\Psi_{-1,\cdot} = 0$,

$$\partial_{x_1} K_3 \Psi_{\ell,s} = \frac{\ell+1}{(2\ell+1)2(d+1)} (\Psi_{\ell+1,s} - \Psi_{\ell+1,s-1}) + \frac{\ell}{(2\ell+1)2(d+1)} (\Psi_{\ell-1,s+1} - \Psi_{\ell-1,s}),$$

$$EK_3 \Psi_{\ell,s} = -\frac{d+\frac{1}{2}}{4d(d+1)} \Psi_{\ell,s-1} + \frac{\Psi_{\ell,s}}{4d(d+2)} + \frac{d+\frac{3}{2}}{4(d+1)(d+2)} \Psi_{\ell,s+1},$$

and $y\partial_y K_3 \Psi_{\ell,s} = EK_3 \Psi_{\ell,s} - x_1 \partial_{x_1} K_3 \Psi_{\ell,s}$;

(e) K_3 , $\partial_{x_1} K_3$, EK_3 and $y\partial_y K_3$ extend to bounded operators $\mathcal{G} \rightarrow \mathcal{L}$ with $\|K_3\| = 4/45$, $\|\partial_{x_1} K_3\| \leq 2R/7$, $\|EK_3\| \leq 6/35$ and $\|y\partial_y K_3\| \leq 6/35 + 2R^2/7$.

Proof. Write $A = \frac{1}{4d(d+1)}$, $B = -\frac{1}{2d(d+2)}$, $C = \frac{1}{4(d+1)(d+2)}$ for the coefficients in (13.9), and $J_\zeta = J_\zeta^\beta$. For $0 \leq k \leq s$ put $u = s - k$; then $\beta + s + k + 1 = d - u + 1$, and (13.6) gives

$$\begin{aligned} \frac{[t^{k+1}]J_{s-1}}{[t^k]J_s} &= \frac{u(u-1)}{(k+1)(\beta+k+1)}, & \frac{[t^{k+1}]J_s}{[t^k]J_s} &= -\frac{u(d-u+1)}{(k+1)(\beta+k+1)}, \\ \frac{[t^{k+1}]J_{s+1}}{[t^k]J_s} &= \frac{(d-u+1)(d-u+2)}{(k+1)(\beta+k+1)}. \end{aligned}$$

Since $u(u-1)(d+2) + 2u(d+1)(d-u+1) + d(d-u+1)(d-u+2) = d(d+1)(d+2)$ identically in u and d ,

$$(13.10) \quad [t^{k+1}]k_{\ell,s} = \frac{[t^k]J_s}{4(k+1)(\beta+k+1)} \quad (0 \leq k \leq s).$$

(a) The coefficient of t^k in $k'_{\ell,s}$ is $(k+1)[t^{k+1}]k_{\ell,s}$. On the other hand, by (13.6),

$$\frac{[t^k]J_s^{\beta+1}}{[t^k]J_s} = \frac{\beta+s+k+1}{\beta+k+1}, \quad \frac{[t^k]J_{s-1}^{\beta+1}}{[t^k]J_s} = -\frac{s-k}{\beta+k+1},$$

so the coefficient of t^k on the right-hand side is

$$[t^k]J_s \frac{\beta+2s+1}{4(d+1)(\beta+k+1)} = \frac{[t^k]J_s}{4(\beta+k+1)}.$$

Both sides have degree s .

(b) For a polynomial $v(t)$, $\Delta(H_\ell v) = v\Delta H_\ell + 2\nabla H_\ell \cdot \nabla v + H_\ell \Delta v = 4H_\ell(tv'' + (\ell + \frac{3}{2})v')$, by (13.5) and $\Delta v(|x|^2) = 4tv'' + 6v'$. The coefficient of t^k in $4(tk'' + (\beta+1)k')$ is $4(k+1)(\beta+k+1)[t^{k+1}]k_{\ell,s} = [t^k]J_s$ by (13.10).

(c) Since $J_\zeta(1) = 1$, $k_{\ell,s}(1) = A + B + C = 0$, and $k'_{\ell,s}(1) = 0$ by (a). Hence $K_3 \Psi_{\ell,s} = 0$ and $\nabla(H_\ell k_{\ell,s}) = k_{\ell,s} \nabla H_\ell + 2k'_{\ell,s} H_\ell x = 0$ on ∂B .

(d) By (13.5), $\partial_{x_1}(H_\ell k) = \frac{2(\ell+1)}{2\ell+1} H_{\ell+1} k' + \frac{2\ell}{2\ell+1} H_{\ell-1}(\beta k + tk')$ and $E(H_\ell k) = H_\ell(\ell k + 2tk')$, with $k = k_{\ell,s}$. The $H_{\ell+1}$ term is given by (a). For the others we expand in the basis $J_\zeta = J_\zeta^\beta$: by (a) and the second identity of Lemma 13.3(b) (with $\beta+1$ in place of β),

$$tk' = \frac{1}{4(d+1)} \left(-\frac{\beta+s}{d} J_{s-1} + \frac{\beta(d+1)}{d(d+2)} J_s + \frac{s+1}{d+2} J_{s+1} \right),$$

where we used $(\beta+s+1)d - s(d+2) = \beta(d+1)$. Adding βk , respectively ℓk , gives

$$\begin{aligned} \beta k + tk' &= -\frac{s}{4d(d+1)} J_{s-1} - \frac{\beta}{4d(d+2)} J_s + \frac{\beta+s+1}{4(d+1)(d+2)} J_{s+1}, \\ \ell k + 2tk' &= -\frac{d+\frac{1}{2}}{4d(d+1)} J_{s-1} + \frac{1}{4d(d+2)} J_s + \frac{d+\frac{3}{2}}{4(d+1)(d+2)} J_{s+1}. \end{aligned}$$

The second line is the formula for EK_3 . By the first identity of Lemma 13.3(b) (with $\beta - 1$ in place of β), $(J_{s+1}^{\beta-1} - J_s^{\beta-1})/(4(d+1))$ has the same expansion as the first line, which gives the formula for $\partial_{x_1}K_3$. The last identity holds because $E = x_1\partial_{x_1} + y\partial_y$ on axisymmetric functions.

(e) We use Lemma 7.2. The operator K_3 preserves ℓ , and its column sum is $|A| + |B| + |C| = 1/(d(d+2))$, which is largest, $4/45$, for $\ell = 0$, $s = 1$. For EK_3 the column sum is $(2d+1)/(4d(d+1))$, decreasing in d , with value $6/35$ at $d = \frac{5}{2}$. For $\partial_{x_1}K_3$ and even ℓ , the normalised column norm is at most $((\ell+1)h_{\ell+1} + \ell h_{\ell-1})/((2\ell+1)(d+1)h_\ell)$. For $\ell = 0$ this is at most $R/(d+1) \leq 2R/7$. For $\ell \geq 2$, Lemma 13.5(b), $\ell^2/(\ell-1) \leq \ell+2$ and $d+1 \geq \ell + \frac{7}{2}$ bound it by $R(2\ell+3)/((2\ell+1)(\ell + \frac{7}{2})) \leq 7R/27.5 < 2R/7$. Finally $\|x_1f\|_{\mathcal{L}} \leq h_1\|f\|_{\mathcal{L}} = R\|f\|_{\mathcal{L}}$ by Lemma 13.8(a). $\square$

The following lemma shows that K_3g is a genuine C^1 function with vanishing Cauchy data. Let $\mathcal{P}_n$ be the space of real polynomials of degree at most n on $\mathbb{R}^3$, with the inner product of $L^2(B)$, and let V_n be the orthogonal complement of $\mathcal{P}_{n-1}$ in $\mathcal{P}_n$.

Lemma 13.11. (a) For $P \in V_n$ and $x \in \overline{B}$, $|P(x)|^2 \leq \frac{(n+1)(n+2)(2n+3)}{8\pi} \|P\|_{L^2(B)}^2$.
 (b) For $\ell \geq 0$ and $s \geq 1$, $\sup_{\overline{B}} |\nabla K_3 \Psi_{\ell,s}| \leq 1$ and $\sup_{\overline{B}} |K_3 \Psi_{\ell,s}| \leq 1$.
 (c) For $g \in \mathcal{G}$, the series $W = \sum g_{\ell,s} K_3 \Psi_{\ell,s}$ converges in $C^1(\overline{B})$; $W = 0$ and $\nabla W = 0$ on ∂B ; $\Delta W = \iota(g)$ in $\mathcal{D}'(B)$; and $\iota(K_3g) = W$, $\iota(\partial_{x_1}K_3g) = \partial_{x_1}W$, $\iota(y\partial_yK_3g) = y\partial_yW$.

Proof. (a) Let $S^4 \subset \mathbb{R}^5 = \mathbb{R}^3 \times \mathbb{R}^2$, with points (x, z) and surface measure $d\zeta$. Parametrising S^4 by $(x, \phi) \mapsto (x, \sqrt{1-|x|^2}(\cos \phi, \sin \phi))$ one finds $d\zeta = dx d\phi$, so $\int_{S^4} f(x) d\zeta = 2\pi \int_B f dx$ for integrable f . Let $Q(x, z) = P(x)$. Its restriction to S^4 is a sum $\sum_{k \leq n} Y_k$ of spherical harmonics of degree k [53, Ch. IV]. Since Q is invariant under rotations of z and the projections onto the spaces of spherical harmonics commute with rotations, each Y_k is the restriction of a harmonic homogeneous polynomial invariant under these rotations, that is, of a polynomial in x and $|z|^2$; as $|z|^2 = 1 - |x|^2$ on S^4 , $Y_k(x, z) = P_k(x)$ for some $P_k \in \mathcal{P}_k$. For $k < n$, $\|Y_k\|_{L^2(S^4)}^2 = \int_{S^4} QY_k d\zeta = 2\pi \int_B PP_k dx = 0$. Hence $Q|_{S^4} = Y_n$ is a spherical harmonic of degree n . For an orthonormal basis (E_i) of the spherical harmonics of degree n on S^4 , $\sum_i |E_i|^2$ is invariant under rotations, hence equal to $N_n/|S^4|$, where $N_n = \binom{n+4}{4} - \binom{n+2}{4} = (n+1)(n+2)(2n+3)/6$ and $|S^4| = 8\pi^2/3$. By the Cauchy–Schwarz inequality, $|P(x)|^2 = |Y_n(x, z)|^2 \leq \frac{N_n}{|S^4|} \|Y_n\|_{L^2(S^4)}^2 = \frac{2\pi N_n}{|S^4|} \|P\|_{L^2(B)}^2$.

(b) Let $n = \ell + 2s$ and $\Xi = K_3 \Psi_{\ell,s}$, a polynomial of degree $n+2$ whose homogeneous components have the parity of n . By Lemma 13.6(a), Ξ is orthogonal to $\mathcal{P}_{n-3}$. For $\varphi \in \mathcal{P}_{n-2}$ and $i = 1, 2, 3$, $\int_B \partial_i \Xi \varphi = - \int_B \Xi \partial_i \varphi = 0$, because $\Xi = 0$ on ∂B . Hence $\partial_i \Xi \in V_{n-1} \oplus V_n \oplus V_{n+1}$. The reflection $x \mapsto -x$ preserves each V_m , and an element of V_m of parity $(-1)^{m+1}$ has no homogeneous part of degree m , hence lies in $\mathcal{P}_{m-1} \cap V_m = 0$; since $\partial_i \Xi$ has parity $(-1)^{n+1}$ and the orthogonal projections onto the V_m commute with the reflection, the component of $\partial_i \Xi$ in V_n vanishes, and $\partial_i \Xi = P_i + Q_i$ with $P_i \in V_{n-1}$, $Q_i \in V_{n+1}$. By (a) and the Cauchy–Schwarz inequality, with $\gamma_m = (m+1)(m+2)(2m+3)/(8\pi)$,

$$|\nabla \Xi(x)|^2 \leq (\gamma_{n-1} + \gamma_{n+1}) \sum_i (\|P_i\|^2 + \|Q_i\|^2) = (\gamma_{n-1} + \gamma_{n+1}) \|\nabla \Xi\|_{L^2(B)}^2.$$

By (c) of Lemma 13.10, (b) of the same lemma and orthogonality, $\|\nabla \Xi\|_{L^2}^2 = - \int_B \Xi \Psi_{\ell,s} = \|\Psi_{\ell,s}\|_{L^2}^2/(2d(d+2))$ with $d = n + \frac{1}{2}$. Since $\gamma_{n-1} + \gamma_{n+1} = (2n+3)(n^2+3n+5)/(4\pi)$ and

$$\|\Psi_{\ell,s}\|_{L^2}^2 = 4\pi/((2\ell+1)(2n+3)),$$

$$|\nabla\Xi(x)|^2 \leq \frac{2(n^2+3n+5)}{(2\ell+1)(2n+1)(2n+5)} \leq 1,$$

because $(4n^2+12n+5) - (2n^2+6n+10) = 2n^2+6n-5 > 0$ for $n \geq 1$. As $\Xi = 0$ on ∂B , integration along a radius gives $|\Xi| \leq 1$ on $\bar{B}$.

(c) By (b), $\sum |g_{\ell,s}| \leq \|g\|_{\mathcal{L}}$ bounds the series in $C^1(\bar{B})$. The partial sums $W_N = K_3 g_N$ are polynomials with $\Delta W_N = g_N$ and vanishing Cauchy data (Lemma 13.10); since $g_N \rightarrow \iota(g)$ in $L^2(B)$ and $W_N \rightarrow W$ in $C^1(\bar{B})$, the Cauchy data and the distributional equation pass to the limit. The coefficient series $K_3 g$, $\partial_{x_1} K_3 g$ and $y \partial_y K_3 g$ converge in $\mathcal{L}$ (Lemma 13.10(e)), hence in $L^2(B)$, to the same limits. $\square$

13.5. The quartic equation. Let

$$b = 7626062690497999 \cdot 2^{-46}, \quad \sigma = 8b,$$

a fixed dyadic rational independent of the varying coefficient c_1 . For $x = (g, c) \in \mathcal{X}$ let $W = K_3 g$, $W_1 = \partial_{x_1} K_3 g$ and $W_2 = y \partial_y K_3 g$, elements of $\mathcal{L}$ by Lemma 13.10(e), and define

$$(13.11) \quad F(g, c) = q_c g + (\partial_{x_1} q_c) W_1 + \left(\frac{\partial_y q_c}{y} \right) W_2 + q_c |p_c|^2 (1 + W), \quad \mathcal{F} = F/b,$$

where the multipliers are given by (13.1), (13.2), the products are those of Lemma 13.8, and $1 = \Psi_{0,0}$.

Lemma 13.12. *$\mathcal{F}$ is a bounded polynomial map of degree four from $\mathcal{X}$ to $\mathcal{Y}$. If $\mathcal{F}(g, c) = 0$ and $\operatorname{Re} \psi'_c > 0$ on $\{|w| < R_0\}$ for some $1 < R_0 < R$, then $U = 1 + \iota(K_3 g) \in C^1(\bar{B})$ is axisymmetric, satisfies (13.4), and $U = 1$, $\nabla U = 0$ on ∂B .*

Proof. By (13.1), $\|r^{j-1} U_{j-1}\|_{\text{ch}} = \sum_{|\nu| \leq j-1, \nu \text{ even}} R^{|\nu|}$, which is at most $j R^{j-1}$ and at most $2R^{j+1}/(R^2-1)$; similarly $\|r^m C_m^{(2)}\|_{\text{ch}} = \sum_{p+p'=m} (p+1)(p'+1) R^{|p-p'|} \leq 2(m+1)R^{m+4}/(R^2-1)^2$. Hence

$$(13.12) \quad \begin{aligned} \|q_c\|_{\text{ch}} &\leq \frac{\|c\|_{\mathcal{E}}}{R}, & \|\partial_{x_1} q_c\|_{\text{ch}} &\leq \frac{2\|c\|_{\mathcal{E}}}{R^2-1}, \\ \left\| \frac{\partial_y q_c}{y} \right\|_{\text{ch}} &\leq \frac{4R\|c\|_{\mathcal{E}}}{(R^2-1)^2}, & \| |p_c|^2 \|_{\text{ch}} &\leq \frac{\|c\|_{\mathcal{E}}^2}{R^2}, \end{aligned}$$

the last by (13.2), Lemma 13.8(c) and the identity $\sum_a |p_a| R^a = R^{-1} \|c\|_{\mathcal{E}}$. Using parts (a) and (b) of Lemma 13.8 and Lemma 13.10(e), it follows that (13.11) is a sum of bounded multilinear maps of degree at most four. Their values are even in x_1 , by the parities recorded after (13.2), so they lie in $\mathcal{L}^{\text{ev}}$, which is contained in $\mathcal{Y}$.

Suppose $\mathcal{F}(g, c) = 0$. By Lemma 13.11(c), $W = \iota(K_3 g) \in C^1(\bar{B})$ has vanishing Cauchy data and $\Delta W = \iota(g)$, and by Lemma 13.8(d) and Lemma 13.6(b),

$$0 = b \iota(\mathcal{F}(g, c)) = q_c \iota(g) + \partial_{x_1} q_c \partial_{x_1} W + \frac{\partial_y q_c}{y} y \partial_y W + q_c |p_c|^2 U \quad \text{a.e. in } B.$$

Off the axis, $\nabla q_c \cdot \nabla U = \partial_{x_1} q_c \partial_{x_1} W + \partial_y q_c \partial_y W$. Let $\varphi \in C_c^\infty(B)$. As q_c is smooth on $\bar{B}$ (Lemma 13.1(a)), $q_c \varphi \in C_c^\infty(B)$ and

$$\begin{aligned} \int_B (q_c \nabla U \cdot \nabla \varphi - q_c |p_c|^2 U \varphi) &= \int_B (\nabla W \cdot \nabla (q_c \varphi) - \varphi \nabla q_c \cdot \nabla U - q_c |p_c|^2 U \varphi) \\ &= - \int_B \varphi b \iota(\mathcal{F}(g, c)) = 0, \end{aligned}$$

using $\int \nabla W \cdot \nabla (q_c \varphi) = - \int \iota(g) q_c \varphi$. $\square$

Nonlinear bounds. Let $x^\circ = (g^\circ, c^\circ) \in \mathcal{X}$ and let $G_0 \geq \|g^\circ\|_{\mathcal{L}}$, $C_0 \geq \|c^\circ\|_{\mathcal{E}}$. Put

$$\begin{aligned} L_2 &= \kappa_R \left[\frac{1}{R} + \frac{2}{R^2 - 1} \cdot \frac{2R}{7} + \frac{4R}{(R^2 - 1)^2} \left(\frac{6}{35} + \frac{2R^2}{7} \right) \right], & L_3 &= \frac{\kappa_R}{R^3}, & L_4 &= \frac{4}{45} \cdot \frac{\kappa_R}{R^3}, \\ c_2 &= \frac{1}{b} \left[\frac{L_2 + 3L_4 C_0^2}{\sigma} + \frac{3C_0(L_3 + L_4 G_0)}{\sigma^2} \right], & c_3 &= \frac{1}{b} \left[\frac{L_3 + L_4 G_0}{\sigma^3} + \frac{3L_4 C_0}{\sigma^2} \right], & c_4 &= \frac{L_4}{b\sigma^3}. \end{aligned}$$

Lemma 13.13. *For $h \in \mathcal{X}$ with $\|h\|_{\mathcal{X}} \leq r$,*

$$\begin{aligned} \|\mathcal{F}(x^\circ + h) - \mathcal{F}(x^\circ) - D\mathcal{F}(x^\circ)h\|_{\mathcal{Y}} &\leq c_2 r^2 + c_3 r^3 + c_4 r^4, \\ \|D\mathcal{F}(x^\circ + h) - D\mathcal{F}(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{Y}} &\leq 2c_2 r + 3c_3 r^2 + 4c_4 r^3. \end{aligned}$$

Proof. Let $\mathbf{b}(c, g) = q_c g + (\partial_{x_1} q_c) W_1(g) + (y^{-1} \partial_y q_c) W_2(g)$ and let $\mathbf{t}(c^1, c^2, c^3)$ be the symmetrisation of $q_{c^1} \operatorname{Re}(p_{c^2} \overline{p_{c^3}})$, so that $b\mathcal{F}(g, c) = \mathbf{b}(c, g) + \mathbf{t}(c, c, c)(1 + K_3 g)$. By Lemma 13.8, (13.12) and Lemma 13.10(e),

$$\|\mathbf{b}(c, g)\|_{\mathcal{Y}} \leq L_2 \|c\|_{\mathcal{E}} \|g\|_{\mathcal{L}}, \quad \|\mathbf{t}(c^1, c^2, c^3) f\|_{\mathcal{L}} \leq L_3 \prod_i \|c^i\|_{\mathcal{E}} \|f\|_{\mathcal{L}},$$

and $\|K_3 \gamma\|_{\mathcal{L}} \leq \frac{4}{45} \|\gamma\|_{\mathcal{L}}$. Write $h = (\gamma, \eta)$; then $\|\gamma\|_{\mathcal{L}} \leq r$ and $\|\eta\|_{\mathcal{E}} \leq r/\sigma$. The terms of degree at least two in h of $b\mathcal{F}(x^\circ + h)$ are $\mathbf{b}(\eta, \gamma)$, $3\mathbf{t}(c^\circ, \eta, \eta) + \mathbf{t}(\eta, \eta, \eta)$, and

$$3\mathbf{t}(c^\circ, c^\circ, \eta) K_3 \gamma + 3\mathbf{t}(c^\circ, \eta, \eta) K_3 g^\circ + 3\mathbf{t}(c^\circ, \eta, \eta) K_3 \gamma + \mathbf{t}(\eta, \eta, \eta) K_3 g^\circ + \mathbf{t}(\eta, \eta, \eta) K_3 \gamma.$$

Their norms are at most $L_2 \frac{r}{\sigma} r$, $3L_3 C_0 \frac{r^2}{\sigma^2} + L_3 \frac{r^3}{\sigma^3}$, and $3L_4 C_0^2 \frac{r}{\sigma} r + 3L_4 C_0 G_0 \frac{r^2}{\sigma^2} + 3L_4 C_0 \frac{r^2}{\sigma^2} r + L_4 G_0 \frac{r^3}{\sigma^3} + L_4 \frac{r^3}{\sigma^3} r$, and collecting powers of r gives the first bound. Each of these terms is the restriction to the diagonal of a multilinear map bounded by a product in which every factor η contributes r/σ and every factor γ contributes r . Differentiating in the direction k with $\|k\|_{\mathcal{X}} \leq 1$ replaces one factor by a component of k , of norm at most $1/\sigma$, respectively 1; hence the derivative of a term of degree m is bounded by m/r times the bound of the term, which gives the second estimate. $\square$

Proposition 13.14. *Let $A: \mathcal{Y} \rightarrow \mathcal{X}$ be bounded, linear and injective, and suppose that $\|A\mathcal{F}(x^\circ)\|_{\mathcal{X}} \leq Y$, $\|I - AD\mathcal{F}(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{X}} \leq Z$, $\|A\| \leq \mathbf{a}$, and that $r > 0$ satisfies*

$$(13.13) \quad Y + (Z - 1)r + \mathbf{a}(c_2 r^2 + c_3 r^3 + c_4 r^4) < 0, \quad Z + \mathbf{a}(2c_2 r + 3c_3 r^2 + 4c_4 r^3) < 1.$$

Then $\mathcal{F}$ has exactly one zero in the closed ball $\bar{B}_{\mathcal{X}}(x^\circ, r)$.

Proof. Let $\mathcal{N}(x) = x - A\mathcal{F}(x)$. For $\|h\|_{\mathcal{X}} \leq r$, $\mathcal{N}(x^\circ + h) - x^\circ = -A\mathcal{F}(x^\circ) + (I - AD\mathcal{F}(x^\circ))h - A(\mathcal{F}(x^\circ + h) - \mathcal{F}(x^\circ) - D\mathcal{F}(x^\circ)h)$, whose norm is less than r by Lemma 13.13 and the first inequality. On the convex ball, $D\mathcal{N}(x^\circ + h) = (I - AD\mathcal{F}(x^\circ)) - A(D\mathcal{F}(x^\circ + h) - D\mathcal{F}(x^\circ))$ has norm less than one by the second inequality, so $\mathcal{N}$ is a contraction of the ball into itself. Its unique fixed point is the unique zero of $\mathcal{F}$ in the ball, since A is injective. $\square$

13.6. The approximate inverse. From now on $M = 180$ and $S = 80$. The exact dyadic centre $x^\circ = (g^\circ, c^\circ)$ consists of 7280 field coefficients with even $\ell \leq 180$ and $1 \leq s \leq 80$, and 91 odd shape coefficients c_j° with $j \leq 181$. These 7371 coefficients are stored in `anc/r3_convex/candidate.json`; their SHA-256 is recorded in the ancillary README. The centre is not assumed to be an exact zero. Write $q^\circ = q_{c^\circ}$, $p^\circ = \sum_a p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, and $W^\circ = K_3 g^\circ$.

Finite and tail coordinates. The *finite inputs* are the 7280 field coordinates (ℓ, s) with $\ell \leq M$ even and $1 \leq s \leq S$, and the 91 shape coordinates c_j with $j \leq M+1$. The *finite rows* are the same 7280 field rows (ℓ, s) and the 91 harmonic rows $\mathcal{T}_n$, $n \leq M$ even. All other inputs and rows are *tail* inputs and rows; tail rows are *harmonic* ($\mathcal{T}_n$, $n \geq 182$) or *nonharmonic* ((ℓ, s) with $s \geq 1$ and $\ell \geq 182$ or $s \geq 81$). Accordingly $\mathcal{X} = \mathcal{X}_f \oplus \mathcal{X}_t$ and $\mathcal{Y} = \mathcal{Y}_f \oplus \mathcal{Y}_t^H \oplus \mathcal{Y}_t^N$. On the shape tail and on the harmonic rows we use normalised coordinates

$$\hat{a}_n = b(n+1)R^{n+1}c_{n+1}, \quad \hat{y}_n = R^n y_n \quad (n \text{ even}),$$

so that the shape part of $\|\cdot\|_{\mathcal{X}}$ is $8 \sum |\hat{a}_n|$ and the harmonic part of $\|\cdot\|_{\mathcal{Y}}$ is $\sum |\hat{y}_n|$.

The principal part on high shape modes. For odd j , the derivative of $b^{-1}q_c|p_c|^2$ at c° in the direction e_j contains the *principal polynomial* $\Pi_j = \frac{2}{b} q^\circ \operatorname{Re}(\overline{p^\circ} j w^{j-1})$, in which q is frozen. On ∂B we have $w = e^{i\theta}$, $\xi = \cos \theta$, $q^\circ = \sum_k q_k e^{ik\theta}$ with $q_k = \sum_{i>|k|} c_i^\circ$ (k even, i odd), and therefore

$$\Pi_j|_{\partial B} = \frac{2j}{b} \sum_{a,k} p_a q_k T_{|j-1-a+k|}(\xi).$$

Let $\operatorname{Me}_j = \frac{2j}{b} \sum_{a,k} p_a q_k \mathcal{T}_{|j-1-a+k|} \in \mathcal{Y}$ be the harmonic element with these boundary values, and define the symbol and its defect

$$s_\delta = \frac{1}{b^2} \sum_{k-a=2\delta} p_a q_k R^{2\delta} \quad (\delta \in \mathbb{Z}), \quad \varepsilon = \sum_{\delta} |s_\delta - \delta_{\delta 0}|.$$

Lemma 13.15. *Let $n \geq 182$ be even. In normalised coordinates, $\operatorname{Me}_{n+1}/(b(n+1)R^{n+1})$ has the entry $\frac{2}{R} s_\delta R^{|n+2\delta|-(n+2\delta)}$ in the harmonic row $|n+2\delta|$, summed over δ . Only $-180 \leq \delta \leq 90$ occur, so the rows $|n+2\delta| \leq M$ occur only for $n \leq 540$, and $n+2\delta \geq -178$. Let T be the part of $\mathbf{M}$ from the shape tail to the harmonic tail rows and $\mathcal{B}$ the part from the shape tail to the finite harmonic rows. Then, on ℓ^1 of the normalised coordinates,*

$$T = \frac{2}{R}(I + E), \quad (E\hat{a})_m = \sum_{n \geq 182} (s_{(m-n)/2} - \delta_{mn}) \hat{a}_n \quad (m \geq 182), \quad \|E\| \leq \varepsilon.$$

If $\varepsilon < 1$, then T is invertible, $T^{-1} = \frac{R}{2} \sum_{k \geq 0} (-E)^k$, $\|T^{-1}\| \leq R/(2(1-\varepsilon))$, and $\|T^{-1}\hat{y} - \frac{R}{2} \sum_{k=0}^L (-E)^k \hat{y}\|_1 \leq \frac{R\varepsilon^{L+1}}{2(1-\varepsilon)} \|\hat{y}\|_1$ for $L \geq 0$.

Proof. The entry of Me_{n+1} in the row $\mathcal{T}_{|n+2\delta|}$ is $\frac{2(n+1)}{b} \sum_{k-a=2\delta} p_a q_k$; multiplying by $R^{|n+2\delta|}$ and dividing by $b(n+1)R^{n+1}$ gives the stated value, since $R^{2\delta} = R^{(n+2\delta)-n}$. As $0 \leq a \leq 180$ and $|k| \leq 180$, $-180 \leq \delta \leq 90$. Hence rows $m \geq 182$ are reached only with $m = n + 2\delta$, and T is $\frac{2}{R}$ times the compression to indices ≥ 182 of the Laurent operator with symbol (s_δ) ; the compression of a convolution has ℓ^1 norm at most the ℓ^1 norm of its kernel. The remaining statements are the Neumann series. $\square$

The operator A. Let J^{fl} be the frozen 7371×7371 binary64 Galerkin matrix and let $\hat{A}$ be its frozen binary64 approximate inverse, regarded as an exact linear map $\mathcal{Y}_f \rightarrow \mathcal{X}_f$. They are reconstructed from a nearby numerical seed; the discrepancy from the true finite Jacobian at the full exact centre is enclosed separately. The boundary symbol of the tail inverse uses the full exact centre. Suppose $\varepsilon < 1$ and

$$(13.14) \quad \eta := \|I - \hat{A}J^{\text{fl}}\|_b < 1,$$

where $\|\cdot\|_b$ is the operator norm for the weights of $\mathcal{X}$ and $\mathcal{Y}$ with σ replaced by b . For $y = y_f + y_t^H + y_t^N \in \mathcal{Y}$ define

$$(13.15) \quad Ay = \hat{A}(y_f - \mathcal{B}T^{-1}y_t^H) + T^{-1}y_t^H + y_t^N,$$

where $T^{-1}y_t^H$ is a shape tail vector and y_t^N is read as a field tail vector with the same coordinates. For a finite row ι with weight w_ι (h_ℓ or R^n) let $\alpha_\iota = \|\hat{A}e_\iota\|_{\mathcal{X}}/w_\iota$, let $\alpha_n^H = \alpha_{\tau_n}$ for $n \leq M$, $\alpha_{\max} = \max_\iota \alpha_\iota$, and

$$\rho_{\mathcal{B}} = \max_{n \geq 182} \sum_{m \leq M} |\hat{\mathcal{B}}_{mn}| \alpha_m^H, \quad A_t = \max \left\{ 1, \frac{R(8 + \rho_{\mathcal{B}})}{2(1 - \varepsilon)} \right\},$$

where $\hat{\mathcal{B}}$ is the matrix of $\mathcal{B}$ in normalised coordinates.

Lemma 13.16. *A: $\mathcal{Y} \rightarrow \mathcal{X}$ is a bounded bijection with $\|A\| \leq \max\{\alpha_{\max}, A_t\}$. It maps every harmonic tail row, normalised, to a vector of norm at most A_t , and every nonharmonic tail row to the identical field tail input. Moreover $A(\text{Me}_j) = e_j$ for every odd $j \geq 183$, and $A\Psi_{\ell,s} = e_{(\ell,s)}$ for every field tail input (ℓ, s) .*

Proof. By (13.14), $\hat{A}J^{\text{fl}}$ and hence $\hat{A}$ are invertible, and T is invertible by Lemma 13.15. The map $(x_f, \hat{a}, x^N) \mapsto (\hat{A}^{-1}x_f + \mathcal{B}\hat{a}) + T\hat{a} + x^N$ is a two-sided inverse of A . We use Lemma 7.2: a finite row e_ι is mapped to $\hat{A}e_\iota$; a nonharmonic tail row to itself; the normalised harmonic tail row $\hat{y} = e_m$ to $T^{-1}e_m$, of shape norm $8\|T^{-1}e_m\|_1 \leq 8R/(2(1 - \varepsilon))$, plus $-\hat{A}\mathcal{B}T^{-1}e_m$, of norm at most $\rho_{\mathcal{B}}\|T^{-1}e_m\|_1$. Finally, $\text{Me}_j = \mathcal{B}e_j + Te_j$ in normalised coordinates, so $A\text{Me}_j = \hat{A}(\mathcal{B}e_j - \mathcal{B}T^{-1}Te_j) + T^{-1}Te_j = e_j$. $\square$

13.7. The defect bounds. We bound $Y = \|A\mathcal{F}(x^\circ)\|_{\mathcal{X}}$ and $Z = \|I - A\mathcal{D}\mathcal{F}(x^\circ)\|$ by normalised column norms. The disjoint field partition is: 7280 finite inputs; 443 exceptional inputs $S < s < S_\ell$ for even $\ell \leq 180$; the 91 radial classes $s \geq S_\ell$; the 14 explicit inputs $(\ell, 1)$ and the 14 radial classes $s \geq 2$ for even $182 \leq \ell \leq 208$; and the uniform angular tail $\ell \geq 210$, $s \geq 1$. Thus there are $443 + 14 = 457$ explicitly checked exceptional field inputs in total. Here S_ℓ equals 93 for $\ell = 0, 2, \dots, 12$; 92 for $14, \dots, 30$; 91 for $32, \dots, 50$; 90 for $52, \dots, 62$; 89 for $64, \dots, 72$; 88 for $74, 76, 78$; 87 for $80, \dots, 90$; 86 for $\ell = 92$; 85 for $\ell = 94$; and 81 for $96, \dots, 180$. The shape inputs are the 310 explicit odd indices $j \leq 619$ and all odd indices $j \geq 621$. These sets exhaust all input columns without overlap.

The residual and the explicit columns. The residual $\mathcal{F}(x^\circ)$ is a polynomial. It is expanded in the basis $\Psi_{\ell,s}$ without truncation, in ball arithmetic, using (13.5) and Lemma 13.3(b) for the multiplications by x_1 and t (which give $x_1\Psi_{\ell,s}$ and $t\Psi_{\ell,s}$ as combinations of at most four, respectively three, Ψ 's), the recurrences $r^{n+1}Q_{n+1} = 2x_1r^nQ_n - tr^{n-1}Q_{n-1}$ for $Q = T, U$ and the three-term recurrence of $C_n^{(2)}$ for the series (13.1), (13.2), and (13.9) and Lemma 13.10(d). Its harmonic part is converted to the coordinates of $\mathcal{Y}$ by the matrix D . Since A has norm at most $\alpha_{\max}$ on $\mathcal{Y}_f$, one on $\mathcal{Y}_t^N$ and A_t on $\mathcal{Y}_t^H$ (Lemma 13.16),

$$Y \leq \alpha_{\max}\|\mathcal{F}(x^\circ)_f\|_{\mathcal{Y}} + \|\mathcal{F}(x^\circ)_t^N\|_{\mathcal{Y}} + A_t\|\mathcal{F}(x^\circ)_t^H\|_{\mathcal{Y}}.$$

For the 457 exceptional field inputs and 310 explicit shape inputs, the column $D\mathcal{F}(x^\circ)e$ is polynomial. The shape columns are expanded completely. For exceptional field columns alone, evaluation retains the shape coefficients only through $j = 121$ and adds the rigorous omitted-tail bound below. In both cases A uses the full centre and the full boundary symbol. It is applied in the form (13.15), with T^{-1} evaluated by 24 terms of the Neumann series of Lemma 13.15 plus the stated remainder bound, and e is subtracted. Let Z_{ex} and Z_{sh} be the resulting maxima.

Put $c^t = (c_j^\circ \mathbf{1}_{j \leq 121})_j$, $C = \|c^\circ\|_{\mathcal{E}}$, $\delta = \|c^\circ - c^t\|_{\mathcal{E}}$, $\kappa_R = 4(1 - R^{-2})^{-1/2} - 1$, and $z = \ell + 2s + \frac{1}{2}$. For a normalised field basis vector the exact three-term inverse and derivative formulas give

$$K_z = \frac{1}{z(z+2)}, \quad D_z = \frac{2R}{z+1}, \quad E_z = \frac{1}{z} + RD_z$$

as upper bounds for the norms of K_3e , $\partial_{x_1}K_3e$ and $y\partial_yK_3e$, respectively. The linear multiplier inequalities (13.12) and the telescoping expansion of the three factors of $q|p|^2$ then give

$$\|(D_g\mathcal{F}(c^\circ) - D_g\mathcal{F}(c^t))e\|_{\mathcal{Y}} \leq \frac{\kappa_R\delta}{b} \left[\frac{1}{R} + \frac{2D_z}{R^2 - 1} + \frac{4RE_z}{(R^2 - 1)^2} + \frac{3C^2K_z}{R^3} \right].$$

Multiplication by the certified full $\|A\|$ bounds the omitted contribution to the X -defect. For these inputs $z \geq 325/2$, giving a common error strictly below 0.007935; the verifier uses the individual z of each column. No truncation enters A or the residual.

Finite field columns. Let $e = e_{(\ell,s)}$ be a finite field input, and let Je and $(D\mathcal{F}(x^\circ)e)_t$ be the components of $D\mathcal{F}(x^\circ)e$ in the finite rows and in the tail rows. Then

$$(I - AD\mathcal{F}(x^\circ))e = (I - \hat{A}J^{\text{fl}})e + \hat{A}(J^{\text{fl}} - J)e - A(D\mathcal{F}(x^\circ)e)_t.$$

The entries of Je are integrals over B of polynomials, namely $\|\Psi_{\ell',s'}\|_{L^2}^{-2} \int_B (D\mathcal{F}(x^\circ)e) \Psi_{\ell',s'}$ for the field rows and, for the harmonic rows, $\sum_{\ell' \leq 540} D_{n\ell'} \|H_{\ell'}\|_{L^2}^{-2} \int_B (D\mathcal{F}(x^\circ)e) H_{\ell'}$; all angular degrees of the output (at most $3M = 540$, because $q^\circ|p^\circ|^2$ has degree at most $2M$ in ξ at fixed r) are integrated before the conversion to Chebyshev coordinates. In spherical coordinates the integrands are even polynomials in ξ of degree at most 1080 and in r of degree at most $(4M + 2S + 2) + 3M + 2 = 1424$. They are integrated exactly by Gauss–Legendre rules with 542 nodes in ξ (exact to degree 1083) and 713 nodes in r (applied to the even extension to $[-1, 1]$, exact to degree 1425), whose nodes and weights are enclosed in ball arithmetic: each nonnegative root is bracketed by a sign change of the Legendre polynomial, the brackets are disjoint, together with their reflections their number equals the degree, and the weights are computed from the enclosures. Let ξ_{fin} bound $\max_e \sum_{\iota} |J_{\iota e}^{\text{fl}} - J_{\iota e}| \alpha_{\iota}^b w_{\iota} / w_e$, where α_{ι}^b is the normalised column norm of $\hat{A}$ in the norm $\|\cdot\|_b$. Converting from $\|\cdot\|_b$ to $\|\cdot\|_{\mathcal{X}}$ costs at most a factor eight, so the first two terms contribute at most $Z_{\text{head}} = 8(\eta' + \xi_{\text{fin}})$, where η' is η increased by explicit allowances for the representation of the weights h_{ℓ} , $b_j R^j$, R^n by binary64 numbers (relative error below 10^{-12} , checked) and for the rounding in their application. The last term is bounded with the output-tail envelope below, whose maximum over the finite field inputs we call $Z_{\text{fin},t}$.

Envelopes for field columns. We define step functions of the angular index L and the radial index ς . For even $\ell \leq M$ and $1 \leq \varsigma \leq S$ let

$$A(\ell, \varsigma) = \max\{1, \alpha_{(\ell', \varsigma')} : \ell' \geq \ell, \varsigma' \geq \varsigma, (\ell', \varsigma') \text{ a finite field row}\},$$

let $A(\ell, \varsigma) = 1$ if $\ell > M$ or $\varsigma > S$, and $A(\ell, \varsigma) = A(\ell + 1, \varsigma)$ for odd ℓ . Let $\alpha_n^H = A_t$ for $n \geq 182$,

$$\beta_\ell = \frac{1}{h_\ell} \sum_n D_{n\ell} R^n \alpha_n^H, \quad \beta^\infty = A_t + \sqrt{2} \sum_{n \leq M} (\alpha_n^H - A_t)_+ R^{n-1200},$$

$B(L) = \max(\{\beta_\ell : \ell \text{ even}, L \leq \ell \leq 1200\} \cup \{\beta^\infty\})$, and $\Gamma(L) = \max\{1, A(L, 1), B(L)\}$. These functions are nonincreasing in L and ς .

Lemma 13.17. *For $\ell \geq L$ (with ℓ even) and $\varsigma \geq 1$, $\|AH_\ell\|_{\mathcal{X}} \leq B(L)h_\ell$ and $\|A\Psi_{\ell, \varsigma}\|_{\mathcal{X}} \leq A(L, \varsigma)h_\ell$.*

Proof. $H_\ell = \sum_n D_{n\ell} \mathcal{T}_n$, and $\|A\mathcal{T}_n\| \leq \alpha_n^H R^n$ by Lemma 13.16; hence $\|AH_\ell\| \leq \beta_\ell h_\ell$. For $\ell > 1200$, since $\sum_n D_{n\ell} R^n = h_\ell$ and by Lemma 13.5(e), $\beta_\ell \leq A_t + \sum_{n \leq M} \sqrt{2} R^{n-\ell} (\alpha_n^H - A_t)_+ \leq \beta^\infty$. The second bound holds by definition for finite rows and because A is the identity on nonharmonic tail rows. $\square$

For $d \geq 0$, $\ell, s \geq 0$ put $\ell_d = \max(0, \ell - d)$, $s_d = \max(1, s - d)$, let X_d be a $\text{Bin}(d, \frac{1}{2})$ random variable, and let $\phi_L(\varsigma) = A(L, \varsigma)$ for $\varsigma \geq 1$ and $\phi_L(\varsigma) = \max\{A(L, 1), B(L)\}$ for $\varsigma \leq 0$. Define

$$\begin{aligned} \mathfrak{B}_d(\ell, s) &= \min \left\{ A(\ell_d, s_d) + (B(\ell_d) - A(\ell_d, s_d))_+ \vartheta_d(\ell, s), \quad \mathbb{E} \phi_{\ell_d}(s - X_d) \right\}, \\ \mathfrak{T}_d(\ell, s) &= \begin{cases} 0, & \ell + d \leq M \text{ and } s + d \leq S, \\ 1 + (A_t - 1) \vartheta_d(\ell, s) \mathbf{1}_{\ell+d > M}, & \text{otherwise,} \end{cases} \end{aligned}$$

with ϑ_d as in Lemma 13.9(b). Both bounds in $\mathfrak{B}_d$ are nonincreasing in ℓ and in s : the first because A, B, ϑ_d are and the expression is nondecreasing in A and in ϑ_d ; the second because ϕ_L is nonincreasing in ς and in L .

Lemma 13.18. *Let $H_a t^k$ be a solid monomial of degree d and $\ell, s \geq 0$ with $a + \ell$ even. Then $\|A(H_a t^k \Psi_{\ell, s})\|_{\mathcal{X}} \leq h_a h_\ell \mathfrak{B}_d(\ell, s)$, and the component $(H_a t^k \Psi_{\ell, s})_t$ in the tail rows satisfies $\|A(H_a t^k \Psi_{\ell, s})_t\|_{\mathcal{X}} \leq h_a h_\ell \mathfrak{T}_d(\ell, s)$.*

Proof. By (13.8) and Lemma 13.7 it suffices to show, for every v with $G_{av} \neq 0$, that

$$\Sigma_v := \sum_{\varsigma} \pi_{\varsigma}^{(v)} \frac{\|A\Psi_{v, \varsigma}\|}{h_v} \leq \mathfrak{B}_d(\ell, s).$$

By Lemma 13.9(a), $v \geq \ell_d$ and $\varsigma \geq s - d$. By Lemma 13.17 and monotonicity, the quotient $\|A\Psi_{v, \varsigma}\|/h_v$ is at most $A(\ell_d, \varsigma) \leq A(\ell_d, s_d)$ for $\varsigma \geq 1$, and at most $B(\ell_d)$ for $\varsigma = 0$. With $\pi_0^{(v)} \leq \vartheta_d(\ell, s)$ this gives the first bound in $\mathfrak{B}_d$. For the second, the quotient is at most $\phi_{\ell_d}(\varsigma)$, the function ϕ_{ℓ_d} is nonincreasing, the index is at least $s - N$ with N as in the proof of Lemma 13.9, and N is stochastically dominated by X_d (Lemma 13.9(c)); hence $\Sigma_v \leq \mathbb{E} \phi_{\ell_d}(s - X_d)$.

For the tail component: by Lemma 13.9(a) the outputs have $v \leq \ell + d$ and $\varsigma \leq s + d$. If $\ell + d \leq M$ and $s + d \leq S$, all of them are finite rows, including the harmonic outputs $H_v = \sum_{n \leq v} D_{nv} \mathcal{T}_n$. Otherwise the tail component consists of nonharmonic tail rows, on which A is the identity, and, only if $\ell + d > M$, of the parts $\sum_{n \geq 182} D_{nv} \mathcal{T}_n$ of harmonic outputs, of norm at most h_v , on which A has norm at most A_t . Weighting with $\pi_0^{(v)} \leq \vartheta_d$ gives the bound. $\square$

At the centre, let $\chi_1 = q^\circ/b - 1$, $\chi_2 = \partial_{x_1} q^\circ/b$, $\chi_3 = y^{-1} \partial_y q^\circ/b$, $\chi_4 = x_1 \chi_3$ and $\chi_5 = q^\circ |p^\circ|^2/b$. For $1 \leq i \leq 5$ let $\tau_i(d)$ be the τ -norm of the homogeneous part of degree d of χ_i . By Lemma 13.8(b), τ_1, τ_2, τ_3 are the Chebyshev norms of the terms of (13.1), with $\tau_1(0) = |c_1^\circ/b - 1|$; $\tau_4(d+1) \leq R \tau_3(d)$; and τ_5 is computed from the expansion of χ_5 in solid monomials. For

$\mathfrak{X} \in \{\mathfrak{B}, \mathfrak{T}\}$ put $\mathfrak{X}^{(i)}(\ell, s) = \sum_d \tau_i(d) \mathfrak{X}_d(\ell, s)$ and, with $\bar{d} = \ell + 2s + \frac{1}{2}$ (the parameter d of Lemma 13.10) and ratios $\rho_{\pm}$,

$$\begin{aligned} \mathfrak{z}\mathfrak{X}(\ell, s; \rho_+, \rho_-, \lambda_+, \lambda_-) &= \mathfrak{X}^{(1)}(\ell, s) + \sum_{i=2,4} \left[\frac{\lambda_+ \rho_+}{2(\bar{d}+1)} (\mathfrak{X}^{(i)}(\ell+1, s) + \mathfrak{X}^{(i)}(\ell+1, s-1)) \right. \\ &\quad \left. + \frac{\lambda_- \rho_-}{2(\bar{d}+1)} (\mathfrak{X}^{(i)}(\ell-1, s+1) + \mathfrak{X}^{(i)}(\ell-1, s)) \right] \\ &\quad + \frac{(\bar{d} + \frac{1}{2}) \mathfrak{X}^{(3)}(\ell, s-1)}{4\bar{d}(\bar{d}+1)} + \frac{\mathfrak{X}^{(3)}(\ell, s)}{4\bar{d}(\bar{d}+2)} + \frac{(\bar{d} + \frac{3}{2}) \mathfrak{X}^{(3)}(\ell, s+1)}{4(\bar{d}+1)(\bar{d}+2)} \\ &\quad + \frac{\mathfrak{X}^{(5)}(\ell, s-1)}{4\bar{d}(\bar{d}+1)} + \frac{\mathfrak{X}^{(5)}(\ell, s)}{2\bar{d}(\bar{d}+2)} + \frac{\mathfrak{X}^{(5)}(\ell, s+1)}{4(\bar{d}+1)(\bar{d}+2)}, \end{aligned}$$

where terms with $\ell-1 < 0$ are omitted. The exact values are $\rho_{\pm} = h_{\ell \pm 1}/h_{\ell}$, $\lambda_+ = (\ell+1)/(2\ell+1)$, $\lambda_- = \ell/(2\ell+1)$.

- Proposition 13.19.** (a) *For a field tail input (ℓ, s) , $\|(I - AD\mathcal{F}(x^\circ))e_{(\ell,s)}\|_{\mathcal{X}}/h_{\ell} \leq \mathfrak{z}\mathfrak{B}(\ell, s)$ (exact parameters), and for fixed ℓ this bound is nonincreasing in s . Hence the radial class of ℓ is bounded by $Z_{\text{rad}}(\ell) = \mathfrak{z}\mathfrak{B}(\ell, S_{\ell})$.*
(b) *The uniform angular class is bounded by $Z_{\text{ang}} = \mathfrak{z}\mathfrak{B}(210, 1; R, \frac{210}{209R}, \frac{211}{421}, \frac{1}{2})$.*
(c) *For a finite field input (ℓ, s) , $\|A(D\mathcal{F}(x^\circ)e_{(\ell,s)})_t\|_{\mathcal{X}}/h_{\ell} \leq \mathfrak{z}\mathfrak{T}(\ell, s)$ (exact parameters).*

Proof. For a field input, by (13.11) and Lemma 13.10(d),

$$D\mathcal{F}(x^\circ)e_{(\ell,s)} = \Psi_{\ell,s} + \chi_1 \Psi_{\ell,s} + \chi_2 \partial_{x_1} K_3 \Psi_{\ell,s} + \chi_3 EK_3 \Psi_{\ell,s} - \chi_4 \partial_{x_1} K_3 \Psi_{\ell,s} + \chi_5 K_3 \Psi_{\ell,s}.$$

For a tail input, $A\Psi_{\ell,s} = e_{(\ell,s)}$ (Lemma 13.16), so $(I - AD\mathcal{F}(x^\circ))e_{(\ell,s)}$ is $-A$ applied to the sum of the five remaining terms. Expanding them with Lemma 13.10(d) and (13.9), and bounding each product of a homogeneous part of χ_i with a Ψ by Lemma 13.18, gives $\mathfrak{z}\mathfrak{B}(\ell, s)h_{\ell}$; the factors $\rho_{\pm}$ convert $h_{\ell \pm 1}$ to h_{ℓ} . Every quantity in $\mathfrak{z}\mathfrak{B}$ is nonincreasing in s (the coefficients decrease in $\bar{d}$, and each $\mathfrak{B}_d$ is nonincreasing), which proves (a). For (b), let $\ell \geq 210$ and $s \geq 1$. By Lemma 13.5(b), $\rho_+ \leq R$ and $\rho_- \leq \frac{210}{209R}$; moreover $\lambda_+ \leq \frac{211}{421}$, $\lambda_- \leq \frac{1}{2}$, $\bar{d} \geq 212.5$, and $\mathfrak{B}_d$ is nonincreasing in both arguments, so every term is bounded by the corresponding term at $\ell = 210$, $s = 1$. For (c), $\Psi_{\ell,s}$ is a finite row, so the tail component of $D\mathcal{F}(x^\circ)e_{(\ell,s)}$ is that of the five remaining terms, and Lemma 13.18 applies with $\mathfrak{T}$. $\square$

For the fourteen near angular degrees $182 \leq \ell \leq 208$, the same bound with the exact ℓ and $s = 2$, monotone in s , covers each infinite radial class; the input $(\ell, 1)$ is included among the 457 explicit exceptional columns. We denote the maximum of these fourteen infinite-class bounds by $Z_{\text{ang, near}}$. Thus the finite field columns are bounded by $Z_{\text{fin}} = Z_{\text{head}} + Z_{\text{fin}, t}$, where $Z_{\text{fin}, t} = \max \mathfrak{z}\mathfrak{T}(\ell, s)$ over the 7280 finite field inputs.

Far shape columns. For a polynomial $f = \sum_a H_a f_a(t) \in \mathcal{L}$ write $f_a = \sum_s f_{a,s} J_s^{a+1/2} = \sum_m f_{a,m}^\sharp (1-t)^m$, and let

$$V_f(L) = \sum_a h_a \Gamma((L-a)_+) \min \left\{ \sum_s |f_{a,s}|, \sum_m |f_{a,m}^\sharp| \frac{m!}{\left(\frac{(L-a)_+ + 1/2}{2} + 1\right)_m} \right\},$$

which is nonincreasing in L . With $\zeta = R^{-2}$, $N \geq 0$ and $J = \lfloor N/2 \rfloor$ put

$$\begin{aligned} \mathsf{T}_f(N) &= \frac{B_0}{2} V_f(N) + \sum_{j=1}^J \frac{B_0 a_j \zeta^j}{2j-1} V_f(N-2j) + \frac{B_0 a_{J+1} \zeta^{J+1}}{(2J+1)(1-\zeta)} V_f(0), \\ \mathsf{U}_f(N) &= \sum_{j=0}^J B_0 a_j \zeta^j V_f(N-2j) + \frac{B_0 a_{J+1} \zeta^{J+1}}{1-\zeta} V_f(0). \end{aligned}$$

Lemma 13.20. *Let $f \in \mathcal{L}$ be a polynomial, $e \geq 0$, and let L, n have the parity for which the products below are even in x_1 . Then $\|A(H_L t^e f)\|_{\mathcal{X}} \leq h_L V_f(L)$. For $n \geq N$, $\|A(r^n T_n(\xi) t^e f)\|_{\mathcal{X}} \leq R^n \mathsf{T}_f(N)$ and $\|A(r^n U_n(\xi) t^e f)\|_{\mathcal{X}} \leq R^n \mathsf{U}_f(N)$. More generally, for every m with $\tau(m) < \infty$ whose homogeneous parts have degree at most d_m , $\|A(m r^n T_n(\xi) t^e f)\| \leq \tau(m) R^n \mathsf{T}_f(N - d_m)$ for $n \geq N \geq d_m$.*

Proof. By Lemma 13.7, $H_L t^e H_a f_a = \sum_k G_{Lak} H_k t^{e+(L+a-k)/2} f_a$ with $k \geq |L-a|$. The $\mathcal{L}$ -coefficients of $H_k t^e f_a$ lie in angular index k , and their ℓ^1 norm in the basis $J^{k+1/2}$ is at most $\sum_s |f_{a,s}|$, by the stochastic steps in the proof of Lemma 13.8(a), and at most $\sum_m |f_{a,m}^\natural| \|(1-t)^m\|_{J^{k+1/2},1}$, by Lemma 13.3(c),(d) (multiplication by t does not increase $\|\cdot\|_{J^\beta,1}$); the latter norm is at most $m!/((L-a)_+ + \frac{1}{2})/2 + 1)_m$ since $k \geq (L-a)_+$. On outputs of angular index at least $(L-a)_+$, A has normalised norm at most $\Gamma((L-a)_+)$ by Lemma 13.17. Summing with $\sum_k G_{Lak} h_k \leq h_L h_a$ gives the first bound. Next, $r^n T_n(\xi) = \sum_j C_{n-2j,n} H_{n-2j} t^j$, so by the first bound, Lemma 13.5(d) and the monotonicity of V_f ,

$$\begin{aligned} \|A(r^n T_n t^e f)\| &\leq \sum_{j \leq n/2} |C_{n-2j,n}| h_{n-2j} V_f(n-2j) \\ &\leq R^n \left(\frac{B_0}{2} V_f(N) + \sum_{j \geq 1} \frac{B_0 a_j \zeta^j}{2j-1} V_f((N-2j)_+) \right), \end{aligned}$$

and the terms with $j > J$ are bounded using $\sum_{j > J} a_j \zeta^j / (2j-1) \leq a_{J+1} \zeta^{J+1} / ((2J+1)(1-\zeta))$. The bound for U_n is the same with v . For a multiplier m we write $m = \sum m_{i,k'} H_i t^{k'}$ with $i \leq d_m$ and use $H_i H_{n-2j} = \sum_k G_{i,n-2j,k} H_k t^{(\cdots)}$ with $k \geq n-2j-d_m$, together with $\sum_k G_{i,n-2j,k} h_k \leq h_i h_{n-2j}$. $\square$

Lemma 13.21. (a) $y^{-1} \partial_y H_\ell = -\sum_{k=\ell-2, \ell-4, \dots \geq 0} (2k+1) H_k t^{(\ell-2-k)/2}$. Hence, for a polynomial $W = \sum_\ell H_\ell w_\ell(t)$, $Z_W = y^{-1} \partial_y W = \sum_\ell ((y^{-1} \partial_y H_\ell) w_\ell + 2H_\ell w'_\ell)$ is a polynomial, and $y \partial_y W = t(1-\xi^2) Z_W$.

(b) $2(1-\xi^2) C_{n-2}^{(2)}(\xi) = \frac{1}{2} (U_n(\xi) + U_{n-2}(\xi)) - n T_n(\xi)$ for $n \geq 2$.

Proof. (a) We have

$$\partial_y(r^\ell P_\ell(x_1/r)) = y r^{\ell-2} (\ell P_\ell(\xi) - \xi P'_\ell(\xi)) = -y r^{\ell-2} P'_{\ell-1}(\xi),$$

and $P'_{\ell-1} = \sum_{k \equiv \ell-2} (2k+1) P_k$ by the classical identities $\xi P'_\ell - P'_{\ell-1} = \ell P_\ell$ and $P'_{k+1} - P'_{k-1} = (2k+1) P_k$. The rest follows from $\partial_y w(t) = 2y w'(t)$ and $y^2 = t(1-\xi^2)$. (b) We have $2C_{n-2}^{(2)} = U'_{n-1}$ and $2\xi U_{n-1} = U_n + U_{n-2}$; differentiating $U_{n-1}(\cos \theta) = \sin n\theta / \sin \theta$ gives $(1-\xi^2) U'_{n-1}(\xi) = \xi U_{n-1}(\xi) - n T_n(\xi)$. $\square$

Let $H^\circ = g^\circ + |p^\circ|^2(1+W^\circ)$, $W_1^\circ = \partial_{x_1} W^\circ$ and $Z^\circ = Z_{W^\circ}$. These are polynomials; their Jacobi coefficients and boundary Taylor coefficients enter V_f . By Lemma 13.10(c), every angular component f_a of W° has a double zero at $t=1$, and every angular component of W_1°

and of Z° vanishes at $t = 1$, so the corresponding Taylor coefficients are exactly zero. Let $q_{(d)}^\circ = c_{d+1}^\circ r^d U_d(\xi)$ be the homogeneous parts of q° , with $\tau(q_{(d)}^\circ) = |c_{d+1}^\circ| \|r^d U_d\|_{\text{ch}}$. Finally, for $N \geq 360$ let

$$P(N) = \frac{2}{b^2 R} \sum_{a,i,\varsigma} |p_a c_i^\circ| R^{\varsigma-a} \left[\sum_{\nu=0}^{J_K} w_\nu^T \Gamma(K-2\nu) \frac{2(m+\nu)}{K-\nu+m+\frac{3}{2}} + \frac{2B_0 a_{J_K+1} \zeta^{J_K+1}}{(2J_K+1)(1-\zeta)} \Gamma(0) \right],$$

where $a \in \{0, 2, \dots, 180\}$, $i \in \{1, 3, \dots, 181\}$, $\varsigma \in \{-(i-1), -(i-3), \dots, i-1\}$, $K = N - a + \varsigma$, $m = a + (i-1-\varsigma)/2$, $J_K = \lfloor K/2 \rfloor$, $w_0^T = B_0/2$ and $w_\nu^T = B_0 a_\nu \zeta^\nu / (2\nu-1)$ for $\nu \geq 1$.

Proposition 13.22. *Let $N = 620$. For every odd $j \geq N + 1$,*

$$\begin{aligned} \frac{\|(I - AD\mathcal{F}(x^\circ))e_j\|_{\mathcal{X}}}{\sigma j R^j} &\leq \frac{1}{8} \left[\frac{\mathsf{U}_{H^\circ}(N)}{b^2(N+1)R} + \frac{\mathsf{U}_{W_1^\circ}(N-1)}{b^2 R^2} \right. \\ &\quad + \frac{\mathsf{T}_{Z^\circ}(N) + (\mathsf{U}_{Z^\circ}(N) + R^{-2} \mathsf{U}_{Z^\circ}(N-2))/(2N+2)}{b^2 R} \\ &\quad \left. + \frac{2}{b^2 R} \sum_a |p_a| R^{-a} \sum_d \tau(q_{(d)}^\circ) \mathsf{T}_{W^\circ}(N-a-d) + P(N) \right] =: Z_{\text{far}}. \end{aligned}$$

Proof. Let $j = n + 1$ with $n \geq N$ even. Differentiating (13.11) in the direction e_j , with $\delta q = r^n U_n$, $\delta \partial_{x_1} q = j r^{n-1} U_{n-1}$, $\delta(y^{-1} \partial_y q) = -2r^{n-2} C_{n-2}^{(2)}$ and $\delta p = j w^n$ by (13.1), gives

$$b D\mathcal{F}(x^\circ) e_j = r^n U_n H^\circ + j r^{n-1} U_{n-1} W_1^\circ - 2r^{n-2} C_{n-2}^{(2)} W_2^\circ + 2q^\circ \text{Re}(\overline{p^\circ} j w^n) (1 + W^\circ),$$

with $W_2^\circ = y \partial_y W^\circ = r^2(1 - \xi^2) Z^\circ$ by Lemma 13.21(a). Since $A M e_j = e_j$ (Lemma 13.16), $(I - AD\mathcal{F}(x^\circ))e_j = -A(D\mathcal{F}(x^\circ)e_j - M e_j)$, and we bound five parts, dividing by the input weight $\sigma j R^j = 8b(n+1)R^{n+1}$.

The first two terms are bounded by Lemma 13.20, using $n+1 \geq N+1$. For the third, Lemma 13.21(b) gives

$$-2r^{n-2} C_{n-2}^{(2)} W_2^\circ = -r^n \left(\frac{U_n + U_{n-2}}{2} - n T_n \right) Z^\circ, \quad r^n U_{n-2} = t r^{n-2} U_{n-2},$$

and Lemma 13.20 applies, since $n/(n+1) < 1$. For the fourth, we write

$$2 \text{Re}(\overline{p^\circ} w^n) = 2 \sum_a p_a t^a r^{n-a} T_{n-a}(\xi), \quad q^\circ = \sum_d q_{(d)}^\circ,$$

and apply the last statement of Lemma 13.20 to each product $q_{(d)}^\circ r^{n-a} T_{n-a} t^a W^\circ$.

The last part is $\frac{2}{b} q^\circ \text{Re}(\overline{p^\circ} j w^n) - M e_j$. Since $T_{n-a} U_{i-1} = \sum_\varsigma T_{n-a+\varsigma}$, with ς as in the definition of P and all indices positive because $n-a-(i-1) \geq N-360 > 0$,

$$\frac{2}{b} q^\circ \text{Re}(\overline{p^\circ} j w^n) = \frac{2j}{b} \sum_{a,i,\varsigma} p_a c_i^\circ r^{K+2m} T_K(\xi), \quad K = n-a+\varsigma, \quad m = a + \frac{i-1-\varsigma}{2},$$

and $M e_j$ is obtained by replacing $r^{K+2m} T_K$ by $\mathcal{T}_K$, which has the same boundary values. Now $r^{K+2m} T_K - \mathcal{T}_K = \sum_\nu C_{K-2\nu,K} H_{K-2\nu}(t^{m+\nu} - 1)$, and by Lemma 13.3(d) the function $H_\ell(t^M - 1)$ has $\mathcal{L}$ -coefficients in angular index ℓ only, with ℓ^1 norm $2M/(\ell + M + \frac{3}{2})$ in the basis $J^{\ell+1/2}$. With Lemmas 13.17 and 13.5(d),

$$\|A(r^{K+2m} T_K - \mathcal{T}_K)\| \leq R^K \sum_{\nu \leq K/2} w_\nu^T \Gamma(K-2\nu) \frac{2(m+\nu)}{K-\nu+m+\frac{3}{2}}.$$

For fixed (a, i, ς) , $R^{K-n} = R^{\varsigma-a}$ and m do not depend on n , while increasing n increases K ; each summand is nonincreasing in K , and the new summands with $\nu > J_K$ that appear for larger K are bounded by $2w_\nu^T \Gamma(0)$, whose sum over $\nu > J_K$ is at most the last term of the bracket. Hence the bracket in $P(N)$, evaluated at $K = N - a + \varsigma$, bounds the corresponding sum for every $n \geq N$. Collecting the five parts proves the claim. $\square$

The program evaluates an upper bound for $P(N)$ by grouping the sum over ν into blocks and bounding each block by its mass times the values of the factors $\Gamma(K - 2\nu)$ and $2(m + \nu)/(K - \nu + m + \frac{3}{2})$, both nondecreasing in ν , at the end of the block.

Collecting the finite, exceptional, radial, near-angular, uniform-angular, explicit-shape and far-shape classes,

$$Z = \max \{ Z_{\text{fin}}, Z_{\text{ex}}, \max_{\ell} Z_{\text{rad}}(\ell), Z_{\text{ang}}, Z_{\text{ang,near}}, Z_{\text{sh}}, Z_{\text{far}} \}$$

bounds $\|I - ADF(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{X}}$.

13.8. Certified bounds and the exact zero. The bounds below are outward roundings of the interval enclosures in `anc/r3_convex/VERIFICATION_RECEIPT.json`. The residual uses the entire polynomial support and the fixed dyadic b , not a moving value of c_1 . The certified finite inverse is nonsingular: the exact dyadic product defect, including a rigorous binary64 dot-product error, is less than 0.000173566. The boundary-symbol defect is less than 0.015579, hence both blocks of A are invertible.

Quantity	Certified bound
$\ A\ $	< 103241.73
$Y = \ A\mathcal{F}(x^\circ)\ _{\mathcal{X}}$	$< 9.358 \cdot 10^{-9}$
Z_{fin} , all 457 exceptional field columns	$< 0.283115, < 0.669070$
$Z_{\text{ang,near}}$ (fourteen radial classes)	< 0.559208
$\max_{\ell} Z_{\text{rad}}(\ell), Z_{\text{ang}}$	$< 0.776701, < 0.767365$
$Z_{\text{sh}}, Z_{\text{far}}$	$< 0.547873, < 0.501461$
$Z = \ I - ADF(x^\circ)\ _{\mathcal{X} \rightarrow \mathcal{X}}$	< 0.776701
c_2, c_3, c_4	$< 3.163938, < 1.304336 \cdot 10^{-5}, < 1.317589 \cdot 10^{-11}$
$r = 10^{-7}$: radii polynomial	$< -9.705 \cdot 10^{-9}$
$r = 10^{-7}$: contraction bound	< 0.8421

TABLE 8. The convex three-dimensional certificate for $M = 180$, $S = 80$, $R = 21/20$ and $\sigma = 8b$. The final Z is the maximum over every field and shape class. The full interval values are in the receipt.

Theorem 13.23. *There is exactly one zero $x^* = (g^*, c^*)$ of $\mathcal{F}$ in $\overline{B}_{\mathcal{X}}(x^\circ, 10^{-7})$.*

Proof. The finite dyadic product and the Toeplitz symbol bounds give an injective A by Lemmas 13.15 and 13.16. The partition above covers all derivative columns and yields $Z < 0.776701$. The verifier evaluates both inequalities (13.13) with the full Arb enclosures of $Y, Z, \|A\|, c_2, c_3, c_4$ at the exact radius 10^{-7} . Their left sides are respectively below $-9.705 \cdot 10^{-9}$ and $0.8421 < 1$, so Proposition 13.14 applies. $\square$

Remark 13.24 (Shape scale). The exact b is held fixed while the shape varies. At the same centre, the scale $\sigma = 3b$ cannot certify this fixed-point operator: the normalised defect of shape column $j = 183$ is strictly greater than 1.15486483728, as recorded in `anc/r3_convex/SCALE3_OBSTRUCTION.json`. This excludes contraction for that operator in that norm; it says nothing about other operators or centres. The scale $8b$ used above gives $Z < 0.776701$ and $c_2 < 3.163938$.

13.9. Reconstruction and convexity. Let $\psi = \psi_{c^*}$, and put $\varepsilon = \|c^* - c^\circ\|_{\mathcal{E}} \leq r/\sigma$. The existence ball projects inside the shape ball of $\mathcal{E}$ -radius one; in fact the projected radius is $68719476736/595786147695156171875 < 1$. Write

$$D_1 = \sum_{j \geq 3} j |c_j^\circ|, \quad D_2 = \sum_{j \geq 3} j(j-1) |c_j^\circ|, \quad H = \sup_{j \geq 3, j \text{ odd}} (j-1) R^{-j}.$$

The last supremum is a finite maximum through $j = 21$, since the ratio of successive odd-index terms is at most one for $j \geq 21$.

Proposition 13.25. *For every c with $\|c - c^\circ\|_{\mathcal{E}} \leq 1$, the map ψ_c is univalent on $|w| \leq 101/100$, its image $D = \psi_c(\mathbb{D})$ is strictly convex and strictly star-shaped, and $c_{19} \neq 0$. More precisely, on the closed unit disc,*

$$(13.16) \quad \operatorname{Re} \left(1 + \frac{w \psi_c''(w)}{\psi_c'(w)} \right) > 0.7878,$$

and $\operatorname{Re} \psi_c' > 106.2027$ on $|w| \leq 101/100$, $\operatorname{Re}(w \psi_c'/\psi_c) > 0.9717$ on $|w| = 1$, and $|c_{19}| > 0.000665$.

Proof. For $|w| \leq 1$ the triangle inequality gives

$$\operatorname{Re} \left(1 + \frac{w \psi_c''}{\psi_c'} \right) \geq 1 - \frac{D_2 + \varepsilon H}{c_1^\circ - D_1 - \varepsilon/R}.$$

The denominator is positive, and the interval computation gives a lower bound > 0.7878989239 at the worst case $\varepsilon = 1$. The same coefficient inequalities on the larger disc give $\operatorname{Re} \psi_c' > 106.2027$ there. Integration along line segments proves univalence; vertical segments give $q_c = \operatorname{Im} \psi_c/y > 0$. The separate boundary estimate gives the stated strict star-shapedness, and the weighted coefficient bound gives $|c_{19}| \geq |c_{19}^\circ| - \varepsilon/(19R^{19}) > 0.000665$.

For $\gamma(\theta) = \psi_c(e^{i\theta})$, its tangent angle derivative is $\operatorname{Re}(1 + w \psi_c''/\psi_c')$, which is positive by (13.16). As the simple tangent winds once, every oriented tangent line supports the interior, so D is strictly convex. It is symmetric under reflection in both axes because ψ_c is odd with real coefficients. The lift $\Theta_c(B)$ is convex exactly when D is: one direction follows by intersecting with a meridian plane; conversely, the vertical sections of D are symmetric intervals, and for X, Y in the solid, convexity of D and $|tX' + (1-t)Y'| \leq t|X'| + (1-t)|Y'|$ show that $tX + (1-t)Y$ lies in the solid for $0 \leq t \leq 1$. $\square$

Lemma 13.26. *Let $\Omega \subset \mathbb{R}^3$ be a bounded domain with real-analytic boundary, and let $u \in C^1(\bar{\Omega})$ satisfy $\int_{\Omega} (\nabla u \cdot \nabla \varphi - u \varphi) = 0$ for all $\varphi \in C_c^\infty(\Omega)$ and $u = 1$ on $\partial\Omega$. Then u extends to a real-analytic function on an open neighbourhood of $\bar{\Omega}$ which satisfies $\Delta u + u = 0$ there.*

Proof. Let $v = u - 1$. For $\varepsilon > 0$ the function $\max(|v| - \varepsilon, 0) \operatorname{sign} v$ is Lipschitz with compact support in Ω and converges to v in $W^{1,2}(\Omega)$ as $\varepsilon \rightarrow 0$, so $v \in W_0^{1,2}(\Omega)$; it is a weak solution of $\Delta v + v = -1$. By global regularity for the Dirichlet problem [26, Theorem 8.13], applied for every k , $v \in W^{k,2}(\Omega)$ for all k , hence $v \in C^\infty(\bar{\Omega})$. By the theorem of Morrey and Nirenberg on analyticity up to the boundary [42] (constant coefficients, analytic right-hand side, analytic boundary and zero Dirichlet data), every point of $\partial\Omega$ has a ball around it to which $v|_{\Omega}$ extends real-analytically; the extension solves $\Delta v + v = -1$ by the identity theorem. Interior analyticity is classical (see also Lemma 10.3). The local extensions are glued to an extension on a neighbourhood of $\bar{\Omega}$ exactly as in the last step of the proof of Lemma 10.4. $\square$

Proof of Theorem 1.1. The estimates of Proposition 13.25 hold for c^* because $r/\sigma < 1$. The proofs of Lemmas 13.1 and 13.2 apply on the smaller disc $|w| < 101/100$, a neighbourhood of the closed unit disc; they give an analytic diffeomorphism Θ_{c^*} near $\bar{B}$, with Jacobian $q_{c^*}|p_{c^*}|^2 > 0$. Let $\Omega = \Theta_{c^*}(B)$, $U = 1 + \iota(K_3 g^*)$, and $u = U \circ \Theta_{c^*}^{-1}$. Since $\mathcal{F}(x^*) = 0$, Lemma 13.12 yields the pulled-back weak Helmholtz equation and zero Cauchy traces of $U - 1$; Lemma 13.2 transfers these to u . The analytic lift across the rotation axis is valid because ψ has odd real coefficients and $q = \text{Im } \psi/y$ is analytic in (x_1, y^2) . Interior analyticity also follows by harmonic lifting or elliptic regularity, and Lemma 13.26 extends u analytically across the boundary.

The positive boundary bound makes Ω strictly star-shaped. The solid-of-revolution argument in Proposition 13.25 makes it convex. The lift of the sphere is a compact real-analytic surface diffeomorphic to S^2 and has the stated $O(2) \times \mathbf{Z}_2$ invariance. A constant solution of $\Delta u + u = 0$ would be zero, contradicting $u = 1$ on the boundary. If Ω were a ball, its symmetries would put its centre at zero, and D would be a disc centred at zero. Schwarz's lemma applied to the normalised conformal map and its inverse would force ψ to be linear, contrary to $c_{19} \neq 0$. This proves the theorem.

13.10. An additional nonconvex example. The earlier three-dimensional construction is independent of the convex one and remains useful as a second exact branch.

Proposition 13.27. *There is a bounded, strictly star-shaped, non-ball domain $\Omega_{\text{old}} \subset \mathbb{R}^3$ with real-analytic boundary diffeomorphic to S^2 and a nonconstant analytic solution of (1.2). It has the same $O(2) \times \mathbf{Z}_2$ symmetry, but is not convex: both principal curvatures at the two poles, with respect to the inner normal, are negative.*

Proof. Apply the preceding analytic lemmas and the contraction criterion with the independent AX3 centre and its parameters $M = 180$, $S = 64$, $R = 21/20$, $b_{\text{old}} = 0\text{x}1.85067\text{a}9\text{afca}2\text{dp}+5$, $\sigma = 3b_{\text{old}}$ and $r = 3 \cdot 10^{-7}$. Its complete verifier and two certificate files are in `anc/r3/`; the certified bounds are displayed in Table 9. The corresponding radius polynomial is $< -4.81 \cdot 10^{-9}$ and contraction bound is < 0.959201 . Its geometry certificate gives $\text{Re}(w\psi'/\psi) > 0.83$ on the unit circle and a nonzero c_{13} , so the same reconstruction gives an analytic, strictly star-shaped non-ball domain and solution. Nonconvexity follows from Proposition 13.28. $\square$

(M, S)	σ	$\ A\ $	Y	Z	r	Geometry
$(180, 64)$	$3b_{\text{old}}$	< 5849.52	$< 2.10029 \cdot 10^{-8}$	< 0.858929	$3 \cdot 10^{-7}$	nonconvex at poles

TABLE 9. The independent AX3 certificate; see `anc/r3/certificate_r3.json`.

Proposition 13.28. *For the AX3 centre $c^{\text{old}, \circ}$, exact rational arithmetic gives*

$$42.81903 < \sum_j j c_j^{\text{old}, \circ} < 42.81904, \quad -7.758015 < \sum_j j^2 c_j^{\text{old}, \circ} < -7.758014.$$

Throughout its certified existence ball, $1 + \psi''(1)/\psi'(1) < -0.181$; hence both principal curvatures at each pole are negative.

Proof. The AX3 shape perturbation has $\mathcal{E}$ norm at most $r/(3b_{\text{old}}) < 2.06 \cdot 10^{-9}$. Its effect on $\psi'(1)$ is at most e/R , and on $\psi'(1) + \psi''(1)$ at most $\sup_{t>0} tR^{-t}e < 7.55e < 1.6 \cdot 10^{-8}$. Thus $0 < \psi'(1) < 42.82$ and $\psi'(1) + \psi''(1) < -7.758$, giving the claimed ratio. The meridian curvature at a pole, with inward normal, is $(1 + \psi''(1)/\psi'(1))/\psi'(1) < 0$; rotation symmetry makes both surface principal curvatures equal to it. The other pole follows by reflection. The exact dyadic check is `anc/r3/check_nonconvex_r3.py`. $\square$

Part 3. Consequences, computations and remarks

14. THE POMPEIU PROPERTY

Proof of Corollary 1.3. Let $n \in \{3, 4, 6, 8, 10, 14\}$, let Ω and u be as in Theorem 1.1 or Theorem 1.2 (proved in Sections 13, 10, 11 and 12), let $\xi \in \mathbb{R}^n$ with $|\xi| = 1$, and let $v(x) = e^{i\xi \cdot x}$, so that $\Delta v + v = 0$. Since u and v are smooth up to the real-analytic boundary and solve the same equation, Green's identity and (1.2) give

$$0 = \int_{\Omega} (u\Delta v - v\Delta u) dx = \int_{\partial\Omega} (u\partial_{\nu}v - v\partial_{\nu}u) dS = \int_{\partial\Omega} \partial_{\nu}v dS = \int_{\Omega} \Delta v dx = - \int_{\Omega} e^{i\xi \cdot x} dx.$$

Thus $\widehat{\mathbf{1}}_{\Omega}$ vanishes on the unit sphere. If $\sigma(x) = Qx + a$ is a rigid motion, then

$$\int_{\sigma(\Omega)} e^{ix_1} dx = e^{ia_1} \int_{\Omega} e^{i(Q^T e_1) \cdot x} dx = 0,$$

and taking real parts, $\int_{\sigma(\Omega)} \cos x_1 dx = 0$. Since $\cos x_1$ is continuous and not identically zero, Ω fails the Pompeiu property. By Theorems 1.1 and 1.2, Ω is a bounded domain, not a ball, whose boundary is real analytic and diffeomorphic to S^{n-1} , so the Pompeiu conjecture fails in $\mathbb{R}^n$; the domain is convex in every listed dimension, by Theorems 1.1 and 1.2. $\square$

15. COMPUTER-ASSISTED VERIFICATION

This section describes, for each dimension, what the scripts check and how the computations are reproduced; Section 15.7 describes the ancillary files. In all dimensions the inequalities are checked in the ball arithmetic of Arb [34], through `python-flint` 0.9.0 [25]: an order relation between balls is true only if it holds for all points of the balls, every check raises an exception when it fails, and NaN or infinite enclosures are rejected. The analytic content of the proofs, namely all lemmas and all tail estimates, is contained in Sections 2–13; the scripts only evaluate the finitely many explicit expressions defined there.

15.1. Dimension four: what the script checks. All finite computations for $n = 4$ are carried out by a single script, `verify_r4.py` (569 lines), which uses only the Python standard library and `python-flint` 0.9.0 [25], the Python interface to the ball arithmetic of FLINT/Arb [34]. Every real number is represented by a ball $[m \pm \varepsilon]$ whose arithmetic is rigorous, including all rounding errors. The 525 centre coefficients are embedded in the script as hexadecimal binary64 strings, which are converted to exact dyadic rationals (exactness is checked); $\rho = 21/20$, $r = 1/50000$ and the rational majorants of Theorem 9.1 are exact. The script performs the following steps.

- (1) It expands $\mathcal{F}(x^\circ)$ and the 525 finite columns of $D\mathcal{F}(x^\circ)$ in ball arithmetic, using (4.3), (5.1) and (6.3); no coefficient is discarded.
- (2) It assembles M_f , computes a ball enclosure of M_f^{-1} , and defines $\widehat{A}$ as the matrix of the (exact, dyadic) midpoints of this enclosure. It then computes $I - \widehat{A}M_f$ in ball arithmetic and verifies (7.5) for the weighted column norm. The matrix $\widehat{A}$ is never treated as an interval-valued operator.
- (3) It computes $\|R\|$, H , the numbers h_n for $M + 2 \leq n \leq N_H = 443$ and the remainder bound of Lemma 7.6, hence A_t and a bound for $\|A\|$.
- (4) It computes Y and Z_f by applying A , in the form (7.6), (7.3), (7.4), to the residual and to the finite columns.
- (5) It computes the 3650 tail g -columns with $n \leq 123$, $s \leq 67$ with the short polynomial $p_{\leq 16}$ and adds the error term of (8.1); it evaluates $Z_{g,\infty}$.

- (6) It computes the 480 shape columns $43 \leq j \leq 1001$ without truncation, checks the hypotheses of Proposition 8.4, and evaluates $U_1, \dots, U_5, U_W$ and $Z_{\text{sh}, \infty}$.
- (7) It evaluates Z, c_2, c_3, C_2, C_3 , checks $Y < 7 \cdot 10^{-6}$, $Z < 37/100$, $C_2 < 4$, $C_3 < 10^{-3}$, and then the two inequalities (7.1) at $r = 1/50000$.
- (8) It checks the geometric inequalities (10.2) and (10.3).

In `python-flint`, an order relation between balls evaluates to true only if it holds for all points of the balls. Every check is an explicit test that raises an exception when it fails (also under `python -O`); maxima over columns are taken over rigorous upper endpoints, and nonfinite enclosures are rejected. After step (4) the script writes an intermediate checkpoint file (`*.base.json`), which carries no status field; only when all checks succeed does it write the certificate file (JSON) with the enclosures of all quantities of Table 2 and the status `PROVED`, and print `PROVED`. A diagnostic mode (`--stage base`) stops after step (4) and never prints `PROVED`.

A second script, `verify_convex.py`, checks that the centre data file coincides with the data embedded in `verify_r4.py`, that the main certificate has status `PROVED` for the radius $r = 1/50000$, and then verifies the bounds of Proposition 10.2, which involve only the finite sums A_1, A_2 and the explicit tail constants E_1, E_2 .

15.2. Dimension four: reproduction. From a directory containing the files `verify_r4.py`, `verify_convex.py` and `data/centre.json`, the certificate is reproduced by

```
mkdir -p logs
nice -n 19 env OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
  MKL_NUM_THREADS=1 python3 verify_r4.py --stage all --bits 128 --jstar 1001 \
  --output logs/certificate_r4_128.json
python3 verify_convex.py --bits 128 \
  --main-certificate logs/certificate_r4_128.json \
  --output logs/convex_128.json
```

The second command prints `PROVED` as its last line, the third prints `PROVED_CONVEX`. The runs use one thread. The distributed certificates are in the subdirectory `certificates/` of `anc/` (Section 15.7); the files written to `logs/` are byte-identical to them. On a single core of a Linux server the 128-bit run took between 163 and 167 seconds of wall-clock time in several independent runs, and the same run at 192 bits (`--bits 192`) took about 198 seconds; the convexity check takes a fraction of a second. Runs under Python 3.10.19 and Python 3.12.3, both with `python-flint` 0.9.0, produced byte-identical certificate files. At 192 bits all inequalities hold with the same displayed bounds, and the finite inverse defect (7.5) is below $7.5 \cdot 10^{-54}$.

15.3. Dimension six. For $n = 6$ the finite computations are split into five stage scripts, which evaluate the quantities of Section 11 in ball arithmetic at 128 bits, in batches, and write interval receipts (JSON files), and a final script that combines the receipts. All scripts use Python, `python-flint` 0.9.0 and, only to read the matrix $\hat{A}$, NumPy.

- (1) `finite_interval_r6.py` expands $\mathcal{F}_6(x^\circ)$ without truncation and computes $A\mathcal{F}_6(x^\circ)$ and Y (receipt `residual_bound_r6.json`); in eight batches covering all 465 finite columns it computes the inverse defect (11.12) and Z_f (`finite_batches_r6.json`). The matrix $\hat{A}$ is read from `inverse_r6_M58_S30.npy`, and its binary64 entries are converted to exact dyadic rationals.
- (2) `tail_inverse_r6.py` computes H , the numbers h_n for $M + 4 \leq n \leq N_H = 458$ and the remainder bound of Lemma 11.12, hence A_t (`tail_inverse_bound_r6.json`).

- (3) `g_tail_interval_r6.py` computes the 705 near tail g -columns in seven batches with the short polynomial $p_{\leq 24}$ and adds the error term of (11.15) (`g_tail_batches_r6.json`); it also evaluates the two far bounds of Lemma 11.13 (`g_far_bound_r6.json`).
- (4) `shape_tail_interval_r6.py` computes the 85 near shape columns $61 \leq j \leq 397$ without truncation, in five batches (`shape_tail_batches_r6.json`).
- (5) `far_shape_r6.py` computes Γ_1, Γ_0 , the expansion of Q° by the connection recurrences, the coefficients $\gamma_{m,t}^W, U_\Gamma, U_W$ and $Z_{\text{sh},\infty}$ for $j_* = 401$, and checks the support condition of Proposition 11.14 (`far_shape_bound_r6.json`).
- (6) `verify_r6.py --stage final` checks the SHA-256 checksums of the centre, of $\hat{A}$, of the seven receipts and of the five stage scripts against values frozen in its source; checks that the batches of each receipt are contiguous and cover all columns of their class; recomputes $\|R\|$ from $\hat{A}$; forms Z and the constants c_2, c_3, c_4 of Lemma 11.7 with $\|A\| \leq 1402.314$; checks the rational majorants of Theorem 11.16 and the inequalities (11.6) at $r = 1/5000$; and checks the geometric inequalities (11.18). Only then does it write the certificate `certificate_r6.json`, with the status `PROVED`, all enclosures, the frozen checksums and its own checksum, and print `PROVED`.
- (7) `verify_convex_r6.py` checks the checksums of `verify_r6.py` and of the centre, and the status, radius, weight and checksums recorded in the certificate, and then verifies the bounds of Proposition 11.19 (`convex_r6.json`, status `PROVED_CONVEX`).

As for $n = 4$, an order relation between balls is true only if it holds for all points of the balls, every check raises an exception when it fails, and maxima are taken over rigorous upper endpoints; a NaN or infinite enclosure, whether computed or read back from a receipt, stops the verification. The script `test_fail_closed_r6.py` tests this behaviour on artificial inputs. A further script, `check_identities_r6.py`, checks in exact rational arithmetic the identities of Lemma 5.1(a)–(d) for all $n \equiv 2 \pmod{4}$, $n \leq 202$, and $1 \leq s \leq 80$ (4080 pairs), reruns the connection recurrences used for Q° in rational arithmetic (450 pairs, all coefficients positive), checks $\|Kg^\circ - (1 - |w|^2)^2 Q^\circ\|_\rho \leq 2.02 \cdot 10^{-31}$ in ball arithmetic, and compares (11.8) with the column algebra of the verifier, exactly for $5 \leq j \leq 2001$ and in ball arithmetic for $5 \leq j \leq 1001$. These identities are proved in Section 11; the script is a consistency check.

The complete six-dimensional verification is reproduced, from the directory `anc/r6/` of the ancillary files, by

```
sha256sum -c SHA256SUMS
bash reproduce_r6.sh
```

The script `reproduce_r6.sh` uses the interpreter given by the environment variable `PYTHON`, if it is set, checks the `python-flint` version, runs `test_fail_closed_r6.py`, regenerates all seven receipts from the centre and $\hat{A}$ with the stage scripts, each batch in a separate single-core process under `nice -n 19` with a time limit of 600 seconds, and compares them byte by byte with the distributed receipts. It then reruns steps (6) and (7) and `check_identities_r6.py` in a scratch directory, compares the resulting `certificate_r6.json`, `convex_r6.json` and `identities_r6.json` byte for byte with the distributed files, and prints `REPRODUCTION COMPLETE`; any mismatch stops it with a nonzero exit code. On one core of a Linux server, with Python 3.10.19 and `python-flint` 0.9.0, the complete reproduction took about three minutes of wall-clock time (3 min 2 s and 3 min 5 s in two independent runs). The individual final steps are

```
python3 verify_r6.py --stage final \
--output certificate_r6.json
```

```
python3 verify_convex_r6.py
python3 check_identities_r6.py --output identities_r6.json
```

which print PROVED, PROVED_CONVEX and IDENTITIES VERIFIED and rewrite the three files byte-identically; they take about two seconds, less than a second and about fifteen seconds, respectively.

15.4. Consistency checks for $n = 4$ and $n = 6$. The following checks are not part of the proof. The algebraic identities of Sections 4, 5 and Lemma 7.3 were verified symbolically for small indices. An independently written floating-point implementation, which computes projections by quadrature instead of the recurrences (4.3) and uses its own approximate inverse, reproduced $\|R\|$, H , Z_f and the largest shape column of Table 2 to the displayed digits. The function u was also evaluated directly in $\mathbb{R}^4$ from a Fourier–Bessel representation $F = \sum_{m \text{ odd}} a_m J_m(r) \sin m\phi$ fitted to the boundary data of the centre: at random points of $\partial\Omega$ the defects $|u - 1|$ and $|\nabla u|$ were below 10^{-10} , and the Fourier transform of $\mathbf{1}_\Omega$, computed by quadrature, was below 10^{-10} in modulus on the unit sphere and between 18 and 28 in modulus on the sphere of radius 0.999.

For $n = 6$, Lemma 11.1 and the averages $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle$ of Section 16.2, for $\ell = 4, 8, 12, 16$, were verified symbolically, and seven exact finite columns of $D\mathcal{F}_6(x^\circ)$ agreed with an independently written quadrature implementation to relative accuracy $1.7 \cdot 10^{-13}$. The Fourier–Bessel solution from which the centre was obtained (Remark 11.17) has Cauchy defects $8.2 \cdot 10^{-14}$ and $2.2 \cdot 10^{-13}$ at 2001 boundary points off the collocation grid, with 20 modes, and a solution with 28 modes agrees with it. Using only its boundary, the first 14 volume integrals over Ω of $O(3) \times O(3)$ -invariant regular solutions of $\Delta v + v = 0$, which vanish when $\widehat{\mathbf{1}}_\Omega$ vanishes on the unit sphere, were computed by quadrature; relative to the corresponding integrals in which the angular integrand is replaced by its modulus, they were between 0 and $3.3 \cdot 10^{-13}$, compared with $4.7 \cdot 10^{-3}$ for a comparison domain that retains only the dominant degree-eight term of the boundary.

15.5. Dimensions eight, ten and fourteen. The computations for $n = 8, 10, 14$ are distributed in the directories `anc/r8`, `anc/r10` and `anc/r14`. They use Python, `python-flint` 0.9.0 [25, 34] and, only to read $\widehat{A}$, NumPy. Each dimension has stage scripts that evaluate the quantities of Sections 12.5–12.7 in ball arithmetic at 128 bits and write interval receipts (JSON files), a final script that combines the receipts, and a separate convexity script.

- (1) For $n = 8$ the algebra of disk polynomials is contained in `verify_r8.py`; the stage scripts are `finite_interval_r8.py` (Y , Z_f , (12.15), $\|R\|$), `tail_inverse_r8.py` (H , h_n for $M + 2m \leq n \leq N_H$, A_t), `g_tail_r8.py` (the 396 near tail g -columns in 20 batches, and the far bounds of Lemma 12.17), `shape_tail_r8.py` (the 29 near shape columns in two batches) and `far_shape_r8.py` (Γ_1 , Γ_0 , Q° , $\gamma_{k,t}^W$ and Proposition 12.18).
- (2) For $n = 10, 14$ the algebra is in `rank2_cap_core.py`, shared by the stage scripts `finite_rank2.py` (for $n = 14$: `finite_batch_rank2.py`, in 26 batches, and `aggregate_finite_r14.py`), `tail_inverse_rank2.py`, `g_tail_rank2.py` (358 columns in 18 batches, resp. 1215 columns in 13 batches), `shape_tail_rank2.py` (31 columns in 4 batches, resp. 10 columns in 8 batches), and `far_shape_rank2.py` for $n = 10$, `far_shape_split_r14.py` for $n = 14$ (the first bound of Proposition 12.18).
- (3) The final scripts `verify_r8.py` (with the option `--stage final`), `verify_r10.py` and `verify_r14.py` check a SHA-256 digest, frozen in their source, of a bundle consisting of the centre, $\widehat{A}$, all stage scripts and all receipts; check that the batches of each receipt class are contiguous and cover all columns of the class; check the support conditions

of Lemma 12.17 and Proposition 12.18; form Z and $c_2, \dots, c_{m+2}$ by (12.9) with the rational ceiling for $\|A\|$; check the rational majorants and (12.10); and check the bounds of Proposition 12.21. Only then do they write a certificate with status **PROVED** and print **PROVED**.

- (4) `verify_convex_r8.py`, `verify_convex_r10.py` and `verify_convex_r14.py` check the status of the main certificate and the checksum of the main verifier, and recompute the bounds of Table 7 (status **PROVED_CONVEX**).
- (5) `check_identities_r8.py` and `check_identities_rank2.py` verify the identities of Lemma 5.1(a),(b) in exact symbolic arithmetic for 12 index pairs, compare (12.12) with the column algebra at 256 bits for three shape indices, check positivity and total mass one of $q_s(n)$, and check in ball arithmetic that all coefficients of $Kg^\circ - (1 - |w|^2)^2 Q^\circ$ are below 10^{-69} in absolute value. These identities are proved in Section 12; the scripts are consistency checks.

As for $n = 4, 6$, the centre coefficients, B and the entries of $\hat{A}$ are converted exactly to dyadic rationals (exactness is checked), and ρ , κ_m , r and all thresholds are exact rationals. An order relation between balls is true only if it holds for all points of the balls, maxima are taken over rigorous upper endpoints, and a NaN or infinite enclosure, computed or read from a receipt, stops the verification. (In a preliminary version of the scripts for $n = 10, 14$, κ_m was a binary64 number, which for $m = 4$ lies $1.2 \cdot 10^{-18}$ below $1/48$; the distributed scripts use κ_m exactly, and the displayed bounds are unaffected.)

In each directory, `bash reproduce_r8.sh` (resp. `reproduce_r10.sh`, `reproduce_r14.sh`) copies the scripts, the centre and $\hat{A}$ into a scratch directory, recomputes every receipt there, each job in a separate single-core process under `nice -n 19`, reruns the final and convexity scripts, and compares every distributed JSON file byte for byte with its recomputed counterpart. The final steps alone are

```
python3 verify_r8.py --stage final \
  --centre center_r8_M45_S24.json --output certificate_r8.json
python3 verify_convex_r8.py
python3 verify_r10.py && python3 verify_convex_r10.py
python3 verify_r14.py && python3 verify_convex_r14.py
```

which print **PROVED** and **PROVED_CONVEX**; each takes less than four seconds on one core. In an independent rerun of all stage scripts, the 106 interval receipts of the three dimensions were reproduced byte for byte, with a total of about 930 seconds of single-core CPU time.

Consistency checks. The following checks are not part of the proof. The script `check_radial_rank2.py` verifies Proposition 12.2 for $\mathfrak{su}(3)$ in exact symbolic arithmetic in explicit matrix coordinates, and for the root data of B_2 and G_2 at 100 random regular points each. An independently written check with explicit matrix realisations of $\mathfrak{so}(5)$ and of $\mathfrak{g}_2 \subset \mathfrak{so}(7)$, as the stabiliser of a generic 3-form, confirmed (12.1) and the harmonicity of ϖ at random regular points to relative accuracy $5.1 \cdot 10^{-13}$. A pointwise evaluation of $\mathcal{H}_m(x^\circ)$ confirmed the sign conventions (relative residuals between 10^{-15} and 10^{-12} , against 2 with the opposite sign). Centred differences at 256 bits reproduced the shape column $j = 2m + 1$ of $D\mathcal{H}_m(x^\circ)$ to weighted accuracy $3.4 \cdot 10^{-12}$, $7.4 \cdot 10^{-16}$ and $2.9 \cdot 10^{-15}$ for $n = 8, 10, 14$; the tail inverse (12.14) was checked against T on four tail shape columns in each dimension (errors below 10^{-62}); the minimal output frequencies and radial indices of the first far tail g -columns were computed directly (for $n = 14$: $126 > M$ and $61 > S$); and 13 artificially corrupted receipts (NaN or infinite values, coverage gaps, bounds above the thresholds) were all rejected by the final scripts.

15.6. Dimension three. The convex $n = 3$ files are in `anc/r3_convex/`. The single-file verifier `verify_r3_convex.py` is frozen; its full SHA-256 is recorded in the ancillary README. It embeds the exact dyadic centre, the stage programs and the C ball-arithmetic accelerator. A fresh isolated replay completed all 17 stages, printed **PROVED**, and wrote the full receipt after 6515.9 seconds with five single-thread workers. A second from-scratch run of the same unmodified verifier, in a separate fresh directory, also printed **PROVED**; its receipt agrees with the first in every mathematical field. The distributed receipts record the centre and matrix hashes, the complete inverse and derivative bounds, the radii inequalities, the shape-ball geometry and the exact coverage counts.

The stages construct the 7371-dimensional finite matrix and certify its dyadic product defect with a fixed-order dot-product error bound; certify the full-centre boundary-symbol inverse; compute the full-support residual; enclose all 7280 finite field columns; compute 457 exceptional field columns with the $j \leq 121$ evaluation truncation and rigorous omitted-tail error; certify 91 radial classes, the fourteen near angular classes and $\ell \geq 210$; check 310 explicit shape columns and every $j \geq 621$; and assemble the nonlinear, geometric and coverage gates. Every strict inequality is checked with Arb balls. Missing columns, stale inputs, nonfinite enclosures or a failed gate prevent **PROVED**.

From an empty scratch directory, reproduction is requested by

```
python3 -O verify_r3_convex.py --workdir <empty dir> \
--output receipt.json --jobs 5
```

with Python 3, NumPy, SciPy, `python-flint` 0.9.0, gcc and 64-bit Linux. The run takes about 1.8 hours on five cores. The present paper edit did not rerun it; its reported numbers are taken from the frozen receipt and the single-file verifier's SHA-256 has been checked locally.

The independent nonconvex AX3 example has its own verifier, two completed certificates and a separate exact rational nonconvexity check in `anc/r3/`. Its computation uses $M = 180$, $S = 64$ and the different centre and shape scale of Proposition 13.27. Its previously certified values are Table 9. They do not enter the convex certificate.

15.7. Availability. All scripts, data and certificates are distributed as ancillary files with the arXiv version of this paper, in the directory `anc/`. The file `anc/README.md` describes the layout, the requirements, the commands and the running times for all dimensions. The four-dimensional files are `verify_r4.py`, `verify_convex.py`, `data/centre.json` and the four certificates in `certificates/`; the six-dimensional files, with a file `RELEASE_README.md` with instructions and the working notes `PROOF_ADDENDUM.md` and `FINAL.md` (the latter in Chinese), are in `r6/`; the files for $n = 8, 10, 14$, each directory with its own `README.md`, are in `r8/`, `r10/` and `r14/`; the convex three-dimensional verifier and its receipts are in `r3_convex/`, and the independent nonconvex example with its two certificates and exact nonconvexity check is in `r3/`; and `check_radial_rank2.py` is the consistency check of Proposition 12.2 described in Section 15.5. The file `anc/SHA256SUMS` lists the SHA-256 checksums of all other files under `anc/`, and each of the subdirectories `r3/`, `r3_convex/`, `r6/`, `r8/`, `r10/`, `r14/` carries its own file `SHA256SUMS`; the integrity of the files is checked by `sha256sum -c SHA256SUMS`, run in `anc/` and in each of these subdirectories. The files for $n = 4$ and $n = 6$ are also available at <https://github.com/aster2024/schiffer-pompeiu-r4> (in `verification/` and `verification/r6/`) and archived on Zenodo with the earlier version [28].

16. REMARKS ON THE MECHANISM, OTHER DIMENSIONS AND CURVED SPACES

The statements in this section are not used in the proofs of Theorems 1.1 and 1.2. Those about the numerical solutions of Section 16.3 are based on numerical computations only and are *not proved*.

16.1. Symmetry classes for which the reduction is exact. Remark 2.5 shows that, among splittings $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with symmetry group $O(d_1) \times O(d_2)$, a reduction to the planar Laplacian without potential exists only for $n \in \{2, 4, 6\}$. The classes used for $n = 4$ and $n = 6$ are the adjoint classes of $\mathfrak{u}(2)$ and $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Remark 12.3), and Section 12 carries out the reduction for all compact Lie algebras of rank two; by Remark 12.22, the dimensions 4, 6, 8, 10 and 14 are the only ones reached by exact planar reductions of adjoint type. The three-dimensional problem admits no such reduction, and Section 13 uses instead a fixed-domain formulation in the unit ball of $\mathbb{R}^3$.

16.2. A transcritical mechanism. Let $n \geq 3$ and let ρ be a positive zero of $J_{n/2}$, so that the unit ball of $\mathbb{R}^n$ carries the radial solution of (1.1) with $\mu = \rho^2$. Consider domains $\{x : |x| < 1 + sY_\ell(x_1/|x|) + O(s^2)\}$ invariant under $O(n-1) \times \mathbb{Z}_2$, where Y_ℓ is the zonal spherical harmonic of even degree ℓ on S^{n-1} , normalised by $Y_\ell(1) = 1$. For $n = 6$ we use instead the class of Section 11, of domains invariant under $O(3) \times O(3)$ and the exchange of the factors; there the invariant spherical harmonics of degree $\ell = 2k$ are the functions $Y_\ell(\omega) = U_k(t)/(k+1)$ of $t = |\omega'|^2 - |\omega''|^2$, and the exchange symmetry requires k to be even. The linearisation of the Schiffer problem at the ball in the direction Y_ℓ is degenerate exactly when $J_{\ell+n/2-1}(\rho) = 0$, and this never happens for integer n and $\ell \geq 2$: for even n this is Bourget's hypothesis, proved by Siegel [51], that Bessel functions of distinct nonnegative integer orders have no common positive zero, and for odd n it follows from a similar transcendence argument based on the Lindemann–Weierstrass theorem. So there is no bifurcation from balls in this class. However, $J_{\ell+n/2-1}(\rho)$ can be small. A formal Lyapunov–Schmidt expansion in the direction Y_ℓ then gives, at second order, a reduced equation of the form

$$-\sigma s + \frac{\rho^2}{2} \frac{\langle Y_\ell^3 \rangle}{\langle Y_\ell^2 \rangle} s^2 + O(s^3, \sigma s^2) = 0,$$

where $\langle \cdot \rangle$ is the average over S^{n-1} and σ is a small quantity proportional to $J_{\ell+n/2-1}(\rho)$. This suggests a non-ball solution with boundary amplitude $s_* \approx 2\sigma \langle Y_\ell^2 \rangle / (\rho^2 \langle Y_\ell^3 \rangle)$.

In the plane the corresponding cubic average $\langle \cos^3 N\phi \rangle$ vanishes, the reduced equation is odd in s , and a sign condition or an additional parameter is needed; compare the relaxed parameter of [10] and the continuation from the constant-flux problem in [14]. For $n \geq 3$ and even ℓ the cubic average does not vanish: for $n = 3$, $\langle P_\ell^3 \rangle / \langle P_\ell^2 \rangle = (2\ell+1) \binom{\ell}{0} \binom{\ell}{0} \binom{\ell}{0}^2 > 0$; for $n = 4$ the zonal harmonics are $U_\ell(\cos \phi)/(\ell+1)$, where $U_\ell(\cos \phi)$ is the character of the irreducible representation of $SU(2) \cong S^3$ of dimension $\ell+1$, and the Clebsch–Gordan rule gives $\langle U_\ell^3 \rangle = \langle U_\ell^2 \rangle = 1$ for even ℓ . For $n = 6$ in the $O(3) \times O(3)$ -invariant class, the invariant measure of S^5 is carried by $t = |\omega'|^2 - |\omega''|^2$ to a multiple of $\sqrt{1-t^2} dt$, for which the U_k are orthonormal after normalisation, and $U_k^2 = \sum_{i=0}^k U_{2i}$ gives $\langle U_k^3 \rangle = \langle U_k^2 \rangle = 1$ for even k , hence $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle = 1/(k+1)$. That quadratic terms make axisymmetric bifurcations in the even-degree representations of $O(3)$ generically transcritical is a classical fact of equivariant bifurcation theory [27, 33]. What we use is only the observation that this classical mechanism applies to the Schiffer problem near a near-resonance, where the planar obstruction disappears. Integrals of products of three spherical harmonics arise in any second-order perturbation analysis near a ball, and we make no claim of novelty for them.

For $(n, \rho, \ell) = (4, j_{2,4}, 6)$, the zero $j_{2,4} \approx 14.79595$ of J_2 is close to the zero $j_{7,2} \approx 14.82127$ of $J_7 = J_{\ell+n/2-1}$, and the formal prediction for the relative Y_6 -amplitude is about $2.4 \cdot 10^{-2}$. The domain of Theorem 1.2 has Y_6 -amplitude about $2.2 \cdot 10^{-2}$ (Remark 1.4); in the meridian plane the Y_6 direction corresponds to the mode $J_7(r) \sin 7\phi$ (Section 2). This is how the example was located. Numerically, Newton iterations started from the ball perturbed by a small multiple of Y_6 converge either back to the ball or to our domain, depending on the size of the perturbation, as expected from a transcritical picture.

For $(n, \rho, \ell) = (6, j_{3,8}, 8)$, the zero $j_{3,8} \approx 28.90835$ of J_3 is close to the zero $j_{10,5} \approx 28.88738$ of $J_{10} = J_{\ell+n/2-1}$, and $J_{10}(j_{3,8}) \approx -3.0 \cdot 10^{-3}$. The formal prediction for the relative amplitude of the degree-eight harmonic is about $7.3 \cdot 10^{-3}$, and the six-dimensional domain of Theorem 1.2 has amplitude about $7.2 \cdot 10^{-3}$ (Remark 1.4). In the meridian plane this direction corresponds to the mode $J_{10}(r) \sin 10\phi$ (Section 11). The centre of Section 11 was located in this way (Remark 11.17).

For $(n, \rho, \ell) = (3, j_{3/2,15}, 12)$ one has $J_{25/2}(j_{3/2,15}) \approx 2.3 \cdot 10^{-3}$, and the additional nonconvex domain of Proposition 13.27 has a dominant degree-twelve harmonic of relative amplitude about $1.5 \cdot 10^{-2}$. The convex domain of Theorem 1.1 instead lies near $(j_{3/2,34}, 18)$, as recorded in Table 1. For the adjoint classes of Section 12 the invariant harmonics and the near-resonances are described in Section 12.4, and the centres were located at the near-resonances of Table 1 (Remark 12.20). In these classes the cubic average does not vanish either: by the proof of Proposition 12.2, the average over S^{n-1} of an $\text{Ad}(G)$ -invariant function is proportional to the integral of its restriction to the unit circle of $\mathfrak{t}$ against $\sin^2(m\theta) d\theta$; the invariant harmonic of degree ℓ restricts to a multiple of $U_{\ell/m}(\cos m\theta)$, and in the variable $m\theta$ this is again the character computation of $SU(2)$, which gives $\langle U_k^3 \rangle = \langle U_k^2 \rangle = 1$ for even $k = \ell/m$, as the enlarged symmetry requires.

16.3. Further numerical solutions in dimensions three and five. With a spectral collocation solver (expansions in exact Helmholtz solutions $r^{1-n/2} J_{\ell+n/2-1}(\sqrt{\mu}r) Y_\ell$, boundary given as a radial graph, Gauss–Newton iteration), we found numerical solutions of (1.1) on non-ball, star-shaped, $O(n-1) \times \mathbb{Z}_2$ -invariant domains near the near-resonances predicted in Section 16.2. For $n = 3$, $\sqrt{\mu} = j_{3/2,15} \approx 48.6741$ (unit ball normalisation) and $\ell = 12$, with amplitude about $1.5 \cdot 10^{-2}$ and off-grid residual about $2 \cdot 10^{-11}$ with 150 modes, such a solution was the starting point for the additional AX3 example of Proposition 13.27, which certifies an exact solution nearby. The following further numerical branches have not been certified:

- $n = 3$, $\sqrt{\mu} = j_{3/2,42} \approx 133.5102$, $\ell = 20$, boundary amplitude about $9.6 \cdot 10^{-4}$. In multiprecision the collocation residual is about 10^{-36} , and the residual off the collocation grid decreases spectrally with the truncation (about 10^{-11} , 10^{-14} , 10^{-17} and 10^{-20} for 120, 160, 200 and 240 modes). A test of the Pompeiu condition that uses only the computed boundary gives $2 \cdot 10^{-12}$, compared with $1.4 \cdot 10^{-2}$ for a comparison domain.
- $n = 5$, $\sqrt{\mu} = j_{5/2,30} \approx 97.3586$, $\ell = 16$, amplitude about $7.1 \cdot 10^{-3}$, and $\sqrt{\mu} = j_{5/2,24} \approx 78.5016$, $\ell = 14$, amplitude about $2.1 \cdot 10^{-2}$; residuals about 10^{-11} , stable under changes of the truncation.

We do not claim that the two bulleted branches exist. By Remark 2.5, a reduction to the planar Laplacian of the type used in Sections 2 and 11 is not available for $n = 3$ and $n = 5$; the formulation of Section 13 applies to the first one, and an analogous formulation is available for $n = 5$ (Section 18).

16.4. Curved spaces. In spheres and hyperbolic spaces the curvature provides a genuine one-parameter family in which radial Neumann eigenvalues and non-radial Dirichlet eigenvalues of

geodesic balls can cross, which allows local bifurcation from geodesic balls. Cao-Labora and Fernández [11] used this to construct contractible Schiffer domains on the half-sphere S^2 , and they pointed out [11, Remark 1.2] that their method generalises to other non-flat spaces, with numerically observed crossings for S^3 , S^4 and H^2 . Earlier examples in spheres are bounded by isoparametric hypersurfaces [46, 50], whose boundaries are not topological spheres in general. The mechanism of Section 16.2 is also relevant in spheres and hyperbolic spaces of dimension $n \geq 3$; we do not pursue curved spaces in this paper.

17. RELATION WITH RIGIDITY RESULTS AND OTHER CLAIMS

We compare our examples with rigidity theorems for (1.1) and for the Pompeiu problem, and with some statements in the literature that they contradict.

Spectral hypotheses. Berenstein [4] for simply connected planar domains, and Berenstein and Yang [5] for bounded Lipschitz domains of $\mathbb{R}^n$ with connected boundary, proved that a domain carrying solutions of (1.1) for an infinite sequence of eigenvalues is a ball. We provide one eigenvalue for each domain; by [5], each of our domains carries solutions for at most finitely many μ . Aviles [3] proved symmetry theorems under spectral hypotheses; in particular, as stated in [18, §1] and [37, (1.4.2)], a planar convex domain carrying a solution of (1.1) with μ not exceeding the seventh Neumann eigenvalue $\mu_7(\Omega)$ is a disc, where $0 = \mu_1(\Omega) < \mu_2(\Omega) \leq \dots$ are the Neumann eigenvalues counted with multiplicity. Deng [18] proved rigidity for planar domains when μ is smaller than the eighth Neumann eigenvalue, and for strictly convex centrally symmetric planar domains when μ is smaller than the thirteenth; and Dai, Liu and Sun [15] proved it in all dimensions when μ equals the second Dirichlet eigenvalue $\lambda_2(\Omega)$. The planar results among these do not apply to our domains. The three-dimensional example has the radial scale $j_{3/2,34} \approx 108.37572$, and the certified inscribed ball of radius 106 puts $\lambda_2(\Omega) < 0.002$; hence its eigenvalue 1 is far above the second-Dirichlet threshold of [15]. The certified bounds of Sections 10, 11, 12 and 13 show that Ω contains the ball of radius r_0 about the origin, with $r_0 = 106, 14.54, 28.69, 20.0, 37.9$ and 87.1 for $n = 3, 4, 6, 8, 10, 14$ (for $n = 3, 8, 10, 14$, the lower bound for $\operatorname{Re} \psi^{*'} on the closed unit disc gives $|\psi^*(w)| \geq \operatorname{Re}(\psi^*(w)/w) \geq r_0$ for $|w| = 1$). By domain monotonicity, $\lambda_2(\Omega) \leq \dots \leq \lambda_{n+1}(\Omega) \leq (j_{n/2,1}/r_0)^2 < 0.15$ in every case; in fact, 1 exceeds more than five hundred Dirichlet eigenvalues of Ω , counted with multiplicity, and hence more than five hundred Neumann eigenvalues, since $\mu_k(\Omega) \leq \lambda_k(\Omega)$ for every k .$

Perturbations of balls. Agranovsky [2] proves rigidity for real-analytic families of planar domains issuing from the disc along which an analytic branch of Pompeiu spectral parameters persists. Canuto [9] proves, in arbitrary dimension, rigidity near a fixed radial Neumann eigenvalue within an eigenvalue-dependent class of perturbations. Kobayashi [37] considers a $C^{1,\alpha}$ family of star-shaped domains $\Omega_t = \{r\omega : 0 \leq r < g(t, \omega)\}$ with $g(0, \cdot) = 1$, and proves that Ω_{t_0} is a ball if the real zero set of the Fourier transform of $\mathbf{1}_{\Omega_{t_0}}$ contains a sphere lying in the ball of radius R about the origin and t_0 satisfies a smallness condition involving constants $\varepsilon(n, R)$ and $\delta(n, R)$ that are not computed [37, Remark 2.4]. Our examples are single domains close to balls, and there is no contradiction with these results. For [37] this can be checked directly. Its smallness condition implies $t_0 B(g) < \varepsilon(n, R) \delta(n, R)$ [37, (4.2.2b), (4.4.1b) and Lemma 4.5], where $B(g)$ [37, (4.2.1)] is at least $\omega_{n-1} \sup_{\omega} |g(t_0, \omega) - 1|/t_0$, with $\omega_{n-1} = |S^{n-1}|$, while by [37, (2.3.2), (2.3.4)] the product $\varepsilon(n, R) \delta(n, R)$ is smaller than the modulus of the Fourier transform of the indicator function of the unit ball at a point between two consecutive zeros of $J_{n/2}$ below R , which is of order $R^{-(n+1)/2}$. Hence the condition forces the relative deviation of $\partial\Omega_{t_0}$ from the unit sphere to be at most of order $R^{-(n+1)/2}$. The numerical comparison in the earlier version gave factors from about 5 for $n = 4$ to more than 10^6 for $n = 14$. We do not use such a numerical factor for the new three-dimensional branch: Kobayashi's theorem

assumes a controlled perturbation family and an additional smallness condition, neither of which is part of Theorem 1.1. Heuristically, the size of such neighbourhoods is limited by the inverse of the linearised operator at the ball, which is large near a near-resonance, and the transcritical picture of Section 16.2 places the non-ball solution at a distance comparable to the small divisors of Table 1.

Convex domains and high frequency. Brown and Kahane [7] showed that a planar convex domain whose minimal width is at most half of its diameter has the Pompeiu property. Dai, Sun, Wei and Zhang [16] prove that a uniformly convex planar domain carrying solutions of (1.1) with sufficiently large eigenvalue, above a threshold depending on the domain, is a disc, and state that the extension to higher dimensions is left for future work; Mondal [41] treats a class of centrally symmetric convex planar domains. These theorems are planar and do not apply to our examples. By Theorem 1.2, convexity alone does not force Schiffer rigidity in $\mathbb{R}^3$, $\mathbb{R}^4$, $\mathbb{R}^6$, $\mathbb{R}^8$, $\mathbb{R}^{10}$ or $\mathbb{R}^{14}$ at every eigenvalue, and a higher-dimensional analogue of the high-frequency results could only hold above a frequency threshold that excludes the eigenvalues of our examples.

Spheres and isoparametric hypersurfaces. The principal orbits of the adjoint representations of $\mathfrak{su}(3)$, $\mathfrak{so}(5)$ and $\mathfrak{g}_2$ on their unit spheres are homogeneous isoparametric hypersurfaces of S^7 , S^9 and S^{13} , with 3, 4 and 6 distinct principal curvatures [31], so domains of these spheres bounded by adjoint orbits are among the isoparametric tubes that fail the Pompeiu property [46, Corollary 7]; see also [50]. These are problems of cohomogeneity one in spheres, reducing to ordinary differential equations, and the boundaries of the domains are not topological spheres. Our domains are Euclidean, of cohomogeneity two, and bounded by topological spheres, and their construction requires the solution of a two-dimensional free boundary problem.

Claims in dimension three and in general dimension. Several published or posted statements are contradicted by Theorem 1.1 and Corollary 1.3. Ramm [47, Theorems 1 and 2] states that a bounded domain in $\mathbb{R}^3$ with C^1 boundary on which (1.1) has a solution with constant 1, or which fails the Pompeiu property, is a ball. Ramm [49, Theorem 1] states that a bounded connected domain in $\mathbb{R}^m$, $m \geq 2$, with smooth boundary, on which an overdetermined problem that includes (1.1) as a special case is solvable, is a ball; [49, Theorem A(a)] states that a bounded connected domain in $\mathbb{R}^3$ with Lipschitz boundary whose indicator function has Fourier transform vanishing on a sphere is a ball; and [49, Theorem C] states the corresponding answer to the Pompeiu problem in $\mathbb{R}^m$. Ramm [48, Theorem B] states the same Fourier transform statement for connected bounded domains in $\mathbb{R}^3$ with C^2 boundary. Chen [12, Theorem 1.1] states, in a scattering formulation of the problem with the Sommerfeld radiation condition, that a star-shaped domain in $\mathbb{R}^3$ containing the origin, with Lipschitz boundary, on which (1.1) has a nonconstant solution with $\mu \geq 1$, is a ball centred at the origin. The domain of Theorem 1.1 satisfies the hypotheses of each of these statements, with $\mu = 1$, and is not a ball; [49, Theorems 1 and C] are also contradicted by the planar counterexamples [10, 14] and by Theorem 1.2. Hence these statements are not correct as stated. We have not tried to identify the step of the respective arguments that fails. Finally, Disser [19, Theorem 11] showed that a bounded $C^{2,1}$ domain of $\mathbb{R}^3$ is a “bad domain” for the fluid-elastic semigroup if and only if it carries a nonconstant solution of (1.1), and conjectured that balls are the only bad domains [19, Conjecture 6]; by Theorem 1.1, this conjecture does not hold.

18. OPEN PROBLEMS

We list some questions left open by our results. Throughout, the domains are bounded, their boundaries are homeomorphic to a sphere, and solutions of (1.1) are nonconstant.

1. *Other dimensions.* Counterexamples are now known in dimensions 2 [10, 14, 21], 3, 4, 6, 8, 10 and 14, and convex ones in dimensions 3, 4, 6, 8, 10 and 14. The reduction of Section 1.4 is tied to the compact Lie algebras of rank two, and hence to the dimensions $2m + 2$ with $m \in \{0, 1, 2, 3, 4, 6\}$. The formulation of Section 13 does not use a planar reduction. For a domain $\Omega = \Theta(B^n) \subset \mathbb{R}^n$ invariant under $O(n - 1)$, where $\Theta(x_1, x') = (\operatorname{Re} \psi(w), \operatorname{Im} \psi(w) x' / |x'|)$ with $w = x_1 + i|x'|$, the computation that leads to (1.7) shows that (1.2) becomes

$$\operatorname{div} (q^{n-2} \nabla U) + q^{n-2} |\psi'|^2 U = 0 \quad \text{in } B^n, \quad U = 1 \quad \text{and} \quad \nabla U = 0 \quad \text{on } \partial B^n,$$

with $q = \operatorname{Im} \psi(w) / \operatorname{Im} w$, and the mechanism of Section 16.2 is available in every dimension $n \geq 3$; numerical solutions for $n = 5$ are described in Section 16. We have not pursued this.

Question 1. Does the Schiffer conjecture fail in every dimension $n \geq 3$, in particular for $n = 5$? Does it fail within the class of convex domains in every dimension $n \geq 3$?

2. *The plane.* The known planar counterexamples are not convex (Section 1). For uniformly convex planar domains, rigidity holds above a domain-dependent eigenvalue threshold [16], and convex planar domains whose minimal width is at most half of their diameter have the Pompeiu property [7]. In the plane the quadratic term of the bifurcation equation vanishes (Section 16.2), and the known constructions perturb a disc by modes of high angular frequency, which tends to destroy convexity.

Question 2. Is every bounded convex planar domain on which (1.1) has a solution a disc? Equivalently [55], does every bounded convex planar domain other than a disc have the Pompeiu property?

3. *Berenstein's conjecture.* The overdetermined Dirichlet problem

$$\Delta u + \mu u = 0 \quad \text{in } \Omega, \quad u = 0 \quad \text{and} \quad \partial_\nu u = \operatorname{const} \neq 0 \quad \text{on } \partial\Omega$$

is conjectured to be solvable only on balls. This fails in the plane [13], and in all dimensions a domain on which it is solvable for infinitely many μ is a ball [5]. For an Ad -invariant domain Ω in a compact Lie algebra of rank two and $\mu = 1$, formula (1.3) turns this problem into the planar problem

$$\Delta F + F = 0 \quad \text{in } D, \quad F = 0 \quad \text{and} \quad \partial_\nu F = c \varpi \quad \text{on } \partial D,$$

for the W -anti-invariant function $F = \varpi u|_{\mathfrak{t}}$ on the Cartan section $D = \Omega \cap \mathfrak{t}$, where $c \neq 0$ is the constant Neumann datum. The methods of this paper apply to this problem as well: with the same reductions and the same kind of computer-assisted contraction argument we have obtained convex non-ball solutions in dimensions four and fourteen, which will be reported separately.

Question 3. Does Berenstein's conjecture fail in every dimension $n \geq 3$, and in particular in dimension three? Does it fail within the class of convex domains in every dimension?

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